

New Mathematics for Marketing Measurement

A Compendium of the Deep-Dive Program

Runs 01–22, 31 August – 7 September 2026

Compiled and reviewed 7 September 2026

`from marketing_measurement/frontier_study/deep_dives/`

This document stitches together twenty-two automated research runs into fifteen self-contained chapters. Each chapter reproduces the background ideas its dive builds on, states the dive’s constructions and results with the dive’s own epistemic labels, tabulates the headline numerical evidence, and closes with a scientific-paper style discussion of impacts and further exploration. Compiler’s reviewer notes (orange boxes) flag what a referee would want checked before believing a claim. Appendix A is the ledger of all 88 open problems the program spawned and their status at compilation.

About this compendium

What was compiled

Between 31 August and 7 September 2026 a scheduled task executed twenty-two “deep dives” into the open problems catalogued by the frontier study of marketing measurement (01_state_of_the_field.md, 02_open_questions.md, 03_mmm_adoption_barriers.md). Each run took one item from a queue, worked it under a fixed depth protocol (formalisation with explicit assumptions, a literature search, at least three construct–attack–refine rounds, at least eight numerical experiments including robustness sweeps and a negative or power control, a clean-room re-derivation of at least one core result by an independent subagent, uncertainty on every headline number, and a devil’s-advocate paragraph against every headline claim), and wrote a markdown report with accompanying code. Every fourth run was an *extension* that pushed the most promising prior dive further. The program’s index file (deep_dives/00_INDEX.md) holds the queue, the backlog of open problems each dive spawned (“BL” items), and a one-line log of each run’s finding.

This compendium was assembled on 7 September 2026 from the twenty-two reports and their code as they stood on disk at that date (the provenance index below records each file’s last-written timestamp). Its purpose is to let a reader review, on paper, what ideas were actually developed—without having to open the dives, the papers they cite, or the index.

How the chapters are organised

There is one chapter per dive, except that a dive and its extension are merged into a single chapter (dives 03+04, 08+09, 13+14, 16+19), and the four dives that form one continuous thread on identification (01, then the Phase D chain 20, 21, 22) are merged into Chapter 1. Chapters otherwise follow the program’s run order. Every chapter has the same skeleton:

Provenance which dives, files, code and frontier-study problem the chapter covers, and which backlog items it opened or closed.

Key ideas the chapter’s central results in a few sentences, with their headline numbers.

Problem and motivation what the dive set out to answer and what the field believed before.

Background the ideas the dive builds on, reproduced in summary so that the chapter is self-contained: for every paper or established result the dive synthesises, its central construction or equation, what it assumes, and what it delivers.

Setup and assumptions the formal model and notation.

Main results the dive’s constructions, theorems and laws, with key derivation steps and the dive’s own epistemic labels (see the legend).

Numerical evidence the headline experiments as tables, with the uncertainty the dive reports, including negative and power controls.

Limitations and the devil’s advocate the dive’s own strongest counterarguments and what the attack-and-refine rounds changed.

Discussion: impacts what changes for practice and for theory.

Further exploration the dive’s next steps, the backlog items it opened, and the compiler’s view of the most promising extension.

Reviewer’s notes the compiler’s own flags, as a referee would write them.

Sources the papers summarised or relied on, with links where the program’s bibliographies had them.

Legend

Results carry the label the dive itself gave them. [\[established in the dive\]](#) marks a result derived in the dive and, where the dive says so, reproduced by its clean-room verifier. [\[transported from the literature\]](#) marks a known result carried into the MMM setting from another literature. [\[numerically established\]](#) marks a result established by simulation rather than proof. [\[conjecture / speculative\]](#) marks speculation the dive labelled as such. [\[refuted within the dive\]](#) marks a claim the dive itself made and then overturned in a later round—these are kept because the refutations are among the more instructive findings. Dive NN refers to run NN; BL NN to backlog item NN in Appendix A. Orange boxes are the compiler’s reviewer notes and are the only content in the chapters that does not come from the dives.

Two program-wide caveats apply to every number in this book. First, all results are from simulation on synthetic data under the model each dive states; none has been run on an advertiser’s data. Second, Dive 21’s verifier found that pseudo-inverses at default tolerance had silently manufactured finite standard errors from singular Fisher matrices; Dive 22 audited dives 20–22 (BL 82) but the FIM-based standard errors in earlier dives (notably 02, 09, 16, 19) have not had that audit. The reviewer notes say where this matters.

Provenance index

Table 1 lists every source dive, its file, the run date recorded in the program log, the file’s last-written timestamp on disk at compilation, its length, and the chapter that covers it. Each dive’s code lives beside it in `deep_dives/code/` under the same stem.

Table 1: Source dives and their dates (UTC). “Run” is the start time recorded in the program log; “Written” is the report file’s last modification time on disk on 2026-09-07. Word counts are of the markdown reports.

Dive	File (<code>deep_dives/</code>)	Run (log)	Written	Words	Ch.
01	01_mmm_identified_set.md	2026-08-31 19:00	2026-08-31 22:08	2,003	1
02	02_experiment_transport_map.md	2026-08-31 21:15	2026-08-31 23:14	2,225	2
03	03_decision_voi.md	2026-09-01 00:21	2026-09-01 00:21	1,582	3
04	04_decision_voi_ext.md (ext. of 03)	2026-09-01 02:30	2026-09-01 01:25	1,982	3
05	05_fusion_loop.md	2026-09-01 05:30	2026-09-01 03:39	2,184	4
06	06_aggregation_privacy.md	2026-09-01 07:45	2026-09-01 07:36	2,582	5
07	07_refresh_stability.md	2026-09-01 12:30	2026-09-01 11:56	2,507	6
08	08_decision_deadband.md	2026-09-01 15:30	2026-09-01 16:55	2,936	7
09	09_decision_deadband_ext.md (ext. of 08)	2026-09-01 20:05	2026-09-01 20:01	3,933	7
10	10_algorithmic_delivery_estimands.md	2026-09-01 23:30	2026-09-01 23:43	3,962	8
11	11_data_archaeology_compiler.md	2026-09-01 21:20 ^a	2026-09-02 03:33	4,024	9
12	12_surrogacy_horizon_bounds.md	2026-09-02 04:50	2026-09-02 11:34	3,640	10
13	13_attribution_impossibility.md	2026-09-02 15:40	2026-09-02 15:40	4,535	11
14	14_attribution_impossibility_ext.md (ext. of 13)	2026-09-02 19:45	2026-09-02 19:44	4,536	11
15	15_calibrated_simulator.md	2026-09-02 23:30	2026-09-02 23:32	4,258	12
16	16_feedback_loop_inference.md	2026-09-03 03:30	2026-09-03 03:44	5,149	13
17	17_human_mechanism_panel.md	2026-09-03 10:30	2026-09-03 14:24	5,021	14
18	18_heavy_tail_efficiency.md	2026-09-03 16:10	2026-09-03 16:01	5,642	15
19	19_feedback_loop_inference_ext.md (ext. of 16)	2026-09-03 14:45 ^a	2026-09-03 21:41	5,489	13
20	20_price_of_identification.md	2026-09-04 09:55	2026-09-04 17:20	5,236	1
21	21_spectral_comb.md	2026-09-06 11:20	2026-09-06 19:17	6,293	1
22	22_price_of_the_prior.md	2026-09-07 16:45	2026-09-07 16:46	6,484	1

^a The log time is earlier than the file’s own timestamps and than the preceding run’s; it is taken as a logging error.

The program’s index file was last written at 2026-09-07 16:47 UTC and lists run 23 (an extension) as the next item in the queue; nothing from run 23 or later is in this compendium.

Contents

About this compendium	i
Introduction: the program and its arc	1
1 The identified set and the price of identification	4
1.1 Problem and motivation	5
1.2 Background: the ideas this chapter builds on	5
1.2.1 Fisher information, the Cramér–Rao bound, local identification and optimality criteria	5
1.2.2 The adstock/Hill MMM and its weak identification (Jin et al., 2017; Chan and Perry, 2017)	6
1.2.3 Emax dose-ranging design theory	6
1.2.4 Frequency-domain system identification	6
1.2.5 Nonlinear distortion and Volterra readouts	7
1.2.6 The cost of learning (Little, 1966; Feit and Berman, 2019)	7
1.2.7 The van Trees inequality and the effective sample size of a prior	7
1.2.8 Priors in MMM practice, and the ridge-with-a-tight-curve observation	7
1.3 Setup and assumptions	8
1.4 Main results	8
1.4.1 A. The identified set and the design recipe (Dive 01)	8
1.4.2 B. The frequency-domain price of identification (Dive 20)	10
1.4.3 C. The n -channel spectral comb (Dive 21)	12
1.4.4 D. The price of the prior (Dive 22)	14
1.5 Numerical evidence	16
1.6 Limitations and the devil’s advocate	19
1.7 Discussion: impacts	19
1.8 Further exploration	20
2 Transporting lift tests into MMM parameters	24
2.1 Problem and motivation	24
2.2 Background: the ideas this chapter builds on	25
2.2.1 Selection diagrams and transportability (Pearl and Bareinboim, 2011, 2014, 2016)	25
2.2.2 The geometric-adstock, Hill-saturation MMM (Jin et al., 2017)	25
2.2.3 What “calibrating to a lift test” means in practice	25
2.2.4 Indirect inference (Gouriéroux, Monfort and Renault, 1993)	26
2.2.5 The adstock transient as a dose-ranging sweep (Chapter 1; Heusch, 2026)	26
2.3 Setup and assumptions	26
2.4 Main results	27
2.4.1 No nonparametric transport: all calibration is parametric	27

2.4.2	The transport identity	27
2.4.3	The operator fix: calibrate on the design, not on a renamed scalar	28
2.4.4	The scalar summary discards identifying power; joint posteriors are free	28
2.5	Numerical evidence	29
2.6	Limitations and the devil’s advocate	30
2.7	Discussion: impacts	30
2.8	Further exploration	30
3	Decision value of information	33
3.1	Problem and motivation	33
3.2	Background: the ideas this chapter builds on	34
3.2.1	Bayesian decision theory and preposterior analysis (Raiffa and Schlaifer, 1961)	34
3.2.2	Test & Roll (Feit and Berman, 2019)	35
3.2.3	Certainty equivalence and the linear-quadratic preposterior identity (Simon, 1956; Theil, 1957)	35
3.2.4	Design criteria: D-, c- and goal-oriented optimality (Attia, Alexanderian and Saibaba, 2018)	35
3.2.5	The MMM model and the experiment operator (dives 01–02)	35
3.3	Setup and assumptions	36
3.4	Main results	36
3.4.1	The quadratic regret identity and the EVPI/EVSI trace formulas (Dive 03)	36
3.4.2	An exact decision-null direction (Dive 03)	37
3.4.3	Information, not optimization (Dive 03)	37
3.4.4	The power plateau and the refuted D-optimality forfeit (Dive 03)	37
3.4.5	The LQC estimator and the re-diagnosis (Dive 04)	38
3.4.6	Anatomy of the plateau: alignment bound and share ceiling (Dive 04)	39
3.4.7	Where the plateau ends (Dive 04)	39
3.5	Numerical evidence	40
3.6	Limitations and the devil’s advocate	40
3.7	Discussion: impacts	41
3.8	Further exploration	42
4	Always-on experiment-to-model fusion: the closed loop	44
4.1	Problem and motivation	44
4.2	Background: the ideas this chapter builds on	45
4.2.1	The extended Kalman filter as a sequential Bayesian updater (Kalman, 1960; Jazwinski, 1970; Ljung, 1979)	45
4.2.2	Random-walk parameter drift (West and Harrison, 1997; Ng et al., 2021)	45
4.2.3	Square-root cadence laws (Harris, 1913; Feit and Berman, 2019)	45
4.2.4	Kalman filtering with intermittent observations (Sinopoli et al., 2004)	46
4.2.5	The ingestion interface and the decision-risk functional (Chapter 2, Chapter 3)	46
4.2.6	Sequential design and greedy portfolios (Foster et al., 2021; Rainforth et al., 2024; Nemhauser et al., 1978)	46

4.3	Setup and assumptions	46
4.4	Main results	47
4.4.1	The renewal cadence law	47
4.4.2	Portfolio ceiling and greedy scheduling	47
4.4.3	Level drains, contrast stalls	48
4.4.4	A good selector and a bad forecaster	48
4.4.5	The trap economy	48
4.4.6	The low-drift trap paradox	49
4.4.7	The E6 architecture	49
4.5	Numerical evidence	49
4.6	Limitations and the devil’s advocate	50
4.7	Discussion: impacts	51
4.8	Further exploration	51
5	Geo aggregation and privacy noise in MMM	54
5.1	Problem and motivation	54
5.2	Background: the ideas this chapter builds on	55
5.2.1	Aggregating nonlinear micro relations (van Garderen, Lee and Pesaran, 2000; Stoker, 1984; Blundell and Stoker, 2005)	55
5.2.2	Differential privacy for aggregated measurement (Delaney et al., 2024; Aksu et al., 2023)	55
5.2.3	Privacy noise as measurement error (Evans and King, 2023, 2025; Gong, 2022; Karwa and Vadhan, 2018)	55
5.3	Setup and assumptions	56
5.4	Main results	56
5.4.1	Closed-form Hill drift laws under share dispersion	56
5.4.2	Stable-share aggregation bias is decision-benign at the operating point	57
5.4.3	Share drift, and the share-calibrated mixture estimator	57
5.4.4	Pricing ε : width law, bounded prize of noise-awareness, granularity frontier	58
5.5	Numerical evidence	58
5.6	Limitations and the devil’s advocate	60
5.7	Discussion: impacts	60
5.8	Further exploration	61
6	Refresh stability and analyst degrees of freedom	63
6.1	Problem and motivation	63
6.2	Background: the ideas this chapter builds on	64
6.2.1	Overlapping windows and the bivariate-normal orthant (Sheppard, 1899; Harri and Brorsen, 2009)	64
6.2.2	Type-S errors, the vibration of effects and the garden of forking paths	64
6.2.3	Power priors, forgetting and why posterior-as-prior double counts	65
6.2.4	MAP, the Laplace approximation and the information ridge (Jin et al., 2017; Chapter 1)	65
6.3	Setup and assumptions	65
6.4	Main results	66

6.4.1	The flip-rate law and the cadence law	66
6.4.2	Naive posterior-as-prior is quantified overconfidence	67
6.4.3	The corrected scheme: increment likelihood with discount $a = \omega$	67
6.4.4	What actually causes MMM refresh flips: rotation, not churn	67
6.4.5	Analyst degrees of freedom: the Janus condition	68
6.4.6	The reporting layer and the protocol	68
6.5	Numerical evidence	68
6.6	Limitations and the devil’s advocate	69
6.7	Discussion: impacts	70
6.8	Further exploration	70
7	The reallocation deadband: when to act on a refreshed model	73
7.1	Problem and motivation	73
7.2	Background: the ideas this chapter builds on	74
7.2.1	Fixed-cost impulse control and the quartic band (Barro, 1972; Dixit, 1991; Stokey, 2009)	74
7.2.2	Multi-product menu costs: balls, ellipsoids and two-threshold policies	74
7.2.3	The Kalman filter on a random walk and Muth’s invariance (Muth, 1960)	75
7.2.4	Discrete monitoring: Broadie–Glasserman–Kou and Gobet–Menozzi	75
7.2.5	Expected-loss rules and decision-framed communication	75
7.3	Setup and assumptions	75
7.4	Main results	76
7.4.1	Estimation noise never enters the band	76
7.4.2	The composite filtered deadband law (scalar)	76
7.4.3	The ellipsoid law (vector)	77
7.4.4	Decision-framed outputs	77
7.4.5	Tiered reallocation under per-channel change costs	77
7.4.6	Cadence separates from the band	78
7.5	Numerical evidence	78
7.6	Limitations and the devil’s advocate	80
7.7	Discussion: impacts	80
7.8	Further exploration	81
8	Creative estimands under algorithmic delivery	84
8.1	Problem and motivation	84
8.2	Background: the ideas this chapter builds on	85
8.2.1	Divergent delivery (Eckles, Gordon and Johnson, 2018; Ali et al., 2019; Braun and Schwartz, 2025; Burtch et al., 2025)	85
8.2.2	Off-policy evaluation with estimated propensities (Hirano, Imbens and Ridder, 2003; Hanna, Niekum and Stone, 2019)	85
8.2.3	Sensitivity bounds: Rosenbaum’s Γ , the marginal Λ model and sharp bounds (Rosenbaum, 2002; Tan, 2006; Kallus and Zhou, 2018, 2020; Namkoong et al., 2020; Dorn and Guo, 2022)	85
8.2.4	Balancing scores and overlap weights (Rosenbaum and Rubin, 1983; Li, Morgan and Zaslavsky, 2018)	86

8.2.5	Simulation–extrapolation (Cook and Stefanski, 1994)	86
8.2.6	Counterfactual logging and the three-arm design (Johnson, Lewis and Nubbemeyer, 2017; Pal and Susarla, 2026)	86
8.3	Setup and assumptions	86
8.4	Main results	87
8.4.1	The divergence-bias identity	87
8.4.2	The cell estimator and the sufficiency theorem	87
8.4.3	Variants, the overlap policy and the logging hierarchy	88
8.5	Numerical evidence	89
8.6	Limitations and the devil’s advocate	90
8.7	Discussion: impacts	90
8.8	Further exploration	91
9	The data-archaeology compiler	94
9.1	Problem and motivation	94
9.2	Background: the ideas this chapter builds on	95
9.2.1	Geo-hierarchical MMM and population-share imputation (Sun et al., 2017; Meridian)	95
9.2.2	Classical and Berkson error; weight shifting (Berkson, 1950; Carroll et al., 2006; Abel, 2017)	95
9.2.3	Hausman specification tests (Hausman, 1978)	95
9.2.4	Prediction-powered inference, matrix calibration, monetary-unit sampling	96
9.3	Setup and assumptions	96
9.4	Main results	96
9.4.1	The imputation leakage law	96
9.4.2	The latent-loading repair, shrinkage, and the leak test	97
9.4.3	Hill response: sign flip across the knee, and what is not identified	98
9.4.4	Taxonomy misclassification: the certify-the-mass law	98
9.4.5	The compiler and the ledger	99
9.5	Numerical evidence	99
9.6	Limitations and the devil’s advocate	100
9.7	Discussion: impacts	100
9.8	Further exploration	101
10	Horizon bounds for long-run effects under weak surrogacy	103
10.1	Problem and motivation	103
10.2	Background: the ideas this chapter builds on	104
10.2.1	The surrogate index (Athey, Chetty, Imbens and Kang, 2019)	104
10.2.2	Sensitivity analysis indexed by a partial R^2 (Cinelli and Hazlett, 2020)	104
10.2.3	Completely monotone kernels, extrapolation, and shape-restricted bounds	104
10.2.4	Chi-bar-square laws at a cone boundary (Chernoff, 1954; Self and Liang, 1987)	105
10.2.5	Known-shape, unknown-scalar de-confounding	105
10.3	Setup and assumptions	105
10.4	Main results	106

10.4.1	The surrogate-index bias identity	106
10.4.2	The completely monotone profile-LR bound indexed by $hl_{\max}$	106
10.4.3	One experiment de-confounds the whole observational lag structure	107
10.5	Numerical evidence	108
10.6	Limitations and the devil’s advocate	109
10.7	Discussion: impacts	109
10.8	Further exploration	110
11	Causal attribution: the impossibility and its escape	113
11.1	Problem and motivation	113
11.2	Background: the ideas this chapter builds on	114
11.2.1	The Shapley value and Harsanyi dividends (Shapley, 1953; Harsanyi, 1963; Grabisch, 1997)	114
11.2.2	Data-driven attribution: Shapley on observational rates (Shao and Li, 2011, and successors)	114
11.2.3	The causal critiques: Berman (2018) and Singal et al. (2022)	115
11.2.4	Multi-channel potential outcomes and the evidence against observational lift	115
11.2.5	Sensitivity analysis: Rosenbaum’s Γ , Tan’s marginal model, robustness values	115
11.2.6	Ghost ads, multi-cell holdouts and optimal design (Johnson, Lewis and Nubbemeyer, 2017; Zhou et al., 2025)	115
11.3	Setup and assumptions	116
11.4	Main results	116
11.4.1	Decomposition identities and the journey-level lemma	116
11.4.2	The impossibility theorem	117
11.4.3	Sensitivity bounds and attribution robustness values	117
11.4.4	The targeting-intensity bias law	118
11.4.5	The design theorem	118
11.4.6	The closed-form estimator and the aliasing law	119
11.4.7	The one over-identifying restriction: add-up test and BLUE	120
11.4.8	The precision–lift frontier and the κ law	120
11.5	Numerical evidence	120
11.6	Limitations and the devil’s advocate	122
11.7	Discussion: impacts	122
11.8	Further exploration	123
12	The calibrated simulator	126
12.1	Problem and motivation	126
12.2	Background: the ideas this chapter builds on	127
12.2.1	Generative agents and synthetic consumers (Park et al., 2023, 2024; Horton, 2023; Brand et al., 2023)	127
12.2.2	Validating simulators against experiments (Hut and Masoero, 2026; Hewitt et al., 2026; Bisbee et al., 2024)	127
12.2.3	Excess variance and prediction intervals (DerSimonian and Laird, 1986; Riley, Higgins and Deeks, 2011)	127

12.2.4	Variance-ratio tests and their power	128
12.2.5	Platt scaling versus latent-scale recalibration (Platt, 1999)	128
12.2.6	Recursive residuals and e-processes (Brown, Durbin and Evans, 1975; Ramdas et al., 2023)	128
12.2.7	Expected value of sample information as the bar (Chapter 3)	128
12.3	Setup and assumptions	128
12.4	Main results	129
12.4.1	The excess-variance certificate and the certification law	129
12.4.2	The counterfactual bar	129
12.4.3	Contamination: the noise-floor test and its blind spot	130
12.4.4	Which recalibration family transports	130
12.4.5	Shift, selection and sequential certification	131
12.5	Numerical evidence	131
12.6	Limitations and the devil’s advocate	132
12.7	Discussion: impacts	132
12.8	Further exploration	133
13	Inference inside the feedback loop	135
13.1	Problem and motivation	136
13.2	Background: the ideas this chapter builds on	136
13.2.1	Confidence sequences and test martingales; why fixed- n intervals fail	136
13.2.2	Least squares under adaptive designs (Lai–Wei, 1982; Abbasi-Yadkori et al., 2011; Hadad et al., 2021; Zhang–Janson–Murphy, 2020)	137
13.2.3	Incomplete learning under certainty equivalence (Keskin and Zeevi, 2014, 2018)	137
13.2.4	Feedback detection and curvature-shaped exploration	137
13.3	Setup and assumptions	138
13.4	Main results	138
13.4.1	A policy-agnostic anytime-valid confidence sequence	138
13.4.2	Smooth feedback is second-order	139
13.4.3	The learning stall and the policy echo	140
13.4.4	Detecting the stall	141
13.4.5	Confounded pseudo-learning: anytime validity is not identification	141
13.4.6	Multichannel geometry, the whitened exploration law and its switch representation	142
13.5	Numerical evidence	143
13.6	Limitations and the devil’s advocate	144
13.7	Discussion: impacts	144
13.8	Further exploration	145
14	The human-mechanism panel: attention, memory, purchase	148
14.1	Problem and motivation	148
14.2	Background: the ideas this chapter builds on	149
14.2.1	Mental availability and category entry points (Sharp, 2010; Romaniuk and Sharp, 2016; Romaniuk, 2023)	149

14.2.2	Attention as a dose (Nelson-Field, 2020; ARF, 2024)	149
14.2.3	Single-source panels (Lodish et al., 1995; Jones, 1995)	149
14.2.4	The unfavourable economics of ad measurement (Lewis and Rao, 2015)	149
14.2.5	Heterogeneity versus state dependence (Heckman, 1981; Kenny and Zautra, 1995)	150
14.2.6	Causal mediation and encouragement designs (Imai, Keele and Yamamoto, 2010; Imai, Tingley and Yamamoto, 2013)	150
14.2.7	The mere-measurement effect (Morwitz, Johnson and Schmittlein, 1993; Chandon, Morwitz and Reinartz, 2005)	150
14.2.8	Adstock, half-life, pseudo-true values, exponential design (Broadbent, 1979; Jin et al., 2017; Box and Lucas, 1959)	150
14.3	Setup and assumptions	151
14.4	Main results	151
14.4.1	The identification ladder	151
14.4.2	Tracking-only decay: heterogeneity masquerading as persistence	152
14.4.3	Sales-fitted adstock is not the memory half-life	152
14.4.4	Design laws	152
14.4.5	Detecting a direct path by timing, not scale	153
14.4.6	The within-person regression is worthless	153
14.5	Numerical evidence	153
14.6	Limitations and the devil's advocate	155
14.7	Discussion: impacts	155
14.8	Further exploration	156
15	Lift under heavy tails: the price of a margin restriction	159
15.1	Problem and motivation	159
15.2	Background: the ideas this chapter builds on	160
15.2.1	The efficiency bound for the ATE (Hahn, 1998; Hirano, Imbens and Ridder, 2003)	160
15.2.2	The unfavourable economics of lift (Lewis and Rao, 2015)	160
15.2.3	Regression adjustment and its heavy-tailed descendants (Deng et al., 2013; Wang and Burke, 2019; Wicker, 2026)	160
15.2.4	Trimming, two-part models and robust means	160
15.2.5	Trimmed Match (Chen and Au, 2022)	161
15.2.6	Latent stratification (Berman and Feit, 2024; Johnson, Lewis and Reiley, 2017)	161
15.2.7	Adaptive estimation of a shift (Stein, 1956; Bickel, 1982; Hodges and Lehmann, 1963)	161
15.2.8	MSE against rollout regret (Parikh et al., 2026)	161
15.3	Setup and assumptions	161
15.4	Main results	162
15.4.1	The margin decomposition of the bound	162
15.4.2	What the restriction costs when it is false	163
15.4.3	The winsorisation calculus	163
15.4.4	The rank-identified basket ratio (Round 3)	163
15.4.5	The blind spot and its guard	164
15.5	Numerical evidence	164

15.6	Limitations and the devil's advocate	166
15.7	Discussion: impacts	167
15.8	Further exploration	167
A	The open-problem ledger	170
B	The program's protocol	173
C	File inventory	174

Introduction: the program and its arc

Where the dives started

The frontier study that the dives answer summarises the state of marketing measurement in 2026 in one organising sentence: *experiments are the ground truth, and everything else is calibrated to them*. The decade's load-bearing negative results—Lewis and Rao's demonstration that purchase variance swamps plausible advertising effects, the eBay null of Blake, Nosko and Tadelis, and Gordon et al.'s finding across hundreds of platform RCTs that even double machine learning on thousands of covariates fails to recover experimental lift—ended the hope that observational statistics could substitute for randomisation. What replaced it is a triangulation architecture: lift RCTs and geo experiments as causal anchors, Bayesian media mix models (MMM) as the cross-channel always-on connective tissue with experiments injected as priors or likelihood terms, and attribution demoted to a fast tactical signal understood to be non-causal. Three forces shaped that settlement: privacy engineering destroyed user-level joins and pushed measurement back to aggregates; open-source tooling (Meridian, Robyn, PyMC-Marketing, GeoLift, Trimmed Match) put the frontier in every analyst's hands; and the delivery algorithms became part of the measurement problem, contaminating the experiments meant to measure them and closing feedback loops between measurement and spend.

The study's second document catalogued the open problems at the *seams* of that architecture: ten engineering problems (E1–E10), ten human-understanding problems (H1–H10), eleven mathematical problems (M1–M11), and three cross-cutting grand challenges—the unified causal measurement system (M2 + M10 + E6), the human-mechanism panel (H1 + H3 + H5), and the calibrated simulator (E9 + H7). A third document explained why most companies have not adopted sophisticated MMM: business-logic barriers (unit economics, incentives, “precision theatre”), mathematical barriers (weak identification, the impossibility of validation, prior elicitation) and operational ones (data archaeology, geo data, talent, experiment operations).

The deep-dive program took those catalogues as a queue. Phase A worked the seams where experiments meet models (M1, M2, M10, E6, M9); Phase B worked stability, trust and adoption mechanics (M6/M4, the decision interface, M3, the operational data problem, M5); Phase C attempted the grand challenges and the frontier's edges (M8, E9/H7, M7, H1, efficiency under heavy tails); Phase D began working the program's own backlog, returning to identification with the tools built in between. Every fourth run extended the most promising prior dive instead of opening a new one.

A map of the chapters

Table 2 lists the chapters with the problem each addresses and its principal dependencies. The chapters are self-contained, but they were not written in isolation: the dives borrowed each other's constructions, and several later dives exist only because an earlier one left a specific question in the backlog. A reader who wants the shortest path through the book's central thread should read Chapter 1 (what a spend path can identify and what it costs), Chapter 2 (how an experiment enters a model), Chapter 3 (what the information is worth), Chapter 7 (when to act on it) and Chapter 13 (what the acting does to the measuring). The remaining chapters branch off that spine.

Recurring ideas across the dives

Reading the twenty-two reports together, a small number of ideas recur often enough to count as the program's findings rather than any single dive's. They are stated here so that the reader can watch for them.

Table 2: Chapter map: the frontier-study problem each chapter addresses, its source dives, and the chapters it most depends on or feeds.

Ch.	Topic	Problem	Dives	Builds on / feeds
1	Identified set of MMM parameters; the price of identification; the spectral comb; the price of the prior	M1, BL1–2, 70, 73, 75	01, 20, 21, 22	feeds 2, 3, 7, 13; uses 7, 9
2	Transport map from lift tests to MMM parameters	M2	02	uses 1; feeds 3, 4, 14
3	Decision-calibrated uncertainty and EVSI	M10	03, 04	uses 2; feeds 4, 7, 11, 12
4	Always-on experiment-to-model fusion	E6	05	uses 2, 3; feeds 7, 13
5	Geo aggregation and privacy noise	M9	06	uses 1, 2
6	Refresh stability and analyst degrees of freedom	M6/M4 (barriers)	07	uses 1; feeds 7, 9
7	The reallocation deadband	M9(comm.), B5, H8	08, 09	uses 3, 4, 6; feeds 9, 10, 12, 13, 15
8	Creative estimands under algorithmic delivery	M3	10	feeds 11
9	The data-archaeology compiler	O1/O3	11	uses 5, 6; feeds 1, 13
10	Long-run effects under weak surrogacy	M5	12	uses 2, 3; feeds 7, 14
11	Impossibility of causal attribution; escape designs	M8	13, 14	uses 8, 3; feeds 4, 7
12	The calibrated simulator	E9/H7	15	uses 3, 7; feeds 15
13	Inference inside the feedback loop	M7	16, 19	uses 1, 3, 4, 7, 9; feeds 1
14	The human-mechanism panel	H1	17	uses 1, 2, 10
15	Lift under heavy tails	M6 (efficiency)	18	uses 7, 12

The decision-optimal policy destroys the information it needs. Chapter 1 shows that the profit-maximising media plan is, to first order, exactly non-identifying, and that budget-neutral variation cannot identify the total marginal value of media at all under a common adstock. Chapter 13 shows that a certainty-equivalent allocation loop without exogenous exploration stalls at an uninformative spend where the model's own marginal ROAS echoes the breakeven target exactly. Chapter 4 finds that belief-driven experiment selection leaves the very channel that is about to trap unprobed. The same mechanism appears in each: what is best to do given current beliefs is what teaches least about them. The escapes are also the same across chapters—exogenous probes with energy off the seasonal harmonics, switch designs, spend floors and coverage constraints, $t^{-1/4}$ exploration—and Chapter 1 prices them in dollars.

Plateaus: design choice is second-order, the statistic is first-order. Chapter 3 finds an EVSI power plateau—among well-powered scalar-readout experiments, which channel, pulse or window to test is decision-irrelevant—and diagnoses it as a ceiling structure. Chapter 1 finds the same plateau in the frequency domain (the information functional is flat within 10% across probe periods from 13 to 182 weeks). Chapter 7 finds that the *shape* of the multichannel inaction region is second-order while using the wrong *statistic* (raw estimates, significance rules, posterior width) costs tens of percent to multiples. The practical reading is that effort spent optimising a design within its admissible class buys little, while effort spent on the estimand, the projection, the filter and the count of experiments buys a lot.

Precisely wrong. Several dives find a sophisticated procedure that is more confident than a naive one and more biased: iROAS used as a prior on marginal ROAS (Chapter 2, IQR seven times tighter, 43% off); naive posterior-as-prior refreshing (Chapter 6, coverage 0.24); pseudo-inverses on singular Fisher matrices (Chapter 1, BL 82); an outcome-selected experiment library that makes a simulator look better than perfect (Chapter 12); a model inside a feedback loop whose marginal ROAS reports the target it was steered to (Chapter 13); a difference-in-means lift estimate that at 200,000 users is worse than not experimenting (Chapter 15). The adoption study called this precision theatre; the dives supply the mechanisms.

Exact identities as the working method. The program’s characteristic move is to find an exact decomposition whose terms name the mechanisms: $iROAS = m_{\infty} \rho C Q$ (Chapter 2); the divergence-bias identity for creative tests (Chapter 8); the leakage law for imputed channels (Chapter 9); the surrogate-index error as bypass tail minus phantom persistence (Chapter 10); Muth’s invariance $\text{Var}(\Delta \hat{z}) = q$ that collapses a partially observed control problem to a fully observed one (Chapter 7); exact comb orthogonality through heterogeneous adstock and the ledger identities through one vector d and one matrix $\tilde{I}$ (Chapter 1); the design theorem for attribution (Chapter 11). Almost all were reproduced by the clean-room verifier to machine precision, which is where the program’s confidence in them comes from.

Every assumption has a price, and the price can be computed. Chapter 15 prices a margin restriction in variance ratios; Chapter 1 prices a shape prior in excitation dollars and an unknown saturation curve as the harmonic-collision penalty; Chapter 11 prices an interaction cap in aliasing and a holdout-only design in n^4 precision loss; Chapter 10 prices a maximum half-life in the width of the identified set; Chapter 5 prices a privacy budget in posterior width. The unifying object is a cost–precision frontier on which a decision-maker chooses, and several chapters conclude that the decision layer of Chapter 7 is where those prices should be paid.

The program corrected itself. Each dive’s later rounds refuted some of its own earlier claims, and the compendium keeps them under the [refuted within the dive] label: the predicted 42% D-optimality forfeit (Chapter 3), the cadence law as a forecaster (Chapter 4), the closed-form confounding factor and the “resolution” account of block length, the closed-form quadrature angle, the “700–2400× worse” fighting claim that was in fact exact singularity, and the admissibility constant (Chapter 1). The reviewer notes add what the dives did not catch.

How to read the numbers

All figures are from simulations under the model stated in each chapter’s setup, with the uncertainty the dive reported; where a dive reported none, the reviewer note says so. Percentages of “media spend” or “revenue” are in the dives’ synthetic units and should be read as orders of magnitude. Where two dives disagree, both numbers are given and the disagreement is flagged. Where a result depends on a Fisher-matrix inversion that has not had the BL 82 audit, the reviewer note names it. The compendium adds no results of its own.

Chapter 1

The identified set of MMM parameters and the price of identification

Provenance

Source dives: Dive 01 *The identified set of MMM parameters (M1)* (run 2026-08-31, file 01_mmm_identified_set.md, written 2026-08-31 22:08 UTC); Dive 20 *The price of identification: the spectrum and the cost of MMM experimental design* (run 2026-09-04, file 20_price_of_identification.md, written 2026-09-04 17:20 UTC); Dive 21 *The n-channel spectral comb* (run 2026-09-06, file 21_spectral_comb.md, written 2026-09-06 19:17 UTC); Dive 22 *Price the prior: what an assumption is worth in excitation dollars* (run 2026-09-07, file 22_price_of_the_prior.md, written 2026-09-07 16:46 UTC). **Code:** code/01_mmm_identified_set.py, code/20_price_of_identification.py, code/21_spectral_comb.py, code/22_price_of_the_prior.py (results code/22_results.json). **Frontier-study problem(s) addressed:** M1 — characterise the sharp identified set of (α, K, S, β) as a functional of the spend path, give formal conditions (frequency content, on/off structure, range coverage) for point versus set identification, and an optimal-design theory for curvature separately from decay; adoption barriers M1 (weak identification) and M7 (prior elicitation). **Backlog items closed:** BL 1, BL 2 (opened by Dive 01, closed by Dive 20); BL 73 (opened by Dive 20, closed by Dive 21); BL 70 (priced by Dive 21); BL 75 (opened by Dive 20, closed by Dive 22); BL 82 discharged by Dive 22 for dives 20–22. **Opened:** BL 74, BL 76, BL 77, BL 78 (Dive 20); BL 79, BL 80, BL 81, BL 82, BL 83, BL 84 (Dive 21); BL 85, BL 86, BL 87, BL 88 (Dive 22).

Key ideas in this chapter

- Spend perturbations of relative amplitude δ leave a likelihood ridge whose smallest Fisher eigenvalue scales as δ^4 (verified slope 3.99): a $\pm 10\%$ wiggle would need about 5.5 million weeks to match one two-level block design. Blocks of half-length $L \approx 3/(1 - \alpha)$ with the adstock transient kept point-identify all four parameters (discarding the transient collapses $\lambda_{\min}$ about $700\times$); annealed budget-neutral designs beat AR(1) spend $5\times$ on the D-criterion.
- Profit cost and Fisher information pass through the same adstock filter, so at matched profit cost the calibrated $\text{se}(\text{mROAS})$ is exactly frequency-invariant (0.0992 across periods 4–182 weeks) and the design frontier is the hyperbola $C \cdot \text{Var}(\hat{c}_1) = \frac{1}{2}\kappa S_\varepsilon(\omega)$. The price of identifying a channel is $L^* = m\sigma\sqrt{\Psi T_r}$, half probing and half residual ignorance: 0.36–1.12% of media spend in probing cost, independent of curvature and adstock.
- Design content lives in the nuisance projection: a probe on a seasonal harmonic carries identically zero information, the Fejér margin rule asks for $\geq 1.15/(\pi\sqrt{\varepsilon}) \approx 1.6$ Fourier bins, week-of-year dummies annihilate every probe whose period divides 52, and the profit-maximising media plan is, to first order, exactly non-identifying.
- Comb tones at distinct DFT bins stay exactly orthogonal through heterogeneous adstock, so the n -channel frontier binds for all channels at once and the price is a channel-count tax (20 channels pay 1.75–5.5% of a fixed budget where one pays 0.09–0.28%). Saturation puts each channel on a harmonic ladder; a collision obeys $\text{VIF} = 1/\sin^2(p\psi_j - \psi_k)$ and phase is a design variable ($15.5\times$ naive, $4.27\times$ closed form, $1.085\times$ numerical optimum), but the penalty deflates to $1.03\times$ once the shape is calibrated. Budget-neutral variation leaves the total marginal value exactly unidentified under a common adstock; synchronised 13-week flighting is exactly singular.

- Everything a shape prior does to a decision functional passes through one vector d and one matrix $\tilde{I}$, and $d = 0$ makes a prior worth exactly zero and harmless. The ladder becomes continuous, $\Psi(u) = 1 + (\Psi_0 - 1)/(1 + u)$, with half its dollar value bought at $u_{1/2} < 3$; a prior is admissible iff $\Delta^2 < 2\tau^2 + \text{sd}_{\text{data}}^2$; the direction law value/Var = $\text{corr}^2 u/(1 + u)$ shows an ROI prior saves 56% of the identification budget against 4.9% for a coefficient prior. The 4-parameter $\Psi_0 = 19.8$ is a $\pm 35\%$ quantity; only the direction-law ratio survives the estimability audit.

1.1 Problem and motivation

The workhorse Bayesian marketing-mix model represents a channel's effect on weekly sales as a saturating function of a geometrically decaying stock of past spend. Its authors report that with weekly national data the likelihood is nearly flat along ridges trading off decay, saturation and effectiveness, so that “prior distributions have a big impact on the posteriors” (Jin et al., 2017), and five equally well-fitting models disagree on optimal allocation by up to 50% (Chan and Perry, 2017). Problem M1 asked for the mathematics: the sharp identified set of (α, K, S, β) as a functional of the spend path $\{x_t\}$; conditions on spend variation—frequency content, on/off structure, range coverage—for point rather than set identification; and an optimal-design theory for learning curvature separately from decay. Behind it sit the adoption barriers M1 (“the data can’t answer the question, so priors do”) and M7 (“the model demands beliefs business users cannot supply”).

Four dives answer M1 as one arc. Dive 01 characterises the identified set locally through the Fisher information and transports dose-ranging design theory into the dynamic setting: the transient of a block design is a free dose-ranging sweep, and information about saturation grows as the fourth power of the excursion. It left two admitted gaps—seasonal nuisance collinearity and the profit cost of probing—which Dive 20 shows to be one question, answered in the frequency domain: cost and information are the same quadratic form in the same filtered signal, so the exchange rate between forgone profit and precision is an exact hyperbola and the price of identification has a closed form. Dive 21 extends the theory to n channels through an orthogonal spectral comb, finds the binding constraint to be nonlinearity and phase rather than frequency, and prices the budget-neutral constraint Dive 19 had found blinding. Dive 22 completes the economics on the Bayesian side: it puts the prior-regularised estimator on the same cost–precision plane, turns Dive 20's discrete knowledge ladder into a continuous function of prior precision, and prices priors on different parameters against one another in excitation dollars. The four share one model and working point but drift in notation; Section 1.3 fixes one notation and the reviewer's notes record where the dives corrected themselves.

1.2 Background: the ideas this chapter builds on

1.2.1 Fisher information, the Cramér–Rao bound, local identification and optimality criteria

For a Gaussian mean model $y_t = \mu_t(\theta) + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, with Jacobian $J_{ti} = \partial \mu_t / \partial \theta_i$, the Fisher information is $I(\theta) = J^T J / \sigma^2$, and the Cramér–Rao lower bound (CRLB) says any unbiased estimator has $\text{Cov}(\hat{\theta}) \succeq I^{-1}$; for a functional $\phi = g^T \theta$, $\text{Var}(\hat{\phi}) \geq g^T I^{-1} g$. The parameter is *locally identified* at θ_0 iff $I(\theta_0)$ has full rank; a near-zero eigenvalue is a likelihood ridge along its eigenvector, identified in principle but with CRLB standard error $\lambda_{\min}^{-1/2}$. Optimal design shapes I through the regressor path: *D-optimality* maximises $\det I$, *c-optimality* minimises $g^T I^{-1} g$ for a chosen functional, and *D_s-optimality* (Atkinson and Cox, 1974) maximises the determinant of the Schur complement $I_{ss} - I_{sc} I_{cc}^{-1} I_{cs}$ when only a block matters. All are local in the unknown θ_0 , the classical circularity of nonlinear design. When a linear nuisance basis Z (intercept, trend, seasonal harmonics) is profiled out, a coefficient with score vector v retains information $\|M_Z v\|^2 / \sigma^2$, $M_Z = \text{Id} - Z(Z^T Z)^{-1} Z^T$: whatever part of v lies in the span of Z is lost.

1.2.2 The adstock/Hill MMM and its weak identification (Jin et al., 2017; Chan and Perry, 2017)

Jin, Wang, Sun, Chan and Koehler (2017) model a KPI as baseline plus, per channel, a carryover transform followed by a shape transform of spend. Carryover is geometric adstock, $a_t = x_t + \alpha a_{t-1}$, so a unit of spend contributes α^l after l periods and constant spend $\bar{x}$ settles at $\bar{a} = \bar{x}/(1 - \alpha)$; the shape is the Hill function $h(a; K, S) = a^S / (a^S + K^S)$ with half-saturation K and slope S , convex below and concave above its inflection; the channel contributes $\beta h(a_t)$ with β the sales ceiling. Fitted by MCMC to weekly national series, the posterior for (α, K, S) was prior-dominated and the likelihood flat along ridges. Chan and Perry (2017) catalogue the data problems—156 observations against 20 or more channels, “insufficient variation in media spend”, collinearity when channels move together—noting that “the data contains little information” about a channel unless it moves independently. Recast’s practitioner note (2026) comes closest to a prescription—flight spend both with and against seasonality, pulse one channel up the week after another—with no mathematics, optimum or cost. These are empirical diagnoses and rules of thumb; none says which variation identifies what, nor what it costs.

1.2.3 Emax dose-ranging design theory

The pharmacological sigmoid Emax model $E(d) = E_0 + E_{\max} d^h / (d^h + ED_{50}^h)$ is the Hill function under another name. Two facts from its optimal-design literature (Dette et al. and the adaptive sigmoid-Emax designs cited by Dive 01) matter here. A locally D-optimal design for the four-parameter model needs at least four distinct dose levels; a two-level experiment cannot identify a four-parameter sigmoid. And *range coverage*: when the largest dose is far below ED_{50} , $E(d) \approx E_0 + E_{\max} (d/ED_{50})^h$ and only the composite $E_{\max}/ED_{50}^h$ is identified—the plateau is extrapolation, and doses must reach the knee to locate it. The static theory sets the dose freely; in the MMM the “dose” a_t is a filtered functional of the spend path.

1.2.4 Frequency-domain system identification

For a linear dynamic system driven by an input with power spectrum $\Phi_u(\omega)$, the Fisher information is an integral of Φ_u against the squared parameter sensitivity of the transfer function (Mehra, 1974; Goodwin and Payne, 1977; Ljung, 1999, ch. 13). Three consequences matter. *Persistent excitation*: an input must carry energy at enough distinct frequencies to excite every parameter direction; a constant input identifies only a gain. *Multisines on the DFT grid*: because information is a moment functional of the spectrum, optimal spectra are realisable as finite sums of sinusoids, and tones at integer numbers of cycles in the record ($\omega = 2\pi k/T$) are exactly orthogonal over the record at any phases; Morelli’s orthogonal optimised multisines for flight testing (NASA, 2021) assign disjoint bin sets to different inputs so that all can be excited at once, and Gevers, Miskovic, Bonvin and Karimi (2006) show that adding an excited input reduces every parameter’s covariance when inputs share parameters. *Leakage*: a rectangular record of length T smears any line by the Fejér kernel

$$F_T(\Delta) = \frac{\sin^2(T\Delta/2)}{T^2 \sin^2(\Delta/2)} \leq \frac{4}{T^2 \Delta^2}, \quad (1.1)$$

the squared normalised Dirichlet kernel, with main lobe $2\pi/T$ and sidelobes decaying as the inverse square of distance; an off-grid tone leaks the fraction $F_T(\Delta)$ of its energy onto a line Δ radians away, an on-grid tone leaks nothing onto other grid lines. Schoukens, Pintelon and Guillaume’s harmonic-suppressed multisines leave chosen lines unexcited so that the energy a nonlinearity puts there can be measured. “Least costly identification” (Bombois et al., 2006) minimises input power subject to a precision constraint, Hjalmarsson (2009) frames excitation energy per unit accuracy as a cost of complexity, and Rivera et al.’s plant-friendly identification bounds the crest factor of the multisine so that a running plant is not disturbed; none of it prices excitation in profit.

1.2.5 Nonlinear distortion and Volterra readouts

A static nonlinearity $f(\bar{a} + B \cos \omega t)$ expanded to third order gives $\frac{1}{4}f''B^2[1 + \cos 2\omega t] + \frac{1}{24}f'''B^3[3 \cos \omega t + \cos 3\omega t] + \dots$: one tone becomes a ladder of lines at $\omega, 2\omega, 3\omega$, the p -th carrying p times the input phase and an amplitude proportional to the p -th derivative. This is the local Volterra description of a weakly nonlinear system and the basis of harmonic-distortion measurement: the second-harmonic line reads the curvature without fitting a curve. It is also a hazard in multi-input excitation, since one input's distortion products can land on another's fundamental.

1.2.6 The cost of learning (Little, 1966; Feit and Berman, 2019)

Little (1966) chose the size of an advertising experiment “to minimize the cost of imperfect information plus the cost of experimentation” in a quadratic response model; the adaptive-control line (Pekelman and Tse, 1980; Kolsarici, Vakratsas and Naik, 2020) makes the probe increment proportional to the posterior variance of effectiveness, and den Boer and Zwart (2014) show that pricing with a controlled dispersion decaying as $t^{-1/4}$ attains $O(T^{1/2+\epsilon})$ regret, by the same second-order Taylor argument as the cost law below. Feit and Berman (2019) give the closed-form profit-maximising test size $n^* \approx \frac{1}{2}\sqrt{N}(s/\sigma)$ and a map from posterior precision to deployment profit—an existing price-of-identification frontier for a two-arm i.i.d. test. Marketing pulsing theory (Sasieni, 1989; Feinberg, 2001) is profit-driven; Feinberg proves no finite pulsing frequency is optimal in continuous time, so a finite probing period must come from discreteness, memory or identification.

1.2.7 The van Trees inequality and the effective sample size of a prior

The CRLB bounds unbiased estimators only. The van Trees (Bayesian Cramér–Rao) inequality in the form of Gill and Levit (1995) bounds the Bayes risk of *any* estimator by $1/(\mathbb{E}_\pi[I(\theta)] + \mathcal{J}(\pi))$, with $\mathcal{J}(\pi)$ the Fisher information of the prior; for a Gaussian prior of precision P the bound is $(I + P)^{-1}$. Morita, Thall and Müller (2008) define a prior's effective sample size by curvature matching; in the normal case a prior of variance $\tilde{\sigma}^2$ is worth $\sigma^2/\tilde{\sigma}^2$ observations; Neuenschwander et al. (2020) show that most other effective-sample-size definitions fail a predictive-consistency axiom. The shrinkage crossover (Casella, 1985; Efron and Morris) says that for $X \sim \mathcal{N}(\theta, \sigma^2)$ with prior $\mathcal{N}(\mu_0, \tau^2)$ the posterior mean beats the MLE in MSE iff $(\theta - \mu_0)^2 < 2\tau^2 + \sigma^2$.

1.2.8 Priors in MMM practice, and the ridge-with-a-tight-curve observation

Meridian (Zhang, Wurm et al., 2024) reparameterises the channel coefficient β_m as a deterministic function of the channel's ROI and the shape parameters so that the prior, often from a lift test, sits on ROI; its documentation notes that the induced prior on β is then not independent of $(\alpha_m, \text{ec}_m, \text{slope}_m)$, that translating an experiment's standard error into a prior “is not a precise formula”, and ships a prior–posterior shift diagnostic without a threshold. LightweightMMM placed $\text{lag_weight} \sim \text{Beta}(2, 1)$ on adstock decay for MCMC stability; Robyn injects lift tests as a third objective with self-declared subjective weights. Dew, Padilla and Shchetkina (2024) show that saturation and time-varying effectiveness are not separately identifiable from observational spend, worst under autocorrelated media, and propose a maximal-separation intervention; Heusch (2026) recovers adstock, saturation and effectiveness structurally from geo experiments and reports a posterior ridge with $\text{corr}(\lambda, \beta) = -0.68$ whose *implied response curve* over the probed range is nevertheless tight. That coexistence of a wide parameter ridge with a tight curve is what Dive 22's residual sensitivity d formalises.

1.3 Setup and assumptions

All four dives use one model, written here in n -channel form with a nuisance basis:

$$y_t = z_t^\top \gamma + \sum_{j=1}^n \beta_j h(a_{jt}; K_j, S_j) + \varepsilon_t, \quad a_{jt} = x_{jt} + \alpha_j a_{j,t-1}, \quad h(a; K, S) = \frac{a^S}{a^S + K^S}, \quad (1.2)$$

with $\theta_j = (\beta_j, K_j, S_j, \alpha_j)$, z_t an intercept, centred linear trend and Fourier harmonics $k \leq 3$ at period 52 (Dive 01 has $z_t \equiv 0$), and ε_t stationary with normalised spectral density $S_\varepsilon(\omega)$, equal to σ^2 for white noise. Adstock is the filter $H_\alpha(\omega) = (1 - \alpha e^{-i\omega})^{-1}$ with memory $\tau = 1/(1 - \alpha)$. The working point is $\theta_0 = (1, 1.5, 2, 0.6)$, $\sigma = 0.05$. Dive 01 sets mean spend $\bar{x} = 1$, so $\bar{a} = 2.5$; dives 20–22 add a margin $m = 2$ and sit at the myopic profit optimum $m\beta h'(\bar{a}^*) = 1 - \alpha$, giving $\bar{a}^* = 2.1804$, $\bar{x}^* = 0.8722$, $h''(\bar{a}^*) = -0.15731$, profit curvature $\kappa \equiv m\beta|h''(\bar{a}^*)| = 0.31463$ and weekly channel profit 0.4854. Local composites are $c_k = \beta h^{(k)}(\bar{a})$, $k = 0, \dots, 3$; the decision functional is the marginal ROAS at the operating point, mROAS = $mc_1/(1 - \alpha)$. The Fisher matrix is I throughout (Dive 01 wrote F); the record length is T (Dive 20 wrote T_w): 156 in dives 01 and 22 (96 for Dive 01's annealer), 364 in dives 20 and 21, the latter two with a 52-week burn-in carrying the periodic extension of the design. T_r is the decision horizon.

Drifting symbols are reconciled as follows. δ is the relative amplitude of a spend perturbation (Dive 01); the prior mis-centring, which Dive 22 also wrote δ , is $\Delta = s_{\text{true}} - s_0$. ρ is the AR(1) residual coefficient (Dive 20); Dive 22's whitened sensitivities ρ_k^2 are written r_k^2 . d is Dive 22's residual shape sensitivity; distance from a nuisance harmonic in Fourier bins is ℓ . Dive 22's frontier constant $A = \frac{1}{2}\kappa S_\varepsilon(\omega)$ is written $\mathcal{A}$, since A_j is a probe amplitude in Dive 21.

Definition 1.1 (Identified set, probing design, ladder). The *sharp identified set* under design x is $\Theta_I(x) = \{\theta : \mu_t(\theta; x) = \mu_t(\theta_0; x) \forall t\}$; *practical identification* is the eigenstructure of $I(x, \theta_0)$. A *probing design* is a mean-zero perturbation u of the operating spend; in dives 20–22 a tone $x_{jt} = \bar{x}_j^*(1 + A_j \cos(\omega_j t + \phi_j))$ on the DFT grid $\omega_j = 2\pi m_j/T$, whose adstock deviation has amplitude B_j and phase $\psi_j = \phi_j + \arg H_j(\omega_j)$. Its *cost* C is the expected profit forgone over the window relative to constant spend at $\bar{x}^*$. The *ladder factor* $\Psi = \text{Var}(\widehat{\text{mROAS}}) / \text{Var}(\widehat{\text{mROAS}} \mid \alpha, \text{shape known})$; Ψ_0 is its value with the shape free.

Definition 1.2 (Prior blocks and ledger objects). Partition $\theta = (c, s)$ into a free block c and a shape block $s = (\alpha, K, S)$ carrying a Gaussian prior of precision P centred at s_0 ; $J = I + P_{\text{full}}$. For $\phi = g^\top \theta$, $g = (g_c, g_s)$, define

$$\tilde{I} = I_{ss} - I_{sc} I_{cc}^{-1} I_{cs}, \quad d = g_s - I_{sc} I_{cc}^{-1} g_c, \quad V_\infty = g_c^\top I_{cc}^{-1} g_c, \quad (1.3)$$

the profile information, the residual shape sensitivity and the shape-known variance; the relative prior strength is u with $P = u\tilde{I}$ (isotropic case) or $u = p(n^\top I^{-1}n)$ for a rank-one prior of precision p along n .

Assumption 1.3 (Model class and locality). Adstock is geometric and the shape exactly Hill; channels are additive; the analysis is local at θ_0 except where a nonlinear Monte Carlo is named; residuals are white unless S_ε appears; the world is stationary with one decision at the horizon and quadratic certainty-equivalent regret, except in Dive 22's deadband section.

1.4 Main results

1.4.1 A. The identified set and the design recipe (Dive 01)

Proposition 1.4 (Constant spend is rank one). *[established in the dive]* If $x_t \equiv \bar{x}$ the mean path is the single number c_0 , Θ_I is the three-dimensional manifold $\{\theta : \beta h(\bar{x}/(1 - \alpha); K, S) = c_0\}$, and I has rank one (eigenvalues $(0, 0, 10^{-10.6}, 10^{5.0})$; CRLB standard errors 10^4 – 10^6).

Law 1.5 (Amplitude law: a δ^4 -slow ridge). *[established in the dive]* Write $x_t = \bar{x}(1 + \delta u_t)$ with u_t bounded and mean zero. Then $a_t = \bar{a}(1 + \delta v_t)$, $v_t = (1 - \alpha) \sum_{j \geq 0} \alpha^j u_{t-j}$, and

$$\mu_t = \beta h(\bar{a}) + \delta \bar{a} \beta h'(\bar{a}) v_t + \frac{1}{2} \delta^2 \bar{a}^2 \beta h''(\bar{a}) v_t^2 + O(\delta^3). \quad (1.4)$$

At $O(1)$ the design reveals c_0 ; at $O(\delta)$ the filter shape (hence α , from any temporally rich u) and c_1 ; only at $O(\delta^2)$ the curvature c_2 . To first order (c_0, c_1, α) are three constraints on four parameters, so the identified set is a one-dimensional ridge through (β, K, S) -space regardless of the frequency content of the perturbation, resolved only by the curvature term:

$$\lambda_{\min}(I) = \Theta(T\delta^4/\sigma^2), \quad \lambda_2(I) = \Theta(T\delta^2/\sigma^2). \quad (1.5)$$

The standard error along the ridge scales as $\sigma/(\delta^2\sqrt{T})$ and the sample size for fixed precision as δ^{-4} : generic point identification (the map $(\beta, K, S) \mapsto (c_0, c_1, c_2)$ is generically invertible) coexists with catastrophic practical weakness, which the dive proposes as the mathematical form of Jin et al.'s ridges. For $x = 1 + \delta \sin(2\pi t/8)$ the log-log slope of the eigenvalues on $\delta \in [0.025, 0.4]$ is 3.990 for $\lambda_{\min}$ (theory 4) and 2.020 for λ_2 (theory 2). A $\pm 10\%$ wiggle at $T = 156$ has $\lambda_{\min} = 0.0028$; matching the block design below ($\lambda_{\min} \approx 99$) would take about 5.5 million weeks.

Proposition 1.6 (Range coverage: below the knee only β/K^S exists). *[transported from the literature]* If $\max_t a_t \ll K$ then $h(a) = (a/K)^S(1 + O((a/K)^S))$, so the mean path depends on (β, K) only through β/K^S and Θ_I contains, to that order, the curve $\{(\beta\lambda^S, \lambda K, S, \alpha) : \lambda > 0\}$.

Scaling the block design to $\max a/K = 0.29$ drops $\lambda_{\min}$ from 10^2 to 10^{-3} , rotates the ridge eigenvector into the (β, K) plane, $(-0.74, -0.66, 0.07, 0.00)$, and a noiseless profile fit reproduces the data ($\text{RSS} \leq 3.4 \times 10^{-4}$) for any $K \in [0.8, 6]$ with β sliding from 0.33 to 15.0 along $\beta/K^S \approx \text{const}$ (0.52, 0.44, 0.42, 0.42). One cannot learn the ceiling from below the knee.

Proposition 1.7 (Block length). *[established in the dive]* For a square wave alternating $x_{\text{hi}}/x_{\text{lo}}$ with half-period L , the steady-state adstock separation between block ends is

$$\Delta a(L) = \frac{x_{\text{hi}} - x_{\text{lo}}}{1 - \alpha} \cdot \frac{1 - \alpha^L}{1 + \alpha^L} \quad (1.6)$$

(two-cycle fixed point). Short blocks attenuate the dose separation, acting like a small δ ; long blocks waste the $O(1)$ -per-event information that the T/L transitions carry about α and curvature. Requiring $\alpha^L \leq 0.05$ gives $L^* \gtrsim 3/\ln(1/\alpha) \approx 3/(1 - \alpha)$.

With $\tau = 2.5$ the D-criterion peaks at $L^* = 8\text{--}10$; $\lambda_{\min}$ is 0.65, 99, 3.1 at $L = 1, 8, 78$. Dive 20 later refuted its own alternative explanation (frequency resolution): L^* stays at 8–10 for $T = 104, 208, 416, 832$, so the settling-time account stands.

Proposition 1.8 (The adstock transient is a free dose-ranging sweep). *[numerically established]* Static theory needs four dose levels for a four-parameter sigmoid, yet a two-level dynamic design is full rank and well conditioned: every transition drags a_t through a continuum of intermediate values along the known exponential path, tracing an arc of the Hill curve. In a two-level design with $L = 13$, keeping only the 84 settled observations gives $\lambda_{\min} = 0.079$; keeping 84 transient-heavy observations gives 53.4 (full series 85.2)—a collapse of about $700\times$ at equal sample size when the transient is removed.

Pipelines that model only steady-state relationships, or geo-test analyses that discard ramp-up weeks, throw away most of the curvature information an experiment bought; Heusch's estimator works because it keeps the transient (Chapter 2 applies this to what an experiment reports).

Construction 1.9 (Annealed budget-neutral design). *[numerically established]* Simulated annealing over sequences on levels $\{0, 0.5, \dots, 2.5\}$ with the mean exactly preserved, maximising $\det(I)^{1/4}$ at $T = 96$, beats three-level blocks by $1.23\times$ and AR(1) lognormal spend by $5.0\times$ on the D-criterion ($3\text{--}4\times$ on standard errors). The solution is interpretable: long full-off runs (35 of 96 weeks at zero, clean decay reads), sustained blocks at 2–2.5 (past the knee) and a minority of intermediate levels (dose ranging). CRLB standard errors $(\beta, K, S, \alpha) = (0.022, 0.065, 0.115, 0.009)$ against $(0.087, 0.190, 0.453, 0.037)$ for AR(1) spend.

The recipe: pulse in blocks of at least $3/(1 - \alpha)$ weeks (6–10 covers most channels); use three or more levels including full-off windows and excursions to $2\text{--}2.5\times$ normal spend so peak adstock reaches the suspected K ; never discard transition weeks; remember that saturation information grows as the fourth power of the excursion, so two big pulses beat fifty small wiggles; anneal the FIM when the schedule is free.

1.4.2 B. The frequency-domain price of identification (Dive 20)

Perturb around the profit optimum with a mean-zero u and let $\tilde{a} = H_\alpha u$ be the adstock deviation.

Theorem 1.10 (Same-filter theorem and the price-of-precision hyperbola). *[established in the dive] Around the optimum the linear profit term vanishes, so the weekly cost is $C_{\text{week}} = \frac{1}{2}\kappa \text{Var}(a) + O(\mathbb{E}\tilde{a}^3)$. The score for c_1 is $\tilde{a}$, so after projecting out Z , $I_{c_1} = \|M_Z \tilde{a}\|^2 / \sigma^2 = T \text{Var}(a)(1 - \nu) / \sigma^2$, ν being the fraction of filtered energy absorbed by Z . Cost and information are the same quadratic form in the same filtered signal; on the clean band, for total window cost C ,*

$$C \cdot \text{Var}(\hat{c}_1) = \frac{1}{2} \kappa \sigma^2, \quad \text{and with coloured residuals} \quad C \cdot \text{Var}(\hat{c}_1) = \frac{1}{2} \kappa S_\epsilon(\omega_{\text{probe}}), \quad (1.7)$$

independently of probing frequency, amplitude and duration.

Halving the standard error costs four times the profit. *Frequency invariance*: probing at 4 or 182 weeks buys identical precision per dollar for the linear effect. *Duration–amplitude invariance*: the energy $E = T_e \text{Var}(a)$ of an excitation lasting T_e weeks is the only design scalar for c_1 , so a 4-week go-dark and a two-year 5% wiggle of equal cost are equally informative about mROAS. *The cheap-channel law*: $I/C = 2/(\kappa_j \sigma^2)$, so flat channels—the ones nobody can pin down—are the cheap ones to probe (verified to 0.2%). *Curvature is the exception*: the score for c_2 is $\frac{1}{2}\tilde{a}^2$, so $I_{c_2} \propto E^2/T_e$; at fixed cost, concentrate. The exact nonlinear cost obeys the second-order law to $\leq 0.5\%$ up to $\pm 50\%$ swings (clean-room: $+0.41\%$ worst over commensurate periods, $+0.72\%$ at 38% amplitude, -3.5% at a degenerate 3-week period); t_3 errors give 0.965 ± 0.023 . The dive concedes the c_1 invariance is nearly a tautology once both sides are the same quadratic form; the content is which parameters break it.

Law 1.11 (Rule L: the Fejér leakage margin). *[established in the dive] A probe Δ radians from a nuisance harmonic loses the fraction $F_T(\Delta)$ of Equation (1.1), summed over harmonics; in bins the envelope is $1/(\pi^2 \ell^2)$, giving a naive margin $\ell \geq 1/(\pi\sqrt{\epsilon})$, which the clean-room found violated when three sidelobes add (loss $0.0576 > 0.05$ at $\ell = 1.42$). Calibrated: keep the probe $\ell \geq 1.15/(\pi\sqrt{\epsilon})$ bins from every nuisance harmonic to lose less than ϵ (≈ 1.6 bins at $\epsilon = 0.05$), or place it on an exact Fourier frequency of the window, where leakage is identically zero at any separation ≥ 1 bin. A probe on a harmonic ($52/k$ weeks) carries identically zero information (matched-cost standard error $10^6\text{--}10^{12}$).*

Proposition 1.12 (Refuted: the closed-form confounding factor). *[refuted within the dive] Round 1 derived the inflation of $\text{Var}(\hat{c}_1)$ under joint estimation of α , $\text{VIF}(\omega) = 1 + (\cos \phi_H + r/|H_\alpha|)^2 / \sin^2 \phi_H$ with $r = (h''/h') \tilde{a}/(1 - \alpha)$ and $\phi_H = \arg(e^{-i\omega} H_\alpha)$, matching the FIM to 0.04–0.8%, and in the long-memory limit a curvature index $\tilde{a}|h''|/h' = |1 - S(1 - 2h(\tilde{a}^*))|$ with a threshold at half-saturation (separable below, confounded above, rising $5.9 \rightarrow 21.2$ as spend goes from $1\times$ to $3\times$ optimal). Monte Carlo (250 replications) gave inflation 1.3 where the formula said 4.6, and 0.98 against 12.1 at a 7-week period—below one, impossible for the quantity claimed.*

The formula prices the slope at a *fixed* reference adstock; the decision functional is the slope at the *fitted* operating point, whose gradient carries an extra term $mc_2 \partial \tilde{a} / \partial \alpha / (1 - \alpha)$ that cancels most of the confounding (corrected gradient $(5, 0, -1.788)$ against the naive $(5, 0, +2.5)$). Individually unidentified quantities combine into a well-determined decision functional, as Chapter 3 found in other coordinates.

Construction 1.13 (The Ψ ladder and the decision frontier). *[numerically established]* With Ψ of Definition 1.1 computed directly,

$$C \cdot \text{Var}(\widehat{\text{mROAS}}) = \left(\frac{m}{1 - \alpha} \right)^2 \frac{1}{2} \kappa S_\epsilon(\omega) \cdot \Psi. \quad (1.8)$$

At matched cost $C = 1.0$ (0.44% of the window's spend), white residuals, the calibrated standard error is 0.0992 at every period from 4 to 182 weeks (clean-room spread 0.004%); Ψ_3 (c_1, c_2, α free) is 17.0, 4.69, 1.97,

1.35, 1.20, 1.08, 10^{27} , 1.06, 1.19 and Ψ_4 (full Hill free) is 3×10^9 , 12.1, 10.7, 10.1, 9.98, 9.87, 10^{16} , 9.94, 10.5 at periods 4, 7, 13, 21.4, 28, 45.5, 52, 91, 182.

For the full Hill model, frequency within the admissible band is nearly decision-irrelevant— Ψ_4 varies by 10% over 13–182 weeks, the plateau of Chapter 3 in the frequency domain—while a harmonic (10^{16}) or a period faster than the memory (3×10^9 at 4 weeks, where carryover is unidentifiable) is catastrophic. The ladder is the lever: from $\Psi \approx 9.9$ to $\Psi = 1$ cuts the standing standard error $\sqrt{9.9} \approx 3.1 \times$ at fixed cost, so a one-time shape-fixing experiment is worth $\sqrt{10}$ on the perpetual budget. Monte Carlo at a 21.4-week period, 10% amplitude, 250 replications: local-3 $\text{se}_{\text{MC}} = 0.1186 \pm 0.0053$ against FIM 0.1134; bounded structural-4 MLE $\text{se}_{\text{MC}} = 0.2022 \pm 0.0091$, bias +0.107 on an mROAS of 1.31, RMSE 0.229 against CRLB 0.311—inside the frontier, the puzzle Dive 22 resolves.

Proposition 1.14 (The noise spectrum restores a frequency optimum). *[established in the dive]* With AR(1) residuals, $S_\varepsilon(\omega) = \sigma^2(1 - \rho^2)/|1 - \rho e^{-i\omega}|^2$ in Equation (1.7): measured inflation at $\omega = 0.2934$ (5 seeds $\times$ 300 replications) is 3.077 ± 0.080 against 3.029 predicted for $\rho = 0.6$, and 2.215 ± 0.057 against 2.185 asymptotic / 2.258 exact finite-sample for $\rho = 0.9$.

MMM residuals are strongly autocorrelated, so the noise floor is low-frequency-heavy and pushes probes fast while parameter separation pushes them slow. Minimising $S_\varepsilon(\omega)\Psi_4(\omega)$ gives optimal periods 84, 84, 84, 12, 11 weeks at $\rho = 0, 0.3, 0.5, 0.7, 0.85$: a sharp crossover from “flight annually-ish” to “quarterly-ish” between 0.5 and 0.7, within a total-loss spread of only $1.0\text{--}2.6\times$ across the band.

Theorem 1.15 (The price of identification). *[established in the dive]* Certainty-equivalent budgeting maps an mROAS error e into a spend error $e(1 - \alpha)^2/\kappa$ and a per-period regret $\text{Var}(\widehat{\text{mROAS}})(1 - \alpha)^2/(2\kappa)$; substituting Equation (1.8), κ and $(1 - \alpha)$ cancel:

$$L(C) = C + \frac{T_r m^2 \sigma^2 \Psi}{4C} \implies C^* = \frac{1}{2} m \sigma \sqrt{\Psi T_r}, \quad L^* = 2C^* = m \sigma \sqrt{\Psi T_r}. \quad (1.9)$$

The minimum price of identifying a channel is margin $\times$ residual sales noise $\times \sqrt{\text{horizon} \times \Psi}$, split exactly half between probing and residual misallocation, independent of curvature, adstock and the duration/amplitude split.

Closed form and numerical argmin agree to 0.4% ($C^* = 1.7289$, $L^* = 3.4578$ at $\Psi = 4.599$, $T_r = 260$; clean-room to six decimals). Over five years the probing cost C^* is 0.36% of media spend fully calibrated ($\Psi = 1$; $L^* = 0.71\%$), 0.37% with the local shape free ($\Psi = 1.06$; 0.73%) and 1.12% with the full Hill free ($\Psi = 9.87$; 2.23%)—an order of magnitude below practitioner rules of thumb for geo holdouts (10–20% of a test region’s budget), which the dive could not source. $L^* \propto m\sigma$ exactly.

Theorem 1.16 (The profit-maximising plan is non-identifying). *[established in the dive]* Let demand seasonality be multiplicative, $s_t = 1 + A_s \cos(2\pi t/52)$, and let spend be set quasi-statically optimally, $ms_t \beta h'(a_t^*) = 1 - \alpha$. Then the spend path lies in the span of the seasonal nuisance basis and carries zero first-order information about c_1 after projection.

At $A_s = 0, 0.15, 0.30, 0.50$ the optimal path has spend coefficient of variation 0, 0.064, 0.134, 0.255 and $I(c_1) = 0.0, 0.0, 0.0, 16.7$ (the residual is Hill nonlinearity leaking into harmonics $k \geq 4$); the same path plus a 10% orthogonal probe gives 2617, 2617, 2617, 2633. A 25% coefficient of variation is more than most advertisers have and identifies nothing, because optimal variation is a deterministic function of the demand signal the model controls for; more history does not help, since the rate of accrual is zero. It is the static counterpart of the certainty-equivalent stall of Chapter 13. The counterweight: log-normal managerial noise at $cv = 10\%$ gives $I(c_1) = 1633 \pm 30$ at cost 0.627, i.e. 2604 information units per profit dollar against 2473 for the designed probe—undesigned variation is not inefficient, it is endogenous and uncontrolled in frequency. The exact zeros need seasonality exactly in the control basis; the verifier noted that under multiplicative seasonality the natural score regressor is $s_t(a_t - \bar{a})$, which the quoted numbers do not use.

The single-channel recipe: never probe on a harmonic; obey Rule L; match the probe to the baseline specification (Table 1.2); keep ≥ 4 cycles in the window and period $\geq 4\tau$; within the admissible set minimise

$S_\epsilon(\omega)\Psi(\omega)$; run a continuous low-amplitude probe for mROAS plus occasional short deep excursions for curvature. On the working example (7-year window, annual seasonality) the best admissible period is 84 weeks, giving $\text{se}(\text{mROAS}) = 0.261$ against 0.381 for 13-week quarterly flighting ($2.1 \times$ less profit forgone at equal precision); a 17-week flight sits 0.36 bins from the third annual harmonic and retains 43% of its information.

1.4.3 C. The n -channel spectral comb (Dive 21)

Channel j gets its own bin m_j . The working point is six channels A–F with $(K, S, \alpha, h(\bar{a}^*)) = (1.5, 2.0, 0.60, 0.68), (2.0, 1.6, 0.30, 0.55), (1.0, 2.4, 0.75, 0.75), (3.0, 1.3, 0.45, 0.60), (0.8, 3.0, 0.20, 0.70), (2.5, 2.0, 0.55, 0.50)$, $m = 2$, β_j solved from the first-order condition so that true $\text{mROAS}_j = 1$ for all; $T = 364$.

Proposition 1.17 (Heterogeneous adstock is free). *[established in the dive] Geometric adstock is linear and time-invariant, hence diagonal in the Fourier basis: on-grid tones at distinct bins produce exactly orthogonal steady-state adstock deviations whatever the channels' memories. Six rates in $[0.20, 0.75]$ at bins (19, 23, 25, 29, 31, 37) give maximum off-diagonal correlation 2.5×10^{-15} (clean-room 7.5×10^{-16}). The only linear leak is the centred trend, whose induced pairwise correlation is the bin-independent constant $-6/(T^2 - 7)$ (clean-room, 12 digits at five window lengths; -4.528678×10^{-5} at $T = 364$, VIF 1.000001), because $\sum_{t < T} t \cos(2\pi m t / T) = -T/2$ for every integer $m \neq 0$. The probe switch-on transient is benign once the baseline is in steady state (coupling 7.5×10^{-3} , VIF 1.0002); starting the channel from $a = 0$ had given 0.39 and was the wrong model of a running advertiser.*

Theorem 1.18 (n -channel frontier). *[established in the dive] Let $v_j = M_Z(H_j \star u_j)$. Then $\text{Var}(\hat{c}_{1j}) = \sigma^2 / \|P_{-j}^\perp v_j\|^2 \geq \sigma^2 / \|v_j\|^2$ with equality iff $v_j \perp \text{span}\{v_k\}_{k \neq j}$, and the cost is separable, $C_j = \frac{1}{2} \kappa_j \|v_j\|^2$. Hence for any design $C_j \cdot \text{Var}(\hat{c}_{1j}) \geq \frac{1}{2} \kappa_j S_\epsilon(\omega_j)$ for every j , with equality iff channel j 's projected excitation is orthogonal to every other's: an orthogonal comb attains the single-channel frontier for all n simultaneously, and every collinearity is a strict Pareto loss.*

The six-channel comb gives $C_j \text{Var}(\hat{c}_{1j}) / (\frac{1}{2} \kappa_j \sigma^2) \in [1.00010, 1.00071]$, exactly 1 to 10^{-12} without the trend; each channel's $\text{se}(\text{mROAS})$ equals its solo value to 1.1×10^{-5} .

Corollary 1.19 (The channel-count tax). *[established in the dive] L^* contains no channel-specific quantity, and $C \cdot \text{Var}$ is invariant to scaling a channel by s (ratio 1.000527 over $s = 0.25$ –4), so the price of identification is the same absolute number for a large and a small channel. Against a fixed six-channel budget over $T_r = 260$, L^* is 0.09%, 0.53%, 1.05%, 1.75% for 1, 6, 12, 20 channels at $\Psi = 1$ and 0.28%, 1.65%, 3.30%, 5.50% at $\Psi = 9.87$. Fragmentation, not scale, makes measurement expensive.*

Theorem 1.20 (Harmonic-collision phase law). *[established in the dive] Expanding $h_j(\bar{a}_j + \tilde{a}_j)$ with $\tilde{a}_j = B_j \cos(\omega_j t + \psi_j)$ gives $\frac{1}{2} c_{2j} \tilde{a}_j^2 = \frac{1}{4} c_{2j} B_j^2 [1 + \cos(2\omega_j t + 2\psi_j)]$ and a third-order line $\propto \cos(3\omega_j t + 3\psi_j)$: channel j occupies the ladder $m_j, 2m_j, 3m_j, \dots$ with phase $p\psi_j$ on the p -th line. In the local Volterra model, if $\omega_k = p\omega_j$,*

$$\text{VIF}(\hat{c}_{1k}) = \text{VIF}(\hat{c}_{pj}) = \frac{1}{\sin^2(p\psi_j - \psi_k)}, \quad (1.10)$$

non-identifying iff $\psi_k \equiv p\psi_j \pmod{\pi}$, exactly orthogonal iff $\psi_k = p\psi_j \pm \pi/2$; only the phase difference matters.

Verified to relative error 3.75×10^{-4} ($p = 2$) and 1.98×10^{-4} ($p = 3$), values sweeping $1.007 \rightarrow 148 \rightarrow 172$, exact to 10^{-10} without the trend (near the pole $d\text{VIF}/d\phi \approx 2500$). A comb therefore need not avoid octaves: capacity is a phase-feasibility problem, the constraints $\psi_k - p\psi_j \equiv \pi/2 \pmod{\pi}$ forming a linear system over $\mathbb{R}/\pi\mathbb{Z}$, always solvable when the collision graph is a forest.

Proposition 1.21 (Refined: the closed-form quadrature angle is the wrong angle). *[numerically established] In the full Hill model every channel-A parameter contributes to the $2\omega_A$ line at a different phase (+0.350 for α , +2.300 for β , -0.841 for S), so the second-harmonic content spans a two-dimensional subspace (singular values 0.768, 0.285) and no single phase for channel B is orthogonal to all of it. At bins 19 and 38, 10% amplitude, $\text{se}(\text{mROAS}_B)$ relative*

to a clean comb is $15.53\times$ at the naive phase $\phi_B = 0$, $7.97\times$ at a random phase (median; IQR 3.7–15.0), $4.27\times$ at the quadrature angle $\phi_B = 0.9650$ and $1.085\times$ at the numerical optimum $\phi_B = 1.5267$; the worst phase is a pole.

The closed form recovers about half the gain in log terms; a line search on the FIM recovers essentially all of it (BL 83). The originally reported “maximum” of $332\times$ was a 49-point-grid artefact (687, 895, 1181 at 401, 4001, 50001 points): never summarise a near-pole function by a grid maximum. On/off flights carry a third harmonic at 32.5% and a fifth at 18.4% of the fundamental, against the saturation 2ω line at 2.0–15.2% over 5–40% amplitudes—larger but cheap (1.12 – $1.23\times$ for a $3\times$ collision, 1.36 – $1.41\times$ for $5\times$) and not phase-defeasible. Non-negativity is a non-issue below 100% amplitude (5.9% and 4.6% harmonics at 130%).

Law 1.22 (Rule R: packing capacity and record length). *[established in the dive]* Bin m is admissible if (i) $m \geq 4$, (ii) $m \leq T/(4\tau_j)$, (iii) rm is not a seasonal bin $pT/52$ for $r = 1, \dots, q$. With $k = T/52$, $rm \equiv 0 \pmod{k}$ reduces to $m \equiv 0 \pmod{k}$ whenever $\gcd(r, k) = 1$, so if k is coprime to 6—a 5-, 7-, 11- or 13-year window—protecting the second and third harmonics costs nothing. Sufficient, not necessary: $k = 2, 3, 9, 15$ also cost nothing because the extra exclusions fall below the $m \geq 4$ floor.

Enumeration ($\tau = 2.5$, seasonal harmonics $p \leq 3$): $|\mathcal{A}| = (20, 20, 20), (30, 30, 30), (51, 51, 51)$ for $q = 1, 2, 3$ at $T = 260, 364, 572$ against $(14, 13, 13), (25, 24, 23), (56, 54, 52)$ at $T = 208, 312, 624$. At $T = 364$: 30 admissible bins, of which the largest subset free of $(m, 2m)$ and $(m, 3m)$ pairs has 20 (exact ILP, reproduced by branch-and-bound and an independent MILP); 14/20/26/33/40/46 phase-free channels at 5/7/9/11/13/15 years. Capacity is not binding for any real advertiser; cost and phase are.

Theorem 1.23 (Budget-neutral rank theorem and escape law). *[established in the dive]* Let $\Phi : c \mapsto \sum_j c_j (H_j \star u_j)$ with $\sum_j u_{jt} = 0$ for all t . With a common adstock, $\Phi(\mathbf{1}) = H \star \sum_j u_j = 0$: the total marginal value $\sum_j c_j$ is unidentified under budget-neutral variation, however many tones or contrast patterns are used (rank 1, 2, 3 of 4 with 1, 2, 4 tones, $\|P_{\text{null}}\mathbf{1}\|/\|\mathbf{1}\| = 1.000$ throughout). With heterogeneous adstock $\Phi(\mathbf{1}) = \sum_j (H_j - \bar{H}) \star u_j$ is generically nonzero and two tones suffice for full rank (the dive first wrote four). For $n = 2$ with $u_2 = -u_1 = -A \cos \omega t$,

$$\text{se}(\hat{c}_{11} + \hat{c}_{12}) = \sigma \sqrt{\frac{2}{T}} \frac{|H_1(\omega) + H_2(\omega)|}{A |\text{Im}(\bar{H}_1(\omega) H_2(\omega))|}, \quad \text{Im}(\bar{H}_1 H_2) = \frac{(\alpha_1 - \alpha_2) \sin \omega}{|1 + \alpha_1 \alpha_2 - \alpha_1 e^{i\omega} - \alpha_2 e^{-i\omega}|^2}, \quad (1.11)$$

from the Gram determinant $(T/2)^2 \text{Im}(\bar{z}_1 z_2)^2$, $z_1 = A H_1$, $z_2 = -A H_2$ (verified $\leq 3.6 \times 10^{-3}$ relative over eight cells; exact without the trend).

The escape is bought with adstock *phase lag*, not gain: $\text{Im}(\bar{H}_1 H_2)$ is the area of the parallelogram spanned by the two transfer functions, which vanishes at DC (every $H_j = 1/(1 - \alpha_j)$ is real) and at Nyquist, so the total marginal value is unidentified at DC however different the memories. For (0.6, 0.3) it is 6.6×10^{-5} at a 364,000-week period, 0.0660 at 364, 0.827 at 19, 0.0987 at 3 and 1.2×10^{-4} at 2 weeks; the matched-cost optimal budget-neutral period is 8.6–20.5 weeks and often sits inside the $4\tau_{\max}$ floor. In frontier units $\Lambda \equiv C \cdot \text{Var} / (\frac{1}{2} \kappa \sigma^2)$, $\Lambda_{\text{neutral}} = |H_1 + H_2|^2 (|H_1|^2 + |H_2|^2) / \text{Im}(\bar{H}_1 H_2)^2$, the total-budget analogue replacing $|H_1 + H_2|$ by $|H_1 - H_2|$, and a single channel’s own c_1 sits at $\Lambda = |H|^2 \approx 3.4$: Table 1.4 is the exchange rate BL 70 asked for.

Construction 1.24 (Model-free harmonic readout). *[established in the dive]* With $Y(\cdot)$ the DFT of the nuisance-projected response,

$$\hat{c}_{2j} = \frac{4}{B_j^2} \text{Re}[Y(2\omega_j) e^{-2i\psi_j}], \quad \hat{c}_{3j} = \frac{24}{B_j^3} \text{Re}[Y(3\omega_j) e^{-3i\psi_j}], \quad \text{se}(\hat{c}_{2j}) = \frac{4\sigma}{B_j^2} \sqrt{\frac{2}{T}}, \quad (1.12)$$

and $|Y(3\omega)|/|Y(2\omega)| = |c_3/c_2|B/6$ is a saturation shape test needing neither β nor a fitted curve.

Noiseless recovery is 0.1–2% for c_2 and 0.5–12% for c_3 ; a 20,000-replication Monte Carlo matches the standard error to 0.05% (0.43380 ± 0.00217 vs 0.43402 ; 0.02711 ± 0.00014 vs 0.02713), the original 400 replications

having been unable to adjudicate. Recovered $|c_3/c_2|$: 0.777 vs 0.781 (A), 1.399 vs 1.403 (C), 1.360 vs 1.397 (E); channel F sits at the inflection where $c_3 = 0$ and the test returns 0.0002—its own null control. The caveat is signal-to-noise: the 2ω line’s SNR is $|c_2|B^2/(4\sigma\sqrt{2/T}) = 0.36$ at a 10% probe, where a DFT-modulus estimator is catastrophically biased (mean -0.563 against a truth of -0.157 ; the phase-known projection is unbiased, -0.1586 at every amplitude). Reaching $z = 2$ on c_2 needs $B^* = \sqrt{8\sigma\sqrt{2/T}/|c_2|} = 23.5\%$ of mean spend—the closed form of “concentrate for curvature” and the cheapest route up the ladder.

Proposition 1.25 (Synchronised flighting is exactly singular). *[established in the dive] (Clean-room correction.) Six channels flighted on a common 13-week cycle at $T = 364$: since $T/28 = 13$, every spend path, adstock, response and Jacobian column is 13-periodic, so all 24 structural columns lie in a 13-dimensional subspace. The full-Hill FIM has rank 19 of 32 (nullity 13, $s_{\min}/s_{\max} = 1.9 \times 10^{-12}$), the calibrated-shape FIM rank 18 of 20, and every channel’s mROAS gradient leaks into the null space ($\|P_{\text{null}}g\|/\|g\|$ up to 0.821 and 0.694). The dive’s first “700–2400 $\times$ penalty” was pinv at default rcond on a singular matrix, pure floating-point noise (63/9.7/202 at 10^{-11} ; 0.26/0.042 at 10^{-6}).*

Synchronised flighting does not degrade multichannel identification; it destroys it—one degree of freedom where the model needs n , the multichannel form of Theorem 1.16. The exactness is an artefact of commensurability; a 13-week flight in a 365-week record is merely very ill-conditioned. The correction produced the program-wide rule BL 82: every FIM-based standard error must carry $s_{\min}/s_{\max}$ and the functional gradient’s null-space leak. Attacks that failed: seasonal sidebands at $\omega_j \pm \omega_s$ change $\text{se}(\text{mROAS}_B)$ by under 2% at seasonal amplitudes 0–0.8; truncating the record to 75% leaves $\text{VIF} = 1.020$; a misspecified α of ± 0.05 leaves the phase-managed ratio at 1.10 ($+0.10$: 1.63 $\times$; $+0.20$: 2.85 $\times$), still far below 15.5 $\times$; off-grid separation (Rule C) gives correlation 0.985 at 0.05 bins, 0.674 at 0.25, about zero at 0.5 and at every integer.

1.4.4 D. The price of the prior (Dive 22)

Dive 22 works in the Gaussian/FIM model with $\theta = (c, s)$ as in Definition 1.2 and penalised MLE $\hat{\theta} = \theta + J^{-1}\xi - J^{-1}P_{\text{full}}(\theta - \theta_{\text{prior}})$, $\xi \sim \mathcal{N}(0, I)$ the score. The functional is mROAS at a fixed observed adstock: at the profit optimum $\text{mROAS} \equiv 1$ for every θ (the first-order condition), so its gradient vanishes identically—Theorem 1.16 in functional form, which broke the dive’s first parameterisation. The anchor is $\bar{a}_0 = 0.85 \bar{a}^*$, giving $\text{mROAS} = 1.290$. Design: $T = 156$, a three-tone comb at periods $\{11, 21, 34\}$ weeks, 10% amplitude each (12% total CV), steady spend 0.872, three-year media spend 136.1 revenue units.

Proposition 1.26 (Dollar exchange rate—a bridge, not a theorem). *[transported from the literature] On the frontier, excitation information is $I_C = C/A$. A prior of precision p on the target gives excitation-equivalent budget $C_{\text{eq}} = A/\text{MSE}$ equal to $A(I_C + p)^2/I_C$ for an oracle ($\Delta \equiv 0$), exactly $C + Ap$ if calibrated ($\mathbb{E}\Delta^2 = 1/p$)—a constant exchange rate of A dollars per unit precision—and capped at A/Δ^2 as $p \rightarrow \infty$ if mis-centred by a fixed Δ , turning negative once $\Delta^2 > 2/p + A/C$ (bracketed at $+0.294/0.000/-0.281$ at $0.9\times, 1\times, 1.1\times$ the threshold).*

The calibrated row is Morita–Thall–Müller’s effective sample size times Dive 20’s marginal cost of information; the cap and sign flip are the shrinkage crossover.

Theorem 1.27 (van Trees attainment identity). *[established in the dive] If Δ is random with $\mathbb{E}[\Delta\Delta^T] = P^{-1}$ and independent of the score, then $\mathbb{E}_\Delta[\text{MSE}(g^T\hat{\theta})] = g^T(I + P_{\text{full}})^{-1}g$ exactly, since $g^T J^{-1} I J^{-1} g + g^T J^{-1} P \mathbb{E}[\Delta\Delta^T] P J^{-1} g = g^T J^{-1} (I + P) J^{-1} g$.*

Monte Carlo at three cells (4×10^5 draws): $z = -0.24, +0.39, -0.11$; clean-room 6.7×10^{-12} over 300 cells. This dissolves Dive 20’s puzzle: the right frontier for a prior-regularised MMM is the van Trees bound $(I + P)^{-1}$, which the penalised MLE attains exactly *provided the prior is calibrated*; “RMSE 0.229 inside CRLB 0.311” was a lower frontier purchased with an assumption.

Theorem 1.28 (The shape-prior ledger). *[established in the dive] With $\tilde{I}, d, V_\infty$ as in Equation (1.3), exactly,*

$$V_{\text{post}}(P) = V_\infty + d^T (\tilde{I} + P)^{-1} d, \quad \text{bias}(P) = -d^T (\tilde{I} + P)^{-1} P \Delta, \quad (1.13)$$

$$V_{\text{samp}}(P) = V_\infty + d^T (\tilde{I} + P)^{-1} \tilde{I} (\tilde{I} + P)^{-1} d, \quad \text{MSE} = V_{\text{samp}} + \text{bias}^2. \quad (1.14)$$

Block-inverting J gives $\Sigma_{ss} = (\tilde{I} + P)^{-1}$, $\Sigma_{cs} = -I_{cc}^{-1} I_{cs} \Sigma_{ss}$, $\Sigma_{cc} = I_{cc}^{-1} + I_{cc}^{-1} I_{cs} \Sigma_{ss} I_{sc} I_{cc}^{-1}$, and $g^\top \Sigma g$ collapses to $V_\infty + d^\top \Sigma_{ss} d$; the bias follows from $g^\top J^{-1} e_s = d^\top (\tilde{I} + P)^{-1}$ and $V_{\text{samp}} = g^\top J^{-1} (J - P_{\text{full}}) J^{-1} g$. Verified to 6.4×10^{-16} (300 random cells) and 4.9×10^{-14} (200 cells); clean-room 6.3×10^{-15} , 6.3×10^{-14} . d measures how much the functional still depends on the shape after the free parameters have re-absorbed all they can. The bias ceiling is $-d^\top \Delta$ however confident the prior (with P nonsingular; a rank-deficient prior pins only $\text{range}(P)$), and if $d = 0$ a shape prior is worth exactly nothing and can do exactly no harm for any P and Δ (12 decimals at $P = 10^9$, $|\Delta| \sim 100$; sampling variance also unchanged)—the formal content of the Heusch/Dew observation.

Law 1.29 (The continuous Ψ ladder). *[established in the dive]* $\Psi(P) = 1 + d^\top (\tilde{I} + P)^{-1} d / V_\infty$; whitening by $\tilde{I}$ and diagonalising,

$$\Psi(P) = 1 + \sum_k \frac{r_k^2}{1 + u_k}, \quad \sum_k r_k^2 = \Psi_0 - 1, \quad \text{and for } P = u\tilde{I}: \quad \Psi(u) = 1 + \frac{\Psi_0 - 1}{1 + u}, \quad (1.15)$$

u_k the eigenvalues of $\tilde{I}^{-1/2} P \tilde{I}^{-1/2}$ and r_k^2 the squared whitened sensitivity in direction k : a prior pays only in directions the functional leaks into.

Exact to 7.9×10^{-16} across $u \in [0, 100]$ on the MMM FIM; precision along a direction with $n^\top \tilde{I}^{-1} d = 0$ leaves Ψ unchanged to 12 decimals at strength 10^8 , while along $\tilde{I}^{-1} d$ it drives $\Psi \rightarrow 1$. The scalar form $\Psi = (1 + u') / (1 + u' - q^2)$, $q^2 = I_{cs}^2 / (I_{cc} I_{ss})$, requires the *free* block to be one-dimensional; otherwise it carries a cosine factor and over-states Ψ by up to 69% (clean-room caveat). On the MMM: $\Psi_0 = 19.804$, $\sqrt{V_\infty} = 0.1187$, $\text{sd}(\text{mROAS}) = 0.5283$ free, $d = (+6.478, -0.229, +0.113)$ against raw $g_s = (+3.226, +0.359, +0.588)$ —re-fitting β amplifies the dependence on adstock—and $r_k^2 = (1.535, 5.256, 12.013)$, all reproduced by the clean-room to 5–6 figures on the exported matrix.

Law 1.30 (Admissibility of a mis-centred prior). *[established in the dive]* For a scalar nuisance and arbitrary g , the penalised estimator beats the flat-prior estimator in frequentist MSE of ϕ iff

$$\Delta^2 < \frac{2}{p} + \frac{1}{\tilde{I}} = 2\tau^2 + \text{sd}_{\text{data}}^2, \quad (1.16)$$

independent of d and V_∞ ($\text{MSE} - V_\infty = d^2 [\tilde{I} + p^2 \Delta^2] / (\tilde{I} + p)^2$ against $d^2 / \tilde{I}$; d^2 cancels); equivalently $k < 2 + u$ with $k = p \mathbb{E}[\Delta^2]$ the factor by which the prior is more mis-centred than it advertises.

The clean-room showed the independence directly (identical root 0.01210018957 for $d \in \{0.01, 6.478, 500\}$, $V_\infty \in \{10^{-4}, 0.0141, 100\}$). As a z-score of the prior–data discrepancy the criterion is $z^2 < (2\tau^2 + \sigma_d^2) / (\tau^2 + \sigma_d^2) \in (1, 2]$: $z < \sqrt{2}$ for a prior much looser than the data, $z < 1$ for one much tighter ($z^* = 1.382 \rightarrow 1.000$ over $u = 0.1 \rightarrow 10^4$). A prior may be 2–5 $\times$ more wrong than it claims and still pay; a calibrated one ($k = 1$) can never hurt, whatever the distribution of Δ . On the MMM ($\tilde{I} = 247.9$, $\text{sd}_{\text{data}}(\alpha) = 0.0635$) the break-even $|\Delta_\alpha^*|$ is 0.291, 0.176, 0.110, 0.082, 0.070, 0.064 at $u = 0.1, 0.3, 1, 3, 10, 100$: an adstock prior centred more than 0.064 from the truth (0.44 weeks of half-life at $\alpha = 0.6$) is worse than no prior if tight, and may be off by 0.29 if loose. Round 2 corrected the constant: the posterior-variance derivation gives $1/p$, the frequentist sandwich $2/p$ ($\Delta^* = 0.132 \rightarrow 0.176$ at $u = 0.3$, $0.073 \rightarrow 0.082$ at $u = 3$), so the Bayesian’s own accounting is 25–33% too pessimistic.

Law 1.31 (Direction law). *[established in the dive]* For a rank-one prior of precision p along any direction n ,

$$\frac{\text{Var}_{\text{flat}} - \text{Var}(p)}{\text{Var}_{\text{flat}}} = \text{corr}^2(\phi, n) \cdot \frac{u}{1 + u}, \quad u = p(n^\top I^{-1} n), \quad (1.17)$$

corr being between $g^\top \hat{\theta}$ and $n^\top \hat{\theta}$ under the sampling distribution; invariant to rescaling n ; exact to 3.3×10^{-16} (clean-room 1.7×10^{-11} over 400 cells).

At matched relative strength the value of a prior is proportional to the squared correlation between the functional and the direction constrained (Table 1.6). The ROI direction has $\text{corr} = 1$ by construction, so nothing beats it at matched u : a derivation of Meridian’s reparameterisation with a price. It also says the coefficient prior most stacks regularise is the least valuable of the four directions, below the adstock prior LightweightMMM set for MCMC stability; pinning α exactly captures 44.5% of the available value, K 7.7%, S 4.4% (subadditive, summing to 0.566).

Proposition 1.32 (Half-value precision and the dollar ladder). *[established in the dive]* A shape prior of strength u is worth $V(u) = m\sigma\sqrt{T_r}[\sqrt{\Psi_0} - \sqrt{\Psi(u)}]$, and $V(u_{1/2}) = V(\infty)/2$ at

$$u_{1/2} = \frac{3s + 1}{s + 3}, \quad s = \sqrt{\Psi_0}, \quad \text{so } u_{1/2} < 3 \text{ always.} \quad (1.18)$$

$u_{1/2} = 1.106, 1.188, 1.472, 1.702, 1.929, 2.385$ at $\Psi_0 = 1.5, 2, 5, 10, 20, 100$, with $u_{90\%}/u_{50\%}$ from 9.0 to 15.5 and $u_{99\%}/u_{50\%}$ from 73 to 220: half the value of any shape prior is bought by a prior no stronger than three times the data’s own shape information, and the last 1% costs 100–220 $\times$ more precision than the first half. Get a shape prior roughly right, never tight. On the MMM ($T_r = 156$): free, $\Psi = 19.80$, $L^* = 5.558$ (4.09% of three-year media spend); $u = u_{1/2} = 1.93$, $\Psi = 7.43$, $L^* = 3.404$ (2.50%); $u = 1$, $\Psi = 10.40$, $L^* = 4.028$ (2.96%); shape known, $L^* = 1.249$ (0.92%). A perfect shape prior is worth 3.17% of three-year media spend, a decent one half of that; the figures sit above Dive 20’s because this design identifies four structural parameters from three tones at $T = 156$. Re-optimising the comb’s amplitudes given the prior improves $\text{sd}(\text{mROAS})$ by only 2.4%, 2.5%, 2.9% at $u = 0, 1, 10$: the value of a prior is design-invariant to second order.

Construction 1.33 (Empirical-Bayes widening). *[numerically established]* The observable is $D = \hat{s}_{\text{flat}} - s_0 = \Delta + \text{noise}$, $\text{Var}(\text{noise}) = 1/\tilde{I}$. Rules: FLAT (ignore the prior); FIXED ($p = 1/\tau^2$); TEST (use p iff $D^2 < 2\tau^2 + 2/\tilde{I}$, the plug-in of Equation (1.16)); EB (widen to $\hat{\tau}^2 = \max(\tau^2, D^2 - 1/\tilde{I})$). At $\tau = \text{sd}_{\text{data}}(\alpha)$: worst-case risk relative to FLAT is 25.4 (unbounded) for FIXED, 1.53 (at $1.38\Delta^*$) for TEST and 1.32 (at $1.54\Delta^*$) for EB; the gain captured when the prior is calibrated is 100%, 42.7% and 66.7%.

EB dominates TEST on both axes (clean-room 1.322 vs 1.528, 66.7% vs 42.7%; stable over $u \in [0.25, 10]$, EB 1.185–1.432 against TEST 1.398–1.590). Do not test a prior and discard it; widen it by the observed discrepancy—a two-line change and the threshold Meridian’s shift check lacks (BL 87).

Proposition 1.34 (The decision layer: bias is invisible to a deadband). *[numerically established]* Under Chapter 7’s policy (act only when $|\hat{z} - x|$ exceeds the band) a constant bias b shifts $\hat{z}$ and, once the band is crossed, x by the same amount, so the trigger statistic is unchanged and a biased estimate never causes churn (excess move cost $\leq 10^{-4}$, against up to 0.204 per period from noise). With π the excess regret from a bias b over that from i.i.d. noise of $\text{sd } b$ (24 seeds $\times$ 6000 periods), $\pi = 0.91 \pm 0.40, 0.583 \pm 0.069, 0.291 \pm 0.015, 0.302 \pm 0.004$ at $b = 0.05, 0.10, 0.20, 0.40$; the act-always power control returns 0.997 ± 0.003 throughout.

Bias is 2–3.4 $\times$ cheaper than equal-MSE variance under a deadband, so the decision-admissible miscentring is $\Delta^2 < (2\tau^2 + \text{sd}_{\text{data}}^2)/\pi$, 1.3–1.9 $\times$ wider in Δ —derived, not verified in the nonlinear Hill loop.

1.5 Numerical evidence

Dive 01 computes finite-difference FIMs at θ_0 over named designs at $T = 156$ (mean spend 1; two-level blocks alternate 1.8/0.2) and checks them against 120 multistart least-squares MLE fits (Table 1.1). The FIM story holds end to end: constant spend leaves the curve unrecoverable (median maximum curve error 0.20, a fifth of the ceiling), the $\pm 10\%$ sine diverges ($\text{RMSE}(K) = 32$), $\text{AR}(1)$ is mediocre ($\text{RMSE}(S) = 0.38$), blocks reach $\text{RMSE}(\beta, K, S, \alpha) = (0.021, 0.087, 0.086, 0.011)$ with curve error 0.012.

Dive 20 works at $T = 364$ with intercept, trend and three annual harmonics, 10% sinusoidal probes, all designs cost-matched by exact expected profit loss. Its clean-room reproduced all eight numbered claims

Table 1.1: Dive 01: practical identification by design at θ_0 , $\sigma = 0.05$, $T = 156$ unless stated. $\lambda_{\min}$ is the smallest FIM eigenvalue; standard errors are CRLB, RMSE from 120 MLE fits (no Monte Carlo standard errors reported).

Design	$\lambda_{\min}(I)$	Precision statement	Exp.
constant spend	0 (rank 1)	CRLB se 10^4 – 10^6 ; MLE curve error 0.20	A, D
sine $\pm 10\%$, period 8	0.0028	$\lambda_{\min} \propto \delta^{3.99}$; MLE RMSE(K) = 32	B, D
2-level blocks, $L = 1 / 8 / 78$	0.65 / 99 / 3.1	D-criterion peaks at $L = 8$ – 10	C
2-level, $L = 13$: settled 84 / transient 84 / all 156	0.079 / 53.4 / 85.2	$\sim 700\times$ collapse without the transient	G
blocks scaled to max $a/K = 0.29$	10^{-3} (from 10^2)	ridge eigenvector $(-0.74, -0.66, 0.07, 0.00)$	E
2-level blocks, MLE	—	RMSE (0.021, 0.087, 0.086, 0.011); curve 0.012	D
annealed, $T = 96$	—	se (0.022, 0.065, 0.115, 0.009); D-crit $5.0\times$ AR(1)	F
AR(1) lognormal, $T = 96$	—	se (0.087, 0.190, 0.453, 0.037)	F

(five constants; the cost law; the frontier 0.099177–0.099181 across periods 4–182; both spectral values and Monte Carlos; four leakage pairs; the argmin; 0.000/0.048/16.65/2616.4 for the non-identifying plan; 662.6 vs 22.16, ratio 29.9, for concentration). Table 1.2 is the dive’s most actionable table and Table 1.3 its economics. The concentration claim was re-explained: the $30\times$ gain from a 28-week burst is not all the E^2/T_e law (which predicts $364/28 = 13$); decomposing $I(c_2)$ into a windowed- 2ω component (277.2, ratio 12.5) and a burst-envelope component (385.4) reveals a window-contrast term that exists only when $T_e < T$ (BL 78).

Table 1.2: Dive 20, E6: probe periods annihilated by each nuisance basis (fraction of c_1 -information retained in parentheses where reported). A flexible baseline is a low-pass nuisance filter with cutoff about twice the knot spacing; week-of-year dummies are a comb filter.

Nuisance basis	Probe periods annihilated
Fourier $k \leq 3$ + trend	52 only (and ≥ 182 partially: 0.85 retained)
Fourier $k \leq 6$	also 13, i.e. quarterly flighting
B-spline baseline, 8 knots	≥ 91 (0.16, 0.001 retained)
B-spline baseline, 52 knots	28 (0.000) and half of 14
week-of-year dummies	every period dividing 52: 4, 13, 26, 52 all 0.000

Table 1.3: The price of identification per channel over five years ($T_r = 260$): Dive 20 §5.3 (one channel, share of its own spend; C^* probing cost, $L^* = 2C^*$ total) and Dive 21 S16 (channel-count tax: L^* for n channels as a share of a fixed six-channel budget—a different denominator).

Knowledge state	Ψ	C^* (1 ch.)	L^* (1 ch.)	L^* , 1 of 6	6 ch.	20 ch.
fully calibrated (α, K, S known)	1.00	0.36%	0.71%	0.09%	0.53%	1.75%
local shape free (c_1, c_2, α)	1.06	0.37%	0.73%	—	—	—
full Hill free (β, K, S, α)	9.87	1.12%	2.23%	0.28%	1.65%	5.50%

Dive 21 runs sixteen sections at $T = 364$ with the six-channel working point (amplitudes 5–40%, collision orders 2, 3, 5, nine adstock pairs) and a nonlinear Monte Carlo (S15: 150 replications $\times$ 4 cost-matched designs, calibrated-shape MMM with β_j, α_j free). Table 1.4 prices budget neutrality and Table 1.5 the collision. The Monte Carlo reframed the dive: two octave collisions cost $1.03\times$ (max 1.29) in median RMSE against the compliant comb, random phases $0.96\times$, the synchronised 13-week flight $11.2\times$ (max 20.2)—the last a lower bound manufactured by the optimiser’s damping and start at the truth, the statistical truth being non-identification. The harmonic-collision penalty is exactly the price of not knowing the saturation curve: it lives inside the Ψ ladder.

Dive 22 runs sixteen experiments, two robustness sweeps and four controls on the three-tone comb at $T = 156$; Table 1.6 is the direction law and Table 1.7 the estimability audit that discharges BL 82 for dives 20–22. Against a fully nonlinear bounded-MLE Monte Carlo ($n = 150$) the FIM is accurate with the shape

Table 1.4: Dive 21: budget-neutral against total-budget probing of the aggregate marginal value at a 13-week period, in frontier units $\Lambda = C \cdot \text{Var} / (\frac{1}{2}\kappa\sigma^2)$; a single channel's own coefficient sits at $\Lambda \approx 3.4$.

(α_1, α_2)	Λ_{neutral}	Λ_{total}	ratio
(0.60, 0.30)	71.5	3.96	18×
(0.80, 0.20)	17.1	4.21	4×
(0.70, 0.50)	84.7	2.85	30×
(0.90, 0.60)	24.8	2.38	10×
(0.65, 0.60)	1 196	2.63	454×
(0.50, 0.45)	2 193	3.57	615×

Table 1.5: Dive 21: the octave collision (bins 19 and 38, 10% amplitude) under the full Hill FIM (S4, converged 4001-point sweep; se of mROAS_B relative to a clean comb) and under the nonlinear calibrated-shape Monte Carlo (S15, median RMSE ratio to the compliant comb; MC standard error about 6%).

Design / phase choice	FIM se ratio (shape free)	MC RMSE ratio (shape calibrated)
naive $\phi_B = 0$	15.53×	—
random phase (median, IQR)	7.97× (3.7–15.0)	0.96× (compliant comb, random phases)
closed-form quadrature $\phi_B = 0.9650$	4.27×	—
numerical optimum $\phi_B = 1.5267$	1.085×	—
comb with two octave collisions	—	1.03× (max 1.29)
synchronised 13-week flight	singular (nullity 13)	11.2× (max 20.2)

pinned (sd(mROAS) 0.1187 vs 0.1236 ± 0.0072) and badly optimistic with it free (0.5283 vs 0.9166 ± 0.0531), so the implied Ψ_0 is 55.0 against the FIM's 19.80: the ledger under-states a shape prior's value $2.8\times$, though the fraction of budget a perfect prior saves, $1 - 1/\sqrt{\Psi_0}$, moves only from 0.775 to 0.865 (BL 88). The linear bias law over-states harm: the exact pseudo-true bias is 0.954, 0.939, 0.908, 0.817, 0.672 of the linear ceiling at $\Delta_\alpha = 0.005, 0.01, 0.02, 0.05, 0.10$, because saturation absorbs a wrong adstock (pseudo-true K collapses from 1.467 to 0.200), relocating the damage to the response curve away from the operating point—where re-allocation reads it, and unpriced (BL 85); a penalised nonlinear Monte Carlo ($n = 200$, $\Delta_\alpha = 0.05$) gives bias -0.150 ± 0.020 vs -0.157 predicted at $u = 1$ and -0.230 ± 0.009 vs -0.285 at $u = 10$. An experiment-derived prior with transport offset $t = 0.10$ in α (Chapter 2) still pays at $u = 1$ (MSE 0.252 vs flat 0.279) and harms at $u = 10$ (0.385). Controls: $d = 0$ gives $\Psi = 1.000000000000$ and bias $\leq 3 \times 10^{-14}$; the Monte Carlo signs the value correctly at $\pm 10\%$ of break-even ($z = -16.2, +18.4$) and is indecisive at it ($z = -1.97$). Ψ_0 is exactly σ -invariant (28.13 at $T = 104$ for $\sigma = 0.02, 0.05, 0.10$ alike), $u_{1/2}$ moves only over $[1.93, 2.04]$ across nine (σ, T) cells, and single-tone designs are near-singular for a four-parameter shape (condition 10^7 – 10^9 , Ψ_0 in the thousands) and “should not be quoted”.

Table 1.6: Dive 22, the direction law on the MMM: value of a rank-one prior at 20% relative standard deviation by the direction it constrains ($T_r = 156$; every entry reproduced by the clean-room to 4–5 significant figures).

Prior placed on	corr(mROAS, ·)	L^* multiplier	budget saved	value, % of 3-yr media spend
coefficient β	+0.309	0.951	4.9%	0.20%
adstock α	+0.756	0.935	6.5%	0.27%
saturation K	−0.340	0.941	5.9%	0.24%
ROI / mROAS	+1.000	0.439	56.1%	2.29%

The two-parameter ladder is a hard number ($\pm 1\%$ at 10^{-3} FIM error); the four-parameter $\Psi_0 = 19.8$ is a $\pm 35\%$ quantity requiring the Fisher matrix to about six significant figures, and a longer record does not fix it. The direction law, a ratio, is robust: the ROI multiplier stays at 0.439 [0.398, 0.512] under 10^{-4} jitter and 0.456 [0.440, 0.473] under 10^{-3} , while the coefficient-prior multiplier degrades to [0.844, 1.160].

Table 1.7: Dive 22 §6.4, the estimability audit (BL 82): Ψ_0 recomputed under relative jitter of every FIM entry (2000 draws per cell; 5–95% range; “fail” = draws with no finite Ψ_0).

Model	$\text{cond}(I)$	Ψ_0	jitter 10^{-6}	jitter 10^{-4}	jitter 10^{-3}
2-param (β, α) , $T = 156$	40.6	13.01	[13.01, 13.01]	[13.00, 13.03]	[12.88, 13.15]
4-param, $T = 156$	3.77×10^5	19.80	[19.77, 19.85]	[12.95, 26.82], 2% fail	[16.57, 19.86]
4-param, $T = 312$	5.10×10^5	23.95	[23.81, 24.10]	[9.50, 37.22], 4% fail	[16.67, 22.12]

1.6 Limitations and the devil’s advocate

Dive 01: no controls, seasonality or trend (repaired by Dive 20); single channel (Dive 21); a local analysis whose D-optimal design depends on the unknown θ , so a Bayesian or minimax design is the production answer; geometric adstock and exact Hill assumed, so the identified set is within-model; a theory of designed, exogenous perturbation that does not fix endogeneity of historical spend (geo differencing does, and the two compose); a δ^4 law at a single operating level. The dive predates the attack-and-refine protocol, so none of its numbers carries an uncertainty.

Dive 20: the hyperbola is second-order in amplitude and asymptotic in record length, exact only for commensurate periods, and prices information rather than any estimator’s RMSE. The price theorem’s concept is Little’s and Feit–Berman’s; only the object is new, and it assumes stationarity and a single terminal decision—drift caps T_r and raises the amortised price, a deadband makes regret piecewise and lowers it, neither computed (BL 76). The plateau holds at a well-run channel near its optimum: the operating-point sweep shows Ψ -type inflation rising from 1.8 to 21 as spend goes from $0.75\times$ to $3\times$ optimal, without a re-run of the full frequency profile. Rule L’s constant is calibrated to three harmonics plus intercept and trend at $T = 364$; more nuisance lines widen it like $\sqrt{\text{lines}}/(\pi\sqrt{\epsilon})$. Probes are sinusoidal, not non-negative and quantised (BL 77); Ψ and $S_\epsilon(\omega)$ are design inputs nobody measures (BL 74); the go-dark comparison rests on unsourced rules of thumb.

Dive 21: additive channels only—synergy puts intermodulation at $\omega_i \pm \omega_j$ and is uninvestigated (BL 79); white residuals, so bin assignment is not yet an optimisation (BL 80); geometric adstock, whose real transfer function at DC makes the budget-neutral null exact, and a delayed-peak kernel may break it (BL 84); a numerical rather than closed-form phase optimum (BL 83); phase management matters “in proportion to $\Psi - 1$ ”; no real calendar audited (BL 81). Dive 22: the four-parameter ladder is numerically fragile; the FIM under-states the prior’s value $2.8\times$; the bias law is linear and conservative, and the relocated curve damage is unpriced; Δ is unobservable and EB’s safety margin inherits any error in $\tilde{I}$; everything assumes the model class is right—Heusch’s observational MMM reporting ROAS $10.6\times$ against a truth of $4.2\times$ with oracle controls is beyond any prior’s help; the decision layer is a reduced form; single channel. Its sharpest self-criticism: the $11.4\times$ ROI-versus-coefficient comparison equalises *relative* spread, not a neutral yardstick across parameters; the scale-free statement is the direction law, and $\text{corr}(\text{mROAS}, \beta) = 0.309$ is a property of this design at this operating point, though the ordering cannot change.

1.7 Discussion: impacts

For practice the arc replaces a diagnosis (“insufficient variation”) with a design theory and a price list. The design content is short: pulse in blocks of about three settling times with the transient kept; push adstock past the knee; put the probe on a DFT bin away from every seasonal harmonic and off every period the baseline spline or dummy set annihilates; give each channel its own bin; set phases by a line search; never synchronise channels on one flying clock; buy the aggregate marginal value with one non-neutral total-budget tone rather than hoping budget-neutral reallocation reveals it. The price content is that a continuous low-amplitude probe costs a fraction of a percent of media spend per channel and the bill scales with the number of channels, not their size—a quantitative form of the barrier that measurement is too

expensive for fragmented mid-size advertisers. The two negative results are the most consequential: the profit-optimal plan under seasonal demand accrues information at rate zero, and synchronised flighting makes the multichannel Fisher matrix exactly singular, with `pinv` manufacturing finite standard errors from it.

For vendors the results land on specific features. Meridian’s ROI reparameterisation is derived with a price: an ROI prior of 20% relative spread is worth $11.4\times$ a coefficient prior of the same spread, and the coefficient priors most stacks regularise are the least valuable direction; its prior–posterior shift diagnostic gets a threshold and a better rule than a test. LightweightMMM’s stability prior on decay sits on the second-best direction. Robyn’s evolutionary search over decay and saturation hyperparameters walks exactly the ridge of Law 1.5. Every vendor’s residual spectrum decides whether the right flighting period is 84 weeks or 11, and the audit of Table 1.7 should accompany any FIM- or Hessian-derived interval. For experimentation platforms, ramp-up weeks are the curvature information (Proposition 1.8) and a 24% probe reads saturation off the second harmonic with no curve fitting (Construction 1.24). For CMOs the message is the ladder: a shape-fixing experiment is worth $\sqrt{10}$ on a perpetual identification budget, and a shape prior worth 3% of three-year media spend is bought half-price at $u \approx 2$.

For theory the chapter connects outward. The transient result underlies Chapter 2’s rank-4 trajectory calibration, and the operator likelihood there is where a shape-fixing experiment enters. The plateau of Chapter 3 reappears twice, in the frequency domain and in the deflation of the collision penalty under a calibrated shape. The non-identifying plan is the static counterpart of the stall of Chapter 13, whose BL 70 the rank theorem prices. The deadband of Chapter 7 makes bias $2\text{--}3.4\times$ cheaper than variance. The leakage machinery of Chapter 9 is the same nuisance projection, and the van Trees identity is the correct frontier whenever a prior is in play.

1.8 Further exploration

The dives’ next steps became fourteen backlog items. From Dive 20: estimate Ψ and S_e from a fitted model (BL 74); a drift-capped horizon turning the one-off budget into a standing probing rate (BL 76); planner-feasible crest-factor-constrained probes (BL 77); the window-contrast term in curvature information (BL 78). From Dive 21: intermodulation under synergy (BL 79); bin assignment as an optimisation with coloured residuals and per-channel Ψ_j (BL 80); audit a real flighting calendar for rank and null-space leak (BL 81); the estimability rule (BL 82); a closed form for the full-model optimal phase from $\partial \arg H / \partial \alpha$ (BL 83); delayed-peak adstock and the DC null (BL 84). From Dive 22: the response-curve estimand for prior pricing, where the ROI prior’s advantage is predicted to shrink because ROI at the current point does not pin the curvature that sets the optimum (BL 85); the multichannel ledger (BL 86); the EB widening rule as a shipped diagnostic (BL 87); the second-order correction to Ψ_0 that would close the 19.8-versus-55 gap (BL 88). Both later dives also ask for a join to Chapter 3’s EVSI: the $\sqrt{10}$ of the ladder, the 3.17% of a perfect shape prior and the $1.09\times$ -versus- $15.5\times$ of phase management are one quantity computed three ways, and reconciling them would close BL 7’s cost side.

In the compiler’s judgement the most promising extension is the one the INDEX also ranks first: compute $\tilde{I}$, d and Ψ_0 with the jitter audit on a real fitted MMM’s Hessian (BL 81+BL 74+BL 86). It needs no new theory, tests the chapter’s two most checkable predictions at once—that common flighting calendars are near-singular and that real advertisers’ d is dominated by adstock—and, since every dollar figure here inherits an unaudited Ψ , it decides whether the price list is a number or a shape. Second is BL 85, which may overturn the ROI-prior ranking: if a wrong adstock is absorbed by saturation, the damage a prior does is read by re-allocation, not by mROAS at the current point, and the direction law has not been evaluated for that functional.

Reviewer’s note (compiler): estimability and the $\pm 35\%$ caveat

What BL 82 means here. Dive 01’s numbers are eigenvalue statements and unaffected, except the CRLB standard errors of Construction 1.9, obtained by inverting $I + 10^{-12} \text{Id}$ with no condition number reported. Dive 20’s Ψ entries of 10^{27} , 10^{16} and 3×10^9 are inverses of singular FIMs and should be read as “singular”. More seriously, Dive 20’s $\Psi_4 \approx 9.87$ —the source of the 1.12%, the $\sqrt{10}$ and the $3.1 \times$ —is a single-tone four-parameter FIM quantity at $T = 364$ that was never audited, while Dive 22 §6.5 states that single-tone designs are near-singular for a four-parameter shape (condition 10^7 – 10^9 , Ψ_0 “in the thousands”) and “should not be quoted”; at $T = 156$ its single-tone Ψ_0 is 2371 at a 13-week period and 1283 at 21 weeks against Dive 20’s 10.7 and 10.1 at $T = 364$. The functionals differ (Dive 22 anchors at $0.85 \bar{\pi}^*$), as do T and amplitude, but the $\Psi_4 \approx 10$ row and everything priced with it should be treated as unaudited. Dive 22’s own $\Psi_0 = 19.8$ is $\pm 35\%$ under 10^{-4} relative FIM error with 2% outright failures, and its nonlinear Monte Carlo gives 55.0; only the direction-law ratios survive. Every absolute dollar figure in Parts B–D is therefore conservative by an unquantified amount (BL 88) and uncertain by at least a third.

Reviewer’s note (compiler): refutations; re-derivations; consistency checks

Refuted and re-derived within the dives. Dive 20: the closed-form VIF (Proposition 1.12); its “resolution” re-explanation of Dive 01’s block length; Rule L’s constant $1 \rightarrow 1.15$; the concentration mechanism (12.5 of 29.9 from the law). Dive 21: $-6/T^2 \rightarrow -6/(T^2 - 7)$; four tones $\rightarrow$ two; the grid maximum $332 \times$; the quadrature angle ($4.27 \times$ vs $1.085 \times$); 700 – $2400 \times \rightarrow$ exactly singular. Dive 22: the admissibility constant $1/p \rightarrow 2/p$; the scalar q^2 form needing $\dim c = 1$; the bias ceiling needing P nonsingular; and a fabricated FIM in the dive’s own verification prompt, caught by the clean-room’s check $g^T I^{-1} g = \Psi_0 V_\infty$ (3.222 vs 0.279).

Plateau versus phase. Dive 20 says frequency within the admissible band is second-order (Ψ_4 flat within 10%); Dive 21 says placing one channel at a multiple of another’s bin, and the phase, is first-order ($15.5 \times$). They are consistent only because both are conditioned on Ψ : the collision penalty deflates to $1.03 \times$ at $\Psi = 1$, and Dive 20’s plateau is a single-channel statement at one operating point whose own E10b sweep shows inflation from 1.8 to 21 across operating points. The practical reading is that both dives’ frequency prescriptions are second-order exactly when the shape is calibrated and first-order otherwise.

Internal inconsistencies. Dive 20 §5.1 calls a shape-fixing experiment “worth a factor of ten in the perpetual identification budget”; $L^* \propto \sqrt{\Psi}$, so it is $\sqrt{10} \approx 3.1$, as the INDEX line and §5.3 say. The $\Psi = 4.599$ used to verify the argmin appears in no table. Dive 21 §3.6 gives the modulus-estimator bias as a mean of -0.563 , §8 and the INDEX as $+0.561$, against a truth of -0.157 . Dive 21 reports the random-phase median as 7.97 (§4.1) and 8.028 (§8); the null-space leak as “up to 0.821 and 0.694” (§5.1), “0.19–0.78” (§8) and “0.19–0.82” (INDEX). The INDEX log line for Dive 21 states Rule R as “free iff $\gcd(T/52, 6) = 1$ ”, whereas the writeup narrowed it to sufficient only. Dive 22’s merged `22_results.json` carries in its `e4` block $\Psi_0 = 3.755$, $d = (+3.00, +0.52, -0.33)$ and condition 6.0×10^4 —the single-tone 26-week design of the `e11` sweep—not the comb numbers the writeup quotes (19.80; $+6.478, -0.229, +0.113$); 3.77×10^5 , which appear in `e6`, `e10`, `e12`, `e14`, `e16`; the `e7` header likewise shows $\bar{I} = 0.7985$ against the writeup’s 247.9, and the power control’s break-even $\Delta^* = 1.938$ (§6.3) is from that other cell, not the MMM’s 0.110. INDEX log times for dives 01, 20 and 21 ($\sim 19:00$, $\sim 09:55$, $\sim 11:20$) precede the files’ last writes (22:08, 17:20, 19:17 UTC) by three to eight hours.

Denominators. Dive 21’s channel-count table divides nL^* by the fixed six-channel budget (7.08 per week $\times 260$; reproduced from the code), so its “1 channel: 0.09%” is not comparable with Dive 20’s “0.36%”, which is C^* (not L^*) over that channel’s own spend; Table 1.3 keeps both. Dive 22’s 0.92–4.09% use $T_r = 156$ and a three-tone design.

Under-supported. Dive 01’s $5.0 \times$ annealing gain and $1.23 \times$ over three-level blocks are single runs; its Monte Carlo has no standard errors; nothing in Part A was independently re-derived. Dive 20’s crossover ($84 \rightarrow 12$ weeks between $\rho = 0.5$ and 0.7) sits inside a total-loss spread of 1.0 – $2.6 \times$. Dive 21’s $0.96 \times$ for random phases is within Monte Carlo error of 1. Dive 22’s deadband widening factor and transport-offset result are derived, not verified in the nonlinear loop, and EB was compared under a Gaussian noise model with the correct $\bar{I}$; the “327-dataset calibration study” it mentions is explicitly unverified and not used here.

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- [24] Program (internal): Dive 02, dives 03/04, 08/09, 11, 16/19; 00_INDEX.md log rows 01, 20, 21, 22 and backlog entries BL 1–BL 2, BL 70, BL 73–BL 88.

Chapter 2

Transporting lift tests into MMM parameters

Provenance

Source dives: Dive 02 *A formal transport map from lift tests to MMM parameters (M2)* (run 2026-08-31, file 02_experiment_transport_map.md, last written 2026-08-31 23:14 UTC). **Code:** code/02_experiment_transport_map.py. **Frontier-study problem(s) addressed:** M2 — write the Pearl–Bareinboim selection diagram for experiment-to-MMM calibration, derive when the experimental functional identifies the model functional, characterise the correction terms (adstock truncation, saturation-point mismatch), and handle several experiments as a joint, correlated posterior over the ROAS vector. **Backlog items closed:** none; **opened:** BL3 (calibration under misspecification), BL4 (a specification test for transport).

Key ideas in this chapter

- In the selection diagram for experiment-to-MMM transport the effect of spend is *not* nonparametrically transportable; every calibration rests on a parametric *shape-invariance* assumption, under which transport is mediated by θ and nothing licenses the scalar map iROAS $\rightarrow$ model ROI.
- Exact identity $\text{iROAS}(d) = m_\infty(\theta) \rho C Q$ with closed-form window capture $C = 1 - \alpha^{P+1}(1 - \alpha^{T_e}) / (T_e(1 - \alpha))$ (verified to 9×10^{-7}): a 4-week test on an $\alpha = 0.8$ channel analysed to test end observes 41% of the sales it caused.
- Naive calibration is *precisely wrong*: iROAS as a prior on marginal ROAS gives -43.0% median bias (theory -43.8%) at the tightest spread of any mode (IQR 0.016), and flips to $+67.3\%$ when the test runs in geos at half the national spend.
- The *operator* likelihood $\hat{L} \sim \mathcal{N}(\mathcal{E}_d(\theta), \text{se}^2)$, which simulates the experiment’s design inside the model, needs no correction factors, halves the uncalibrated curve error, wrecks no fits and has $65\times$ lower median decision loss than the ROI-prior mode.
- A scalar iROAS carries rank-1 Fisher information; the per-period trajectory of the same experiment carries rank 4 (standalone SE 0.20 on α); joint correlated posteriors over many experiments come free from entering each as a likelihood row on the shared θ .

2.1 Problem and motivation

An incrementality experiment measures a finite-horizon intention-to-treat effect of a specific spend change, at a specific operating point on the response curve, in a specific subpopulation, over a specific window. An MMM parameter—an average or marginal return on spend—is a steady-state structural quantity for the target population. Calibration practice equates the two: the experiment’s incremental ROAS (iROAS) and its standard error become the mean and standard deviation of a prior on the model’s ROI. Practitioners document the symptom (“reconciliation headaches rather than trust-building calibration” when the experiment’s KPI, geography or window differ from the model’s; warnings that experiment-derived priors “transfer bias across non-comparable populations”) without the mechanism. Problem M2 asked for the mechanism: instantiate the Pearl–Bareinboim transportability calculus, derive the exact map between the experimental and model functionals, characterise the correction terms, and say how several experiments should enter one

posterior.

Dive 02 answers all four parts in the single-channel adstock/Hill model of Chapter 1: a negative theorem, an exact three-factor identity with a closed form for its largest factor, a Monte Carlo demonstration that the standard shortcut is biased by tens of percent while *appearing* more certain than any alternative, and a constructive fix—calibrate on the experiment’s design, not on a renamed scalar—that is the marketing form of indirect inference. The dive predates the program’s attack-and-refine protocol (see the reviewer’s note).

2.2 Background: the ideas this chapter builds on

2.2.1 Selection diagrams and transportability (Pearl and Bareinboim, 2011, 2014, 2016)

Transportability asks when a causal relation learned in a source population Π (where an experiment ran) can be carried to a target Π^* (where only observational data exist). Starting from the causal diagram G the two share, a *selection diagram* adds a selection node $S_i \rightarrow V_i$ for every variable whose generating mechanism may differ between populations; the absence of an S -node asserts invariance. Writing $P^*(v) = P(v \mid s^*)$, the target effect $P^*(y \mid do(x))$ is *transportable* if it is uniquely computable from the source experimental distribution $P(y \mid do(x))$ and target observational data $P^*(v)$. The simplest sufficient condition is *s-admissibility*: if pre-treatment covariates Z satisfy $(Y \perp S \mid X, Z)$ in $G_{\bar{X}}$ (arrows into X removed) then

$$P^*(y \mid do(x)) = \sum_z P(y \mid do(x), z) P^*(z), \quad (2.1)$$

the *transport formula*: experimental conditional effects re-weighted by the target’s covariate distribution. The complete characterisation (Pearl and Bareinboim, 2011, 2014) is that $P^*(y \mid do(x))$ is transportable if and only if $P(y \mid do(x), s)$ can be reduced by do-calculus to an expression in which S appears only in do-free terms; the 2016 PNAS paper extends this to fusing many sources. The negative side matters here: if an S -node points directly into the outcome Y and nothing observed screens it off, no derivation exists and the effect is not transportable without parametric assumptions.

2.2.2 The geometric-adstock, Hill-saturation MMM (Jin et al., 2017)

The Bayesian MMM modern packages descend from models a KPI as baseline plus, per channel, a *carryover* transform followed by a *shape* transform of spend x_t . Carryover is geometric adstock, $a_t = x_t + \alpha a_{t-1}$ (a unit of spend contributes α^l after l periods); the shape is the Hill function

$$h(a; K, S) = \frac{a^S}{a^S + K^S}, \quad (2.2)$$

with half-saturation K and slope S , convex below its inflection $a^+ = K((S-1)/(S+1))^{1/S}$ and concave above; the channel contributes $\beta h(a_t)$ with β the sales ceiling. Jin et al. show that with weekly national data the likelihood is flat along ridges in (α, K, S) and posteriors are prior-dominated (Chapter 1 gives the mathematics). At steady spend $\bar{x}$ the adstock settles at $\bar{a} = \bar{x}/(1-\alpha)$; the business consumes the *average ROAS* $\beta h(\bar{a})/\bar{x}$ and the *steady-state marginal ROAS* $m_\infty = \beta h'(\bar{a})/(1-\alpha)$, the long-run sales per extra unit of sustained spend.

2.2.3 What “calibrating to a lift test” means in practice

Meridian (Google) follows Zhang, Wurm et al. (2024): the model is reparameterised so the prior sits not on the coefficient β_m but on the channel’s ROI (incremental KPI over the data window divided by spend), β_m becoming a deterministic function of ROI and the other parameters; a lift experiment supplies the mean and standard deviation of a lognormal ROI prior (an mROI variant, the return to a small spend increase, is also offered). The documentation concedes “there is no single formula to translate an experiment result into a

prior". *Robyn* (Meta; Runge et al., 2024) is a ridge MMM whose adstock and saturation hyperparameters come from an evolutionary search over a Pareto front; calibration enters as a third objective, the discrepancy (MAPE.LIFT) between the model's decomposed channel contribution over the experiment's date window and the experiment's absolute lift. *PyMC-Marketing* holds the strongest prior art: a lift test with baseline x and change Δx enters the posterior as a Normal likelihood on the *static saturation secant* $\beta[s(x + \Delta x) - s(x)]$ —exact on curvature, but with no adstock dynamics, window truncation or population selection (Ordutz's simulation study works in the same spirit).

2.2.4 Indirect inference (Gouriéroux, Monfort and Renault, 1993)

When a structural model can be simulated but its likelihood is awkward, indirect inference fixes an *auxiliary* statistic $\hat{\beta}$, defines the *binding function* $b(\theta)$ as the value that statistic takes on data simulated from the model at θ , and sets $\hat{\theta} = \arg \min_{\theta} (\hat{\beta}_{\text{obs}} - b(\theta))^T \Omega (\hat{\beta}_{\text{obs}} - b(\theta))$. The auxiliary quantity is never *equated* with a structural parameter, even one of the same name; the model is asked what value of the statistic it would produce, and that is matched. Under misspecification the estimator converges to a pseudo-true value.

2.2.5 The adstock transient as a dose-ranging sweep (Chapter 1; Heusch, 2026)

Dive 01 found that a two-level spend design identifies all four parameters through its *transitions*: each step drags a_t along a known exponential path, tracing an arc of the Hill curve, and discarding transient observations collapses the smallest Fisher eigenvalue about $700\times$. Heusch (2026) recovers adstock, saturation and effectiveness structurally from the temporal variation a geo experiment induces by the same fact. Proposition 2.10 applies the observation to what an experiment *reports*.

2.3 Setup and assumptions

The model is that of Chapter 1: $y_t = \beta h(a_t; K, S) + \varepsilon_t$, $a_t = x_t + \alpha a_{t-1}$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, $\theta = (\beta, K, S, \alpha)$, working point $\theta_0 = (1, 1.5, 2, 0.6)$, national mean spend $\bar{x} = 1$ (so $\bar{a} = 2.5$), $\sigma = 0.05$. Subscripts $\star$ and e mark the target domain $\Pi_\star$ (national, business-as-usual spend) and the experiment domain Π_e (a subpopulation over a window).

Definition 2.1 (Design and functionals). A design is $d = (\bar{x}_e, \{\delta_t\}_{t < T_e}, W = T_e + P)$: baseline spend $\bar{x}_e$ in the experiment geos, a pulse δ_t for T_e periods, and a window of W periods extending P periods past the pulse. The *experiment functional* is the model-implied total incremental KPI over the window,

$$\mathcal{E}_d(\theta) = \beta_e \sum_{t \in W} [h(a_t(x_e + \delta)) - h(a_t(x_e))], \quad \text{iROAS}(d) = \frac{\mathcal{E}_d(\theta)}{\sum_t \delta_t}, \quad (2.3)$$

where x_e is the baseline spend path (constant at $\bar{x}_e$) and adstock starts at its steady state $\bar{x}_e / (1 - \alpha)$; the experiment reports an estimate $\hat{L}$ of $\mathcal{E}_d$ with standard error se . The *model functionals* are $m_\infty(\theta) = \beta_\star h'(\bar{a}_\star) / (1 - \alpha)$ and the average ROAS $\beta_\star h(\bar{a}_\star) / \bar{x}_\star$. Δa_t and Δh_t denote the adstock and Hill differences the pulse induces.

Assumption 2.2 (Shape invariance). (K, S, α) are shared between Π_e and $\Pi_\star$; only β (and seasonality/demand) may differ.

Assumption 2.3 (Internal validity and model family). $\hat{L}$ is unbiased for the intention-to-treat effect under $do(X)$ (no contamination or spillover); adstock is geometric and, for the closed form of C , the pulse is uniform.

2.4 Main results

2.4.1 No nonparametric transport: all calibration is parametric

The shared graph is $X_t \rightarrow A_t \rightarrow Y_t$ with an unobserved demand state $U \rightarrow X_t, U \rightarrow Y_t$ (severed by $do(X)$ in Π_e), plus three selection nodes: $S_\beta \rightarrow Y$ (effectiveness differs by geo), $S_x \rightarrow X$ (operating point differs), $S_g \rightarrow Y$ (seasonality and demand differ).

Proposition 2.4 (Non-transportability). *[established in the dive] In this selection diagram the causal effect of spend on sales is not nonparametrically transportable from Π_e to Π_* : no do-calculus derivation of $P^*(y \mid do(x))$ from $P(y \mid do(x))$ and $P^*(v)$ exists.*

S_β points directly into the outcome mechanism and nothing observed screens it off, so $P(y \mid do(x), s)$ cannot be made s -free. The dive calls this “direct from established theory, but worth stating sharply” because of its consequence: every experiment-to-MMM calibration rides on a parametric invariance assumption, usually the implicit Assumption 2.2. Under it the experiment is informative about θ and all transport is θ -mediated (experiment $\rightarrow$ likelihood on $\theta \rightarrow$ any target functional). Nothing licenses a scalar map from iROAS to a model ROI; the identity below prices that shortcut.

2.4.2 The transport identity

Theorem 2.5 (Transport identity). *[established in the dive] For any design d and any θ ,*

$$\text{iROAS}(d) = m_\infty(\theta) \cdot \underbrace{\rho}_{\text{selection}} \cdot \underbrace{C(\alpha; T_e, P)}_{\text{window capture}} \cdot \underbrace{Q(\delta, \bar{a}_e; \theta)}_{\text{curvature/secant}}, \quad (2.4)$$

with $\rho = \beta_e h'(\bar{a}_e) / (\beta_* h'(\bar{a}_*))$, $C = (1 - \alpha) \sum_{t \in W} \Delta a_t / \sum_t \delta_t$ and $Q = \sum_t \Delta h_t / (h'(\bar{a}_e) \sum_t \Delta a_t)$.

The identity is exact because Q absorbs the remainder by definition; its content is the closed forms and the naming of three mechanisms. The derivation is two lines: $\mathcal{E}_d = \beta_e h'(\bar{a}_e) Q \sum_t \Delta a_t$ and $\sum_t \Delta a_t = C \sum_t \delta_t / (1 - \alpha)$ by the definitions, so $\text{iROAS} = \beta_e h'(\bar{a}_e) C Q / (1 - \alpha) = m_\infty \rho C Q$.

Lemma 2.6 (Window capture, closed form). *[established in the dive] For a uniform pulse of length T_e and a window of $T_e + P$ periods,*

$$C(\alpha; T_e, P) = 1 - \frac{\alpha^{P+1} (1 - \alpha^{T_e})}{T_e (1 - \alpha)}. \quad (2.5)$$

C is the fraction of the pulse’s infinite-horizon adstock response that falls inside the window ($T_e = 1$ gives the folklore $1 - \alpha^{P+1}$); it follows by summing each impulse’s in-window contribution $(1 - \alpha^{W-s}) / (1 - \alpha)$ over $s < T_e$. The magnitudes are the practical point (Table 2.2): a 4-week test on an $\alpha = 0.8$ channel analysed only through test end has $C = 0.41$ —it observes 41% of the incremental sales it caused; at $\alpha = 0.9$, $C = 0.23$. Common practice (short tests, high-carryover channels, no post-period) bakes a 2.5–4 $\times$ understatement of the long-run effect into the “ground truth” before any modelling starts.

Proposition 2.7 (Curvature/secant factor). *[established in the dive] To second order in the pulse,*

$$Q \approx 1 + \frac{h''(\bar{a}_e)}{2h'(\bar{a}_e)} \cdot \frac{\sum_t \Delta a_t^2}{\sum_t \Delta a_t}, \quad (2.6)$$

a transient-weighted generalisation of the static secant, which is the $\alpha \rightarrow 0$ case.

The sign of $Q - 1$ is that of $h''(\bar{a}_e)$: above the inflection $a^\dagger$ the experiment under-states the marginal slope ($Q < 1$); below it, as for a test in low-spend geos, it over-states ($Q > 1$). The expansion is accurate to under 2% for $\delta/\bar{x} \leq 0.25$ but fails for big tests (at $\bar{a}/K = 1.67$, $\delta = \bar{x}$: exact $Q = 0.51$ versus 0.17), so large pulses need exact evaluation. The selection factor ρ is an effectiveness ratio times an operating-point slope ratio: even with $\beta_e = \beta_*$, a test in geos with $\bar{a}_e < \bar{a}_*$ has $\rho > 1$, sitting on a steeper part of the curve.

Example 2.8 (Three numbers a practitioner might equate). At θ_0 with a +50%, 8-week test analysed to test end ($P = 0, \rho = 1$): iROAS = 0.219, $m_\infty = 0.389$, average ROAS = 0.735—a spread of $3.4\times$ in a fully standard setting. By arithmetic from these figures $CQ = 0.563$, and $C = 0.816$ from Equation (2.5), so $Q = 0.69$: truncation and secant contribute comparably.

2.4.3 The operator fix: calibrate on the design, not on a renamed scalar

Construction 2.9 (Operator likelihood). [\[established in the dive\]](#) (Novel for MMM; indirect inference in spirit.) Add to the MMM posterior the likelihood term

$$\hat{L} \sim \mathcal{N}(\mathcal{E}_d(\theta), \text{se}^2), \quad (2.7)$$

where $\mathcal{E}_d(\theta)$ of Equation (2.3) simulates the experiment’s actual design inside the model at every posterior draw: baseline path, delta path, adstock transient, window truncation, experiment-geo operating point.

No correction factors are needed because nothing is transported scalar-to-scalar; C , Q and ρ are computed implicitly and exactly at each θ . In indirect-inference terms the experiment’s estimator is the auxiliary statistic and $\mathcal{E}_d$ its binding function; PyMC-Marketing’s secant likelihood is the static ($\alpha = 0$, full-window, no-selection) special case. The cost is that the experiment report must carry its design, not just (iROAS, se): the *calibration header* of Table 2.1, every field of which the experimenter already has. Reporting the total lift with the design attached makes the result transport-safe; reporting the trajectory upgrades a rank-1 constraint to rank 4 (Proposition 2.10).

Table 2.1: The calibration header: design metadata an experiment must ship for $\mathcal{E}_d(\theta)$ to be computable (Dive 02 §4).

Field	Content and purpose
channel, kpi	model channel id; KPI must match the model’s y
baseline_spend_path	per-period spend in the experiment geos (not just the mean): fixes $\bar{a}_e, \rho$
delta_spend_path	per-period pulse δ_t : fixes Q and, with the window, C
window, post_period_len	start, end, P : fixes the truncation
population	geo ids or share-of-country weights, baseline spend share: fixes selection
estimate	$(\hat{L}, \text{se})$ as total incremental KPI, <i>not</i> iROAS
trajectory (optional)	per-period treated-minus-control with se_t : rank 1 becomes rank 4

2.4.4 The scalar summary discards identifying power; joint posteriors are free

Proposition 2.10 (Rank of the experimental information). [\[numerically established\]](#) At θ_0 , for an 8-week pulse ($\delta = 0.5, \bar{x}_e = 1$) with 6 post-periods and per-period noise 0.05 on the differenced treated-minus-control series, the Fisher information from the full trajectory has rank 4, eigenvalues $(2.7 \times 10^{-5}, 0.059, 7.8, 577)$, and a standalone Cramér–Rao SE of 0.20 for α ; the Fisher information from the scalar iROAS of the same experiment has rank 1.

A single number constrains one direction in θ -space and carries no separate information about α ; the trajectory shows ramp-up and decay (hence α) and its transient traces an arc of the Hill curve. The dive’s corollary: k scalar summaries constrain k directions, so scalar calibration under shape invariance needs at least three well-separated designs plus the observational level to point-identify (β, K, S, α) ; trajectory calibration needs one experiment.

Corollary 2.11 (Joint, correlated posterior). [\[established in the dive\]](#) Because each experiment enters as a likelihood row Equation (2.7) on the shared θ , the posterior over any vector of derived ROAS quantities is automatically jointly correlated: the joint posterior M2 asks for is a consequence of refusing to convert experiments into independent per-channel scalar priors.

Two experiments on one channel at different deltas jointly constrain (K, S) in a way two independent ROI priors cannot express—they would simply conflict, and Section 2.5 shows what conflict does.

2.5 Numerical evidence

Experiment A compares Equation (2.5) with brute-force simulation of an infinitesimal pulse ($\delta = 10^{-6}$) on the grid of Table 2.2— $\alpha \in \{0.3, 0.6, 0.8, 0.9\}$ and six (T_e, P) pairs spanning $T_e = 1$ to 13 and $P = 0$ to 26—with maximum discrepancy 9×10^{-7} . *Experiment B* compares exact Q with Equation (2.6) at $\bar{a}/K \in \{0.5, 1, 1.67, 2.5\}$ and $\delta/\bar{x} \in \{0.1, 0.25, 0.5, 1\}$, giving the accuracy statements after Proposition 2.7. *Experiment D* computes the Fisher matrices of Proposition 2.10 by finite differences of the model-implied 14-period trajectory and of its sum (whose variance is W times the per-period variance).

Table 2.2: Window-capture factor $C(\alpha; T_e, P)$ from Equation (2.5) on the Experiment A grid, where the closed form matches simulation to 9×10^{-7} . Columns are (pulse length T_e , post-period P) in weeks.

α	(1, 0)	(4, 0)	(8, 0)	(8, 4)	(8, 26)	(13, 8)
0.3	0.700	0.894	0.946	1.000	1.000	1.000
0.6	0.400	0.674	0.816	0.976	1.000	0.998
0.8	0.200	0.410	0.584	0.830	0.999	0.951
0.9	0.100	0.226	0.359	0.580	0.959	0.778

Experiment C is the head-to-head of calibration modes. Each of 60 replications draws a weakly identifying national series ($T = 104$ weeks, AR(1) log-spend with about $\pm 10\%$ variation, $\sigma = 0.05$) and one well-powered experiment ($\bar{x}_e = 1$, $\delta = 0.5$, $T_e = 8$, $P = 0$, $se = 5\%$ of the true lift). Four modes are fitted by multistart least squares, the calibration entering as one extra standardised residual: *uncalibrated*; *naive-avg* (Meridian-style, $\widehat{iROAS}$ centring a prior on average ROAS $\beta h(\bar{a})$); *naive-marg* ($\widehat{iROAS}$ centring a prior on m_∞); *operator* (Equation (2.7)). Metrics: median relative bias and IQR of $\hat{m}_\infty$; median maximum error of the fitted response curve over $x \in [0.05, 2.5]$ (units of the sales ceiling); share of “wrecked” replications (curve error above 0.5); median profit loss from allocating at the fitted optimum (margin 3).

Table 2.3: Experiment C, matched operating point: 60 Monte Carlo replications, $T = 104$, one experiment with $se = 5\%$ of lift. Medians; the dive reports no Monte Carlo standard errors.

Mode	m_∞ median bias	m_∞ IQR	med. max curve err.	wrecked reps	med. profit loss
uncalibrated	−7.4%	0.108	0.262	3%	0.0057
naive-avg	+4.0%	0.301	0.228	30%	0.1368
naive-marg	−43.0%	0.016	0.352	12%	0.0212
operator	−3.8%	0.093	0.124	0%	0.0021

Four findings follow. First, theory predicts the naive bias to within a point: with $\rho = 1$, $CQ\rho - 1 = -43.8\%$ against -43.0% observed. Second, naive calibration is *precisely wrong*: naive-marg has the tightest spread of any mode (IQR 0.016, about $7\times$ tighter than the uncalibrated fit) centred 43% off the truth—calibration bought confidence, not accuracy, the quantitative form of the adoption study’s “precision theatre”, produced here by the sophisticated method. Third, naive-avg creates likelihood *conflict*, not bias: the $\widehat{iROAS}$ (0.219) contradicts the observational sales level $\beta h(\bar{a}) = 0.735$ that 104 weeks of data pin down; the data win on the level but the fit escapes along the identification ridge, 30% of replications end with a wrecked curve, and median profit loss is $65\times$ the operator’s. This is the mechanism behind the reconciliation headaches: the two evidence sources are made to disagree by the transport error itself. Fourth, the operator dominates everywhere.

The *operating-point mismatch* variant reruns the design with the test in geos at 50% of national spend ($\bar{x}_e = 0.5$, $\delta = 0.25$). Naive-marg flips to +67.3% bias (theory +72.5%): same pipeline, opposite sign,

depending on where the test happened to run. The operator’s median curve error *improves* to 0.048 (from 0.124 matched): through the correct map an off-operating-point experiment is more informative, because it probes the curve where it bends; through the naive map it is more toxic.

2.6 Limitations and the devil’s advocate

The dive’s own counterarguments are five. *Shape invariance is an assumption, not a result*: if (K, S, α) differ between the experiment geos and the nation the operator transports incorrectly too; unlike the naive map it makes the assumption explicit and testable (two experiments in different strata over-determine the shared shape), but hierarchical β_g only partially absorbs S_β and nothing absorbs an S_K node without multi-stratum experiments (BL 4). *Misspecification transfers*: if the true adstock is delayed-peak and the model geometric, the operator matches the experiment by distorting θ to pseudo-true values; naive calibration is also wrong in that case (BL 3). *Internal validity is taken as given*: contamination, spillover and control-market disruption are upstream. *The closed form for C assumes geometric adstock and a uniform pulse*; non-uniform pulses need the per-impulse sum, and sales-lag dynamics beyond adstock are not modelled. *Experiment C uses MAP-style point fits*, not posteriors; the “precisely wrong” IQR should be re-confirmed with MCMC widths, expected to be starker. No attack-and-refine round changed a result, because none was run.

2.7 Discussion: impacts

For practice the message is that the object a lift test should hand to an MMM is a *design plus a total*, not a number. Every current mechanism loses something: Meridian’s ROI prior transports a scalar into a parameter measuring a different quantity (Example 2.8 shows a $3.4\times$ gap at the working point); Robyn’s MAPE.LIFT compares a windowed contribution but as an objective to trade off rather than a likelihood; PyMC-Marketing’s secant is the operator with $\alpha = 0$ and no window. The operator is a small change to any of them—a forward simulation of the design at each draw—and the calibration header is the interface change that enables it. The identity also lets an experimentation platform report what it has *not* measured, since C follows from its own window and the channel’s retention.

For CMO decision-making the “precisely wrong” finding is the sharper lesson: naive-marg is the mode a careful analyst would prefer on the evidence in front of them—the tightest interval—and it is the most biased. Conversely, the mismatch variant turns a governance liability (tests run wherever politics allows) into an asset under the correct map.

For theory the chapter settles M2’s first three parts in the parametric setting and shows the fourth to be free. Outward links: Chapter 3 uses $\mathcal{E}_d(\theta)$ as the forward map of its value-of-information design chooser and finds the average-ROI level that Meridian-style priors chiefly move nearly worthless for fixed-budget allocation, while Dive 04 finds the rank-4 trajectory information does *not* project onto the cross-channel contrasts that decision needs (decision-value index $0.29 \rightarrow 0.30$): the trajectory identifies the channel, not the allocation. Chapter 4 ingests experiments through the operator in a closed loop; Chapter 5 proposes de-biasing nationally fitted (K, S) before operator transport (BL 17); Chapter 14 would transport a panel-measured memory kernel through the same likelihood (BL 61).

2.8 Further exploration

Of the dive’s six next steps, two became backlog items: the misspecification study (BL 3: delayed-peak truth fitted with a geometric operator, measure the pseudo-true bias, try a sandwich or inflated-se likelihood as a patch) and the specification test (BL 4: two experiments in different geo strata plus a shared-shape model, with a posterior-predictive check as a formal test of shape invariance—the field has no specification test for calibration at all). The others are an `add_experiment_operator()` prototype against PyMC-

Marketing’s API; a multi-channel operator able to express the cross-channel contamination (treated-geo budget reallocations) a scalar iROAS cannot; feeding $\mathcal{E}_d$ to the VOI chooser (done in Chapter 3); and an analytic account of the naive-avg conflict-to-ridge-escape mechanism.

In the compiler’s judgement the most promising extension is BL 4 combined with the trajectory header: if every experiment ships its per-period trajectory, any two experiments on one channel over-determine (K, S, α) and their posterior-predictive discrepancy tests shape invariance at no extra experimental cost, turning the chapter’s one unfalsifiable premise into a routine diagnostic. Second is the multi-channel operator, since Chapter 7 and Chapter 13 show allocation decisions live in cross-channel contrasts the single-channel identity does not reach.

Reviewer’s note (compiler)

Protocol. Dive 02 predates the depth standard adopted “as of run 03” (attack-and-refine rounds, clean-room re-derivation, uncertainty on every headline). Nothing here has been independently re-derived within the program, and Table 2.3 reports medians over 60 replications with no standard errors: “30% wrecked” is 18 replications, “12%” is 7, “3%” is 2, and the operator’s “0%” is a bound of roughly one in sixty. “Within a point” (−43.0% versus −43.8%) has no standard error on the median.

The rank-4 claim is nominal. The smallest trajectory eigenvalue, 2.7×10^{-5} , is 4.7×10^{-8} of the largest; the code declares rank by the threshold $10^{-8} \times$ the largest eigenvalue, so the fourth direction clears it by a factor of about five, and the CRLB along that eigenvector is $\approx 1/\sqrt{2.7 \times 10^{-5}} \approx 190$. The honest reading is “rank 3 strong plus one very weak direction”. The 0.20 SE for α is plausible (the weak direction lives in (β, K, S) , the level ridge an experiment cannot see) but comes from inverting a matrix of condition number $\sim 2 \times 10^7$ without the $s_{\min}/s_{\max}$ audit BL 82 later made mandatory for FIM-based standard errors. The “three well-separated designs” corollary is a counting heuristic, not a proved statement.

Decision relevance was later softened. Dive 04 found trajectory readouts “worthless” for fixed-budget cross-channel allocation (κ 0.29 $\rightarrow$ 0.30); Proposition 2.10 concerns identification of one channel, not decision value.

Prior-art survey. The current Meridian documentation (checked 2026-09-07) describes a `CalibrationBuilder` that adjusts experiment results for spend scale, duration and recency before forming the ROI prior; the dive does not mention it. It is not the design-simulating operator, but a reader should check how much of C and Q it implements before crediting “never stated in the MMM calibration literature” with full force. The indirect-inference lineage is asserted, not surveyed.

Consistency checks passed. The INDEX log line agrees with the writeup on every number; $65\times$ is $0.1368/0.0021 = 65.1$; “7 $\times$ tighter” is $0.108/0.016 = 6.75$ against the uncalibrated mode ($5.8\times$ against the operator); $m_\infty = 0.389$, average ROAS 0.735 and $C = 0.816$ reproduce from θ_0 . The mismatch variant’s “theory +72.5%” and the operator’s 0.048 appear in the text only, with no table. The log timestamp ($\sim 21:15$) precedes the file’s last write (23:14 UTC) by two hours without explanation.

Sources

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- [12] Frontier study (internal): the open-questions catalogue `02_open_questions.md`, entry M2, and the adoption-barriers catalogue `03_mmm_adoption_barriers.md`, entries O7, B5 and M5.
- [13] Program (internal): Dive 01, file `01_mmm_identified_set.md`, sections 3.2 and 3.5; Dive 04, file `04_decision_voi_ext.md`, finding F4.

Chapter 3

Decision-calibrated uncertainty and the EVSI power plateau

Provenance

Source dives: Dive 03 *Decision-calibrated uncertainty & value of information (M10)* (run 2026-09-01, file 03_decision_voi.md); Dive 04 *EXTEND of dive 03: the anatomy and the edge of the EVSI power plateau (M10, BL5, BL6)* (run 2026-09-01, file 04_decision_voi_ext.md). **Code:** code/03_decision_voi.py, code/04_decision_voi_ext.py (the extension imports the model, posterior and experiment operator of the original). **Frontier-study problem(s) addressed:** M10 — “(a) tractable Bayes-optimal budget allocation under a joint posterior over multi-channel response curves — a stochastic program with rank-correlated uncertainty; (b) value-of-information theory for the next experiment: which geo test, at what spend delta, maximally reduces expected budget-decision regret? Test & Roll solves the two-arm case; the multi-channel, curve-level problem is open.” **Backlog items closed:** BL 5, BL 6 (opened by Dive 03, resolved by Dive 04); BL 7 partially advanced; **opened:** BL 8–BL 10.

Key ideas in this chapter

- Under a fixed budget the Bayes-optimal allocation beats the plug-in allocation by 0.027% of revenue, while perfect knowledge of the response curves is worth 1.76% (exact EVPI 0.0420 ± 0.0036 on revenue 2.381): information, not optimization, is the asset.
- A common multiplicative error in every channel’s effectiveness parameter is *exactly* decision-null under a fixed budget; decision value lives only in cross-channel contrasts of marginal ROAS and in curvature.
- A rank-1 decision-aligned EVSI score is correct where its linear-quadratic assumptions hold (Pearson 0.945 against nested Monte Carlo at quarter posterior width), but its prediction that D-optimal design selection forfeits 42% of decision value is refuted: well-powered scalar-readout designs tie within MC resolution (the “power plateau”), while underpowered designs lose about 85% at 16σ .
- Dive 04’s LQC estimator (exact importance-weighted conditioning, quadratic decision layer) reproduces nested-MC rankings exactly (Spearman 1.000) for a 39-design grid in 4.7 s; the plateau is a ceiling structure $\kappa(d) \leq \rho^2(d) \lambda_1 / \text{tr } A$ with per-channel share ceilings $\kappa_j \leq s_j$, verified numerically.
- The plateau ends across experiment *count* (dual experiment +47%) and via a budget-neutral *switch* design aligned with the top decision contrast (+22%), both MC-confirmed; oversized switches collapse nonmonotonically (κ 0.47 $\rightarrow$ 0.21), trajectory readouts are worthless for allocation, and the conjecture that a free budget rehabilitates D-optimality is refuted.

3.1 Problem and motivation

A marketing-mix model delivers a joint posterior over channel response curves, and the posterior is only worth what it changes about the budget. Problem M10 asks two things of it. Part (a) is a stochastic program: practitioners maximize revenue against the posterior-*mean* curves (the plug-in allocation), whereas the

Bayes-optimal allocation maximizes *expected* revenue under the full posterior; with saturating curves, a simplex constraint and rank-correlated uncertainty these can differ. Part (b) is an information problem: of all the geo tests one could run next — which channel, what spend delta, how long, with or without a post-period — which most reduces the expected regret of the budget decision? Test & Roll (Feit and Berman, 2019) answers a two-arm version of (b) in closed form; the multi-channel, curve-level version was open.

The field’s implicit belief was that both halves matter comparably, and that (b) is an optimal-experimental-design problem: pick the test that most shrinks the posterior, presumably by a D-optimal criterion of the kind Chapter 1 develops. Dive 03 priced both halves in one linear-quadratic frame and found the emphasis inverted: the Bayes-versus-plug-in refinement is a rounding error, the information is worth sixty times more, and the natural decision-aligned design score ranks designs correctly only where the frame is exact. Outside that regime, exact preposterior Monte Carlo shows that every adequately powered single-channel pulse delivers about the same decision value — a *power plateau*. Dive 04 explained the plateau (a ceiling structure in a two-dimensional decision spectrum), replaced the failing closed form with an estimator that ranks designs as nested Monte Carlo does at a fraction of the cost, and found the two ways the plateau ends: more simultaneous readouts, or one readout realigned with the top decision contrast by a budget-neutral switch experiment. The always-on scheduler of the frontier study’s grand challenge (E6 + M10) needs exactly such a scorer, built around the decision’s sufficient statistic — the marginal-ROAS vector — rather than parameter information.

3.2 Background: the ideas this chapter builds on

3.2.1 Bayesian decision theory and preposterior analysis (Raiffa and Schlaifer, 1961)

Let θ be the unknown state with prior π , b an action and $R(b, \theta)$ the payoff; the Bayes action is $b^* = \arg \max_b \mathbb{E}_\pi[R(b, \theta)]$. The *expected value of perfect information* is the expected gain from learning θ before acting,

$$\text{EVPI} = \mathbb{E}_\theta[\max_b R(b, \theta)] - \max_b \mathbb{E}_\theta[R(b, \theta)], \quad (3.1)$$

which equals the expected regret of the incumbent action. The *expected value of sample information* of an experiment d producing data ℓ is

$$\text{EVSI}(d) = \mathbb{E}_\ell[\max_b \mathbb{E}_{\theta|\ell}[R(b, \theta)]] - \max_b \mathbb{E}_\theta[R(b, \theta)], \quad (3.2)$$

so $0 \leq \text{EVSI}(d) \leq \text{EVPI}$. Computing it is a *preposterior* analysis: integrate over data not yet observed and, for each hypothetical data set, re-solve the decision under the corresponding posterior. The brute-force estimator is nested Monte Carlo (Heath and Baio, 2018): draw (θ_i, ℓ_i) from the prior predictive (outer loop), approximate the posterior given ℓ_i (inner loop), optimize and evaluate; the cost is outer $\times$ inner $\times$ one optimization, and the inner maximum carries a winner’s-curse bias unless the sample that chooses the action is separated from the sample that evaluates it.

Health economics, which prices clinical trials, built cheaper estimators. Strong, Oakley, Brennan and Breeze (2015) observe that for a *finite* action set $\mathbb{E}[R(a, \theta) \mid \ell]$ is a smooth function of a low-dimensional summary of ℓ , fit it by nonparametric regression across the prior sample, and average the maxima of fitted values — no inner loop. Heath, Manolopoulou and Baio (2019) match moments of the preposterior distribution of the conditional payoff across sample sizes; the Gaussian-approximation family (Li, Jalal and Heath, 2024, for payoffs nonlinear in the parameters) replaces conditioning by a linear-Gaussian shrinkage of the decision-relevant summary. All were built for discrete treatment choices, none for a continuum of budget allocations.

3.2.2 Test & Roll (Feit and Berman, 2019)

Feit and Berman recast A/B test sizing as profit maximization. Two arms with mean responses $m_1, m_2 \sim \mathcal{N}(\mu, \sigma^2)$ a priori and response noise s are tested on n customers each out of N ; the better-looking arm is rolled out to the remaining $N - 2n$. Maximizing expected total profit gives, for symmetric priors,

$$n^* = \sqrt{\frac{N}{4} \left(\frac{s}{\sigma}\right)^2 + \left(\frac{3}{4} \left(\frac{s}{\sigma}\right)^2\right)^2} - \frac{3}{4} \left(\frac{s}{\sigma}\right)^2, \quad (3.3)$$

typically far below the hypothesis-testing sample size: the test is priced by the profit forgone on the test cells against the expected improvement of the roll-out decision. It is the canonical “EVSI minus cost” calculation in marketing, for two arms and a scalar mean; the dives treat it as the two-arm case of M10(b) and the shape the power–cost frontier (BL7) must take.

3.2.3 Certainty equivalence and the linear-quadratic preposterior identity (Simon, 1956; Theil, 1957)

In a linear-quadratic (LQ) control problem — linear dynamics, quadratic cost, Gaussian uncertainty — the optimal control depends on the state only through its conditional mean (*certainty equivalence*), and the expected cost of not knowing the state is an additive term $\frac{1}{2} \text{tr}(Q\Sigma)$, with Q the cost curvature and Σ the covariance of the estimation error. Two consequences carry to allocation: if the payoff is locally quadratic in the decision and the uncertainty enters the first-order conditions linearly, the Bayes action equals the plug-in action to first order; and EVPI reduces to a trace of the decision curvature against the covariance of the decision-relevant statistic, EVSI to the reduction of that trace produced by the experiment’s covariance update. Dive 03 instantiates both on the budget simplex; Dive 04 keeps the quadratic decision layer but makes the conditioning exact — the “LQ-conditional” (LQC) estimator.

3.2.4 Design criteria: D-, c- and goal-oriented optimality (Attia, Alexanderian and Saibaba, 2018)

Classical optimal design scores an experiment by the Fisher information it adds. For a scalar readout $\hat{L} \sim \mathcal{N}(\mathcal{E}_d(\theta), se^2)$ linearized as $g = \nabla_{\theta} \mathcal{E}_d(\bar{\theta})$, the Gaussian posterior covariance updates by the rank-1 formula

$$\Sigma' = \Sigma - \frac{\Sigma g g^T \Sigma}{se^2 + g^T \Sigma g}, \quad (3.4)$$

so the D-criterion (total information gain, log det) becomes $\frac{1}{2} \log(1 + g^T \Sigma g / se^2)$ and the c-criterion (variance reduction of one target functional $c^T \theta$) becomes $(c^T \Sigma g)^2 / (se^2 + g^T \Sigma g)$. D-optimality is what Chapter 1 uses to design for identification of the full parameter vector. Goal-oriented OED (Attia, Alexanderian and Saibaba, 2018) argues that when the experiment serves a downstream quantity $Q = P\theta$ the design should minimize $\text{tr}(P\Sigma'P^T)$ instead; GoBOED (Go, Qian and Yoon, 2026) pushes this to a differentiable convex decision layer. The dives’ decision-aligned score is of this family with $P = D$, the map from parameters to marginal ROAS, weighted by a matrix Ω of rank $J - 1$: only two of twelve parameter directions are scored. The chapter’s central negative finding is that every criterion built from the linearization g misranks designs once the posterior is MMM-wide.

3.2.5 The MMM model and the experiment operator (dives 01–02)

The dives inherit the response model of Chapter 1 and the experiment forward map of Chapter 2: weekly revenue $y_t = \sum_j \beta_j h(a_{jt}; K_j, S_j) + \varepsilon_t$ with geometric adstock $a_{jt} = x_{jt} + \alpha_j a_{j,t-1}$ and Hill saturation $h(a; K, S) = a^S / (a^S + K^S)$. A geo experiment on channel j applies a uniform spend pulse δ for T_e weeks from steady

state and reads out the incremental revenue accumulated over $T_e + P$ weeks (the post-period P captures the adstock tail):

$$\mathcal{E}_d(\theta) = \beta_j \sum_{t=1}^{T_e+P} [h(a_{jt}^1; K_j, S_j) - h(a_{jt}^0; K_j, S_j)], \quad (3.5)$$

with a^0, a^1 the adstock paths without and with the pulse. Chapter 2 showed that such a scalar readout constrains one direction of channel j 's four parameters while the per-week trajectory carries rank-4 information; this chapter asks how much of either reaches the allocation decision.

3.3 Setup and assumptions

Throughout, $J = 3$ channels, $\theta = (\beta_j, K_j, S_j, \alpha_j)_{j=1}^3 \in \mathbb{R}^{12}$, noise $\sigma = 0.05$. The truth θ_0 is channel 1: (1.0, 1.5, 2.0, 0.6), channel 2: (0.8, 1.0, 1.8, 0.3), channel 3: (1.3, 2.5, 2.2, 0.8), baseline spends $\bar{x} = (1.0, 0.8, 1.2)$, total budget $B = 3$. (Dive 03's writeup numbers channels 1–3; Dive 04's uses the code's 0-based labels ch0–ch2; this chapter uses 1–3 throughout.) The decision is the steady-state allocation

$$R(b, \theta) = \sum_j \beta_j h\left(\frac{b_j}{1 - \alpha_j}; K_j, S_j\right), \quad b^* = \arg \max_{b \geq 0, \mathbf{1}^\top b = B} \mathbb{E}_\pi[R(b, \theta)], \quad (3.6)$$

all values of information being in expected steady-state revenue per period. Marginal ROAS is $m_j(b, \theta) = \beta_j h'(b_j / (1 - \alpha_j)) / (1 - \alpha_j)$.

Assumption 3.1 (Laplace posterior from an observational fit). $\theta \sim \mathcal{N}(\theta_0, \Sigma)$ with $\Sigma = (\mathcal{J}^\top \mathcal{J} / \sigma^2 + \Lambda_{\text{prior}})^{-1}$, where $\mathcal{J}$ is the Jacobian of the 104-week mean path under correlated AR(1) spends ($\pm 10\%$ swings around $\bar{x}$ with a shared demand cycle) and Λ_{prior} a weak prior of 50% relative scale, truncated to the admissible region ($\beta_j > 0.05, K_j > 0.1, S_j > 0.3, 0 < \alpha_j < 0.95$). Widths are MMM-realistic: marginal relative standard deviations 11–49%, maximum cross-channel correlation 0.71.

Assumption 3.2 (Scalar readout with assumed power). A design $d = (j, \delta, T_e, P)$ returns $\hat{L} \sim \mathcal{N}(\mathcal{E}_d(\theta), se^2)$ with $se = 0.03\sqrt{T_e + P}$ (window-accumulated noise at 0.03 per week), assumed rather than derived from a geo-count power model. The grid is $j \in \{1, 2, 3\}, \delta \in \{0.25, 0.5, 1\} \times \bar{x}_j, T_e \in \{4, 8\}, P \in \{0, 8\}$ (36 designs); “powered” means $\delta \geq 0.4\bar{x}_j$ and $T_e = 8$ (12 designs).

Assumption 3.3 (Correct specification, centered posterior). The model is correctly specified and the posterior is centered at the truth; every EVSI here is a value of calibration, not of model checking.

The linear-quadratic objects are the incumbent optimum $\bar{b}$ at θ_0 ; curvatures $H_j = -\partial m_j / \partial b_j > 0$ at $(\bar{b}, \theta_0)$; the sensitivity $D = \partial m / \partial \theta \in \mathbb{R}^{J \times 12}$; the decision-sufficient statistic $u(\theta) = m(\bar{b}, \theta) \in \mathbb{R}^J$; and the readout gradient $g = \nabla_\theta \mathcal{E}_d(\theta_0)$. Exact EVSI is nested Monte Carlo with importance-weighted conditioning on a shared posterior sample: for a simulated outcome ℓ , weights $w_i \propto \exp\{-\frac{1}{2}(\mathcal{E}_d(\theta_i) - \ell)^2 / se^2\}$ approximate the posterior, the allocation is optimized on the even-indexed half of the sample and evaluated on the odd half (split-sample debiasing), and outer draws use common random numbers (CRN) across designs. Dive 03 used 4000 inner draws with 200–600 outer; Dive 04 6000 inner and 240 outer.

3.4 Main results

3.4.1 The quadratic regret identity and the EVPI/EVSI trace formulas (Dive 03)

Proposition 3.4 (LQ regret of the incumbent allocation). *[established in the dive] At an interior optimum of Equation (3.6) the marginal ROAS are equalized, $m_j(\bar{b}_j, \theta) = \mu$. Perturbing θ by $d\theta$ shifts the marginal-ROAS vector*

by $u = D d\theta$; solving the displaced first-order conditions on the simplex with curvatures H_j gives the regret of holding $\bar{b}$,

$$\text{Regret} \approx \frac{1}{2} u^\top \Omega u, \quad \Omega = H^{-1} - \frac{H^{-1} \mathbf{1} \mathbf{1}^\top H^{-1}}{\mathbf{1}^\top H^{-1} \mathbf{1}}, \quad H = \text{diag}(H_j), \quad (3.7)$$

and hence

$$\text{EVPI} \approx \frac{1}{2} \text{tr}(\Omega D \Sigma D^\top), \quad \text{EVSI}(d) \approx \frac{1}{2} \frac{g^\top \Sigma D^\top \Omega D \Sigma g}{se^2 + g^\top \Sigma g}. \quad (3.8)$$

The derivation is three lines: the displaced optimum satisfies $u_j - H_j db_j = d\mu$ with $\mathbf{1}^\top db = 0$, so $db = H^{-1}(u - d\mu \mathbf{1})$ and $d\mu = \mathbf{1}^\top H^{-1} u / \mathbf{1}^\top H^{-1} \mathbf{1}$; the gain from moving is $\sum_j (u_j db_j - \frac{1}{2} H_j db_j^2) = \frac{1}{2} u^\top \Omega u$. Taking expectations gives the EVPI trace; substituting Equation (3.4) for $\Sigma - \Sigma'$ gives the EVSI form, the exact linear-Gaussian preposterior value of a scalar readout. Two facts follow: $\Omega \mathbf{1} = 0$, so a shift raising every channel's marginal ROAS equally has no decision value, only cross-channel *contrasts* count, and rank $\Omega = J - 1 = 2$; and the EVSI score is a decision-aligned c-type criterion computable in microseconds per design, to be set against the D-score $\frac{1}{2} \log(1 + g^\top \Sigma g / se^2)$.

3.4.2 An exact decision-null direction (Dive 03)

Proposition 3.5 (Common effectiveness error is decision-null). *[established in the dive] Under a fixed total budget, a common multiplicative error $\beta_j \rightarrow \beta_j(1 + s)$, any $s > -1$, leaves $\arg \max_b R(b, \theta)$ unchanged: revenue is linear in β , so the shock factors out of the objective. The statement is exact, not second order.*

Corollary 3.6. *The industry-default calibration target — the average ROI level, which Chapter 2 argues is what Meridian-style iROAS-to-ROI priors chiefly move — is nearly worthless for fixed-budget allocation; decision value lives in contrasts and curvature. The null breaks when total budget is also chosen, because the level then sets the spend margin.*

3.4.3 Information, not optimization (Dive 03)

[numerically established] The two halves of M10 have very different prices (Table 3.1). The Bayes and plug-in allocations differ by a few percent per channel, but the expected-revenue gap is 6.4×10^{-4} , 0.027% of the baseline revenue 2.381; exact EVPI from per-draw oracle re-optimization is 0.0420 ± 0.0036 , 1.76% of revenue — sixty-fold larger. The Bayes refinement is a rounding error; the experiment scheduler is the asset, inverting M10's own emphasis. The quadratic formula gives 0.0151, and shrinking Σ by $s \in \{1, \frac{1}{2}, \frac{1}{4}\}$ takes MC/quadratic from 3.37 to 1.69 to 1.27 while the quadratic scales exactly as s^2 : the LQ theory is correct locally, and at MMM width about two thirds of the value is higher-order curvature it cannot see. Proposition 3.5 is confirmed with CRN: a 30%-sd common β shock leaves exact EVPI within noise, an equal-magnitude contrast shock (channel 1 up, channel 3 down) raises it 55%, and with a free budget the common shock adds real value, as predicted.

3.4.4 The power plateau and the refuted D-optimality forfeit (Dive 03)

The closed form Equation (3.8) was tested against nested MC on eight designs at two widths (Table 3.2). At quarter width (the LQ regime) it is right: Pearson 0.945, Spearman 0.93, level ratio 1.16. At full width it fails as a ranker: levels understated about $2.2\times$, Spearman 0.33. *[refuted within the dive]* On the 36-design grid the closed form predicts that decision-aligned selection beats D-optimality decisively — Kendall $\tau(\text{EVSI}, \text{D-score}) = 0.23$, the D-best design forfeiting 42% of the EVSI-best design's value (17% at quarter width). Exact nested MC refutes the forfeit at both widths: the two designs tie, paired difference -0.00037 ± 0.00067 at full width and $(-2 \pm 13) \times 10^{-6}$ at quarter width, where the D-best design beats its own rank-1 prediction (0.000157 against 0.000121).

Law 3.7 (EVSI power plateau, scalar single-channel pulses). *[numerically established]* Among adequately powered single-channel pulse designs, the choices practitioners agonize over — channel, pulse size, post-period — move exact EVSI by less than the MC resolution ($\approx 10\%$) at both realistic and tight posterior widths, while signal-to-noise determines decision value almost entirely: a tenth-size pulse, a $6\times$ -noise readout and a tiny two-week test score 0.00108, 0.00134 and 0.00079 against 0.00721 for the strong design (paired difference 0.00612 ± 0.00038 , about 16σ). One well-powered experiment recovers 15–18% of exact EVPI.

Dive 03 reconciled this with Chapter 1: design determines identification of the full parameter vector (there, $\lambda_{\min}$ swings by $700\times$), but the decision consumes only $J - 1$ contrast directions, which almost any powered readout informs through the posterior’s correlations. Its diagnosis of *why* the gradient criteria fail — a large pulse is “nonlinearly informative” beyond the direction g , so no gradient-based score can see it — was refuted by the extension (Section 3.4.5). A greedy scheduler with rank-1 updates and the LQ score runs channel 3, then 2, then 1, cutting residual quadratic EVPI to 69%, 51% and 36% before saturating (34%, 31%): diminishing returns after one experiment per channel.

3.4.5 The LQC estimator and the re-diagnosis (Dive 04)

Construction 3.8 (LQC: exact conditioning, quadratic decision layer). *[transported from the literature]* The obstruction to fast EVSI was the inner *optimization* per outer draw, not the inner integration. By Proposition 3.4 the decision consumes θ only through $u(\theta) = m(\bar{b}, \theta)$, with value $\frac{1}{2}\Delta u^\top \Omega \Delta u$. With a shared posterior sample $\{\theta_i\}$, precompute u_i and readout means $L_i = \mathcal{E}_d(\theta_i)$; for each simulated outcome ℓ form self-normalized weights $w_i \propto \exp\{-\frac{1}{2}\|(L_i - \ell)/se\|^2\}$ and set

$$\widehat{\text{EVSI}}_{\text{LQC}}(d) = \mathbb{E}_\ell \left[\frac{1}{2} (\hat{u}_\ell^e - \bar{u})^\top \Omega (\hat{u}_\ell^o - \bar{u}) \right], \quad \hat{u}_\ell^{e/o} = \sum_{i \in \text{even/odd}} w_i u_i, \quad (3.9)$$

the even/odd cross-product removing the self-normalized-IS noise bias. The cost is $O(n)$ vector operations per outer draw, and multivariate readouts (trajectories, simultaneous experiments) enter through the weight kernel for free. $\text{EVPI}_{\text{LQ}} = \frac{1}{2}\mathbb{E}[(u - \bar{u})^\top \Omega (u - \bar{u})]$ is the same functional at perfect information; $\kappa(d) = \text{EVSI}(d)/\text{EVPI}_{\text{LQ}}$ is the *capture fraction*.

This is Strong–Oakley regression EVSI transported to a continuum of actions: they regress per-action net benefit for finitely many actions; LQC regresses the decision-sufficient statistic and closes the continuum analytically through Ω . Against nested MC on seven CRN-matched grid designs (Table 3.3) LQC has Pearson 0.974 and Spearman 1.000 with median MC/LQC = 1.33 (the LQ conservatism factor at this width); the rank-1 closed form on the same designs has Pearson 0.31, Spearman 0.32 and level ratio 2.4.

Proposition 3.9 (u -space Gaussian score). *[established in the dive]* Linear-Gaussian conditioning in u -space with exact nonlinear sample moments gives

$$\text{EVSI}_{\text{gauss-}u}(d) = \frac{1}{2} c^\top \Omega c, \quad c = \frac{\text{Cov}(u, L)}{\sqrt{\text{Var}(L) + se^2}}, \quad (3.10)$$

with $\text{Cov}(u, L)$ and $\text{Var}(L)$ sample moments of the shared draw; on five powered designs it matches LQC to within -2% to $+6\%$.

[refuted within the dive] The consequence is a re-diagnosis. The rank-1 θ -space formula fails not because conditioning is non-Gaussian but because its ingredients $g^\top \Sigma D^\top$ are *linearizations of the two forward maps*: over an MMM-realistic posterior, $\text{Cov}(L, u) \neq g^\top \Sigma D^\top$. With exact sample moments, Gaussian conditioning is already enough. This resolves BL6 negatively and better than asked — no Hessian or sigma-point machinery, the correction is a covariance estimate — and explains why every g -based criterion misranks MMM experiments.

3.4.6 Anatomy of the plateau: alignment bound and share ceiling (Dive 04)

Let $A = \Omega^{1/2} \text{Cov}(u) \Omega^{1/2}$ with eigenvalues $\lambda_1 \geq \lambda_2, \lambda_3 = 0$ (since $\Omega \mathbf{1} = 0$): the *decision-relevant spectrum*, with $\text{tr } A = 2 \text{EVPI}_{\text{LQ}}$ and principal contrasts c_k in u -space.

Law 3.10 (Alignment bound and share ceiling). *[established in the dive]* For any scalar readout $\text{Var}(\mathbb{E}[u \mid \ell])$ has rank one, and under Equation (3.10) Cauchy–Schwarz gives

$$\kappa_{\text{scalar}}(d) \leq \rho^2(d) \frac{\lambda_1}{\lambda_1 + \lambda_2}, \quad (3.11)$$

where ρ^2 is the squared correlation of the standardized readout with its best-aligned contrast, with equality iff the readout is aligned with c_1 . Under an independent-channels posterior a channel- j readout can move only u_j , giving the sharper share ceiling

$$\kappa_j \leq s_j = \frac{\Omega_{jj} \text{Var}(u_j)}{\text{tr}(\Omega \text{Cov } u)}. \quad (3.12)$$

For Equation (3.11), write $\Omega^{1/2}c = \text{Cov}(w, z)$ with $w = \Omega^{1/2}u$ and z the standardized readout, and bound $\|\text{Cov}(w, z)\|^2 = \max_{\|v\|=1} \text{Cov}(v^\top w, z)^2 \leq \rho^2 \lambda_1$; Equation (3.12) follows because $\text{Cov}(u, L) = e_j \text{Cov}(u_j, L)$ and $\text{Cov}(u_j, L)^2 \leq \text{Var}(u_j) \text{Var}(L)$. The plateau is therefore a *ceiling structure*: powered designs sit near their ceilings, and the ceilings of the conventional design family happen to be comparable ($s_j \approx 0.25\text{--}0.43$). *[refuted within the dive]* Dive 04’s own opening conjecture — plateau $\Leftrightarrow$ effective rank of A near 1 — is rejected by its numbers: $\lambda = (0.0257, 0.0182)$, effective rank $(\sum \lambda)^2 / \sum \lambda^2 = 1.94$, top-contrast share 0.59. Decision uncertainty is genuinely two-dimensional; the plateau comes from *correlation spreading*: every powered readout loads on both contrasts (channel 1: $\text{Corr}(L, c_1^\top u) = +0.78$, $\text{Corr}(L, c_2^\top u) = +0.28$), so all single-channel κ land in a compressed band. Two escapes follow: add readouts (κ of a k -readout design bounded by the top- k share sum — *[conjecture / speculative]* stated without proof, BL 9), or realign one readout with c_1 . Since $c_1 = (1, -0.33, -0.67)$ is a *reallocation* direction, no single-channel test can align with it, but a budget-neutral switch experiment ($+\delta$ on one channel, $-\delta'$ on another, one readout) can.

3.4.7 Where the plateau ends (Dive 04)

[numerically established] Four findings (Table 3.3), on Dive 03’s posterior unless stated. First, the plateau is a band, not a line: among the 12 powered designs LQC max/min is 1.65 (κ 0.22–0.36); Dive 03’s tie was real but an artifact of dueling two mid-band designs. Second, trajectory readouts are worthless for this decision (κ 0.29 $\rightarrow$ 0.30; 0.19 $\rightarrow$ 0.19 on a rank-2 posterior): the rank-4 within-channel information of Chapter 2 does not project onto cross-channel contrasts. Third, simultaneous two-channel designs break the ceiling as the share law predicts: the baseline dual (channels 3 and 2, $\delta = \frac{1}{2}\bar{x}$) has $\kappa = 0.52$ against 0.29 scalar; on the variance-equalized independent-channel posterior duals reach $\kappa = 0.48\text{--}0.65$ against 0.35 for the best single, and nested MC confirms +47%. The “ $> 2\times$ ” regime Dive 03 asked about exists across experiment *count*, never within single-scalar designs. Fourth, the switch experiment — +0.5 on channel 1 and –0.6 on channel 3 (each half its channel’s baseline), one scalar readout — achieves $\text{Corr}(L, c_1^\top u) = 0.83$ and $\kappa = 0.47$ against 0.36 for the best single-channel design and the 0.59 bound; the sign-reversed switch is identical (0.47); nested MC gives +22% over the best conventional design (LQC predicted +31%).

Proposition 3.11 (Nonmonotone switch collapse). *[numerically established]* Doubling the switch to $(+1.0, -1.2)$ collapses κ from 0.47 to 0.21 (MC 0.00556, about –45% against the best conventional design): large opposed pulses drive the two Hill responses into asymmetric saturation and the readout decorrelates from the contrast it was built to measure. Switch designs must be sized, not maxed (BL 8).

[refuted within the dive] Finally, Dive 03’s conjecture that a free total budget rehabilitates D-optimality and ROI-level calibration fails. With $\Omega_{\text{free}} = v_m \text{diag}(1/H)$ (no null direction along $\mathbf{1}$; v_m converts revenue to profit so that $\bar{b}$ remains optimal), EVPI_{LQ} rises $4.0\times$ because the level direction is live, yet rankings barely

move (Spearman 0.86 with the fixed-budget ranking), the best design is unchanged, and the powered band is *tighter* (max/min 1.26): every powered readout carries level information, and the level is the easiest direction to inform, so it differentiates designs even less.

3.5 Numerical evidence

The simulation design is that of Assumptions 3.1 and 3.2: one Laplace posterior from a 104-week collinear observational fit, a fixed truth, scalar (in Dive 04 also trajectory, dual and switch) readouts through Equation (3.5), and nested MC with split-sample debiasing and CRN outer draws.

Table 3.1: Dive 03: valuing the two halves of M10 (expected steady-state revenue per period; baseline revenue 2.381).

Exp.	Quantity	Value
E2	Plug-in allocation b_{plug}	(1.027, 0.919, 1.054)
E2	Bayes allocation b_{Bayes} (4000 draws)	(0.988, 0.941, 1.071)
E2	Expected-revenue gap, Bayes – plug-in	6.4×10^{-4} (0.027%)
E2	Exact EVPI (400 oracle draws)	0.0420 ± 0.0036 (1.76%)
E2	Quadratic EVPI, Equation (3.8)	0.0151
E2b	MC/quadratic at Σ scale $1, \frac{1}{2}, \frac{1}{4}$	3.37, 1.69, 1.27
E3	Exact EVPI: baseline / +common β shock (30% sd)	0.0444 / 0.0474
E3	Exact EVPI: +contrast shock (ch. 1 up, ch. 3 down)	0.0689 (+55%)
E3	Free budget: baseline / +common shock	0.304 / 0.414

Table 3.2: Dive 03: rank-1 closed form versus exact nested MC and the plateau duels (E4, E5, E7). A = EVSI-best design (ch. 3, $\delta = \frac{1}{2}\bar{x}_3$, $T_e = 8$, $P = 8$; in the quarter-width duel the code uses $\delta = \bar{x}_3$); B = D-best design (ch. 2, $\delta = \bar{x}_2$, $T_e = 8$, $P = 0$). Controls at full width: tenth-size pulse (ch. 3, $\delta = 0.06$, 8+8); A with $6\times$ noise; tiny-short (ch. 1, $\delta = 0.1$, $T_e = 2$).

Quantity	Full width	Quarter width
Closed form vs MC (8 designs): Spearman	0.33	0.93
Closed form vs MC (8 designs): Pearson / level ratio MC:closed	— / ≈ 2.2	0.945 / 1.16
Kendall τ (EVSI, D-score), 36-design grid, closed form	0.23	0.84
Predicted forfeit of D-best design, closed form	42%	17%
MC EVSI, design A	0.00721 ± 0.00040	0.000155
MC EVSI, design B (rank-1 prediction at $\frac{1}{4}$: 0.000121)	0.00758 ± 0.00055	0.000157
Paired A – B	-0.00037 ± 0.00067	$(-2 \pm 13) \times 10^{-6}$
Controls: tenth-size / $6\times$ noise / tiny-short	0.00108 / 0.00134 / 0.00079	—
Paired strong – weak	0.00612 ± 0.00038	—

Two controls anchor the instrument: the power cliff (weak designs near 0.001 against 0.006–0.009 for powered ones) is reproduced by both dives, with a 16σ paired separation in Dive 03, and the sign-reversed switch returning the identical κ checks that the switch’s value comes from alignment with c_1 , not the sign of the pulse. The only within-band MC comparison in Dive 04 (F3, LQC-best against LQC-worst channel) came back a tie with the order mildly reversed, which the dive reads as orderings of about $1.2\times$ being below both methods’ resolution: only cross-band claims are MC-established.

3.6 Limitations and the devil’s advocate

Both dives are explicit that every result lives inside one frame: a correctly specified model, a Laplace posterior centered at the truth, steady-state per-period revenue, one three-channel truth, and an assumed readout

Table 3.3: Dive 04: LQC validation, plateau anatomy and the edge of the plateau. $\kappa = \text{EVSI}/\text{EVPI}_{\text{LQ}}$; MC entries are nested MC (240 outer $\times$ 6000 inner, CRN). Channel vectors are ordered (1, 2, 3).

Item	Result	Comparator
F1: LQC vs MC, 7 designs: Pearson / Spearman / MC:LQC	0.974 / 1.000 / 1.33	rank-1: 0.31 / 0.32 / 2.4
F1: controls: weak pulse / $6\times$ noise / tiny-short	0.00106 / 0.00131 / 0.00066	powered $\approx$ 0.006–0.009
F1b: powered band (12 designs), LQC max/min	1.65 (κ 0.22–0.36)	MC band ends 0.00701, 0.00929
F2: $\lambda_{1,2}$; effective rank; $\lambda_1 / \text{tr } A$	(0.0257, 0.0182); 1.94; 0.59	$c_1 = (1, -0.33, -0.67)$
F2b: gauss- u vs LQC, 5 powered designs	–2% to +6%	
F3: block-diagonal posterior, κ_j vs ceiling s_j	$(0.26, 0.23, 0.29) \leq (0.32, 0.25, 0.43)$	MC best / worst: 0.00806 / 0.00842
F3: variance-equalized, κ_j vs s_j	$(0.35, 0.31, 0.19) \leq (0.41, 0.34, 0.25)$	channel ratios 1.25, 1.85
F4: trajectory readout, κ (baseline; rank-2)	$0.29 \rightarrow 0.30$; $0.19 \rightarrow 0.19$	worthless
F4: dual (ch. 3 & 2, $\frac{1}{2}\bar{x}$), baseline, κ	0.52	scalar 0.29
F4: dual MC, equalized posterior	0.01497 ± 0.00106	single 0.01015 ± 0.00084 (+47%)
F5: free budget: EVPI_{LQ} ; Spearman; band	$\times 4.0$; 0.86; max/min 1.26	best design unchanged
F6: switch (+0.5, –0.6) on ch. 1, 3: corr, κ	0.83, 0.47 (reversed 0.47)	best single 0.36; bound 0.59
F6: switch MC	0.01231 ± 0.00106	best conv. 0.01011 ± 0.00086 (+22%)
F6: oversized switch (+1.0, –1.2)	$\kappa = 0.21$; MC 0.00556	–45% vs best conventional

noise $se(d)$. The last is the strongest objection. Because the cost of power is never modeled, the plateau says only that among experiments *granted* adequate power the design does not matter, nothing about whether one should buy that power — Test & Roll’s territory and BL 7’s. The same omission undercuts the escapes: a dual experiment is plausibly about twice the cost of a single one, which would erase its +47%, and a switch experiment requires the two channels’ geo assignments to coincide, which real ad platforms only partially allow.

The instrument has known limits. The truncated Laplace posterior is not the true posterior of the observational fit; nested MC, though split-sample-debiased, retains the importance-sampling approximation of the conditional posterior; per-period units need a horizon and discounting to become dollars. LQC inherits the LQ decision layer: its levels are about 25% conservative at full width (MC/LQC 1.33) and its within-band orderings ($\lesssim 1.3\times$) cannot be confirmed by MC at feasible outer counts. The attack-and-refine rounds changed two results materially: Dive 03’s “nonlinear conditioning bonus” gave way to Dive 04’s moment-error diagnosis, and Dive 04’s opening effective-rank conjecture was withdrawn when the spectrum came back with effective rank 1.94, Law 3.10 being the replacement. A devil’s advocate adds that the effective-rank and share numbers are properties of this truth; the dives’ reply is that the *laws* — alignment bound, share ceiling, count-additivity — are the portable content, of which only the first two are proved.

3.7 Discussion: impacts

For practice the ordering of the two halves of M10 is the headline. An MMM vendor (Meridian, Robyn, PyMC-Marketing) that invests in a Bayes-optimal allocator under the joint posterior is buying 0.03% of revenue; the same posterior used to *schedule* the next experiment addresses a 1.8% opportunity, of which one well-powered test recovers roughly a sixth. Proposition 3.5 sharpens what that test must measure: under a fixed budget, calibration that moves every channel’s ROI level together — the commonest use of a lift test today — has no decision value; the value is in cross-channel contrasts and curvature, which is why Law 3.10 makes a *reallocation* pulse the most valuable single readout and why trajectory readouts, whatever they do for identification (Chapter 2), buy nothing here. For experimentation platforms the plateau is liberating and cautionary at once: among single-channel pulses, agonizing over channel, delta and post-period is wasted effort provided the test is powered, whereas underpowering costs 85% of the value, and an oversized switch is worse than a conventional pulse.

For theory the chapter delivers a design criterion that is not a Fisher-information criterion. The rank-1 score Equation (3.8) is exactly the goal-oriented c-optimality one would write down, and it is wrong at

MMM widths for a reason unrelated to non-Gaussianity: linearized covariances $g^\top \Sigma D^\top$ are poor estimates of $\text{Cov}(L, u)$ when the forward maps are Hill curves under a 50%-wide posterior. Proposition 3.9 shows the fix is an estimated covariance, and Construction 3.8 makes it cheap enough to sit inside a scheduler — the production scorer for the always-on loop of Chapter 4 and Chapter 13. Later chapters consume the machinery: the regret loss Equation (3.7) prices ranking flips in Chapter 6 and caps in Chapter 15; Chapter 11 joins its κ law to EVSI to size a lift budget; Chapter 13 asks whether switch experiments are the cheapest escape from a stalled certainty-equivalent loop.

3.8 Further exploration

The dives' next steps are concrete. BL 8 asks for the collapse point of Proposition 3.11 analytically — the δ at which $\text{Cov}(L_{\text{switch}}, c_1^\top u)$ turns over as a function of Hill curvature at the two operating points, with the conjectured rule that δ^* is where the two channels' second-order lift terms are equal and opposite. BL 9 is the k -experiment portfolio law: if $\kappa(\text{portfolio})$ is approximately the sum of captured spectral shares, greedy share-hunting is a scheduler, and Dive 03's greedy loop should be rebuilt with LQC as scorer. BL 7 remains the power–cost frontier: replace $se(d)$ with a geo-count power model and charge duals and switches their true cost, with +47% and +22% the margins to beat. BL 10 tests LQC under misspecification, since $u(\theta)$ is computed under the fitted model. The fifth step is to prove count-additivity and the equality conditions of Law 3.10.

In the compiler's judgement the most promising extension is BL 7 carried out in Test & Roll's currency: express each design's $se(d)$ through the geos and weeks it consumes, charge the forgone profit of the test cells as Feit and Berman do, and maximize LQC EVSI minus that cost. This is the multi-channel Test & Roll that M10 literally asked for, it uses only machinery the dives already validated, and it settles the question every finding leaves hanging — whether the +22% of a budget-neutral switch, which by construction costs no extra spend, survives once power is priced, in which case the switch becomes the recommended first experiment for a fixed-budget advertiser.

Reviewer's note (compiler)

- **Two different “fractions of EVPI”.** Dive 03 says one powered experiment recovers 15–18% of *exact* EVPI (0.042); Dive 04's κ for the same designs (0.22–0.36) is normalized by EVPI_{LQC} , whose baseline value the writeup never states (implied ≈ 0.021 from $\text{MC}/\text{LQC} = 1.33$, $\kappa = 0.36$ and MC 0.01011). The denominators differ by about $2\times$; $\kappa = 0.47$ is not 47% of the 1.8% of revenue.
- **The +22% switch margin is the weakest MC claim.** F6 reports marginal errors only (0.01231 ± 0.00106 vs 0.01011 ± 0.00086 , about 1.6σ unpaired); the paired CRN error that Dive 03 gave for its duels is absent, as is any error on the oversized-switch value 0.00556. The dual's +47% is about 3.6σ even unpaired. LQC predicted +31% and MC delivered +22%; the gap is within LQC's stated conservatism but undiscussed.
- **“Budget-neutral” is approximate.** The switch is +0.5 on channel 1 and -0.6 on channel 3 (each half of its channel's baseline), net -0.1 in spend units.
- **Not reproducible from the final scripts.** Dive 03's quarter-width grid statistics ($\tau = 0.84$, forfeit 17%) are quoted in E5, but the script's main block runs E5 at full width only; the quarter-width duel's designs are not named in the writeup, and in the code design A there is ch. 3 at $\delta = \bar{x}_3$, not $\frac{1}{2}\bar{x}_3$. The “ $\sim 1000\times$ cheaper” figure for LQC appears in the INDEX log line only; the writeup gives 4.7 s for 39 designs and no nested-MC timing.
- **Within-band reversal.** F3's MC returned the LQC-worst channel *above* the LQC-best (0.00842 vs 0.00806); the dive rightly calls it a tie, but “Spearman 1.000 vs nested MC” (F1, 7 designs) is agreement on cross-band ordering only. The INDEX line for Dive 03 (“channel/pulse/window choice is decision-irrelevant”) overstates what survives Dive 04's $1.65\times$ band.
- **Dates and citations.** Both dive headers say 2026-08-31 while the INDEX logs 2026-09-01 (00:21 and 02:30; the assignment gives 01:25 for Dive 04). Dive 03 attributes the Value in Health nested-MC tutorial to “Strong & Oakley et al.”; it is Heath and Baio (2018). Dive 04 calls PMC3980703 an “earlier open version” of Strong et al. (2015); it is the 2014 EVPPI regression paper. Corollary 3.6's claim about what Meridian-style ROI priors “chiefly move” is inherited from Dive 02, not tested here. No pinv/FIM (BL 82) issue arises: Σ is a ridge-regularized

inverse, not a pseudo-inverse.

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- [12] Internal: Dive 01 (FIM and D-criterion design machinery, Chapter 1); Dive 02 (operator forward map $\mathcal{E}_d$, rank-1 versus rank-4 readouts, Chapter 2); frontier study 02_open_questions.md M10; 03_mmm_adoption_barriers.md B5, H8.

Chapter 4

Always-on experiment-to-model fusion: the closed loop

Provenance

Source dives: Dive 05 *Always-on experiment-to-model fusion: the closed loop (A5 / E6)* (run 2026-08-31/09-01 over two sessions — the first built the system and completed stages setup-s4 before being interrupted, the second recovered the saved state and ran s5-s9, the adversarial rounds and the writeup; file 05_fusion_loop.md, written 2026-09-01 03:39 UTC). **Code:** code/05_fusion_loop.py (imports the plant and design machinery of 03_decision_voi.py and 04_decision_voi_ext.py). **Frontier-study problem(s) addressed:** E6 — a system “that continuously schedules geo/budget-split experiments chosen to maximally reduce MMM posterior uncertainty, then folds results back automatically — is a described ambition, not a shipped system”; the engineering realisation of M10. **Backlog items closed:** none (BL 9 advanced); **opened:** BL 11 (belief-independent risk ledgers), BL 12 (the low-drift trap paradox), BL 13 (filter-class dependence of trap economics).

Key ideas in this chapter

- The E6 loop exists as a construction: a 12-parameter extended Kalman filter with random-walk drift, weekly revenue updates, a nine-design menu ingested by moment matching through the operator likelihood of Chapter 2, and a decision-risk ledger $\rho_t = \frac{1}{2} \text{tr}(\Omega D \Sigma_t D^T)$ that decides when to launch.
- The renewal cadence law $\tau^* = \sqrt{c/(r(1/\kappa - \frac{1}{2}))}$ gives 40 weeks at baseline — the folk “one test per channel per quarter”, derived — and is verified as a *selector* (best cadence tracks τ^* at 3/3 drift rates; $\sqrt{c}$ exponent 0.457 ± 0.042) but refuted as a *forecaster*: loss is nearly flat in drift (exponent ≈ 0.1 – 0.5 against 1.0).
- The flatness is filter pathology, not observation drainage: an exact linear-Gaussian clean-room loop gives drift exponent 1.01, while the EKF’s trap rate *rises* as drift falls (17/48 at $q = 0.0075$ against 5–7/48 at $q \geq 0.015$) — process noise is regularisation.
- In-loop experiment value is mostly insurance: never-experiment traps 15/32 runs, any powered stream 8–10/32; mean loss $0.161 \pm 0.047 \rightarrow 0.088 \pm 0.027 \rightarrow 0.031 \pm 0.013$ with a 15% spend floor, while non-trapped runs of every policy sit at 0.014–0.020.
- Belief-driven components fail when needed: the ledger under-states realised regret $13\times$ in traps, and belief-scored design selection leaves a channel unprobed in 5/32 runs, all five of which trap. Coverage constraints, minimum cadence and spend floors beat scoring.

4.1 Problem and motivation

Lift-test calibration of an MMM is manual and point-in-time: a geo test runs when budget and politics allow, its result becomes a prior, and the prior sits there while the world drifts. E6 asks for the system nobody has shipped — one that continuously tracks the model’s beliefs, decides *when* to launch a calibration experiment and *which*, and folds the result back automatically. The dive found no closed-loop scheduler in production: the nearest practice is Recast’s time-bounded experiment priors that re-widen after a test (a hand-rolled drift model) plus prioritisation by widest confidence interval, Haus’s LLM-assisted test suggestions and LiftLab’s

analyst-driven programmes. Ingestion mechanics exist in isolation: Google’s calibration of ROI priors from lift tests (Zhang et al., 2024; the Meridian interface) and Heusch’s (2026) structural estimation of the full parameter vector from geo experiments. Cadence guidance is folklore — about one test per channel per quarter — and nobody derives it. The case for anchoring is sharp: the 2026 endogenous-spend benchmark finds algorithmic spend endogeneity inflating observational ROAS about $2.5\times$, so an unanchored model drifts *and* is biased.

Dive 05 composes the separately existing parts on the nonlinear plant of Chapters 1 to 3 with a moving truth, derives the loop’s operating law, and then spends four attack-and-refine rounds and a clean-room re-derivation learning that the law is right about *when* but wrong about *how much*, that most of an experiment’s in-loop value is protection against a failure the belief cannot see, and that the scheduler’s own intelligence produces that failure.

4.2 Background: the ideas this chapter builds on

4.2.1 The extended Kalman filter as a sequential Bayesian updater (Kalman, 1960; Jazwinski, 1970; Ljung, 1979)

For a Gaussian state with $x_{t+1} = Fx_t + w_t$, $w_t \sim \mathcal{N}(0, Q)$, and $y_t = Hx_t + v_t$, $v_t \sim \mathcal{N}(0, R)$, the Kalman filter is the exact Bayesian recursion on the posterior (m_t, Σ_t) : predict $(m, \Sigma) \rightarrow (Fm, F\Sigma F^T + Q)$; update with innovation $e = y - Hm$, $S = H\Sigma H^T + R$, gain $K = \Sigma H^T S^{-1}$, $m \leftarrow m + Ke$, $\Sigma \leftarrow \Sigma - KH\Sigma$. The extended Kalman filter (EKF) keeps the recursion for a nonlinear observation $y = g(x) + v$ by using the Jacobian $\nabla g(m_t)$ for H — a first-order Gaussian approximation with no optimality guarantee. As a *parameter* estimator (state = parameters, $F = I$) it is sequential Gauss–Newton with a prior; Ljung (1979) showed it can converge to wrong values or diverge, and injecting fictitious process noise so that Σ never collapses is a remedy as old as Jazwinski (1970). In an MMM the state is $\theta = (\beta_j, K_j, S_j, \alpha_j)_j$; the observation $y_t = \sum_j \beta_j h(a_{jt}; K_j, S_j)$ is nonlinear in (K, S, α) and, through the adstock recursion $a_{jt} = x_{jt} + \alpha_j a_{j,t-1}$, in the whole spend history, so the Jacobian needs the sensitivity $s_{jt} = \partial a_{jt} / \partial \alpha_j = a_{j,t-1} + \alpha_j s_{j,t-1}$, carried alongside the adstock.

4.2.2 Random-walk parameter drift (West and Harrison, 1997; Ng et al., 2021)

The random walk $\beta_{t+1} = \beta_t + q \zeta_t$ is the dynamic-linear-model default for a non-stationary coefficient: the variance of any fixed estimate grows linearly, $\text{Var}(\beta_{t+k} | \mathcal{F}_t) = \text{Var}(\beta_t | \mathcal{F}_t) + kq^2$, and the Kalman predict step implements it by adding q^2 to the diagonal each period — which is also the filter’s *forgetting*; with $q = 0$ the posterior contracts forever and learning stops. Uber’s time-varying-coefficient MMM (Ng et al., 2021; coefficients as kernel-weighted latent knot values) is the production form; Recast’s re-widening priors are the same idea hand-rolled.

4.2.3 Square-root cadence laws (Harris, 1913; Feit and Berman, 2019)

Harris’s economic order quantity balances a fixed cost K per batch against a holding cost h per unit-time at demand rate D : average cost $KD/Q + hQ/2$, minimised at $Q^* = \sqrt{2KD/h}$; in time units ($\tau = Q/D$) the cost is $c/\tau + g\tau/2$ and $\tau^* = \sqrt{2c/g}$. The structure is generic — a fixed cost per event against a penalty accruing linearly between events, the sawtooth averaging half its peak — and the renewal-reward theorem (long-run cost equals expected cycle cost over expected cycle length) extends it to random cycles, as in age-replacement maintenance. Feit and Berman’s Test & Roll is the decision-theoretic cousin on the sample-size axis: with N customers, arm means from a $\mathcal{N}(\mu, \sigma^2)$ prior and response noise s , the profit-maximising test size is $n^* = \sqrt{\frac{N}{4}(s/\sigma)^2 + (\frac{3}{4}(s/\sigma)^2)^2} - \frac{3}{4}(s/\sigma)^2$, a square-root law that prices a test by profit rather than power. The dive’s law adopts that spirit on the time axis with decision risk as the holding cost.

4.2.4 Kalman filtering with intermittent observations (Sinopoli et al., 2004)

When each observation arrives only with probability λ , the error covariance follows a random Riccati recursion whose expectation is bounded by the modified iteration

$$P_{t+1} = FP_tF^T + Q - \lambda FP_tH^T(HP_tH^T + R)^{-1}HP_tF^T, \quad (4.1)$$

the usual recursion with the correction scaled by the arrival rate. Sinopoli et al. prove a critical rate λ_c exists with $\mathbb{E}[P_t]$ bounded if and only if $\lambda > \lambda_c$, $\lambda_c \geq 1 - 1/\alpha_{\max}^2$ for the largest unstable eigenvalue of F . For a random-walk state ($F = I$) any positive rate keeps the covariance bounded, but its steady level scales inversely with λ : width is a function of experiment frequency. Law 4.4 is Equation (4.1) projected onto the decision-relevant directions, with deterministic arrivals every τ weeks and a fixed fractional correction κ .

4.2.5 The ingestion interface and the decision-risk functional (Chapter 2, Chapter 3)

An experiment enters the belief through the *operator likelihood* of Chapter 2: the design d is simulated inside the model, $\mathcal{E}_d(\theta)$ being the model-implied windowed incremental revenue, and $\hat{L} \sim \mathcal{N}(\mathcal{E}_d(\theta), \text{se}^2)$; no scalar iROAS becomes a prior, window truncation and curvature are handled implicitly, and switch and dual designs use the same map. The decision layer is Chapter 3's: with $u = m(\bar{b}, \theta)$ the marginal-ROAS vector at the belief-optimal allocation, the linear-quadratic (LQ) regret from holding it is $\frac{1}{2}u^T\Omega u$, $\Omega = H^{-1} - H^{-1}\mathbf{1}\mathbf{1}^TH^{-1}/\mathbf{1}^TH^{-1}\mathbf{1}$ (H the diagonal of curvatures), and $\Omega\mathbf{1} = 0$: only cross-channel *contrasts* carry value. With $D = \partial m / \partial \theta$, $\text{EVPI} \approx \frac{1}{2} \text{tr}(\Omega D \Sigma D^T)$; a design's capture fraction $\kappa(d) = \text{EVSI}(d) / \text{EVPI}_{\text{LQ}}$ is estimated by the gauss- u score $\frac{1}{2}c^T\Omega c$, $c = \text{Cov}(u, L) / \sqrt{\text{Var}(L) + \text{se}^2}$, from a shared belief sample; the *decision spectrum* is $A = \Omega^{1/2} \text{Cov}(u) \Omega^{1/2}$, eigenvalues $\lambda_1 \geq \lambda_2, \lambda_3 = 0$.

4.2.6 Sequential design and greedy portfolios (Foster et al., 2021; Rainforth et al., 2024; Nemhauser et al., 1978)

Bayesian experimental design maximises an expected utility, classically the expected information gain $\mathbb{E}_{y|d}[\text{KL}(p(\theta | y, d) || p(\theta))]$, and sequential design chooses each d given the history. Deep Adaptive Design amortises this into a policy network mapping past (design, outcome) pairs to the next design, trained offline on a contrastive lower bound of total information gain so that deployment needs no posterior computation; Rainforth et al. survey the estimators and decision-utility variants. The dive's scheduler is the decision-aligned counterpart with an explicit Gaussian belief, a fixed menu and a myopic score. For *sets* of experiments, greedy selection is $(1 - 1/e)$ -optimal for monotone submodular set functions (Nemhauser, Wolsey and Fisher, 1978); Gaussian information gain is submodular, but decision value under a rank-2 Ω need not be, hence the dive's audit.

4.3 Setup and assumptions

The plant is Chapter 3's with a moving truth: $J = 3$ channels, $y_t = \sum_j \beta_{jt} h(a_{jt}; K_j, S_j) + \varepsilon_t$ with geometric adstock, Hill saturation and $\sigma = 0.05$; θ_0 is channel 1: (1.0, 1.5, 2.0, 0.6), channel 2: (0.8, 1.0, 1.8, 0.3), channel 3: (1.3, 2.5, 2.2, 0.8); baseline spends $\bar{x} = (1.0, 0.8, 1.2)$, budget $B = 3$. Effectiveness drifts as $\beta_{j,t+1} = \beta_{jt} + q \xi_{jt}$, baseline $q = 0.015$ per week (glossed by the dive as about 20% per year). Each week the budget is reallocated to the belief-optimal simplex point (projected gradient ascent, optional per-channel floors); spends carry 3% AR(1) jitter; *regret* is oracle steady-state revenue at the true θ_t minus the realised allocation's. A run's *loss* is mean regret over weeks 50– T plus total experiment cost over T ($T = 400$; 500 for cadence grids and scaling sweeps).

Construction 4.1 (The loop under test). A Gaussian belief (m_t, Σ_t) over the 12-dimensional θ , initialised at Chapter 3's Laplace posterior Σ_0 from a 104-week observational fit, is propagated by an EKF: predict by

adding q_f^2 to the β diagonals (optionally inflated by an innovation-based factor in $[1, 40]$); update weekly on y_t with the closed-form gradient including the $\partial a / \partial \alpha$ recursion. The *menu* holds nine designs from Chapter 3: six single-channel pulses ($\delta = \frac{1}{2}\bar{x}_j$ and $\bar{x}_j$, $T_e = 8$, $P = 8$), two budget-neutral switches (+0.5 on channel 1 with -0.6 on channel 3; +0.4 on channel 2 with -0.5 on channel 3) and one dual (channels 3 and 2, cost $2c$). A launched experiment delivers 16 weeks later, $\hat{L} = \mathcal{E}_d(\theta_{\text{true}}) + \text{se} \xi$ with $\text{se} = 0.03\sqrt{T_e + P}$, at cost c (baseline 0.15). *Ingestion* is by moment matching: $n = 1500$ belief draws are pushed through $\mathcal{E}_d$, $\text{Cov}(\theta, L)$ and $\text{Var}(L)$ estimated, and the Kalman-style update $m \leftarrow m + \text{Cov}(\theta, L)(\text{Var}(L) + \text{se}^2)^{-1}(\hat{L} - L)$, $\Sigma \leftarrow \Sigma - \text{Cov}(\theta, L)(\text{Var}(L) + \text{se}^2)^{-1} \text{Cov}(L, \theta)$ applied — the sampled analogue of Meridian-style calibration, no refit. The *ledger* $\rho_t = \frac{1}{2} \text{tr}(\Omega D \Sigma_t D^\top)$ at the current allocation, evaluated every two weeks, drives the scheduler.

Policies: *never*; *cadence*: τ (every τ weeks, round-robin menu); *trigger* ($\rho_t \geq p^*$, round-robin); *trigger-pp* (ledger at the probe point $\max(b_t, 0.15\bar{x})$); *trigger-lqc* (design by gauss- u score per unit cost); *trigger-lqcg* (the same with a coverage guard forcing a probe of any channel untested for $G = 100$ weeks or dark — spend below $0.2\bar{x}_j$ — for 30); *trigger-rand*. One experiment is in flight at a time; seeds share common random numbers across policies. A *realistic start* draws the initial belief mean from $\mathcal{N}(\theta_0, \Sigma_0)$; a *clean start* sets it to θ_0 .

Assumption 4.2 (Drift, filter, experiments). Only β drifts; $q_f = q$ except in the misspecification study; the filter treats the belief-driven spend as exogenous. Readout noise is assumed rather than derived from a geo-count power model; cost is identical across designs except the dual’s $2c$; latency is fixed at 16 weeks.

Definition 4.3 (Trap). A run is *trapped* if its loss exceeds 0.05 (orderings checked at 0.03 and 0.10). Loss is bimodal across seeds, so medians and trap counts are the stable statistics; means carry standard errors over seeds.

4.4 Main results

4.4.1 The renewal cadence law

Between experiments the covariance grows by $Q = \text{diag}(q^2)$ on the β entries, so the ledger grows linearly at rate $r = \frac{1}{2} \text{tr}(\Omega D Q D^\top) = r_u q^2$, $r_u = 0.208$ for this plant.

Law 4.4 (Renewal cadence law). *[established in the dive]* If decision risk grows at rate r and each experiment removes a fraction κ of current risk at cost c , the steady cycle of period τ has average loss $L(\tau) = r\tau(1/\kappa - \frac{1}{2}) + c/\tau$, whence

$$\tau^* = \sqrt{\frac{c}{r(1/\kappa - 1/2)}}, \quad \ell^* = 2\sqrt{cr(1/\kappa - 1/2)}, \quad p^* = \sqrt{\frac{cr}{\kappa(1 - \kappa/2)}}, \quad (4.2)$$

p^* the ledger level at which a threshold policy should fire; periodic and threshold policies achieve the same ℓ^* under deterministic growth. All three formulas were independently re-derived and confirmed to about 10^{-5} by the clean-room verification agent.

The derivation is the EOQ sawtooth with an imperfect reset. With ρ_- the risk just after an experiment, steady state requires $\rho_- = (1 - \kappa)(\rho_- + r\tau)$, so $\rho_- = r\tau(1/\kappa - 1)$; the cycle average is $\rho_- + r\tau/2$, adding c/τ and differentiating gives τ^* , and the peak $r\tau^*/\kappa$ is p^* . With $\kappa = 1$ this is Harris’s formula with r as holding rate; an imperfect experiment raises the effective holding rate from $\frac{1}{2}$ to $1/\kappa - \frac{1}{2}$. At baseline ($q = 0.015$, $c = 0.15$, planning $\kappa = 0.4$) $\tau^* = 40$ weeks — the folk cadence derived — and, by arithmetic from the stated constants, $\ell^* \approx 0.0075$, $p^* \approx 0.0047$; at $q = 0.0075$ and 0.03 , $\tau^* = 80$ and 20 .

4.4.2 Portfolio ceiling and greedy scheduling

Proposition 4.5 (Top- k ceiling and near-submodularity). *[numerically established]* For any set A of menu readouts, $\kappa(A)$ is at most the sum of the top- $|A|$ eigenshares of the decision spectrum: 0/129 subsets of size 1–3 violate it. The

set function is nearly submodular — 21/324 diminishing-returns checks fail, worst late/early gain ratio 1.068, all involving switch-pair complementarities — and greedy/optimal = 1.000, 0.926, 1.000 at $k = 1, 2, 3$; three experiments capture 90.4% of EVPI_{LQ} .

This makes Chapter 3's count-additivity conjecture (BL 9) concrete for this plant: greedy is safe without the Nemhauser guarantee, and beyond three experiments under 10% of EVPI remains — which is why the loop's economics turn on drift and traps rather than precision.

4.4.3 Level drains, contrast stalls

Proposition 4.6 (Level/contrast split). *[numerically established] In the no-experiment loop (24 seeds, $q = 0.015$) the revenue stream drains the decision-null level risk $12\times$ (0.033 to 0.0026) while the contrast risk ρ_t stalls at a floor of about 0.018, roughly 80% of static EVPI_{LQ} .*

This is Chapter 3's null theorem in motion: near-constant spends inform $\mathbf{1}^\top u$, which Ω annihilates. *[refuted within the dive]* The dive first read the flat-in-drift loss below as this stall — observational drainage capping risk regardless of q . The clean-room loop showed otherwise; the split is directionally right but is not the mechanism.

4.4.4 A good selector and a bad forecaster

Law 4.7 (Selector, not forecaster). *[numerically established] Across a $4\times$ drift range the empirical best cadence tracks the law's argmin (80 / 40 / 10–20 weeks against $\tau^* = 80/40/20$ at $q = 0.0075/0.015/0.03$: 3/3), and the cost half holds (fitted c -exponent 0.457 ± 0.042 against 0.5). *[refuted within the dive]* The drift half fails in the nonlinear loop: the q -exponent of minimum loss is 0.47 ± 0.08 on the 9-cell, 16-seed grid and 0.1–0.2 on a 48-seed refit (medians 0.0150/0.0149/0.0202/0.0178 across $q = 0.0075\text{--}0.06$), against 1.0. The level is off by $1.4\text{--}4.0\times$ (mean 2.2 over the 12-cell grid), consistent with Chapter 3's $2\text{--}3\times$ LQ conservatism plus delivery delay and $\kappa_{\text{actual}} < \kappa_{\text{plan}}$.*

The verification agent's independent linear-Gaussian closed loop — an exact Kalman filter with the same cadence economics — reproduces the law: q -exponent 1.01, ℓ^* within 15–20%, c -exponents 0.43–0.50, argmin within $2\times$. So the flattening is not observation drainage, which a linear loop would show too; it is a property of the EKF. The dive's own caveat: the 48-seed bootstrap se (± 0.4) cannot exclude 0.5 and excludes 1.0 only marginally. What is solid is that loss is far flatter in drift than any linear theory allows, and that the flatness co-occurs with the trap-rate inversion.

4.4.5 The trap economy

Proposition 4.8 (Experiments as insurance). *[numerically established] From realistic starts (32 seeds) the never policy traps 15/32 runs; any powered experiment stream cuts this to 8–10/32, robust to the trap threshold. Among non-trapped runs every policy lands at regret 0.014–0.020. Mean loss: never 0.161 ± 0.047 , trigger 0.088 ± 0.027 , trigger plus a 15% spend floor 0.031 ± 0.013 ; a 30% floor alone, with no experiments, reaches 0.051 ± 0.011 .*

Proposition 4.9 (The ledger cannot see the failures it must prevent). *[numerically established] In trapped runs the belief-computed risk under-states realised regret about $13\times$ (late regret 0.230 against $\rho = 0.018$, 12 seeds, trap-dominated mean): the belief is confidently wrong, so a threshold policy does not fire when most needed.*

Proposition 4.10 (Belief-scored selection self-traps). *[numerically established] Trigger-lqc leaves some channel entirely unprobed in 5/32 runs and all five trap: $\mathbb{P}(\text{trap} \mid \text{zero-probe}) = 1.00$ against 0.19 otherwise; round-robin never leaves a channel unprobed (0/32). With the coverage guard zero-probe runs vanish and performance matches round-robin, after which scoring adds nothing: trigger 0.088, trigger-rand 0.090, trigger-lqcg 0.104, all $\pm$ about 0.03.*

The marginal precision that Chapter 3's EVSI machinery prices is real but small in-loop; the money is in keeping the belief out of a dark corner — a channel believed worthless, allocated nothing, therefore never

observed. The scheduler exploits its own misbelief: such a channel scores near zero, is never tested and never corrected, the sequential, self-inflicted form of Chapter 3’s plateau lesson. Round 2 caught a related blindness: at a corner $b_j \approx 0$, Hill’s $h'(0) = 0$ zeroes the j th column of D and hides the dark channel from ρ altogether, hence the probe-point ledger. In-loop, *which* powered experiment matters even less than statically; *whether, when* and *coverage* pay.

4.4.6 The low-drift trap paradox

Conjecture 4.11 (Process noise is regularisation). *[numerically established] (observation) [conjecture / speculative] (mechanism) The EKF’s trap rate rises as drift falls: 17/48 at $q = 0.0075$ against 5–7/48 at $q \geq 0.015$, and the zero-drift, zero-Q loop is itself trap-prone (never-policy median loss 0.29: no process noise, no forgetting, no escape). The proposed mechanism is that the predict step’s q_f^2 is what keeps Σ_t from collapsing around a wrong mean; the formal statement, the trap basins and the minimal floor or cadence guaranteeing escape are open (BL 12).*

Consistently, filter misspecification is asymmetric (Table 4.3): $q_f = q/4$ is catastrophic, $q_f = 4q$ merely wasteful. Set filter drift high, not low.

4.4.7 The E6 architecture

Table 4.1 is the dive’s design document, components in order of measured value. Weekly: predict with drift and inflation; update on revenue; ingest any delivered readout; reallocate at the belief mean subject to floors; evaluate the ledger at the probe point; launch the next powered design if $\rho_t \geq p^*$, or the cadence backbone says a test is due, or the coverage constraint names a stale or dark channel — whichever comes first, never later than $\tau_{\max} \approx 1.5\tau^*$.

Table 4.1: The E6 architecture (Dive 05 §6): components in order of measured value.

#	Component	Role and evidence
1	Drift-aware filter	generous process noise plus innovation inflation ($q_f = q/4$ costs 0.243; bursts $0.378 \rightarrow 0.289$ with inflation)
2	Spend floors	10–15% of historical spend per channel; belief-independent insurance at near-zero cost ($0.088 \rightarrow 0.031$)
3	Cadence backbone	launch from $\tau^*(c, r, \kappa)$ with planning-time constants; the belief may accelerate ($\rho \geq p^*$), never postpone
4	Coverage constraint	every channel probed within $G \approx 2\tau^*$ weeks, dark channels within about 30; a hard constraint, not a score bonus
5	Design menu	powered designs only (weak menu $\approx$ never); prefer switch/dual for higher κ ; round-robin an acceptable default
6	Moment-matching ingestion	through the operator map; no weekly refit; indistinguishable from the linearised update at this noise
7	Risk ledger ρ_t	reporting and acceleration only, never sole gatekeeper (13× under-statement in traps)

4.5 Numerical evidence

All runs use Construction 4.1 with common random numbers across policies. The *policy duel* (stage s4) runs seven policies from realistic starts over 32 seeds at baseline $(q, c) = (0.015, 0.15)$ with p^* from Equation (4.2); the *floor control* (s9) reruns never and trigger with floors. The *cadence grids* (s3, s3b) sweep $\tau \in \{15, 25, 40, 60, 90, 140, 220\}$ from clean starts over 24 seeds, s3b with spend jitter cut to 0.4% to check that the observational stream is not doing the experiments’ work; rounds 4–5 added grids at $q = 0.0075$ and 0.03.

The *scaling sweep* (s5) runs the trigger policy over $q \in \{0.0075, 0.015, 0.03\} \times c \in \{0.05, 0.15, 0.45\}$, 16 seeds per cell, clean starts, later refit with 48 seeds and a $q = 0.06$ cell; its numbers are those of Law 4.7. Stages s6–s8 are the ingestion comparison (20 paired seeds), robustness and controls (16 seeds each; Table 4.3). The instrument discriminates: the weak menu is indistinguishable from never while powered menus roughly halve loss, and the zero-drift control fires only to drain the initial posterior. Trap counts, not means, carry the ordering in Table 4.2: means have standard errors a third of their size because loss is bimodal.

Table 4.2: Policy duel from realistic starts, 32 seeds with common random numbers, baseline $(q, c) = (0.015, 0.15)$. Loss is mean regret plus amortised cost, mean $\pm$ standard error over seeds; trap = loss > 0.05 .

Policy	Mean loss	Traps	Note
never	0.161 ± 0.047	15/32	non-trapped runs 0.014–0.020 for every policy
trigger (round-robin)	0.088 ± 0.027	8–10/32	fires 16–22 times under drift
trigger-rand	$0.090 (\pm \text{about } 0.03)$	8–10/32	random menu design
trigger-lqc (scored)	—	8–10/32	5/32 leave a channel unprobed; all five trap
trigger-lqcg (scored, guard)	$0.104 (\pm \text{about } 0.03)$	8–10/32	zero-probe runs vanish
never + 30% floor	0.051 ± 0.011	—	no experiments at all
trigger + 15% floor	0.031 ± 0.013	—	best configuration

Table 4.3: Robustness and controls, trigger policy unless stated, 16 seeds (ingestion: 20 paired seeds); mean $\pm$ standard error.

Condition	Result	Reading
$q_f = q/4$	0.243 ± 0.072	over-confidence: few triggers, traps
$q_f = 4q$	0.111 ± 0.050 , 24 experiments	merely wasteful
burst drift (rare ± 0.25 jumps)	0.378; 0.289 with inflation	smooth-drift trigger breaks; inflation recovers part
weak menu (tenth-size pulses)	0.199 ± 0.081 vs never 0.192	power cliff reproduced in-loop
cadence ≥ 90 weeks	$\approx$ never	
zero drift ($q = 10^{-6}$)	about 7 launches vs 16–22	drains initial uncertainty only; never-policy median 0.29
moment vs linearised ingestion	median +0.0007; mean $+0.012 \pm 0.027$	no resolvable difference

4.6 Limitations and the devil’s advocate

A single EKF is not full MCMC: trap rates are partly a property of this filter, and a weekly re-fit-from-scratch Bayesian MMM would trap differently — probably less, at about $1000\times$ the compute — so the insurance economics should be rerun against a NUTS refit before the absolute trap rates are believed (BL 13). Drift is on β only; cost is exogenous (BL 7’s power model is unbuilt); latency is fixed; one experiment is in flight; $J = 3$; the trap threshold is a chosen scalar; near-submodularity is unproven; means at 16–32 seeds carry wide intervals. The compiler adds two: the observational stream is adaptively collected, which the EKF treats as exogenous — Chapter 13 shows a certainty-equivalent allocator on this plant stalls at an uninformative spend, and the coverage constraint here is the exogenous exploration that chapter finds necessary; and the law’s linear-growth premise is bounded above by the never policy’s risk floor (Proposition 4.6), so at high drift ℓ^* can exceed what any policy loses. What the rounds changed: round 2 added the probe-point ledger, the coverage guard and the jitter and floor controls; round 3 replaced trap-contaminated means by trap anatomy, refuted the drift half of the law and supplied the missing zero-drift baseline; round 4 revised the *attribution* of the flatness from drainage to filter pathology without changing the construction; round 5 confirmed and stopped.

4.7 Discussion: impacts

For practice the ordering of Table 4.1 is the result. A vendor building E6 on Meridian, Robyn or PyMC-Marketing should invest first in the filter’s drift terms and a spend floor, second in a cadence backbone with a coverage constraint, and last in design scoring. The counter-intuitive parts are actionable: never turn drift terms off in a production filter (the zero- Q loop traps), set them high rather than low, treat the ledger as accelerator and report rather than gatekeeper, and enforce coverage as a constraint. Experimentation platforms gain a derived answer to “how often”: τ^* with planning-time constants reproduces the quarterly folklore and says how cadence should move with cost ($\sqrt{c}$), but should not be asked to forecast the loss it buys. For CMO decision-making, the in-loop value of a calibration test is mostly insurance against a confidently wrong model, and the model’s own uncertainty report is least trustworthy exactly when the insurance is needed — the adoption study’s precision-theatre barrier (B5) produced by the filter rather than the analyst.

For theory the chapter turns E6 from an ambition into a construction with a law, a verified portfolio bound and three named pathologies, and locates the law’s validity: exact for linear-Gaussian loops, a selector only for the nonlinear filter. Chapter 7 (Dive 09) advanced the cadence thread from the refresh side — refresh interval and reallocation band separate exactly, and overlap-correlated refresh errors move the optimal interval from weekly to quarterly, the same square-root economics under a different noise structure. Chapter 13 (Dive 16) supplies the stall this loop would suffer without exploration and proposes an “information clock stopped” statistic for the coverage trigger. Chapter 6 (Dive 07) rediscovers trap insurance from the refresh side — no refresh scheme recovers from a structural break quickly, so breaks are what calibration experiments are for — and shares the NUTS budget that BL 13 and BL 19 both need. Later dives reuse the scheduler as a template (BL 49, BL 53, BL 76).

4.8 Further exploration

The dive’s next steps became three backlog items. BL 12, the sharpest, is trap theory: why does EKF trap probability rise as drift falls; formalise process noise as regularisation, characterise the basins and derive the minimal floor or cadence guaranteeing escape. BL 11 asks for belief-independent risk ledgers — worst-case regret over a trust region around the belief, or a held-out predictive-residual monitor — as the trigger in place of ρ . BL 13 is the NUTS-refit comparison at matched compute. Two steps stay in the dive’s list: BL 7’s per-design cost $c(d)$, then redo the trap economics; and portfolio scheduling under overlap with Proposition 4.5 as the concurrency planner.

In the compiler’s judgement the most promising extension is BL 11 and BL 12 taken together, because they attack one object from two sides: a held-out residual monitor is precisely a statistic that does not trust the belief mean, and Chapter 13’s stall statistic is a candidate with a derivable null distribution. A trigger that fires on evidence the filter cannot rationalise away would demote the cadence backbone and coverage constraint to fallbacks and give the law’s selector role the forecaster it lacks. Second is BL 13, since the whole insurance economics — and so the ordering of Table 4.1 — is conditional on the EKF being the filter.

Reviewer’s note (compiler)

Provenance. Stages setup-s4 were recovered from a pickled state produced by an interrupted earlier run; the reader cannot verify that the code on disk produced that state. The INDEX log line is timed ~05:30 UTC, about two hours *after* the file’s write time (03:39), unexplained. The log line agrees with the writeup on every number checked (0.457 ± 0.042 , 3/3, 1.01, 15/32 $\rightarrow$ 8–10, 0.031 vs 0.161, 5/5).

Results not in the code file. `05_fusion_loop.py` holds one cadence grid, $\tau \in \{15, 25, 40, 60, 90, 140, 220\}$ at $q = 0.015$, and the 9-cell, 16-seed sweep. The round-4/5 grids at $q = 0.0075$ and 0.03, the 48-seed refit, the $q = 0.06$ cell, the trap anatomy, the never-policy zero-drift baseline (s8 runs only the trigger policy) and the clean-room linear loop are not on disk; “best cadence 80 / 40 / 10–20” names values absent from the only grid in the file, so “argmin tracks 3/3” rests on runs the reader cannot inspect.

The drift exponent moved. 0.47 ± 0.08 (round 3) became 0.1–0.2 (48 seeds) and is quoted as “ $\approx 0.1\text{--}0.5$ ” in F1; the dive concedes the bootstrap se (± 0.4) cannot exclude 0.5. Read the refutation as “far flatter than 1.0”, not a measured exponent. Two mechanical contributions are not separated from the trap story: at $q = 0.06$ the law’s $\tau^* = 10$ weeks is below the loop’s 16-week latency, so that cell cannot realise the law’s cadence; and if the 48-seed row is at $c = 0.15$ its $q = 0.06$ median (0.0178) lies below the law’s $\ell^* = 0.030$, ratio 0.59, outside the stated 1.4–4.0 (the row’s c is not stated, and the “12-cell grid” does not match the 9-cell sweep in the code).

Small counts and glosses. “All five trap” is five events and “0.19 otherwise” about 5/27; “scoring adds nothing” (0.088 / 0.090 / 0.104, all $\pm$ about 0.03) is a non-significance at 32 seeds, not an equivalence; the $13\times$ under-statement is a trap-dominated mean over 12 seeds; the ingestion null rests on 20 paired seeds. “ $q = 0.015/\text{wk} \approx 20\%/\text{yr}$ sd” does not follow: $0.015\sqrt{52} = 0.108$ absolute, 8–14% of $\beta_i \in \{0.8, 1.0, 1.3\}$. $\kappa_{\text{plan}} = 0.4$ is a planning constant; Chapter 3’s measured κ for the menu’s designs run 0.22–0.36 for the powered pulses, 0.47 for the switch and 0.52 for the dual.

Labels and citations. Proposition 4.5’s ceiling is labelled numerical, but within the gauss- u approximation it has a two-line proof: $\text{Cov}(u, L) \text{Var}(L)^{-1} \text{Cov}(L, u)$ is positive semidefinite of rank at most $|A|$ and Loewner-below $\text{Cov}(u)$, so its Ω -weighted trace is bounded by the top- $|A|$ eigenvalue sum of A (Ky Fan); the open part is the near-submodularity. The dive’s “Zhang et al. 2021 PNAS, inference under adaptive allocation” resolves to Hadad, Hirshberg, Zhan, Wager and Athey (2021); no result depends on it.

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- [20] Frontier study (internal): the open-questions catalogue `02_open_questions.md` (E6, M10) and the adoption-barriers catalogue `03_mmm_adoption_barriers.md` (B5).
- [21] Program (internal): Dive 02 (operator map); Dive 03 (E6 greedy scheduler, LQ regret, null theorem); Dive 04 (LQC and gauss- u scorers, switch and dual designs, count-additivity conjecture); Dive 07 §2 (T4 and the break-recovery finding); Dive 09 §3.5; Dive 16 next step 2; INDEX backlog entries BL 7, BL 9, BL 11–BL 13, BL 19, BL 49, BL 53, BL 76.

Chapter 5

Geo aggregation and privacy noise in MMM

Provenance

Source dive: Dive 06 MMM under aggregation and privacy noise (M9) (run 2026-09-01, file 06_aggregation_privacy.md, written 07:36 UTC; INDEX log stamp ~07:45). **Code:** code/06_aggregation_privacy.py (entry points drift_laws, drift_noise, dp, frontier [b]). **Frontier-study problem addressed:** M9 — identification and estimation when covariates aggregate over heterogeneous units (ecological bias through nonlinear Hill transforms, the “Jensen-gap bias in β ”) and outcomes carry calibrated DP noise of known mechanism; nobody had derived the corrections or priced privacy in ROAS posterior width. Touches adoption barrier O3 (geo-level data is a step-change harder to obtain). **Backlog items closed:** none; **opened:** BL 14–BL 17.

Key ideas in this chapter

- Uneven geo spend shares (coefficient of variation c) make the national Hill curve a mixture whose fitted half-saturation drifts by $\Delta \log K = +c^2/2$ (s -independent; exact at all orders for symmetric two-point shares) and whose local slope flattens to $s_{\text{eff}} = s - c^2 s^3 H(1 - H)$.
- The M9 alarm that the Jensen gap biases β is refuted in its decision-relevant form: with stable shares the operating-point marginal-ROAS bias is $\leq 0.4\%$ even with heterogeneous, targeted geos; the bias sits in the shape ($K + 13\%$, $s - 0.25$ at $c = 0.5$) and in deep counterfactuals ($+28\%$ at $4\times$ spend).
- The aggregation bias that does reach the operating point is share *drift*: $+10.9\% \pm 3.0$ (up to $+20.9\% \pm 2.7$) on mROAS at $1\times$, which a share-calibrated mixture estimator using national outcomes plus the advertiser’s own geo spend cuts to $+0.3\% \pm 2.2$.
- Known Laplace(b) noise on outcomes inflates posterior width by exactly $\sqrt{1 + 2b^2/\sigma^2}$; an exact noise-aware likelihood recovers at most a factor 2 in information over this fix (ARE limit 50%); national-level DP is effectively free.
- Geo-granularity release beats national release iff the per-cell noise scale satisfies $b \lesssim \sigma_{\text{geo}} \sqrt{G}$; beyond that, coarse release plus the mixture estimator wins.

5.1 Problem and motivation

Post-cookie marketing mix models consume data that have been aggregated over people and places and then deliberately noised. Two worries hide in that sentence. The first is *ecological*: an MMM fits a saturating (Hill) response to national spend, but sales arise geo by geo, each geo at its own point on the curve because spend is spread unevenly, and a nonlinear function of an average is not the average of the function. The frontier study (M9) phrased this as a Jensen-gap bias in the effectiveness coefficient β . The second is *privacy*: the aggregated measurement APIs that replaced user-level attribution (Google’s Attribution Reporting API, Meta’s and Apple’s aggregated reporting, clean rooms with DP checks) add calibrated noise of a *known* mechanism to every released cell, and nobody had priced the privacy parameter ϵ in the width of a ROAS posterior.

The two worries meet in one design choice: at what granularity should an API release outcomes? Geo-week cells carry the cross-sectional variation that identifies saturation, but each carries the same noise scale against a signal G times smaller; national-week cells are barely touched by the noise but reintroduce ecological bias. Dive 06 set out to derive closed-form corrections for the Hill-shape bias under share dispersion, to price ε , and to put the two on one axis. Along the way it refuted its own planned indictment of naive national fitting, and found that the operating-point bias comes from share *instability* rather than *dispersion* and can be removed without geo-level outcomes.

5.2 Background: the ideas this chapter builds on

5.2.1 Aggregating nonlinear micro relations (van Garderen, Lee and Pesaran, 2000; Stoker, 1984; Blundell and Stoker, 2005)

Let units $g = 1, \dots, G$ obey $y_g = f(x_g; \theta) + \varepsilon_g$ with f nonlinear, while the analyst sees only $Y = \sum_g y_g$ and $X = \sum_g x_g$. Jensen's inequality is the elementary statement: if f is concave on the range of the x_g then $\sum_g f(x_g) \leq Gf(X/G)$, with a gap of order $\frac{1}{2}f''(\bar{x}) \text{Var}_g(x)$; for an S-shaped f , as Hill is when $s > 1$, the sign depends on where the units sit relative to the inflection. Van Garderen, Lee and Pesaran (2000) make this precise: no aggregate model in aggregate variables alone reproduces the aggregated nonlinear micro relation in general; the best aggregate predictor is the conditional expectation of $\sum_g f(x_g)$ given aggregate information, and it depends on the cross-sectional distribution of the regressors. Stoker (1984) supplies the identification counterpart: writing the aggregate as $\bar{f}_t = \int f(x) dP_t(x)$, the micro f is recoverable from a time series of aggregates only if the family $\{P_t\}$ is *complete*, rich enough that distinct micro functions cannot yield the same aggregates; when P_t is frozen up to scale (stable geo shares) the aggregate identifies only the mixture $u \mapsto \int f(wu) dP(w)$, never f . Blundell and Stoker (2005) survey the two escapes, modelling the distribution or collecting micro data; Sun et al. (2017) and the Meridian documentation take the second: a geo-level hierarchical Bayesian MMM partially pools Hill and adstock parameters across geos, and cross-geo variation in spend intensity is what pins down saturation.

5.2.2 Differential privacy for aggregated measurement (Delaney et al., 2024; Aksu et al., 2023)

A randomized release M is ε -differentially private if for every pair of datasets D, D' differing in one user's records and every output set S , $\mathbb{P}(M(D) \in S) \leq e^\varepsilon \mathbb{P}(M(D') \in S)$. The Laplace mechanism releases $f(D) + \eta$, $\eta \sim \text{Lap}(b)$ with density $e^{-|\eta|/b}/(2b)$ and variance b^2 , at scale $b = \Delta/\varepsilon$ where Δ is the L_1 sensitivity of f (the most one user can move it; $\Delta = 1$ for a count with one contribution per user); the Gaussian mechanism substitutes $N(0, \sigma_{\text{DP}}^2)$ noise and delivers the relaxed (ε, δ) guarantee. Sequential composition says repeated queries on the same users add their ε 's; *parallel* composition says queries on disjoint user sets each enjoy the full budget, so G geo-week cells can carry the same per-cell scale b as one national-week cell at the same ε — the premise of this chapter's frontier. Deployed advertising systems are of the “known mechanism” type: the Attribution Reporting API bounds each user's total contribution, sums contributions into aggregation-key buckets, and adds discrete Laplace noise of a published scale to every bucket of the summary report (Aksu et al., 2023, optimize the allocation of the contribution budget across buckets); Delaney et al. (2024) formalize the attribution rule and contribution bounding that make the guarantee hold. The analyst knows the exact noise distribution on every number she receives.

5.2.3 Privacy noise as measurement error (Evans and King, 2023, 2025; Gong, 2022; Karwa and Vadhan, 2018)

Known DP noise is measurement error of known distribution, and the classical/Berkson distinction says where it bites. Classical error, $W = X + U$ with U independent of the true X , attenuates a regression slope

by the reliability $\text{Var } X / (\text{Var } X + \text{Var } U)$; Berkson error, $X = W + U$ with U independent of the observed W , leaves a linear slope unbiased and only inflates variance. Noise on the *outcome* of a linear-in-parameters model is the benign case: least squares stays unbiased and the residual variance rises by $\text{Var } U$. Noise on a *covariate* (SKAN- or ARA-style noisy attributed conversions used as inputs) is classical error through a nonlinear link, scoped out as BL 14. Evans, King, Schwenzfeier and Thakurta (2023) correct estimates from DP-protected data by treating the added noise as measurement error of known variance, and Evans and King (2025) extend this to nonlinear transformations. Gong (2022) puts the mechanism into the likelihood, $p(z | \theta) = \int p(z | y) p(y | \theta) dy$, and uses Monte Carlo EM or approximate Bayesian computation for *exact* inference from the noised release. Karwa and Vadhan (2018) price privacy for a normal mean: an ϵ -DP confidence interval carries a privacy term of order $\sigma / (n\epsilon)$ against the sampling term $\sigma / \sqrt{n}$, so at fixed ϵ privacy is asymptotically free. Estimators are compared below by Fisher information for a location family with density f , $I = \int (f'/f)^2 f$ ($1/\sigma^2$ for a Gaussian, $1/b^2$ for a Laplace); asymptotic relative efficiency (ARE) is the ratio of two estimators' asymptotic informations.

5.3 Setup and assumptions

Geos $g = 1, \dots, G$, weeks $t = 1, \dots, T$. Geo spend x_{gt} passes through geometric adstock $a_{gt} = x_{gt} + \alpha a_{g,t-1}$ and the Hill transform $H(a; K, s) = a^s / (a^s + K^s)$ with half-saturation K and slope s . The micro model is

$$y_{gt} = \frac{B}{G} + \frac{\beta}{G} H(a_{gt}; K, s) + \varepsilon_{gt}, \quad \varepsilon_{gt} \sim N(0, \sigma_{\text{geo}}^2), \quad (5.1)$$

so β is national incremental sales at full saturation. Shares w_g ($\sum_g w_g = 1$) have cross-geo coefficient of variation c . Under stable shares $x_{gt} = w_g X_t$ and, adstock being linear, $a_{gt} = w_g A_t$ exactly with A_t the adstock of national spend. The national modeler sees X_t and $Y_t = \sum_g y_{gt}$ (noise sd $\sigma_{\text{nat}} = \sigma_{\text{geo}} \sqrt{G}$) and fits $Y_t = B + \beta H(A_t; \tilde{K}, \tilde{s}) + e_t$; tildes mark pseudo-true (population least-squares) national parameters. With $u = a/K$, $\text{logit } H = s \log u$ exactly: the Hill family is linear in (logit, log) coordinates. Defaults: $G = 20$, $T = 104$, $\alpha = 0.5$, $K = 1$, $s = 2$, $\beta = 100$, $B = 500$, $\sigma_{\text{geo}} = 1$ ($\sigma_{\text{nat}} \approx 4.47$), block-pulse national spend (levels uniform on $[8, 45]$ in 4–8-week blocks, the identifying design of Chapter 1), lognormal shares. Estimands are the shape (K, s) — the transport currency of Chapter 2 — and the marginal ROAS $\text{mROAS}(X^*) = d(\text{incremental sales})/dX$ at steady-state spend X^* , at the operating point ($1 \times$ mean spend) and at $0.25 \times$ – $4 \times$. A DP curator may release outcomes with additive $\text{Lap}(b)$ noise per cell, $b = \Delta/\epsilon$ known. Penalized fits use Meridian-style weak priors identical across estimators ($\log s \sim N(\log 1.5, 0.5^2)$, $\log K \sim N(\log \text{median}(a), 1)$, $\log \beta \sim N(\log 4 \text{sd}(Y), 1)$, $\text{logit } \alpha \sim N(0, 1)$), maximized to a MAP with Laplace (Hessian) widths.

Assumption 5.1 (Exogenous shares). Shares are fixed over time or follow an exogenous AR(1) log-share process (coefficient 0.9, innovation sd 0.15–0.30, renormalized weekly), uncorrelated with geo-level demand shocks.

Assumption 5.2 (Known mechanism, parallel composition). DP noise is additive Laplace on outcomes only, with b known and the same per cell whether cells are national-week or geo-week.

Assumption 5.3 (Single channel, correct family). One media channel, additive Gaussian micro noise, a Hill micro response of the family the modeler fits, no baseline covariates.

5.4 Main results

5.4.1 Closed-form Hill drift laws under share dispersion

Under stable shares the national curve is the fixed mixture $\bar{H}(u) = \mathbb{E}_w[H(wu)]$, w of mean one and variance c^2 after rescaling by $1/G$. Expanding $H(wu)$ to second order in $w - 1$ with $dH/d \log u = sH(1 - H)$ and

$d^2H/d(\log u)^2 = s^2H(1-H)(1-2H)$ gives $\bar{H} = H + \frac{1}{2}c^2H(1-H)\psi(u) + O(c^4)$,

$$\psi(u) = s \frac{(s-1) - (s+1)u^s}{1+u^s}, \quad (5.2)$$

and through the logit, $\logit \bar{H} = s \log u + \frac{1}{2}c^2\psi(u) + O(c^4)$: the whole distortion sits in ψ .

Law 5.4 (Hill drift laws). *[established in the dive]* Let $s_{\text{eff}}(u) = d \logit \bar{H} / d \log u$ be the local slope of the national curve and u^* its half-saturation point. To leading order in c ,

$$s_{\text{eff}}(u) = s - c^2s^3H(u)(1-H(u)) + O(c^4), \quad (5.3)$$

$$\Delta \log \bar{K} = \log u^* = +\frac{1}{2}c^2 + O(c^4), \quad \text{independent of } s, \quad (5.4)$$

and β is unbiased at leading order. Flattening is maximal at half-saturation, $\Delta s = -c^2s^3/4$. For symmetric two-point shares $w \in \{1-c, 1+c\}$ the half-saturation law is exact at all orders: $\log u^* = -\frac{1}{2} \log(1-c^2)$.

The slope law is one derivative: with $v = u^s$, $d\psi/dv = -2s^2/(1+v)^2$, so $\frac{1}{2}c^2 d\psi/d \log u = -c^2s^3v/(1+v)^2 = -c^2s^3H(1-H)$. The half-saturation law follows from $\psi(1) = -s$ for every s , whence $s \log u^* = \frac{1}{2}c^2s$ and the slope cancels — the source of the s -independence. The exact two-point result, found by the clean-room agent, is elementary: $\bar{H}(u^*) = \frac{1}{2}$ requires $H((1-c)u^*) + H((1+c)u^*) = 1$, which with $p = ((1-c)u^*)^s$, $q = ((1+c)u^*)^s$ reduces to $pq = 1$, i.e. $u^{*2}(1-c^2) = 1$. The same check located the slope law's validity boundary, $c^2s^2 \gtrsim 1$ (at $s = 8$, $c = 0.5$ it predicts $s_{\text{eff}} = -24$ against an actual 0.64), outside marketing grids ($s \leq 4$, $c \lesssim 0.3$ at the level of designated market areas), though where c exceeds 0.8 across US DMAs the laws are first-order guides only. Verification (E1b, symmetric two-point shares, $G = 4000$, $s \in \{1.5, 2, 3\}$): $\max |s_{\text{eff}} - \text{prediction}| \leq 5\text{--}10\%$ of the drop scale $c^2s^3/4$ at $c = 0.3$, and $\log u^*$ matches $c^2/2$ to three decimals at $c \leq 0.2$. Noiseless projected fits with lognormal shares (E2, $T = 520$) give at $c = 0.5$ $\bar{K} = 22.48 \pm 2.08$ on the national scale ($GK = 20$ at $c = 0$) against the law's 22.62, $\bar{s} = 1.75 \pm 0.07$ against the design-averaged Equation (5.3), and $\hat{\beta} \in [99.0, 100.5]$ across the c -sweep.

5.4.2 Stable-share aggregation bias is decision-benign at the operating point

Proposition 5.5 (Benign projection). *[numerically established]* With stable shares the least-squares projection of the fixed mixture $\bar{H}$ onto a single Hill curve matches its level and slope on the design's support, so the pseudo-true national mROAS at the operating point is biased by -0.0% to -0.1% for c up to 0.8, and by at most 0.4% with geo-heterogeneous half-saturations (K_g lognormal, $cv \ 0.5$) or targeted shares $w_g \propto 1/K_g$ (E4). The M9 alarm that the Jensen gap biases β is *[refuted within the dive]* in this decision-relevant form.

The bias lives elsewhere: in the parameters ($\bar{K} + 13\%$, $\bar{s} - 0.25$ at $c = 0.5$, as Law 5.4 predicts) and in deep counterfactuals, where a single Hill and the mixture separate (mROAS at $4\times$: $+10.3\% \pm 3.0$ at $c = 0.3$, $+28.3\% \pm 7.7$ at $c = 0.8$; at $0.25\times$: -3 to -7%). Both matter for uses that consume (K, s) directly (lift-test transport as in Chapter 2, “headroom to saturation” narratives) and both now have a formula; neither matters for the modal national decision, a small reallocation at current spend.

5.4.3 Share drift, and the share-calibrated mixture estimator

When shares move over time the national curve is a *moving* mixture no single Hill can track, and bias invades the operating point. The dive's reframe: aggregation bias in practice is a share-instability problem, and its cure needs no geo outcomes.

Construction 5.6 (Share-calibrated mixture estimator). *[established in the dive]* Fit the national outcome to the geo-spend mixture,

$$Y_t = B + \frac{\beta}{G} \sum_{g=1}^G H(\text{adstock}_\alpha(x_{g,\cdot})_t; K, s) + e_t, \quad (5.5)$$

using only Y_t and the advertiser's own geo spend x_{gt} — always observable, never privacy-exposed. Same parameter count as the naive fit; the estimand is the micro curve rather than its projection.

The construction is exact in the pseudo-true limit (0.0% bias at both drift levels) and coincides with the naive fit to four decimals when $c = 0$ (negative control E8a). Under realistic noise it cuts the drift bias at $1 \times$ from $+10.9\% \pm 3.0$ to $+0.3\% \pm 2.2$ (Table 5.1). A control with equal mean shares ($c = 0$) but drift sd 0.30 isolates the mechanism: the naive bias is *larger* there ($+20.9\% \pm 2.7$); the drift, not the static dispersion, does the damage.

5.4.4 Pricing ε : width law, bounded prize of noise-awareness, granularity frontier

Law 5.7 (DP width law). *[established in the dive]* With known $\text{Lap}(b)$ noise added to outcomes of natural noise sd σ , sampling and posterior widths of linear-in- β fits scale as $\sqrt{\sigma^2 + 2b^2}$; the inflation factor $\sqrt{1 + 2b^2/\sigma^2}$ is the whole price of ε for a national release.

This is the outcome-noise case of the measurement-error taxonomy: least squares stays unbiased and the residual variance gains $\text{Var Lap}(b) = 2b^2$. Its content is numerical: weekly national conversions with natural noise sd ~ 100 and $\Delta = 1$, released at $\varepsilon = 0.1$ per week, give $b = 10$ and inflation $\sqrt{1 + 200/10^4} \approx 1.01$; national-level DP is effectively free.

Proposition 5.8 (Bounded prize for exact noise-aware likelihoods). *[established in the dive]* Compare Gaussian least squares with variance inflated to $\sigma^2 + 2b^2$ against maximum likelihood under the exact convolved density $N(0, \sigma^2) * \text{Lap}(b)$. Their asymptotic relative efficiency for a location parameter is

$$\text{ARE}(b/\sigma) = \frac{1/(\sigma^2 + 2b^2)}{I_{\text{exact}}(\sigma, b)}, \quad \text{ARE} \rightarrow 1 \ (b/\sigma \rightarrow 0), \quad \text{ARE} \rightarrow \frac{1}{2} \ (b/\sigma \rightarrow \infty), \quad (5.6)$$

so the exact-inference machinery of Gong and of Evans–King buys at most a factor 2 in information over the two-line fix “inflate σ^2 by $2b^2$ ”.

The limit is transparent: as $b \gg \sigma$ the convolution tends to a Laplace, whose location MLE has information $1/b^2$ while least squares has $1/(2b^2)$. The dive computed I_{exact} by integrating the score of the FFT-convolved density (Table 5.2); the clean-room implementation gave 51.9% at $b/\sigma = 30$ and 50.6% at 100. The dive calls this the first such bound stated for the DP–MMM setting.

Law 5.9 (Privacy–aggregation frontier). *[numerically established]* The curator chooses between national-week cells (little DP damage, ecological limits) and geo-week cells (G times more cells and full cross-sectional identification, but per-cell signal β/G against the same per-cell noise b). Release at geo granularity while $b \lesssim \sigma_{\text{nat}} = \sigma_{\text{geo}}\sqrt{G}$; beyond that, coarse release plus the share-calibrated mixture wins. Finer panels are more privacy-fragile, consistent with per-cell $\text{SNR} \propto 1/(Gb)$: at $G = 50$ the geo release already trails at $b = 3$ (16.6% vs 10.7% RMSE for the mixture).

Cross-geo dispersion is thus the enemy of the national fit (the drift laws) and the friend of the geo fit, which it spreads across operating points — exactly what identifies the curve.

5.5 Numerical evidence

All experiments use the setup defaults ($G = 20$, $T = 104$, $\sigma_{\text{geo}} = 1$, block-pulse spend, lognormal shares, identical weak priors); “pseudo-true” rows are noiseless $T = 520$ fits isolating the projection bias.

Two further E7 facts interpret Table 5.3: RMSE at $2 \times$ and RMSE of $\hat{s}$ show the same crossover, and the geo release recovers $s \approx 1.94$ – 1.97 against a prior centred at 1.5 while the national fits stay prior-dominated at every b . The round-3 penalized comparison at stable shares gave mixture $s = 1.99 \pm 0.12$ against naive 1.90 ± 0.15 — the pseudo-true bias direction, but only just.

Table 5.1: Share drift (AR(1) log-share wiggle) and the mixture estimator: bias of mROAS at $1\times$ with $\pm\text{sem}$; RMSE across seeds. E5 pseudo-true (naive errors at $2\times$ reach $\pm 65\%$); E5n realistic noise, 24 seeds; R4 the mechanism control.

Experiment	Shares	Naive national fit	Mixture estimator
E5 pseudo-true, drift 0.15	$c = 0.5$	$+5.3\% \pm 4.5$	0.0%
E5 pseudo-true, drift 0.30	$c = 0.5$	$+10.1\% \pm 8.4$	0.0%
E5n noisy, drift 0.30	$c = 0.5$	$+10.9\% \pm 3.0$ (RMSE 18.4%)	$+0.3\% \pm 2.2$
E5n fresh 24 seeds	$c = 0.5$	$+10.0\% \pm 3.7$ (RMSE 20.6%)	$-2.9\% \pm 3.7$
R4 control, drift 0.30	$c = 0$	$+20.9\% \pm 2.7$	—
RMSE range, drift present		18.4–24.9%	10.7–18.5% ($\times 1.7$ – 2.3)
E8a control, no drift	$c = 0$	estimators coincide to four decimals	

Table 5.2: Pricing ε . Left: width law on a fixed design and identified linear sub-fit (E6c, ratio of $\text{sd}(\hat{\beta})$ to its $b = 0$ value, 400 reps; clean-room confirmed at 2.2% MC precision). Right: asymptotic efficiency of variance-inflated Gaussian LS relative to the exact convolved-density MLE (E6b).

b/σ_{nat}	Measured ratio	$\sqrt{1 + 2b^2/\sigma^2}$	b/σ	LS efficiency
1	1.70	1.73	0.25 / 0.5	100% / 98.9%
2	3.00	3.00	1 / 2	90.5% / 75.5%
4	5.82	5.74	4 / 8	63.7% / 57.0%
			30 / 100 (clean room)	51.9% / 50.6%
			∞	50% (limit)

Table 5.3: Privacy-aggregation frontier (E7): RMSE of mROAS at $1\times$, $c = 0.5$, 6 seeds per cell, per-cell $\text{Lap}(b)$ noise, $\sigma_{\text{nat}} \approx 4.47$. Six-seed cells carry roughly $\pm 30\%$ relative MC error; $b = 3$ is a tie within error. National arms use $\sigma^2 = 20 + 2b^2$, the geo arm $1 + 2b^2$.

b	National naive	National mixture	Geo release
0	12.5%	12.9%	3.6%
1	13.2%	13.7%	6.4%
3	15.5%	16.1%	13.5%
10	27.4%	27.4%	38.2%
30	53.9%	50.9%	129.6%

5.6 Limitations and the devil’s advocate

The attack rounds changed the story materially. In round 2 the unpenalized national fits, naive and mixture alike, ran to a degenerate ridge ($\hat{\beta} = 5972 \pm 2910$ against a true 100: $\beta \rightarrow \infty, K \rightarrow \infty$ with $\beta(a/K)^s$ finite mimics a power curve), so at $T = 104$ the national likelihood cannot pin the Hill shape at all and the ecological question is second-order to identification. The planned Jensen-gap indictment collapsed at the operating point (Proposition 5.5). And the first width-law experiment showed $\text{sd}(\hat{\beta})$ falling as DP noise rose ($20.3 \rightarrow 12.1$ against a predicted $5.7\times$ rise) — prior shrinkage masquerading as precision — which forced the fixed-design, identified-sub-fit re-test of Table 5.2. Round 4 (fresh seeds, the $c = 0$ drift control, the $G = 50$ check, a prior-centre sweep) confirmed round 3 without revision.

Four negatives bound the scope. At realistic national SNR the drift laws are subdominant to sampling error and prior pull: 24-seed naive fits give $s = 1.98 \pm 0.08$ where the pseudo-true value is 1.75, a finite-sample upward pull cancelling the drift by coincidence, and E8b confirms the s -drift is undetectable in one realistic fit; the laws matter for long panels, for refresh averaging (bias persists while noise averages; Chapter 6) and for transport. National shape recovery is prior-dominated either way (E9a): centring the s -prior at 1.0 or 3.0 drags the mixture posterior to 1.77 or 2.90 — the mixture moves the *target* to the right place but cannot conjure identification the national likelihood lacks, whereas geo data at small b can. The mixture does not dominate: at stable shares its $4\times$ -extrapolation RMSE (1.03) is worse than the naive fit’s (0.67). And the K -drift law is design-local (E9b): at $\alpha = 0.7$ the design sits deep in saturation, the crossing is unvisited, and the measured drift (0.116) overshoots the law (0.064), which holds at $\alpha = 0.3$ and 0.5.

The dive’s own devil’s advocate adds that benignness is proven only for single-channel, correctly specified, additive-noise worlds; that under drift the “true” benchmark is mROAS at mean shares, so a few points of naive bias could be convention, though the mixture faces the same benchmark and lands near zero; that exogenous AR(1) drift is the kindest case and spend chasing demand would confound both estimators (BL 15); that the geo arm of the frontier uses variance-inflated LS rather than the exact likelihood, which by Proposition 5.8 costs at most $\sqrt{2}$ in width and could shift the crossover left by $\lesssim 1.4$ without erasing the regime change; that parallel composition fails for event-level budgets split across cells; and that the ARE bound is asymptotic and location-only. The optimizer is MAP plus Laplace, not NUTS, and Chapter 4’s lesson on filter-class artifacts suggests re-checking prior domination under MCMC.

5.7 Discussion: impacts

For practice the chapter reverses the intuitive ordering of worries. A vendor fitting a national Hill curve (Robyn, PyMC-Marketing, Meridian in national mode) need not fear stable geographic dispersion for the everyday decision of nudging spend at the current operating point — the projection bias is a few tenths of a percent — but must fear it whenever (K, s) are consumed directly: calibrating to a geo lift test through the operator of Chapter 2, quoting headroom to saturation, planning a budget doubling. For those uses Law 5.4 gives an analytic de-biasing that costs nothing but the geo spend shares. The more consequential message is that the aggregation bias which *does* reach the operating point comes from shares that move, and Construction 5.6 removes it with data every advertiser already holds — a direct response to barrier O3: enterprises that cannot produce geo-level *outcomes* can still calibrate the national likelihood on geo-level *spend*, with no extra parameters and no privacy exposure.

For privacy engineering the results calibrate effort. National-week releases are essentially free at any plausible ϵ , and the exact-likelihood corrections of the DP-inference literature cannot repay more than a factor 2 in information over variance inflation — worth having when $b \gg \sigma$, ignorable otherwise. The frontier gives advertisers something concrete to ask of an aggregation API: cells fine enough to identify saturation but no finer than $b \approx \sigma_{\text{geo}} \sqrt{G}$ allows, with the mixture estimator as fallback when the API will not bargain. For theory the (logit, log) linearization of the Hill family is the reusable object: it turns the van Garderen–Lee–Pesaran wedge into two one-line laws and locates the s -independence of the half-saturation drift in the identity $\psi(1) = -s$. Stoker’s completeness condition explains the whole pattern: stable shares

freeze the cross-sectional distribution, so the aggregate identifies only the mixture (benign for its own slope, wrong for (K, s)); drifting shares move the distribution, but in a way no single Hill can absorb.

5.8 Further exploration

The dive's next steps are the four backlog items it opened. BL 14 treats DP noise on *covariates*, the SKAN/ARA case: known discrete-Laplace noise on inputs through the Hill link, its Berkson-versus-classical decomposition, corrected estimating equations, and whether the 50% ARE bound has an analogue; the literature gap is confirmed unclaimed. BL 15 replaces exogenous AR(1) drift with spend-chasing-demand allocation to see how much of the mixture estimator's advantage survives endogeneity (links BL 3/BL 10). BL 16 makes the frontier a mechanism-design statement: given ε and a decision loss, derive the optimal release granularity G^* in closed form from per-cell $\text{SNR} \propto 1/(Gb)$. BL 17 applies the drift laws as analytic de-biasing of nationally fitted (K, s) before the transport of Chapter 2. A full-MCMC check of prior domination shares a compute budget with BL 13 and BL 19.

In the compiler's judgement the most promising extension is BL 16 coupled with BL 15: a closed-form $G^*(\varepsilon, \text{loss})$ would turn a six-seed rule of thumb into a negotiating position with API vendors, and the endogenous-drift test decides whether the mixture estimator is a fix or only a fix for the simulator. The all-orders-exact half-saturation law also deserves a proof for general share distributions, being the one piece of the chapter that looks like a theorem rather than an expansion.

Reviewer's note (compiler)

(1) *Code versus writeup*. The file on disk implements the width-law test (E6c) with 40 seeds and a design re-simulated per seed, whereas the writeup reports a "fixed design" and "400 reps"; the round-3 fixed-design variant is not in the file. The `drift_noise` entry point calls `e3b.fitmap`, a module not on disk, so the E5n headline is not reproducible from the shipped code alone. (2) *The mixture estimator's pair of numbers*. The writeup gives "+0.3% $\pm$ 2.2 / -2.9% $\pm$ 3.7 across the drift scenarios" without naming which is which; Table 5.1 assigns them to the main and fresh-seed E5n runs (the compiler's reading), and the INDEX one-liner quotes only the favourable +0.3%. (3) *Projected-fit support for the K law is weak on its own*. $\tilde{K} = 22.48 \pm 2.08$ at $c = 0.5$ lies about 1.2 standard errors above the no-drift value 20; the law's real support is the pointwise E1b check and the clean-room proof, and exactness is proven only for symmetric two-point shares. (4) *Frontier resolution*. Six-seed cells with $\pm 30\%$ MC error support the $\times 3.5$ –10 gaps at $b \in \{0, 30\}$ but not a precise crossover; the rule $b \lesssim \sigma_{\text{geo}} \sqrt{G}$ rests on one grid plus one $G = 50$ check, and the geo arm is handicapped by up to $\sqrt{2}$ through variance-inflated LS. (5) *"National DP is free" is a units statement*: the $\times 1.01$ example assumes a count outcome with $\Delta = 1$ and $\sigma \approx 100$; revenue outcomes with large per-user sensitivity change b/σ . (6) *Novelty of the ARE bound*. The 50% limit is the textbook Laplace-versus-Gaussian location-information ratio; the dive claims only first statement for DP-MMM, and it is asymptotic and location-only. (7) Beyond item (2) no disagreement between writeup and INDEX line was found (file stamp 07:36 vs log $\sim$ 07:45 UTC), and no number changed between rounds without a stated reason.

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- [13] Frontier study, *02_open_questions.md* (M9) and *03_mmm_adoption_barriers.md* (O3); Dive 01 (identifying designs), Dive 02 (transport consumes K, s), Dive 03/Dive 04 (decision-loss framing).

Chapter 6

Refresh stability and analyst degrees of freedom

Provenance

Source dives: Dive 07 *Refresh stability and analyst degrees of freedom* (B1: *barriers M6/M4*) (run 2026-09-01, file 07_refresh_stability.md, last written 2026-09-01 11:56 UTC; INDEX log entry ~12:30). **Code:** code/07_refresh_stability.py (base stages s1-s11 plus merged attack modules attack1-attack7). **Frontier-study problem(s) addressed:** queue item B1, i.e. adoption-barrier catalogue entries M6 (“refresh instability: the ROAS that flipped”) and M4 (“model multiplicity and analyst degrees of freedom”) of 03_mmm_adoption_barriers.md, with open-questions entry H8 (pre-registered decision rules) nearest; the open-questions codes M4/M6 (auction interference, heavy tails) are unrelated. **Backlog items closed:** none; **opened:** BL 18 (rotation theory of refresh decorrelation), BL 19 (posterior mean versus MAP flip rates), BL 20 (optimal hysteresis deadband, later resolved by Dive 08, Chapter 7).

Key ideas in this chapter

- For asymptotically linear estimators on overlapping windows, adjacent refreshes correlate at the overlap fraction $\rho = \omega = (W - \Delta)/W$ and a ranking flips with probability $F(z, \rho) = 2[\Phi(-z) - \Phi_2(-z, -z; \rho)]$, verified to ± 0.016 (clean-room ± 0.015); annualised flips scale as $\Delta^{-1/2}$, so refreshing *more* often shows *more* churn at identical model quality.
- The nonlinear MMM breaks the law informatively ($\rho \approx 0.46\text{--}0.63 \ll 0.875$): optimizer, warm start, functional and noise level are exonerated, an exactly linear “anchored” estimator still gives $\rho = 0.568$ (decomposition predicts 0.620), and the cause is rotation of the information matrix’s near-null eigenspace with each window’s realised spend.
- Naive posterior-as-prior on a moving window counts overlapping data once per window: width collapses as $1/\sqrt{k+1}$ (0.327 versus $1/3$ at $k = 8$), 90% coverage falls to 0.24 in the MMM, and RMSE *rises* ($0.51 \rightarrow 0.66$).
- CPPD (increment likelihood, discount $a = \omega$, variance floor, drift test) gives flips 0.12 versus 0.28 for fresh refits at coverage 0.91 and the best static log-RMSE (0.48); freezing coefficients is genuinely best under noise domination (RMSE 0.59) but carries a structural break as permanent bias.
- Analyst degrees of freedom amplify flips only under the *Janus condition*—near-tied specifications disagreeing in sign on the reported gap—where random selection at a 30% switch rate multiplies flips $5.8\times$ (0.295 versus 0.051); sign-agreeing ties *reduce* flips (0.204 versus 0.279).
- A pre-registered 1σ reporting deadband cuts reported flips 78% ($0.25 \rightarrow 0.057$) with the wrong-order rate unchanged ($0.518 \rightarrow 0.518$).

6.1 Problem and motivation

A marketing-mix model is refitted every month or quarter on a window that slides forward, and its channel rankings change. The adoption study records this as the single most visible trust-destroyer for executives: a Robyn deployment in which a channel’s ROAS went from 0.75 to 0.2 over five refreshes, driven partly by

the moving window; Robyn issues #890 and #507 in which refreshes zero out channels; Meridian’s warning that “even a relatively small amount of new data may have a noticeable effect”. The industry’s quiet fix is to freeze media coefficients between refreshes, which Recast calls “effectively just lying about the model results”; Robyn’s `robyn_refresh` bounds hyperparameters near the last run, stability by construction. The principled alternative—carry the previous posterior forward as the next prior—is named and left open by Meridian’s documentation: “setting priors that match an old MMM’s results effectively counts the old data twice”.

Dive 07’s literature search confirmed two gaps: no one publishes a *measured* flip rate, and no stability-constrained refresh scheme with a formal guarantee exists anywhere. The dive assembles the unassembled machinery around five deliverables: a flip-rate law, a diagnosis of what drives refresh instability in nonlinear MMMs, an overlap-corrected sequential updating scheme, a characterisation of when analyst degrees of freedom amplify instability, and a pre-registrable reporting protocol. Its sharpest finding is negative for the folk theories: refresh flips in a weakly identified MMM are a *geometry* problem, not an optimizer, seed or noise problem, so better data will not fix them.

6.2 Background: the ideas this chapter builds on

6.2.1 Overlapping windows and the bivariate-normal orthant (Sheppard, 1899; Harri and Brorsen, 2009)

Two standard normal variables with correlation ρ fall in the negative quadrant with probability given by Sheppard’s formula, $\mathbb{P}(U < 0, V < 0) = \frac{1}{4} + \arcsin(\rho)/(2\pi)$, so they *disagree* in sign with probability $\frac{1}{2} - \arcsin(\rho)/\pi$: $\frac{1}{2}$ at $\rho = 0$, 0 at $\rho = 1$. For a general threshold the joint tail is the bivariate normal distribution function $\Phi_2(h, k; \rho) = \mathbb{P}(U \leq h, V \leq k)$, non-elementary but routine to evaluate.

Harri and Brorsen study regressions whose observations are W -period sums of an underlying series (multi-period returns, moving averages): two observations Δ periods apart share $W - \Delta$ underlying shocks and their errors correlate at $(W - \Delta)/W$, a moving-average structure standard errors must account for. The same arithmetic applies to any estimator that is, to first order, an average of per-observation scores: an OLS coefficient on W stationary observations is $\hat{\theta} \approx \theta + W^{-1}M^{-1}\sum_t x_t \varepsilon_t$ with $M = \mathbb{E}[x_t x_t^T]$, so two windows sharing $W - \Delta$ observations have $\text{Cov}(\hat{\theta}_k, \hat{\theta}_{k+1}) = ((W - \Delta)/W) \text{Var}(\hat{\theta}_k)$: the overlap fraction $\omega = (W - \Delta)/W$ is the adjacent-refresh correlation of any asymptotically linear estimator with exogenous regressors and independent noise.

6.2.2 Type-S errors, the vibration of effects and the garden of forking paths

Gelman and Carlin (2014) define the *Type-S* error of an estimate as the probability that it has the wrong sign and the *Type-M* error as its expected exaggeration, both computed from the true effect and the standard error; when the true effect is small relative to the standard error the sign of any single estimate is close to a coin toss. Patel, Burford and Ioannidis (2015) fit every combination of 13 adjustment covariates (8,192 specifications) to 417 exposure variables predicting mortality in NHANES and call the spread of estimates across specifications the *vibration of effects*; 31% of variables showed a *Janus effect*, named for the two-faced god: estimates on both sides of the null depending on the specification. Gelman and Loken (2013) supply the epistemology, the *garden of forking paths*: an analyst who chooses window length, controls or transformations after seeing the data has performed a data-contingent analysis, invalid even without conscious fishing because a different data set would have led down a different path; pre-registration fixes the path in advance. In MMM the forks are the modelling window, the seasonality terms, the adstock prior “camp” (short versus long carryover), and the choice of one model from Robyn’s Pareto front, where the `decomp.RSSD` criterion rewards attribution close to spend shares.

6.2.3 Power priors, forgetting and why posterior-as-prior double counts

The *power prior* of Ibrahim and Chen (2000) discounts historical data D_0 by raising its likelihood to a power, $\pi(\theta \mid D_0, a_0) \propto L(\theta \mid D_0)^{a_0} \pi_0(\theta)$, $a_0 \in [0, 1]$: $a_0 = 1$ pools fully, $a_0 = 0$ ignores the past, and for a Gaussian posterior with precision P_1 the effect is the scaling $a_0 P_1$. The control-theory version is *exponential forgetting* (Kulhávy and Kárný, 1984): before each update the posterior is flattened towards an alternative density, $p_{t|t-1}(\theta) \propto p_{t-1|t-1}(\theta)^\lambda p_{\text{alt}}(\theta)^{1-\lambda}$, which for Gaussians blends precisions, $\lambda P_{t-1} + (1-\lambda)P_{\text{alt}}$. Smídl and Gustafsson (2012) estimate λ online and read its drops as change detection; Masegosa et al. (2017) put a hierarchical prior on the power; Pawel et al. (2023) show normalised power priors always discount. Reznik (2026) derives the discount minimising predictive log-loss, $a_0^* = d/(2N_0\bar{D}_0 + d)$, with d the parameter dimension, N_0 the historical sample size and $\bar{D}_0$ the mean Kullback–Leibler divergence between historical and current data-generating distributions: compatible data pool in full, any discrepancy caps the borrowed information.

Sequential Bayes is exact only on *disjoint* increments, $p(\theta \mid D_1, D_2) \propto p(\theta \mid D_1) p(D_2 \mid \theta, D_1)$. Taking window k 's posterior as the prior and multiplying by the likelihood of the *whole* window $k+1$ re-counts the $W - \Delta$ shared observations; Du (2026) shows such factorised updates reproduce the joint posterior only under conditional independence and deviate with the correlation strength, while conditional-likelihood updates stay consistent. Kessler et al. (2023) show exact sequential updating fails under misspecification, where each increment's pseudo-true value moves. Both are the everyday condition of an MMM refresh.

6.2.4 MAP, the Laplace approximation and the information ridge (Jin et al., 2017; Chapter 1)

A MAP fit maximises the log posterior; the Laplace approximation replaces the posterior by $\mathcal{N}(\hat{\theta}, H^{-1})$, H the Hessian of the negative log posterior at $\hat{\theta}$, which for Gaussian-noise nonlinear regression in Gauss–Newton form is $H = J^T J / \sigma^2 + P_0$ with J the Jacobian of the mean function and P_0 the prior precision. One Gauss–Newton step from an anchor θ_a gives an estimator *linear in the data*,

$$\hat{\theta} = \theta_a + \left(\frac{J^T J}{\sigma^2} + P_0 \right)^{-1} \left[\frac{J^T (y - \mu(\theta_a))}{\sigma^2} + P_0(m_0 - \theta_a) \right], \quad (6.1)$$

m_0 the prior mean. Along eigenvectors of $J^T J$ with large eigenvalues the data term dominates; along near-null eigenvectors the prior term does, however small σ is. Chapter 1 showed that in the geometric-adstock, Hill-saturation MMM of Jin et al. (2017) the likelihood is flat along ridges trading adstock decay, saturation shape and effectiveness, so with weekly national data the near-null eigenspace is several dimensions wide. This chapter's addition: the *orientation* of that eigenspace depends on the window's realised spend path, and so rotates from refresh to refresh.

6.3 Setup and assumptions

The data-generating process is a weekly national MMM with $J = 3$ channels,

$$y_t = c_0 + \sum_{j=1}^3 \beta_j \text{Hill}(\text{adstock}(x_{jt}; \alpha_j); K_j, s_j) + \gamma_c \cos \frac{2\pi t}{52} + \gamma_s \sin \frac{2\pi t}{52} + \varepsilon_t, \quad (6.2)$$

$\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ i.i.d., geometric adstock $a_t = x_t + \alpha a_{t-1}$, $\text{Hill}(a; K, s) = a^s / (a^s + K^s)$, exogenous AR(1)-lognormal spend (autocorrelation 0.5, coefficient of variation 0.35). Truth: $\beta = (1.20, 1.00, 0.55)$, $K = (1.00, 1.20, 0.80)$, $s = (1.8, 1.6, 2.0)$, $\alpha = (0.55, 0.35, 0.70)$, mean spend $(1.0, 0.9, 0.6)$, $c_0 = 2$, $\sigma = 0.25$. The 15 parameters $(c_0, \gamma_c, \gamma_s, \log \beta_j, \log K_j, \log s_j, \text{logit } \alpha_j)$ are estimated by MAP under weak independent Gaussian priors on the transformed scale (sds 3, 1, 1, 1.5, 1.0, 0.5, 2.0), with Laplace covariance. The design is deliberately MMM-realistic: weakly identified, with log-ROAS estimation noise of about 0.4–0.9.

Definition 6.1 (Refresh, overlap, flip). Refresh k fits the window of $W = 104$ weeks ending at $t_k = W + k\Delta$; the *overlap fraction* is $\omega = (W - \Delta)/W$ ($\omega = 0.875$ at the quarterly $\Delta = 13$). The reported quantity is the window ROAS $\hat{R}_j = \hat{\beta}_j \overline{\text{Hill}(\hat{a}_{jt})} / \bar{x}_j$ over the window. A *ranking flip* for the pair (i, j) is a reversal of $\text{sign}(\hat{R}_i - \hat{R}_j)$ between adjacent refreshes. Writing $D_k = \hat{R}_{i,k} - \hat{R}_{j,k}$, the *signal* is $z = |\mathbb{E}D_k| / \text{sd}(D_k)$ and $\rho = \text{Corr}(D_k, D_{k+1})$. Unless stated otherwise the reported pair is channels 1 and 2, which are near-tied ($z \approx 0.2$).

Definition 6.2 (Refresh schemes). *IND*: refit each window from the base prior. *NPP*: previous Laplace posterior as prior, likelihood of the full new window. *FRZ*: media parameters frozen at the first fit, baseline and seasonality refitted. *CPP*: likelihood of the Δ new observations only, adstock warmed on the full spend history, prior = (previous posterior) ^{a} $\otimes$ (base prior) ^{$1-a$} with $a = \omega$. *CPPD*: CPP plus a per-parameter prior-sd floor of $0.3 \times$ the base prior sd and a predictive drift test (chi-square of the Δ new standardised residuals at nominal level 0.01, parameters at the previous MAP) that drops a to 0.2 when it fires.

Assumption 6.3 (Regime of the flip law). For Law 6.5 the estimator is asymptotically linear in the data, the regressors are exogenous and stationary, the noise is independent, and (D_k, D_{k+1}) is bivariate normal with common mean and variance.

Assumption 6.4 (Simulation scope). Known σ ; MAP/Laplace throughout; one DGP family; static truth except in the drift (log-random-walk β , step 0.01) and break (β_1 cut to 45% at week 156) regimes; no endogenous spend.

6.4 Main results

6.4.1 The flip-rate law and the cadence law

Law 6.5 (Flip-rate law). *[established in the dive]* Under Assumption 6.3, $\rho = \omega$ and the probability that the pair's ranking flips between adjacent refreshes is

$$F(z, \rho) = 2[\Phi(-z) - \Phi_2(-z, -z; \rho)], \quad F(0, \rho) = \frac{1}{2} - \frac{\arcsin \rho}{\pi}. \quad (6.3)$$

With $\delta = \mathbb{E}D_k$, $s_D = \text{sd}(D_k)$, $U = (D_k - \delta)/s_D$, $V = (D_{k+1} - \delta)/s_D$ and $z = \delta/s_D \geq 0$, a flip from positive to negative has probability $\mathbb{P}(U < -z, V > -z) = \Phi(-z) - \Phi_2(-z, -z; \rho)$, the reverse flip the same by symmetry; at $z = 0$ Sheppard's formula gives the closed form. The falsifiable content is the orthant form plus $\rho = \omega$ (z uses the empirical sd, which the dive notes is self-referential). In the linear probe (OLS, $\Delta \in \{4, 13, 26, 52, 104\}$, 400 replications of 8 refreshes) ρ_{emp} was within 0.01 of ω and F within 0.005–0.016 of prediction; a clean-room agent re-derived the formula and confirmed it at 40,000 replications (F within 0.005–0.015; its ρ ran 0.837 against $\omega = 0.875$, attributed to finite-sample design variation, the mechanism of Section 6.4.4).

Corollary 6.6 (Cadence law). *[established in the dive]* For a null pair the annualised number of visible flips is

$$\frac{52}{\Delta} F\left(0, 1 - \frac{\Delta}{W}\right) \approx \frac{52}{\pi} \sqrt{\frac{2}{W\Delta}} \propto \Delta^{-1/2}, \quad (6.4)$$

using $\arcsin(1 - x) = \pi/2 - \sqrt{2x} + O(x^{3/2})$.

More frequent refreshes generate *more* observed ranking churn per year at identical model quality: quarterly to weekly multiplies visible flips by $\sqrt{13} \approx 3.6$ in principle. The clean-room round forced a softening: measured at $\Delta = 4$ versus $\Delta = 26$ the ratio is 2.26 ± 0.04 against 2.50 from the exact law (the asymptote overstates by about 10% at small Δ), and at $\Delta \leq 2$ empirical flips run *below* prediction. The practical reading stands: a CMO buying monthly refreshes is buying more apparent instability, by arithmetic rather than model failure.

6.4.2 Naive posterior-as-prior is quantified overconfidence

Proposition 6.7 (Double counting). *[established in the dive] Under NPP with Gaussian conjugacy and a stationary design, the posterior precision after k refreshes is $P_k \approx P_0 + (k + 1)P_W$ with P_W the full-window likelihood precision, so the reported width collapses as $1/\sqrt{k + 1}$ while the information actually available stays that of one window.*

Each observation is counted once per window containing it. The exact-conjugate linear simulation gives width ratio 0.327 at $k = 8$ against $1/3$ (clean-room 0.326–0.328); nominal 90% coverage collapses to 0.52–0.56 in the linear model and 0.24 in the nonlinear MMM while flips drop to 0.036–0.048. This is the mechanics of stability by freezing beliefs: Robyn-refresh-style anchoring buys calm with collapsing honesty, and RMSE simultaneously *rises* ($0.51 \rightarrow 0.66$ in log-ROAS) because the frozen belief locks in early-window errors.

6.4.3 The corrected scheme: increment likelihood with discount $a = \omega$

Construction 6.8 (CPP and CPPD). *[established in the dive] Condition refresh k on the Δ new observations only, with adstock warmed on the full spend history, under the Gaussian prior with precision $aP_{k-1} + (1 - a)P_0$ and precision-weighted mean, $a = \omega$. Add a per-parameter variance floor and a predictive drift test with a pre-registered escalation (CPPD).*

The discount is derived. With P_Δ the precision of one increment, the recursion $P_k = P_\Delta + aP_{k-1} + (1 - a)P_0$ has steady state

$$P^* = P_0 + \frac{P_\Delta}{1 - a} = P_0 + \frac{W}{\Delta} P_\Delta \approx P_0 + P_W \quad \text{at } a = \omega, \quad (6.5)$$

the honest moving-window precision: the forgetting factor that reproduces a W -week window is exactly the overlap fraction, connecting moving-window practice to the power-prior and forgetting literature in one line. Exact-conjugate verification: steady-state width within 5% of the moving-window width, coverage 0.97–0.98. The clean-room agent derived the conservatism: the geometric weights ω^m spread the information over $(1 + \omega)/(1 - \omega)$ increments, i.e. $2W - \Delta$ observations, so the true sd is about $\sqrt{W/(2W - \Delta)} \approx 0.73 \times$ the reported one, and CPP is variance-better than a hard window at equal claimed width (linear-probe RMSE 0.166 versus 0.233). In the nonlinear MMM Laplace propagation still over-concentrates (coverage 0.80 in the nine-refresh static run); the floor at $0.3 \times$ base prior sd restores coverage 0.91 at flips 0.07–0.08 against IND's 0.26, and without it coverage is 0.59 with a 23% trap rate (final log-ROAS error above 0.69)—Chapter 4's low-drift trap paradox in refresh clothing. The drift test fires at 0.14 (well-specified) and 0.17 (delayed-peak adstock truth) against nominal 0.01, inflated by ignoring parameter uncertainty. Round 2 found that resetting the adstock transient inside increments wrecks the scheme (RMSE $0.71 \rightarrow 0.55$ after warming on spend history), re-confirming Chapter 1's transient result.

6.4.4 What actually causes MMM refresh flips: rotation, not churn

In the nonlinear MMM Law 6.5 fails informatively: measured $\rho = 0.46\text{--}0.63 \ll \omega = 0.875$, roughly doubling flip rates.

Proposition 6.9 (Refuted churn hypothesis). *[refuted within the dive] Round 1 proposed $\rho = \omega(1 - h)$ with h the variance share from optimizer-idiosyncratic multi-basin convergence, seemingly supported by a multistart probe (h up to 0.40). Round 2 killed it: cutting σ threefold left flips unchanged, and a multistart from 10 dispersed initialisations converged to identical ROAS ($sd \leq 5 \times 10^{-4}$); the MAP is unique, the probe's dispersion having come from a few multimodal outlier windows.*

Proposition 6.10 (Rotation mechanism). *[numerically established] The anchored estimator Equation (6.1) (one Gauss–Newton step from the window-0 MAP, exactly linear in y) still shows $\rho = 0.568$. Its exact decomposition $\hat{\theta}_k = \mathbb{E}[\hat{\theta}_k | X] + C_k J_k^T \varepsilon_k / \sigma^2$ with $C_k = (J_k^T J_k / \sigma^2 + P_0)^{-1}$, applied to the ROAS gap, gives $\rho_{\text{pred}} = (\rho_\varepsilon V_\varepsilon + \rho_X V_X) / (V_\varepsilon + V_X) = 0.620$: the σ -independent design-wobble term V_X —the prior-shrunk component in the near-null eigenspace, which swings as that eigenspace rotates with each window's realised spend—carries 21% of the*

variance at $\rho_X = 0.41$, and the noise term correlates at only $\rho_\varepsilon = 0.676 < \omega$ because shared noise enters through rotating projections.

The attack rounds eliminated the folk explanations one by one (Table 6.1); most telling, a ninefold cut in σ leaves $\text{sd}(D) \approx 0.38$ unchanged, so flip dispersion is noise-floor-independent, and Chapter 1-style block pulsing restores the correct pseudo-true sign ($z: 0.22 \rightarrow 1.10$) without restoring the correlation ($\rho \leq 0.31$). Refresh instability in weakly identified MMMs is a geometry problem: the identified ridge’s thin directions rotate with the spend realisation. Better data will not fix it; only information in the thin directions—priors carried across refreshes, calibration experiments (Chapter 2), identifying spend perturbation—will, which is also why CPP works: the carried prior fills the rotating null space.

6.4.5 Analyst degrees of freedom: the Janus condition

Proposition 6.11 (Janus condition). *[numerically established] Per-refresh specification selection adds ranking churn if and only if the near-tied specification set is sign-heterogeneous on the decision quantity. (a) Deterministic selection (AIC over $W \in \{78, 104, 130\} \times \text{seasonality on/off}$) picks one specification 100% of the time and adds nothing. (b) Random tie-breaking among specifications within 2 AIC units whose pseudo-true answers agree in sign (tight short- versus long-carryover adstock prior camps, logit- α means ∓ 1.5 , $\text{sd } 0.5$, $\times W \in \{104, 130\}$) reduces flips, 0.204 ± 0.024 versus 0.279 ± 0.027 for the pre-registered broad-prior fit. (c) When near-tied specifications disagree in sign on the reported gap, random selection at a 30% switch rate amplifies flips $5.8\times$ at matched signal (0.295 versus 0.051), approximately $F \approx \pi_{\text{switch}} \cdot \mathbb{P}(\text{spec sign dominates}) + \text{noise floor}$.*

Results (a) and (b) are refutation-grade evidence *against* the naive M4 story that any analyst arbitrariness destabilises: strong camp priors stabilise more than arbitrariness destabilises. M4 bites exactly in the sign-disagreeing case—the documented Robyn Pareto-front situation, Patel et al.’s 31% Janus rate—which is what `decomp.RSSD`-style multi-objective fronts produce; pre-registration of the specification has value there and only there.

6.4.6 The reporting layer and the protocol

Construction 6.12 (Pre-registered hysteresis). *[numerically established] Report a ranking change only when the new estimated gap crosses a published deadband of h standard deviations of the gap in the new direction, while the underlying posterior keeps updating. On IND estimates (the worst case) at $h = 1$, reported flips fall from 0.25 to 0.057 (−78%) with the wrong-order rate statistically unchanged, $0.518 \rightarrow 0.518$ ($n = 160$).*

For near-null pairs the reported ranking was a coin toss either way; hysteresis does not hide information, it stops narrating the coin, and unlike coefficient freezing it is honest because the rule is pre-registered at the reporting layer while the posterior keeps updating. The assembled deliverable for B1 is the protocol: (1) pre-register the specification, frozen, with re-selection only at scheduled annual rebuilds—mandatory when candidate specifications disagree in sign, cheap insurance otherwise; (2) CPPD updating with a pre-registered escalation when the drift test fires (floor drop or experiment trigger, never a silent refit); (3) hysteresis reporting at $h \approx 1$ with the deadband published; (4) a cadence chosen knowing flips per year scale as $\Delta^{-1/2}$, the refresh deck reporting $F(z, \rho)$ *expected* flips beside realised ones: “3 flips expected, 3 observed” instead of “the model changed its mind again”.

6.5 Numerical evidence

All nonlinear experiments use $W = 104$, $\Delta = 13$ and the DGP of Equation (6.2); flip rates are means over adjacent-refresh pairs with binomial standard errors. Table 6.1 collects the probes behind Proposition 6.10; each changes one thing and none restores $\rho \rightarrow \omega$.

Table 6.1: Exonerating the folk explanations for MMM refresh decorrelation (channels 1–2 gap, $\omega = 0.875$, Dive 07 T5 and controls).

Probe	Finding	Verdict
Multistart, 10 dispersed inits	identical ROAS, $\text{sd} \leq 5 \times 10^{-4}$	MAP unique; not optimizer
Cold versus warm start	ρ 0.70 versus 0.60	not path dependence
ROAS at fixed reference path	ρ 0.601 versus 0.589	not the functional
σ cut $9\times$	$\text{sd}(D) \approx 0.38$ unchanged	not SNR
Block-pulsed identifying design	z 0.22 $\rightarrow$ 1.10, $\rho \leq 0.31$	sign fixed, correlation not
Anchored linear estimator	$\rho = 0.568$; decomposition predicts 0.620	rotation of near-null eigenspace
Linear big-gap control	flip 0.007 (10/1400)	instrument reads ≈ 0 under strong ID

The negative control matters: no tested *nonlinear* configuration reached the linear floor, itself a statement about MMM-realistic identification. The regime duel (Table 6.2) runs $n = 40$ replications per regime over 12 refreshes, scoring flips, log-RMSE against the current truth over the last four refreshes, 90% coverage of $\log \hat{R}_1$, the trap rate (final $|\log \text{error}| > 0.69$) and, in the break regime, the final absolute log error; $\pm \text{SE}$ is about 0.03–0.08, with seed sweeps at two extra seeds where a range is given.

Table 6.2: Regime duel (Dive 07 T7): $W = 104$, $\Delta = 13$, 12 refreshes, $n = 40$ per regime. Static columns: flip rate / log-RMSE / 90% coverage. Drift: trap rate. Break: end error / trap rate. Ranges are seed sweeps.

Scheme	flip	log-RMSE	coverage	drift trap	break end-error	break trap
IND (refit fresh)	0.28	0.89	0.72	0.60	0.83	0.55
NPP (posterior-as-prior)	0.04	1.15	0.24	—	1.18	0.70
FRZ (freeze media)	0.01	0.59	0.93	0.24 ± 0.03	0.58 ± 0.03	0.36–0.47
CPPD	0.12	0.48	0.91	0.10 ± 0.05	0.60 ± 0.05	0.20–0.44

CPPD dominates or ties wherever honest schemes compete: well under half IND’s flips, the best static RMSE, calibrated coverage, the lowest drift trap rate. Two findings cut against the dive’s own scheme. First, *freezing is genuinely good in the estimation-noise-dominated regime*: under static and mild-drift truth FRZ’s RMSE (0.59) beats refitting fresh (0.89), because refitting re-rolls the rotation lottery every quarter—this is *why* practitioners freeze. Its cost is structural: a break becomes permanent bias behind stale-but-wide intervals (coverage 0.93 on a number that is no longer an estimate). Second, on breaks CPPD $\approx$ FRZ (0.60 versus 0.58); a seed-7 advantage (0.49 versus 0.66) did not survive the sweep. At this identification level no refresh scheme recovers from a break quickly, because 13-week increments barely re-estimate; breaks are what calibration experiments are for (Chapter 4’s trap-insurance result, rediscovered from the refresh side). Round 4’s consistency check: CPPD’s own flips (0.08–0.12) match the law at its measured adjacent correlation, $F(z, \rho \approx 0.95) \approx 0.09$.

6.6 Limitations and the devil’s advocate

The dive’s counterarguments, section by section. Against the laws: z uses the empirical sd , so the law is self-referential as stated; $\rho = \omega$ holds only in the linear, strongly identified regime it was derived for; the cadence law counts *visible* flips, and frequent refreshes each carry less per-flip surprise. Against CPP/CPPD: the floor (0.3) was tuned in-sample; the drift test is a crude chi-square at a fudged threshold; coverage claims lean on Laplace approximations twice over; FRZ matches CPPD on break recovery, so a frozen model with a scheduled annual rebuild might match CPPD at lower complexity (untested); and the linear conservatism (coverage 0.97) means CPP over-reports width by about $1.4\times$, which a sceptic may call miscalibration too. Against the rotation mechanism: the decomposition was computed for the anchored estimator, not the full MAP, leaving about 0.05 (0.62 versus 0.57) unexplained; σ -independence was shown at two values of σ ; and one DGP family, where time-varying-coefficient models or MCMC posterior means

could change the geometry (BL 19). Against the Janus result: the $5.8\times$ is a synthetic-normal experiment with the sign-disagreeing tie engineered, whereas the MMM-fitted version produced camps that agreed. Against hysteresis: $h = 1$ delays reporting of true regime change by construction, and the flat wrong-order metric owes something to the pair being near-null. Beyond Assumption 6.4, the functional is a ranking rather than budget-decision regret (Chapter 3’s machinery would sharpen “flip” into “reallocation flip”), the specification menu is small, and spend is exogenous (BL 15).

6.7 Discussion: impacts

For vendors the chapter turns three pieces of folklore into arithmetic. Meridian’s double-counting warning becomes Proposition 6.7, with a quantified cost (coverage 0.24 at $k = 8$) and a fix that is a few lines in any Gaussian-propagation code path: condition on the increment, blend precisions at $a = \omega$, floor the variance. Robyn’s `robyn_refresh` bounding and coefficient freezing sit on the same axis as NPP—stability purchased with honesty—with the concession that under noise domination freezing has the best RMSE of any scheme tested, which explains its persistence better than the “lying” critique does. And the rotation result reframes the problem for every package: the instability is in the identified set’s geometry, so the levers are identifying spend variation (Chapter 1), experiments constraining the thin directions (Chapter 2) and carried priors, not more careful optimisation or more data of the same kind.

For CMO decision-making the cadence law is the most immediately usable statement: monthly refreshes show about $\sqrt{3}$ times the ranking churn per year of quarterly ones from the same model. The Janus condition tells a measurement lead *when* to pay for pre-registration: whenever candidate specifications disagree in sign on the quantity being decided, the normal condition on a Pareto front and the rare one among nested windows.

For theory the chapter supplies the correlation structure later chapters consume. Chapter 7 derives the optimal deadband h^* from flip cost against staleness cost (resolving BL 20) and finds the optimal cadence jumps from weekly to quarterly once refresh errors carry this chapter’s overlap correlation; its field case, $\rho \approx 0.5$ from rotation, sits between the independent and ω -correlated limits. Chapter 4’s trap paradox reappears as the no-floor trap rate, and Chapter 13 is where spend chasing last refresh’s estimates would couple the rotation wobble into the design itself.

6.8 Further exploration

The dive’s next steps are five. BL 18 asks for the rotation theory: a law $\rho = \omega \mathbb{E}[\cos \Theta]$ with Θ the rotation of the near-null eigenspace between windows and $\mathbb{E}[\cos \Theta]$ derived from the spend-process spectrum, making Proposition 6.10 predictive ex ante and closing the loop with Chapter 1’s Fisher-information design theory. BL 19 (merging BL 13) asks whether full MCMC integration over the ridge restores $\rho \rightarrow \omega$ and cuts flips; one NUTS budget would serve this, Chapter 4’s filter-class question and Chapter 5’s prior-domination check. BL 20 asked for decision flips—the duel rescored on budget-reallocation regret with an optimal h^* —and was resolved by Dive 08. The remaining two are an online optimal discount replacing the fixed $a = \omega$ with Reznik’s a_0^* under a Masegosa-style hierarchical prior, and endogenous-spend refreshes (BL 15).

In the compiler’s judgement the most promising extension is BL 18, because it would make the chapter’s central number ($\rho \approx 0.5$) computable before any refresh is run: the spend spectrum is observable, the information matrix’s eigenstructure follows from it, and a closed-form $\mathbb{E}[\cos \Theta]$ would let a vendor quote an expected flip rate for a client’s actual media plan and price the spend perturbation that would raise ρ . Second is BL 19, the one most likely to *change* a result: a posterior mean averages over the rotating ridge rather than sitting at one point on it.

Reviewer's note (compiler)

Log line versus writeup. The INDEX log claims CPPD has the “best RMSE in every regime”; Table 6.2 shows FRZ ahead on break end-error (0.58 ± 0.03 versus 0.60 ± 0.05), which the writeup itself calls a null after the seed sweep demoted a seed-7 advantage (0.49 versus 0.66). “Halves flips vs refitting” is 0.12 versus 0.28, i.e. 0.43. Both round in the scheme's favour.

Baseline coverage. IND's 90% coverage is 0.72 in the static regime, so Laplace intervals under-cover even for honest independent refits; CPPD's “calibrated” 0.91 is partly the in-sample-tuned $0.3 \times$ floor compensating for Laplace narrowness. Two no-floor CPP coverages appear unreconciled: 0.80 (T4, nine refreshes) and 0.59 (no-floor trap probe, twelve refreshes).

Evidence not in the archived code. The merged code contains the linear probes, NPP/CPP/CPPD, the duel, the hysteresis, both MMM-fitted specification-selection experiments, the anchored estimator and its decomposition. It does not contain the synthetic-normal Janus experiment behind the $5.8 \times$ (0.295 versus 0.051) headline, the ninefold- σ probe (the archived power controls use $\sigma = 0.08$ against 0.25, a $3.1 \times$ cut; the round history says “ $\sigma/3$ ”), the seed sweeps, or the 40,000-replication clean-room run. The duel function defaults to $n = 60$ where the writeup reports $n = 40$, and the hysteresis $n = 160$ does not match the archived s_{10} (50 replicates $\times$ 11 transitions). That script also scales the deadband by the *empirical* sd of the gap across replicates, not the posterior sd the protocol prescribes (“ $h \approx 1$ posterior-sd”); the 78% figure is for the former. Treat the $5.8 \times$ as the dive's report of an unarchived script.

Arithmetic and scope. -78% is $1 - 0.057/0.25 = 0.772$, consistent only if 0.25 is rounded from about 0.26; $\rho_{\text{pred}} = 0.676 \times 0.79 + 0.41 \times 0.21 = 0.620$, $\sqrt{104/195} = 0.730$ and the exact cadence ratio 2.50 all reproduce. The clean-room's $\rho = 0.837$ against $\omega = 0.875$ is explained post hoc as design variation, consistent with Proposition 6.10 but untested in the linear probe. Laplace covariances enter every coverage number and none has had the BL82 singular-value audit, though the anchored decomposition inverts the prior-regularised $J^T J / \sigma^2 + P_0$ rather than a pseudo-inverse.

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Chapter 7

The reallocation deadband: when to act on a refreshed model

Provenance

Source dives: Dive 08 *Decision-framed uncertainty outputs: the filtered deadband law (B2, absorbs BL20)* (run 2026-09-01, file `08_decision_deadband.md`, last written 2026-09-01 16:55 UTC); Dive 09 *EXTEND of dive 08: the multichannel, budget-coupled deadband (B3)* (run 2026-09-01, file `09_decision_deadband_ext.md`, last written 2026-09-01 20:01 UTC). **Code:** `code/08_decision_deadband.py`, `code/09_decision_deadband_ext.py`. **Frontier-study problem(s) addressed:** M9 (adoption barriers: uncertainty communication — honest wide intervals are heard as “the model doesn’t know”), B5 (precision theatre: a pointy wrong number beats a right interval in the meeting), H8 (decision-making under measurement conflict: no pre-registered decision rules exist for acting on a model). **Backlog items closed:** BL 20 (by Dive 08), BL 21 (opened by Dive 08, resolved by Dive 09); **advanced:** BL 22; **opened:** BL 23, BL 24 (from Dive 08), BL 25–BL 28 (from Dive 09).

Key ideas in this chapter

- The Kalman-filtered deviation of the optimum from the incumbent allocation is a random walk with the *true* drift variance q whatever the refresh noise v (Muth’s identity; vector form $\text{Cov}(\Delta\hat{u}) = Q$ proved), so the partially observed problem reduces *exactly* to fully observed impulse control; posterior width enters only through shrinkage and a policy-independent floor $\gamma P/2$.
- Scalar law: act when the filtered deviation leaves $h^* = (12Kq/\gamma)^{1/4} - 0.5826\sqrt{q}$, DP-verified within 0.4% for $K \geq 30$. The wrong statistic is first-order: raw estimates $2.6\times$ worse at $v = 10$; the $1.96\sqrt{P}$ significance rule $2.1\times$ worse at $v = 1$ and right at $v = 10$ only by coincidence; the myopic “expected gain $> K$ ” rule has exponent $\frac{1}{2}$ instead of $\frac{1}{4}$.
- With n channels under a budget the ergodic inaction region is an *exact* ellipsoid $\{\hat{u}^\top M \hat{u} < 1\}$, $4M \text{tr}(QM) + 8MQM = \Sigma/K$, solved by whitening and within 0.01% of a 2-D DP optimum; but shape is second-order (per-channel scalar bands lose $\sim 1\%$ at $m = 2$, $\leq 4.7\%$ at $m = 8$) while the statistic is first-order (χ^2 CI rule $+31$ – 107% , posterior-width band $+45\%$, unfiltered $3.7\times$).
- Per-channel change costs make partial moves optimal below the residual distance $r_c = R^* \sqrt{1 - \sqrt{1 - c/K_{\text{full}}}}$ to a transfer line (tiered heuristic within 0.65% of DP, saving up to 12.5%); cadence and band separate exactly (Δ^* independent of K , elasticity $2/3$ in refresh cost), but under the overlap-correlated refresh errors of Chapter 6 the optimal interval jumps from weekly to quarterly.
- Expected gain $\frac{\gamma}{2}\hat{z}^2$, probability of regret $\Phi(-|\hat{z}|/(2\sqrt{P}))$ and a derived tolerance α^* give a pre-registerable ACT/HOLD protocol; in nonlinear Hill transports it beats act-always by $6.5\times$ (two channels) and $7\times$ (four) and the posterior-width band by $3\times$.

7.1 Problem and motivation

The adoption catalogues locate one of MMM’s sharpest failures at the interface rather than in the mathematics. Honest posteriors under collinearity are wide; a dashboard reading “ROAS 4.27” beats a model reading “1.2–3.8” in the meeting (B5); a wide interval is heard as “the model doesn’t know” (M9); and no organisation

has a protocol saying when a refreshed recommendation should be acted on (H8). Chapter 6 ended with the same hook from the other side: a 1σ reporting hysteresis cut reported flips by 78% at no measured accuracy cost, but nobody could say what the right hysteresis was (BL 20). Problem B2 asked for the *interface layer*: outputs in decision units (probability of regret, expected loss of reallocation, “the answer at your risk tolerance”) and a decision rule an organisation could pre-register.

Dive 08 makes this a control problem. An organisation runs allocation x_c ; the true optimum drifts; each refresh delivers a noisy estimate of the deviation; operating off-optimum costs a quadratic flow loss; reallocating costs a fixed K bundling execution, coordination and the credibility charge of visibly changing the recommendation (Dietvorst et al., 2015: algorithms lose trust faster than humans when seen to err; Burgeno and Joslyn, 2020: forecast inconsistency costs trust). What should the model report each refresh, and when should the organisation act? The answer is a deadband on the *filtered* deviation whose closed-form width does not depend on the posterior width at all. Dive 09 extends it to n channels sharing a budget (BL 21), to per-channel change costs and to the refresh cadence. Both dives ran attack-and-refine rounds with clean-room re-derivation; Dive 08’s post-writeup prior-art audit relabelled its continuous-time core as rediscovered (see the reviewer’s note).

7.2 Background: the ideas this chapter builds on

7.2.1 Fixed-cost impulse control and the quartic band (Barro, 1972; Dixit, 1991; Stokey, 2009)

The menu-cost literature studies a state z_t (a price gap, a cash balance) that drifts as a Brownian motion with variance rate q , a quadratic flow loss, and a fixed cost K to reset z to zero. Because the cost is fixed, small deviations are never worth correcting and the optimal policy is a band: act when $|z|$ reaches h . Dixit’s treatment uses renewal-reward: for Brownian motion started at zero and absorbed at $\pm h$, Dynkin’s formula with $f(z) = z^2/q$ gives the cycle length $\mathbb{E}[\tau] = h^2/q$ and with $f(z) = z^4/(6q)$ gives $\mathbb{E} \int_0^\tau z_t^2 dt = h^4/(6q)$, so with flow loss $\frac{\gamma}{2}z^2$ the average cost per unit time is

$$g(h) = \frac{\gamma h^4/(12q) + K}{h^2/q} = \frac{\gamma h^2}{12} + \frac{Kq}{h^2}, \quad h^* = \left(\frac{12Kq}{\gamma}\right)^{1/4}, \quad g(h^*) = \sqrt{\frac{Kq\gamma}{3}}. \quad (7.1)$$

The exponents are transparent: flow loss scales as h^2 and switching cost as K divided by the h^2/q time a Brownian path needs to traverse the band, so they balance at $h^4 \propto Kq/\gamma$ (Barro’s constant is 6 when the loss is written Bz^2). With a cost proportional to the move, $\kappa|z|$, the switching term is $\kappa q/h$ and the band scales as $(6\kappa q/\gamma)^{1/3}$: the exponent identifies the cost structure. Stokey’s monograph collects these results; all assume the state is *perfectly observed*.

7.2.2 Multi-product menu costs: balls, ellipsoids and two-threshold policies

Alvarez and Lippi (2014) solve the n -product firm: n independent price gaps, each Brownian with variance σ^2 , and one menu cost ψ that changes all prices at once. Symmetry makes the inaction region a ball in $y = \sum_i \text{gap}_i^2$ with small-cost threshold $\bar{y} = \sqrt{2(n+2)\sigma^2\psi/B}$ for flow loss $B \sum_i \text{gap}_i^2$, adjustment rate $N = n\sigma^2/\bar{y}$ and price-change kurtosis $3n/(n+2)$; nothing anisotropic is in closed form. Altarovici, Muhle-Karbe and Sonner (2015), for a multi-asset portfolio with a fixed transaction cost, show the leading-order (small-cost) no-trade region is an ellipsoid $\{\rho^T M \rho < 1\}$ in the deviation ρ from the frictionless target, with M solving $4M \text{tr}(AM) + 8MAM = \Sigma$ (A the target’s diffusion covariance, Σ the loss curvature). Bonomo, Carvalho, Kryvtsov, Ribon and Rigato (2023) add a per-product cost to the per-event cost and, with a continuum of products, obtain a two-threshold policy: one constant decides *when* to review prices, a second *which* prices to change given a review.

7.2.3 The Kalman filter on a random walk and Muth's invariance (Muth, 1960)

The local-level model is $z_{k+1} = z_k + w_k$, $y_k = z_k + \varepsilon_k$ with $w_k \sim \mathcal{N}(0, q)$, $\varepsilon_k \sim \mathcal{N}(0, v)$ independent. Its steady-state Kalman filter has posterior variance P solving $P(P+q) = qv$, i.e. $P = (-q + \sqrt{q^2 + 4qv})/2$, gain $\lambda = (P+q)/(P+q+v)$ and filtered mean $\hat{z}_k = \hat{z}_{k-1} + \lambda(y_k - \hat{z}_{k-1})$, an exponentially weighted average of the observations. Muth's result is that the optimal forecast is itself a random walk whose innovations carry the variance of the *permanent* shock: $\text{Var}(\Delta \hat{z}) = \lambda^2(P+q+v) = (P+q)^2/(P+q+v) = q$, since $q(P+q+v) = q(P+q) + P(P+q) = (P+q)^2$. Noise lowers the gain and leaves a stationary residual $z - \hat{z}$ of variance P but never changes the variance rate of the estimate. Baley and Blanco (2019) apply a Dixit-type quartic band to a Kalman-filtered markup gap in a menu-cost model with infrequent large shocks; there the band widens with current uncertainty because uncertainty cycles, and they credit the constant-uncertainty case, where “the information friction has no effects in steady state”, to the appendix of Alvarez, Lippi and Paciello (2011), whose main text studies costly *perfect* observation and finds a review interval scaling as $(\text{cost})^{1/2}$. General partially observed impulse control (Mazziotto et al., 1988) has existence theorems but no closed forms.

7.2.4 Discrete monitoring: Broadie–Glasserman–Kou and Gobet–Menozzi

A Brownian path checked only every Δ time units overshoots a continuous barrier before the crossing is seen. Broadie, Glasserman and Kou (1997) show that a barrier monitored discretely at H behaves like a continuous barrier shifted by $\beta\sigma\sqrt{\Delta}$, $\beta = -\zeta(\frac{1}{2})/\sqrt{2\pi} \approx 0.5826$ (ζ the Riemann zeta function); Howison and Steinberg (2007) obtain the constant by matched asymptotics. Gobet and Menozzi (2010) generalise to stopped diffusions in d dimensions: shift the boundary inward along its unit normal n by $\beta\sqrt{n^\top Q n}$ per step when the per-step covariance is Q . Here refreshes are the monitoring instants, so a continuous-time band must be *narrowed* by $\beta\sqrt{q}$.

7.2.5 Expected-loss rules and decision-framed communication

Experimentation practice already frames some decisions in loss units: Stucchio's VWO SmartStats ships a variant when its expected loss falls below a “threshold of caring”, and Feit and Berman's Test & Roll (2019) sizes an A/B test by maximising the expected profit of test-then-deploy in closed form. Both are static — no drift, no adjustment cost — and their thresholds are elicited, not derived. Joslyn and LeClerc (2012) show that probabilistic formats improve incentivised decisions and buffer trust against forecast error; Burgeno and Joslyn (2023) that probabilistic framing mitigates the trust penalty of inconsistency. No MMM tool ships an act/hold statistic with a derived threshold (Recast reports intervals and simulations, Meridian a point-optimal budget with credible intervals, PyMC-Marketing CVaR *objectives*); Pathak, Jeunen and Lambert (2026) build regret distributions retrospectively, as an audit.

7.3 Setup and assumptions

Definition 7.1 (Scalar reallocation problem). Let $z_k = x^*(\theta_k) - x_c$ be the deviation of the optimum from the incumbent allocation at refresh k . Between refreshes $z_{k+1} = z_k + w_k$, $w_k \sim \mathcal{N}(0, q)$, q the per-refresh variance of the drifting optimum. Refresh k delivers $y_k = z_k + \varepsilon_k$, $\varepsilon_k \sim \mathcal{N}(0, v)$, v the posterior variance in allocation units. Operating off-optimum costs $\frac{\gamma}{2}z^2$ per period, γ the curvature of expected profit at the optimum. Acting sets $x_c \leftarrow x_c + \hat{z}_k$ (so $z \leftarrow z - \hat{z}_k$, $\hat{z} \leftarrow 0$) at fixed cost K ; the objective is long-run average cost. $P, \lambda, \hat{z}$ are the steady-state Kalman quantities.

Definition 7.2 (Vector problem under a budget). With n channels and $\sum_i z_i = 0$, let B ($n \times m$, $m = n - 1$) be an orthonormal basis of the budget tangent space and $u = B^\top z$. Then $u_{k+1} = u_k + w_k$, $w \sim \mathcal{N}(0, Q)$; $y_k = u_k + \varepsilon_k$, $\varepsilon \sim \mathcal{N}(0, V)$; flow loss $\frac{1}{2}u^\top \Sigma u$ with $\Sigma = B^\top \Gamma B$, Γ the Hessian of expected profit at the

incumbent allocation; a reallocation resets $\hat{u}$ to zero at cost K (or $K_0 + c \times \text{\#channels}$ changed in Section 7.4.5). The steady-state filter has posterior covariance P , predictive $\bar{P} = P + Q$ and gain $G = \bar{P}(\bar{P} + V)^{-1}$.

Assumption 7.3 (Slow Gaussian drift, known constants). The optimum drifts as a random walk (strong mean reversion is a scope condition, Section 7.6); drift and refresh noise are Gaussian and the filter is correctly specified; (q, v, γ, K) or (Q, V, Γ, K) are known; the optimum is interior; the criterion is ergodic (undiscounted).

7.4 Main results

7.4.1 Estimation noise never enters the band

Theorem 7.4 (Filtered increments carry the true drift variance). *[transported from the literature] (scalar: Muth, 1960; rediscovered by Dive 08) [established in the dive] (vector form, Dive 09). In the scalar problem $\text{Var}(\Delta \hat{z}) = q$ exactly for every v . In the vector problem $\text{Cov}(\Delta \hat{u}) = \bar{P}(\bar{P} + V)^{-1} \bar{P} = \bar{P} - P = Q$ exactly for every V , by the Riccati fixed point $P = \bar{P} - \bar{P}(\bar{P} + V)^{-1} \bar{P}$.*

The vector algebra was checked symbolically (residual 4×10^{-14}) and by Monte Carlo over 4×10^5 steps with anisotropic 3-D Q and V (maximum relative error 0.2%). The identity is not tied to white noise: for any correctly specified stationary linear-Gaussian observation structure $\hat{z}$ is a martingale and $z - \hat{z}$ is stationary while $\text{Var}(z)$ grows as qk , forcing $\text{Var}(\Delta \hat{z}) = q$; with AR(1) refresh noise ($\rho = 0.5$) an augmented-state filter measures increment variance 1.000 where a misspecified white-noise filter measures 0.82.

Proposition 7.5 (Exact separation). *[established in the dive] Acting subtracts the known quantity $\hat{z}$ from z , so the filter is unaffected by the control and $\mathbb{E}[\frac{\gamma}{2} z^2 \mid \text{data}] = \frac{\gamma}{2} (\hat{z}^2 + P)$ with P policy-independent. The partially observed problem is exactly the fully observed impulse control of the martingale $\hat{z}$ (a random walk of rate q by Theorem 7.4) plus the floor $\gamma P/2$; in $\mathbb{R}^m$, of $\hat{u}$ plus $\frac{1}{2} \text{tr}(\Sigma P)$.*

This exactness is special to the reset-by- $\hat{z}$ action structure. The corollary that names the chapter: *the band cannot depend on v* . Posterior width matters — through the shrinkage $\lambda < 1$ that slows $\hat{z}$ and through the floor — but not through where the trigger sits.

7.4.2 The composite filtered deadband law (scalar)

Law 7.6 (Composite filtered deadband). *[established in the dive] (Dive 08; continuous-time core [transported from the literature] from Dixit and Baley–Blanco). Applied to $\hat{z}$ and independent of v ,*

$$h^* = \left(\frac{12Kq}{\gamma} \right)^{1/4} - 0.5826 \sqrt{q}. \quad (7.2)$$

With $h_c = h^* + \beta \sqrt{q}$ the continuous band, the minimum average cost is $\sqrt{Kq\gamma/3} + \gamma P/2$ up to a discrete-time offset (Conjecture 7.9); the act rate is q/h_c^2 ; the demanded expected gain is $G^* = \frac{\gamma}{2} h^{*2}$ (continuous limit $\sqrt{3\gamma Kq}$; the square root of the switching cost); with proportional cost $\kappa |\hat{z}|$ the band is $(6\kappa q/\gamma)^{1/3} - \beta \sqrt{q}$.

The first term is Equation (7.1) on the reduced problem, the second the Broadie–Glasserman–Kou shift for a state monitored once per refresh; in calendar units q/γ is refresh-rate-free, so only the $\beta \sqrt{q}$ term shrinks with faster refreshing. The law’s first version used Barro’s constant 6 and no shift, and DP thresholds drifted from 0.79 to 1.12 of “theory” across K ; after both corrections the DP/law ratio is 0.994–1.000 for $K \geq 30$, and a free fit of the shift constant gives $\beta = 0.599 \pm 0.015$ against 0.5826. A round-2 apparent refutation — DP log-log slopes of h against K of 0.295 (fixed cost) and 0.42 (proportional) against $\frac{1}{4}$ and $\frac{1}{3}$ — turned out to be the composite law’s own prediction: with the additive $-\beta \sqrt{q}$ term the effective slopes over those ranges are 0.292 and 0.40, and the proportional-cost level matches (ratio 0.996 at $\kappa = 100$).

7.4.3 The ellipsoid law (vector)

Theorem 7.7 (Exact ergodic inaction ellipsoid). *[established in the dive] (Dive 09; the matrix equation is [transported from the literature] from Altarovici et al., where it is a small-cost asymptotic). For the reduced problem of Definition 7.2 the continuation region is the ellipsoid $\{u : \Phi(u) = u^\top M u < 1\}$ with*

$$4M \operatorname{tr}(QM) + 8MQM = \Sigma/K, \quad g = 2K \operatorname{tr}(QM), \quad (7.3)$$

exactly (modulo the standard verification theorem), with relative value function $W(u) = K[2\Phi - \Phi^2]$.

Inside the region the ergodic HJB equation is $\frac{1}{2} \operatorname{tr}(Q \nabla^2 W) + \frac{1}{2} u^\top \Sigma u = g$; with the ansatz, $\frac{1}{2} \operatorname{tr}(Q \nabla^2 W) = K[2(1 - \Phi) \operatorname{tr}(QM) - 4u^\top M Q M u]$, so it holds identically in u iff the quadratic forms match, which is Equation (7.3). The quasi-variational inequality follows directly: value matching $W = K$ on $\Phi = 1$; smooth fit $\nabla W = 2K(1 - \Phi)Mu = 0$ there; $W \leq K$ everywhere; outside, $\frac{1}{2} u^\top \Sigma u = K[4u^\top M Q M u + 2\Phi \operatorname{tr}(QM)] \geq g$ whenever $\Phi \geq 1$ because $MQM \succeq 0$. The LQ ergodic problem has no wealth effects, which is why the portfolio literature’s leading-order ellipsoid is here the whole answer.

Construction 7.8 (Closed form by whitening). *[established in the dive] Let $S = Q^{1/2} \Sigma Q^{1/2} / K = U \operatorname{diag}(s) U^\top$. Then $M = Q^{-1/2} \tilde{M} Q^{-1/2}$ with $\tilde{M} = U \operatorname{diag}(m) U^\top$, where $m_i(4T + 8m_i) = s_i$ and $T = \sum_i m_i$: one scalar root-find in T . In one dimension this returns Equation (7.1); for isotropic $Q = qI$, $\Sigma = \gamma I$ it returns Alvarez and Lippi’s radius $R^* = (4(m + 2)Kq/\gamma)^{1/4}$. K and Q enter only through KQ .*

Discrete refreshes are handled by the Gobet–Menozzi normal shrink $\beta \sqrt{n^\top Q n}$, which gives the closed-form trigger

$$\text{act iff } \Phi(\hat{u}) + 2\beta \sqrt{\hat{u}^\top M Q M \hat{u} / \Phi(\hat{u})} \geq 1; \quad (7.4)$$

a clean-room implementation found the plain $\Phi \geq 1$ trigger 6.7% worse, so the shift is detected, not cosmetic.

Conjecture 7.9 (Discrete-time cost offset). *[conjecture / speculative] Charging the flow cost at the start of each period rather than integrating it gives $g_{\text{disc}} \approx 2K \operatorname{tr}(QM) + \frac{1}{2} \operatorname{tr}(\Sigma P) - \kappa \operatorname{tr}(\Sigma Q)$ with $\kappa = \frac{1}{4} - \beta^2/4 = 0.165$; measured $\kappa = 0.154\text{--}0.164$ in one dimension (q from 1 to 1/8) and 0.158 in the 2-D case B, trending toward 0.165 as step/band $\rightarrow 0$. Supported to about 3%, not derived (BL 27).*

7.4.4 Decision-framed outputs

Proposition 7.10 (Probability of regret and derived tolerance). *[established in the dive] With $z \mid \text{data} \sim \mathcal{N}(\hat{z}, P)$ the gain from moving by $\hat{z}$ is $\gamma \hat{z}(z - \hat{z}/2)$, so the probability that the move hurts is $\Phi(\cdot)$ here the standard normal distribution function, not the quadratic form of Theorem 7.7)*

$$\text{PoR} = \Phi\left(-\frac{|\hat{z}|}{2\sqrt{P}}\right), \quad \text{PoR}_{\text{vec}} = \Phi\left(-\frac{\frac{1}{2}\hat{u}^\top \Sigma \hat{u}}{\sqrt{\hat{u}^\top \Sigma P \Sigma \hat{u}}}\right), \quad (7.5)$$

with expected gain $\hat{G} = \frac{\gamma}{2} \hat{z}^2$ (vector $\frac{1}{2} \hat{u}^\top \Sigma \hat{u}$). In the scalar case PoR is monotone in $|\hat{z}|$, so “act when $\text{PoR} \leq \alpha^$ ” with $\alpha^* = \Phi(-h^*/(2\sqrt{P}))$ is identical to the deadband policy: the risk tolerance is derived from (K, q, γ, v) , not elicited.*

Monte Carlo: scalar PoR predicted 0.20711, observed 0.20725; vector 0.30281 versus 0.30327 ($n = 2 \times 10^6$). In the vector case PoR is not monotone in the trigger statistic unless Σ , P and M are proportional, so Dive 09 has the organisation register M and report PoR as a companion, not as the trigger. α^* is where v re-enters the reported numbers: the same h^* is a tolerance of 0.8% at $v = 1$ and 27% at $v = 100$ ($K = 30$, $q = \gamma = 1$), and 0.13 in the Hill transport of Section 7.5.

7.4.5 Tiered reallocation under per-channel change costs

Proposition 7.11 (Residual-distance threshold). *[numerically established] (structure DP-verified, constant approximate). With cost $K_0 + c \times (\# \text{channels changed})$ and $n = 3$ (2-D tangent plane), a pairwise transfer $i \leftrightarrow j$ moves $\hat{u}$*

along the projected direction of $e_i - e_j$ (three lines at $0^\circ, 60^\circ, 120^\circ$) at cost $K_0 + 2c$ against $K_0 + 3c$ for a full reset. The DP with both action types chooses the pairwise move iff the perpendicular distance r from $\hat{u}$ to the nearest transfer line is below

$$r_c = R^* \sqrt{1 - \sqrt{1 - c/K_{\text{full}}}}, \quad R^* = (4(m+2)K_{\text{full}}q/\gamma)^{1/4}, \quad (7.6)$$

obtained by requiring the residual left behind to cost less than the saved channel: $W(r) = K_{\text{full}}(2\Phi - \Phi^2) < c$ with $\Phi = r^2/R^{*2}$.

In the DP the maximum residual among pairwise states and the minimum among full-reset states coincide to two decimals (1.63/1.64, 2.12/2.12), so the threshold is sharp; the formula gives 1.61 and 2.00 for $c = 6.67$ and 10 at $K_{\text{full}} = 30$, about 5.5% low, which the clean-room re-derivation softened to “ $\approx$ ”. The tiered heuristic (full-cost ellipsoid trigger; pairwise-cost trigger on the along-line component when $r < r_c$; pairwise iff $r < r_c$) is the deadband analogue of Bonomo et al.’s two-threshold policy.

7.4.6 Cadence separates from the band

Proposition 7.12 (Cadence separation). *[established in the dive] In calendar time with drift rate q , refreshes every Δ at cost c_r and per-refresh noise v , the band being v -invariant and (after the monitoring shift) Δ -invariant, the cost rate separates as*

$$\text{cost}(\Delta, h) = \underbrace{\sqrt{Kq\gamma/3}}_{\text{band: } K \text{ only}} + \underbrace{\gamma P(\Delta)/2 + c_r/\Delta + \kappa'\gamma q\Delta}_{\text{cadence: no } K}, \quad P(\Delta) = \frac{-q\Delta + \sqrt{q^2\Delta^2 + 4q\Delta v}}{2}, \quad (7.7)$$

with $\kappa' \approx 0.075\text{--}0.09$. Hence Δ^* is independent of K and h^* of c_r . For small Δ and independent refresh noise, $P \approx \sqrt{q\Delta v}$ and

$$\Delta^* \approx \left(\frac{4c_r}{\gamma\sqrt{qv}} \right)^{2/3}, \quad (7.8)$$

elasticity $2/3$ in refresh cost (measured 0.68–0.72 over $c_r \in [0.01, 10]$) against $\frac{1}{2}$ for costly perfect observation, and $\Delta^* \propto v^{-1/3}$ (measured -0.3 to -0.44): noisier refreshes should be run more often.

Proposition 7.13 (The $2/3$ law is an independent-noise bound). *[refuted within the dive] (as a general cadence rule; Dive 09 round 2). With refresh-error correlation $\rho(\Delta) = 1 - \Delta/W$ ($W = 104$ weeks, the overlapping-window arithmetic of Chapter 6) and a correctly specified augmented filter, the floor barely moves with Δ (4.38–4.41 at $K = 30$, $q = 1/13$ per week, $v = 10$) because more refreshes of the same window information do not average the error; Δ^* moves from weekly to 8–13 weeks for every refresh cost tried ($c_r \in \{0.05, 0.2, 0.8\}$), and a white filter under correlated noise adds 12–25%. The general rule is $\Delta^* = \arg \min_{\Delta} [c_r/\Delta + \gamma P_{zz}(\Delta; q, v, \rho(\Delta))/2]$, P_{zz} the steady-state filtered variance of z in the augmented (z , noise) filter.*

7.5 Numerical evidence

Dive 08 ran 15 experiments and Dive 09 16, with standard errors from seed replication and DP results grid-converged at $d = 0.2/0.1/0.05$. *Law verification.* Scalar: DP thresholds match Equation (7.2) within 0.4% for $K \geq 30$ across $q \in \{0.25, 1\}$ (worst case $K = 1$: -3.6%); $\text{Var}(\Delta\hat{z}) = q$ on a 3×3 (q, v) grid within 0.5%; the empirically optimal band stays at 3.6–3.8 ($h^* = 3.77$) while v spans 0 to 100, i.e. noise standard deviation up to $2.7\times$ the band; the floor increments predict the cost at $v = 1/10/100$ as 3.31/4.35/7.76 against observed 3.31/4.35/7.78 from a $v = 0$ level of 3.004; act rate $q/h_c^2 = 0.0526$ versus 0.0518–0.0521. A clean-room subagent re-derived the identity (residual 10^{-14}) and re-implemented DP and simulation (thresholds 3.760/7.160 versus 3.773/7.163; floors $+0.3116 \pm 0.0067$ and $+4.7573 \pm 0.0233$ versus $+0.309/ +4.756$). Vector: isotropic $m = 2$ scale-sweep argmin at 1.00 for $v \in \{1, 10, 100\}$; DP radius 4.05 versus law 4.10; the anisotropic comparison of Table 7.1, reproduced clean-room at 7.0384/7.0390/7.0389 across grids (gap

0.011–0.013%). *Controls.* $K \rightarrow 0$ collapses the DP band to ~ 0.3 (the step scale) and the ellipsoid semi-axes from 4.95/2.98 to 1.57/0.94; the $(q + v)$ -scaled alternative is rejected at 141σ ; the DP shape instrument separates axis ratio 1.73 from 1.00; the CI-versus-protocol gap at $v = 1$ is 380 SE.

Table 7.1: Dive 09 E3b: 2-D average-cost DP optimum against the zero-tuned Riccati ellipsoid with closed-form monitoring shrink, and the best-scaled alternative shapes (percent above DP); $K = 30$.

Configuration	DP	Ellipsoid+shrink	loss-ellipsoid	drift-Mahal.	sphere	box
A: $Q = I, \Sigma = \text{diag}(1, 4)$	7.4496	7.4503 (+0.01%)	+0.37%	+5.2%	+5.2%	+6.4%
B: $Q \text{ corr } 0.7, \Sigma = \text{diag}(1, 4)$	7.0383	7.0391 (+0.01%)	+0.46%	+11.0%	+4.5%	+5.0%
C: $Q = \text{diag}(1, 0.1), \Sigma \text{ corr } 0.8$	3.3034	3.3034 (+0.00%)	+0.38%	+17.0%	+6.3%	+8.4%

The DP regions' second-moment axis ratios (1.73, 1.72, 2.29) match the *shrunk* ellipsoid rather than the continuous one (1.60/1.66/2.15), as does the orientation (-171.5° versus -166.4° ; 140.9° versus 143.5°). Table 7.2 carries the practical message. Under noise, filtering is most of the value (raw triggering $2.6\times$ worse at $v = 10$ scalar, $3.7\times$ vector). The significance rule's width scales with estimation noise, which has nothing to do with the drift/cost economics: it is near-optimal at $v = 10$, where $1.96\sqrt{P} = 3.22 \approx h^*$ by accident, and $2.1\times$ worse at $v = 1$. The myopic "act when $\frac{\gamma}{2}\hat{z}^2 > K$ " rule has threshold $\sqrt{2K/\gamma} = 7.75$, the wrong exponent, and waits about twice too long. The "posterior-width" ellipsoid (Riccati solved with $Q + V$) costs +5% at $v = 1$ and +45% at $v = 10$. Shape, by contrast, is cheap: per-channel scalar bands with their own (q_i, γ_i) lose 1.0%/0.7%, and in the isotropic dimension sweep at $v = 1$ (sphere law 3.31/5.75/7.71/10.89/14.66 at $m = 1/2/3/5/8$) the best box loses +0.1/ +1.2/ +1.8/ +2.3/ +2.3%, the box at the scalar-law width +0.0/ +1.3/ +2.2/ +3.5/ +4.7%, and the χ^2 CI rule +107/ +74/ +59/ +44/ +31%. Scale is flat too: $\text{cost}(mh^*)/\text{cost}(h^*) = 1.64, 1.17, 1.00, 1.26, 1.86$ at $m = 0.5, 0.7, 1, 1.5, 2$, and a $2\times$ error in K or Q costs 3.4–3.8% in the vector case.

Table 7.2: Policy duels: long-run average cost $\pm$ SE over seeds. Scalar block (Dive 08): $K = 30, q = \gamma = 1, 8$ seeds. Vector block (Dive 09): case B of Table 7.1, $K = 30, 6$ seeds. Act-always is reported without SE in both dives.

Policy	$v = 1$	$v = 10$
<i>Scalar</i>		
filtered band h^* (protocol)	3.307 ± 0.006	4.354 ± 0.006
filtered band, width from $q + v$	3.414 ± 0.005	5.494 ± 0.010
raw band on y (no filter)	3.609 ± 0.005	11.388 ± 0.011
myopic: act when $\frac{\gamma}{2}\hat{z}^2 > K$	6.338 ± 0.017	7.387 ± 0.020
CI rule: act when $ \hat{z} > 1.96\sqrt{P}$	6.864 ± 0.006	4.468 ± 0.010
act-always	30.31	31.35
<i>Vector, case B</i>		
ellipsoid + shrink on filtered $\hat{u}$ (protocol)	8.444 ± 0.005	13.212 ± 0.006
box of per-channel scalar bands (own q_i, γ_i)	8.532 ± 0.004	13.308 ± 0.009
Riccati with $Q + V$ ("posterior width" band)	8.904 ± 0.005	19.159 ± 0.020
CI rule: $\hat{u}^T P^{-1} \hat{u} > \chi^2_2(0.95)$	10.881 ± 0.004	14.370 ± 0.016
ellipsoid on raw y (no filter)	11.736 ± 0.008	48.521 ± 0.012
act-always	31.406	36.189

Tiering (Dive 09 E4–E4c, $Q = \Sigma = I$): pairwise moves save 2.0/6.3/12.5/7.7% over full resets for $(K_0, c) = (20, 3.3)/(10, 6.7)/(0, 10)/(15, 15)$; the tiered heuristic is +0.07/ +0.41/ +0.65/ +0.38% from DP while the full-reset ellipsoid alone is 2–14% worse; the clean-room pairwise DP gave a 12.58% saving and threshold 2.1221/2.1224. *Cadence* (E7, E10): the band argmin sits at $h_c - \beta\sqrt{q\Delta}$ for all $\Delta \in [0.25, 4]$; the Δ argmin is identical for $K = 30$ and 120 at every c_r ; Equation (7.7) is within 0.5% of simulation at $\Delta \leq 2$ (5.588 versus 5.611 ± 0.01); clean-room costs at $\Delta = 1/2/4$ (5.606/5.643/6.153) match the dive's (5.611/5.645/6.113), with the flag that dropping the $O(\gamma q \Delta)$ term picks $\Delta = 2$ instead of 1.

Nonlinear MMM transports. [numerically established] (one configuration each; flat basin). Dive 08 E10: two Hill channels, log-random-walk effectiveness drift $\eta = 0.02$, posterior noise $\tau = 0.10$, reallocation cost 0.02, budget 2; (γ, q_x, v_x) calibrated numerically at the operating point give $h^* = 0.064$ (3.2% of budget) and $\lambda = 0.17$ ($v_x/q_x \approx 30$). True-regret flow per refresh: protocol 0.0032 ± 0.0008 ; act-always 0.0209 ± 0.0002 ($6.5\times$ worse); myopic 0.0059 ± 0.0010 ; an oracle band sweep has its flat argmin at $0.75h^*$ (0.0030 ± 0.0003), indistinguishable from h^* . Dive 09 E8: four concave Hill channels, budget 1, same η , τ and per-event cost, (Γ, Q, V) calibrated at the operating point ($v/q \approx 26$, Kalman gains 0.17–0.19, semi-axes 3.0–4.8% of budget), 8 paired seeds $\times$ 4000 periods: protocol 0.00299 ± 0.00056 per period; sphere $0.97\times$, box $1.04\times$, CI $0.95\times$ (all within one paired SE); $Q + V$ ellipsoid $2.96\times$; myopic $4.1\times$; unfiltered $5.6\times$; act-always $7.0\times$; paired oracle-scale differences at 0.8/1.0/1.25 of -0.00016 ± 0.00016 , 0, $+0.00065 \pm 0.00027$.

7.6 Limitations and the devil’s advocate

The dives’ own counterarguments: *policy class* — v -invariance is proved for band-on- $\hat{z}$ policies given reset-by- $\hat{z}$ actions, with no proof that a non-band policy cannot do better (symmetry and convexity make it very plausible). *Stacked asymptotics* — small-cost band, monitoring shift and LQ loss; the free-fit $\beta = 0.599 \neq 0.5826$ shows residual structure and the cost level carries the unproved Conjecture 7.9. *Mean reversion* — an OU optimum with $\phi \geq 0.98$ per refresh leaves the argmin at h^* , but $\phi = 0.9$ widens it about 17% (free reversion adds option value to waiting), though acting still beats never acting (2.08 versus 2.63). *Noise structure* — misspecified white filtering costs $+2.6\%$ at $\rho = 0.5$, $v = 10$, not swept to extreme ρ ; t_3 refresh noise costs $+8.9\%$ scalar and $+7.9\%$ vector with the linear filter, the scalar band unmoved and the vector argmin at scale 0.9 (BL 23). *Parameters must be estimated* — q is confounded with refresh noise in observed jitter (the rotation wobble of Chapter 6 would inflate a naive $\hat{q}$ and widen the band spuriously), γ comes from a possibly misspecified fitted curvature, and K is an organisational quantity nobody measures (BL 22); the fourth root blunts this (a $2\times$ error in K , q or γ moves h^* by 19% and costs $\lesssim 5\%$) without removing it. *Trust as a stock* — if chummy recommendations degrade K itself, it is a different control problem (BL 24). Further: near a bound the region is truncated and the reset target is not zero; (Γ, Q, V) were not re-calibrated as effectiveness drifts (BL 26); r_c is derived from the full-reset value function for $n = 3$ only (BL 25); discounting changes the region, second-order at business rates; everything is Gaussian.

Against v -invariance the natural misreading is “uncertainty doesn’t matter”; it matters through λ (0.17 in the transports) and the floor, just not through the band, and that is the deliverable. Against the duel, rankings depend on parametrisation and the CI rule tied twice ($v = 10$ scalar, $v/q \approx 26$ vector); the point is the wrong scaling variable, not the size of a gap. Against the ellipsoid, the dives’ own reframing is that the decisive content is the filter, the scale (KQ product, $(m+2)^{1/4}$ growth) and the tiering, not the ellipse. Against the framing, an organisation may reject deadbands (“why pay for a model we mostly don’t act on?”), which is B5 again, but with inaction now carrying a derived price.

7.7 Discussion: impacts

For practice the proposal is an interface change, not a modelling change. At engagement start an organisation registers the loss-curvature source (the model Hessian at the incumbent allocation), the $\hat{q}$ estimation procedure (refresh-history variance decomposition with estimation noise removed), the K elicitation (execution cost plus an agreed credibility charge), the derived h^* , G^* and α^* — or, for several channels, the tangent basis, Γ , $\hat{Q}$ and M , or per-channel bands with their documented $\leq 5\%$ penalty — the per-line-item cost c with its r_c , the filter’s noise-correlation structure, and the cadence from the floor-with-correlation rule (expecting quarterly, not monthly, under overlapping windows). Each refresh the model reports three lines: expected annualised gain from moving to the filtered target, the probability the move hurts, and ACT-FULL / ACT-PAIR(i, j) / HOLD against the registered threshold; intervals go to an appendix. HOLD reads “the expected gain does not yet cover the registered cost of switching”: inaction becomes a priced option value of waiting, not an admission of ignorance. This meets B5 and M9 in the only way consistent with honesty —

the wide posterior never reaches the meeting as a naked interval but multiplied by the decision it implies — and supplies H8’s missing pre-registered rule. Meridian, Robyn and PyMC-Marketing already have the ingredients (a Hessian at the recommended budget, a posterior covariance in allocation units, a refresh history); missing are the filter across refreshes and the derived threshold. For CMO decision-making the sharpest lessons are that the intuitive rules are wrong in exponent, not in constant — significance thresholds scale with the noise, expected-loss thresholds with $K^{1/2}$ rather than $K^{1/4}$ — and that noisier models should be refreshed more, not less, unless the refreshes reuse the same window.

For theory the chapter contributes an exactness statement (the LQ ergodic partially observed problem separates, and its region is exactly an ellipsoid), a discrete composite formula verified to 0.4% and 0.01%, and an honest deflation: what the field would argue about (shape) is second-order, what it does by default (the statistic) is first-order. Outward links: Chapter 6 supplies the refresh-error correlation $\rho \approx 0.5$ and the hysteresis this chapter prices; Chapter 4’s cadence law $\tau^* = \sqrt{c/(r(1/\kappa - \frac{1}{2}))}$ is the other half of a unified inaction theory that Proposition 7.12 begins; Chapter 1 (Dive 22) later used this band to show that a constant estimation bias shifts $\hat{z}$ and x_c equally and is therefore *invisible* to the trigger — a biased model never churns, only a noisy one. Later chapters wire the protocol to other triggers (BL 49, BL 56, BL 66, BL 72) and replace its scalar EVSI bar with the ellipsoid loss (BL 53).

7.8 Further exploration

Dive 08’s next steps became BL 21 (done here), BL 22 (estimate q under rotation wobble with the anchored-estimator decomposition of Chapter 6, and a revealed-preference elicitation for K), BL 23 (a Huberised or Student filter to recover part of the +9% heavy-tail loss), BL 24 (trust as a state degraded by acting-and-regretting) and a one-page field template. Dive 09 opened BL 25 (general- n tiering: a greedy “add channels while the marginal W -reduction exceeds c ” rule against a lattice DP for $n = 4-5$, connected to Bonomo et al.’s continuum policy), BL 26 (adaptive M : whether re-calibrating (Γ, Q, V) each refresh makes the trigger matrix itself churn — the instability of Chapter 6 in new clothes), BL 27 (prove $\kappa = \frac{1}{4} - \beta^2/4$ by matched asymptotics) and BL 28 (estimate $\hat{\rho}(\Delta)$ and $\hat{q}$ jointly from refresh history and run the floor-with-correlation cadence rule end-to-end on the nonlinear MMM, where $\rho \approx 0.5 \ll \omega$ makes frequent refreshes more valuable than the overlap arithmetic suggests).

In the compiler’s judgement the most promising extension is BL 28 merged with BL 22: every constant the protocol consumes except K is a second moment of the refresh history, and the cadence answer swings by an order of magnitude on one of them, so one anchored decomposition of refresh jitter into drift, white estimation noise and window-overlap correlation would make h^* , Δ^* and the filter field-estimable at once. Second is BL 24, the one item that could change the *form* of the answer: if trust is a stock, the credibility charge depends on the history of regretted moves and the band should widen after a visible error — the dynamics Dietvorst et al. measured, which a fixed K cannot represent.

Reviewer’s note (compiler)

Novelty. Dive 08’s same-day audit relabelled its §3.1–3.2 from “derived here” to “rediscovered; known in continuous time” (Muth, 1960; Alvarez–Lippi–Paciello, 2011, appendix; Baley–Blanco, 2019), keeping the discrete composite formula, the floor decomposition and the MMM/PoR layer as “plausibly unclaimed ($\sim 75\%$ confidence)”. Dive 09’s ellipsoid equation is Altarovici et al.’s with a change of notation; the dive concedes this and claims only exactness, the whitening closed form and the verification. In Baley–Blanco proper the band *does* depend on uncertainty because uncertainty cycles; the invariance is the constant-uncertainty limit. Neither literature search is reproduced; “no marketing/OR application found” rests on a subagent.

The scalar cost formula is off at the level. Dive 08 states the minimum average cost as $\gamma h_c^2/12 + Kq/h_c^2 + \gamma P/2$, which at $K = 30, q = \gamma = 1$ is $3.162 + \gamma P/2$; the observed $v = 0$ cost is 3.004 and the “predicted 3.00/3.31/4.35/7.76” series is anchored on that observed level, so only the floor *increments* were verified (the clean-room checked increments too). The 5% level gap is exactly Dive 09’s discrete-time offset $-\kappa \operatorname{tr}(\Sigma Q) \approx -0.165$ ($3.162 - 0.165 = 2.997$), which Dive 09 presents as new without noting that it corrects Dive 08’s formula. Nothing downstream

depends on the level.

Numbers not to over-read. The CI rule “ties” twice by coincidence of scale — at $v = 10$ scalar ($1.96\sqrt{P} = 3.22$ against $h^* = 3.77$; still 2.6% worse) and in the 4-channel transport ($v/q \approx 26$); Dive 08’s 2-channel transport did not run it; the ranking among filtered shapes in the 4-channel transport is within one paired SE, and both transports are single configurations calibrated at the initial point. The tiered rule is $n = 3$ only with a constant 5.5% low. Dive 09 states the correlated-noise cadence as “8–13 weeks” in §4 and “1 to 13 weeks” in §5. The κ conjecture is supported at 3%, and the effective-exponent reconciliation (0.292/0.40 versus 0.295/0.42) is a fit over a chosen K range. Act-always has no SE in the scalar duel.

INDEX consistency. Both log lines agree with the writeups on every headline ($6.5\times$, $7\times$, $3\times$, 0.01%, 0.012%, 12.5%, 0.65%, +45%, $3.7\times$); “ χ^2 CI rule +30–107%” is the isotropic sweep’s +31–107% (case B gives +29%/ +9%). The Dive 08 log timestamp (~15:30) precedes the file’s last write (16:55 UTC), consistent with the post-writeup audit but unstated. No FIM/pinv computation enters either dive, so BL 82 does not apply; the transports’ Γ is a finite-difference Hessian, which BL 70 later notes presumes the budget-direction total value known.

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Chapter 8

Creative estimands under algorithmic delivery

Provenance

Source dives: Dive 10 *Estimands under algorithmic delivery: cross-score logging identifies the fixed-policy creative effect* (B4 / M3) (run 2026-09-01, file `10_algorithmic_delivery_estimands.md`, last written 2026-09-01 23:43 UTC).
Code: `code/10_algorithmic_delivery_estimands.py`. **Frontier-study problem(s) addressed:** M3 (queue item B4) — define and identify “the effect of creative A versus B under a *fixed* delivery policy π ” from data generated under adaptive per-arm policies π_A, π_B , when the logging policy is a black-box auction/pacing system with unlogged propensities; and say what partial logging suffices. **Backlog items closed:** none; **opened:** BL 29 (sharp segment-level Λ -bounds), BL 30 (ratio-balancing specification test), BL 31 (fixed-policy estimands under budget-coupled arms and frequency).

Key ideas in this chapter

- The fixed-policy creative effect $\tau(\pi) = \mathbb{E}[\pi(X)(\delta_A(X) - \delta_B(X))]$ and the A/B intention-to-treat contrast obey the exact identity $\tau(\pi_B) - \text{ITTDiff} = \mathbb{E}[(\pi_B - \pi_A)\delta_A]$: the A/B test’s bias is delivery divergence times effect heterogeneity, and the optimizer manufactures both at once.
- Creative rankings flip sign with the policy: in a CTR regime the A/B test reports $+7.99$, the effect of A in B’s audience is -1.69 , in A’s own audience $+16.68$ (all $\times 10^{-4}$; clean-room ± 0.1).
- Bucketed per-creative scores (m_A, m_B) logged for every user in every arm—no propensities—are a sufficient logging statistic under *ratio balancing*, a theorem with an exact error decomposition; cross-score cells recover $\tau(\pi_B)$ to 1% of the divergence bias where single-score cells land at -3.4 and $+16.4$.
- Two identical creatives, one scored more noisily, produce a spurious A/B winner of 5.48 ± 0.03 that the estimator nulls to 0.075 ± 0.002 ; the overlap policy $\pi_{\text{ov}} = 2\pi_A\pi_B / (\pi_A + \pi_B)$ costs no precision (sd 0.95 versus 0.92 for ITTDiff) and is nearly immune to cell coarseness.
- Negative results: advertiser-segment Λ -bounds are vacuous ($\Lambda_{95} = 5.70$, width 66 on an effect of scale 8); noisy logs attenuate (reliability 0.91 loses 18%, SIMEX recovers 82%); arm-asymmetric δ -correlated shocks are the theorem’s boundary; conversion-outcome tests are noise-dominated until $N \approx 10^8$, CTR tests bias-dominated from $N \approx 10^5$.

8.1 Problem and motivation

A platform A/B test of two ad creatives randomizes users to arms, then lets each arm’s delivery system—auction, pacing and a per-creative predicted-response model—decide whom to actually show the ad. The arms therefore reach different audiences, systematically: the optimizer for creative A finds the users it predicts will respond to A, the optimizer for B finds B’s. The platform’s “A beats B” compares A on A’s audience with B on B’s, conflating creative quality with the algorithm’s audience choice. Braun and Schwartz named this *divergent delivery*; Meta’s audit of 181,890 tests confirmed its ubiquity; and problem M3 recorded that “the effect of creative A versus B” is ill-defined when exposure is mediated differently per arm, that operational mitigation exists but identification theory does not, and that the formal problem is off-policy

evaluation with a black-box, non-stationary logging policy whose propensities are never written down.

The relevant counterfactual for a creative decision is not “what happened under two optimizers” but “if I ship A instead of B the algorithm will re-target, and creative quality is the part that transports”. Dive 10 formalises that as the effect under a *fixed* delivery policy π , derives an exact identity for the A/B test’s bias, shows that the fixed policy can reverse a creative ranking, and answers the logging question constructively: not the propensities but the two per-creative *scores* that drive them, bucketed and logged for every user in every arm, identify $\tau(\pi)$ —a log field rather than the extra experimental arm of the only prior formal fix. The chapter also records what does not work and prices the regime in which any of it is detectable.

8.2 Background: the ideas this chapter builds on

8.2.1 Divergent delivery (Eckles, Gordon and Johnson, 2018; Ali et al., 2019; Braun and Schwartz, 2025; Burtch et al., 2025)

Eckles, Gordon and Johnson (2018) pointed out, against a psychological-targeting field study, that a platform’s delivery optimization makes exposure within an ad arm non-random, so arm-level outcome differences are not creative effects. Ali et al. (2019) supplied the mechanism: under maximally inclusive targeting, Facebook’s relevance layer skews who sees an ad by the ad’s content alone, because the delivery model predicts per-user response to each creative. Braun and Schwartz (2025) drew the consequence for experiments: two *identical* public-service ads in separate arms drifted apart in realised audience (57.0% versus 50.8% female) from delivery noise alone, so an A/B contrast is a total effect entangling audience selection with creative response. Burtch, Moakler, Gordon, Zhang and Hill (2025) quantified it across 181,890 Meta A/B tests and 3,204 lift tests: lift tests (one live arm versus a holdout) show no audience imbalance, A/B tests (two separately optimized live arms) do, and harmonizing campaign configuration reduces but never eliminates it. None defines the estimand a creative decision needs or says what to log to identify it.

8.2.2 Off-policy evaluation with estimated propensities (Hirano, Imbens and Ridder, 2003; Hanna, Niekum and Stone, 2019)

Off-policy evaluation (OPE) values a target policy π from data logged under a behaviour policy π_0 by reweighting each logged outcome with $\pi(a | x) / \pi_0(a | x)$; the estimator is unbiased when π_0 is known and positive wherever π is. Hirano, Imbens and Ridder (2003) showed that inverse-propensity weighting with a nonparametrically *estimated* propensity attains the semiparametric efficiency bound and beats weighting by the true propensity, because the estimate corrects chance imbalance; Hanna, Niekum and Stone (2019) transported this to reinforcement learning (“regression importance sampling”). Both require the estimated propensity to condition on everything the behaviour policy used: estimated from a coarser set it is a marginal, and reweighting by it is biased unless the omitted variables are irrelevant to the outcome. The dive’s cell estimator is an estimated-behaviour-policy estimator whose conditioning set is chosen by a theorem.

8.2.3 Sensitivity bounds: Rosenbaum’s Γ , the marginal Λ model and sharp bounds (Rosenbaum, 2002; Tan, 2006; Kallus and Zhou, 2018, 2020; Namkoong et al., 2020; Dorn and Guo, 2022)

When propensities are unknown one bounds how far they stray. Rosenbaum’s model says two units with the same observed covariates have treatment odds within a factor Γ , $\Gamma^{-1} \leq \pi_i(1 - \pi_j) / (\pi_j(1 - \pi_i)) \leq \Gamma$; a finding is reported as the Γ at which it stops being significant. Tan’s (2006) *marginal sensitivity model* bounds instead the odds ratio between the true propensity $e(x, u)$ and the nominal $e(x)$ by Λ , so the true importance weight lies in a known interval around the nominal one. Kallus and Zhou (2018, 2020) made this operational for policy evaluation and learning—the worst-case policy value is a fractional linear program over weights

in the Λ -box, solved by sorting outcomes—and Namkoong et al. (2020) extended it to sequential decisions. Dorn and Guo (2022) showed those bounds are not sharp: the sharp bound puts the extreme weights on units above or below the $\Lambda / (1 + \Lambda)$ quantile of the outcome given covariates (“quantile balancing”). The dive’s segment bound is the elementary member of this family, with the within-segment delivery *ratio* as the unknown odds ratio; its vacuity motivates BL 29.

8.2.4 Balancing scores and overlap weights (Rosenbaum and Rubin, 1983; Li, Morgan and Zaslavsky, 2018)

A *balancing score* $b(X)$ is any function with $X \perp D \mid b(X)$; the propensity score is the coarsest, and conditioning on any balancing score removes confounding when treatment is unconfounded given X . The dive’s ratio-balancing condition is weaker: only the *ratio* of the two arms’ exposure probabilities must be constant given the logged variable. Li, Morgan and Zaslavsky (2018) introduced *overlap weights*: weight treated units by $1 - e(x)$ and controls by $e(x)$, targeting the population tilted by $e(x)(1 - e(x))$ —the units both arms could plausibly reach—with bounded weights and minimum asymptotic variance among balancing weights. The dive’s overlap policy is the delivery analogue: the harmonic mean of the two arms’ exposure probabilities, whose weights against either arm are at most 2.

8.2.5 Simulation–extrapolation (Cook and Stefanski, 1994)

When a covariate is observed as $W = X + U$ with known error variance σ_u^2 , SIMEX adds *further* error $W_b(\lambda) = W + \sqrt{\lambda} \sigma_u \varepsilon_b$ for $\lambda \in \{0, 0.5, 1, 1.5, 2\}$, recomputes the estimator, averages over b , fits a curve in λ and extrapolates to $\lambda = -1$, the error-free point; its bias is whatever the extrapolant (usually quadratic) misses. The dive applies it to the cells: noise is added to the logged scores, cells re-formed, the cell estimator extrapolated.

8.2.6 Counterfactual logging and the three-arm design (Johnson, Lewis and Nubbe-meyer, 2017; Pal and Susarla, 2026)

Ghost ads made lift measurement cheap by logging, for control users, the moment the platform *would have* shown the ad—the algorithm’s decision recorded without the exposure. Pal and Susarla (2026) address divergent delivery with a third arm in which the delivery algorithm sees creative A ’s metadata while users see the control creative: the total A/B effect decomposes into a natural direct effect (the creative) and a natural indirect effect (the re-targeting), point-identified without sequential ignorability; they also show that adjusting on the realised audience is biased, the audience being a post-treatment mediator. The design needs the platform to build an arm; the dive’s proposal is in the ghost-ads spirit—log the algorithm’s inputs for everyone instead.

8.3 Setup and assumptions

Users have covariates X , rich on the platform side, coarsely visible to the advertiser. Potential outcomes are $Y(0)$ (unexposed) and $Y(1, a)$ for creative $a \in \{A, B\}$, with creative effects $\delta_a(X) = \mathbb{E}[Y(1, a) - Y(0) \mid X]$. Arms $Z \in \{\text{control}, A, B\}$ are randomized across users. Within arm a the platform exposes a user with an unlogged probability $\pi_a(X)$, a function of its per-creative score $m_a(X)$ (predicted response), of pacing and competition, and of anything else it uses; D is the realised exposure indicator.

Definition 8.1 (Estimands). $\text{ITT}_a = \mathbb{E}[\pi_a(X)\delta_a(X)]$ is the effect of creative a under its own delivery (what the test measures) and $\text{ITDiff} = \text{ITT}_A - \text{ITT}_B$. The *fixed-policy creative effect* is

$$\tau(\pi) = \mathbb{E}[\pi(X)(\delta_A(X) - \delta_B(X))], \quad (8.1)$$

with the natural choices $\pi = \pi_B$ (“A in B’s audience”), $\pi = \pi_A$, and the *overlap policy* $\pi_{\text{ov}} = 2\pi_A\pi_B/(\pi_A + \pi_B)$.

π_B is a legitimate fixed policy—the platform would deliver the winner with its own optimizer, so $\tau(\pi_B)$ answers “would swapping the creative for the same audience help?”—and π_{ov} is the audience both optimizers would reach.

Assumption 8.2 (Design and exclusion). Arms are randomized; creative identity affects Y only through exposure, so $Y(0)$ does not depend on the arm; effects are additive in δ_a with single exposure; per-arm delivery is independent (no shared budget or pacing).

Assumption 8.3 (Ratio balancing). For a logged discrete cell variable $C = C(X)$, within every cell c either (a) the delivery ratio $\pi_B(X)/\pi_A(X)$ is constant, or (b) $\delta_A(X)$ is constant, or more generally $\text{Cov}(\pi_B/\pi_A, \delta_A | c) = 0$ under the π_A -weighted measure.

Condition (a) holds whenever $\pi_a(X) = g_a(m_A(X), m_B(X))$ —delivery depends on X only through the two logged scores—and is weaker than Rosenbaum–Rubin balancing: unlogged shocks that hit both arms’ logits identically (pacing, time of day, competition common to both creatives) cancel in the ratio.

8.4 Main results

8.4.1 The divergence-bias identity

Theorem 8.4 (Divergence-bias identity). *[established in the dive]* For any fixed policy π , ITTdiff is identified by the randomized design and

$$\tau(\pi_B) - \text{ITTdiff} = \mathbb{E}[(\pi_B - \pi_A)\delta_A], \quad \tau(\pi) - \text{ITTdiff} = \mathbb{E}[(\pi - \pi_A)\delta_A] - \mathbb{E}[(\pi - \pi_B)\delta_B]. \quad (8.2)$$

The proof is one line: $\tau(\pi_B) = \mathbb{E}[\pi_B\delta_A] - \text{ITT}_B$ and $\text{ITTdiff} = \mathbb{E}[\pi_A\delta_A] - \text{ITT}_B$. The content is the reading: the A/B test’s bias for the fixed-policy estimand is a covariance between the delivery difference and the heterogeneity of the *new* creative’s effect. If δ_A is constant the bias is exactly $\mathbb{E}[\pi_B - \pi_A]\delta_A$, a pure reach difference of known sign; if $\pi_A = \pi_B$ there is no bias whatever the heterogeneity. Both are required, and this is precisely what the algorithm manufactures: it diverges *because* it detects heterogeneity in predicted response. Verified to float precision (residual 10^{-19} ; clean-room 10^{-19} to 3×10^{-19}).

8.4.2 The cell estimator and the sufficiency theorem

Let C be a logged discrete function of X with cell masses p_c . Within cell c define the reach $r_a(c) = \mathbb{E}[\pi_a | c]$ (identified from exposure logs), the cell ITT $\text{ITT}_a(c) = \mathbb{E}[\pi_a\delta_a | c]$ (identified from the arm-versus-holdout contrast within c), and the reach-normalized effect $\tilde{\delta}_a(c) = \text{ITT}_a(c)/r_a(c) = \mathbb{E}_{\pi_a}[\delta_a | c]$, the π_a -weighted mean of δ_a in the cell.

Construction 8.5 (Cross-score cell estimator). *[established in the dive]* With $\pi(c) = \mathbb{E}[\pi | c]$ ($r_B(c)$ for target π_B ; $2r_Ar_B/(r_A + r_B)$ for π_{ov}),

$$\hat{\tau}(\pi) = \sum_c p_c \pi(c) [\tilde{\delta}_A(c) - \tilde{\delta}_B(c)]. \quad (8.3)$$

Proposition 8.6 (Exact error decomposition). *[established in the dive]* For target π_B ,

$$\hat{\tau} - \tau(\pi_B) = - \sum_c p_c \left\{ \mathbb{E}[\pi_B\delta_A | c] - r_B(c) \mathbb{E}_{\pi_A}[\delta_A | c] \right\} = - \sum_c p_c r_A(c) \text{Cov}_{\pi_A}(\pi_B/\pi_A, \delta_A | c). \quad (8.4)$$

The B-term of Equation (8.3) is exact because $\mathbb{E}[\pi_B\delta_B | c]$ is directly identified; all error sits in the cross term $\mathbb{E}[\pi_B\delta_A | c] = r_A(c)\mathbb{E}_{\pi_A}[(\pi_B/\pi_A)\delta_A | c]$, and $\mathbb{E}_{\pi_A}[\pi_B/\pi_A | c] = r_B(c)/r_A(c)$ gives the covariance form. The decomposition reproduced the estimator’s error to three decimals in experiment E5: the term is the whole story.

Theorem 8.7 (Ratio-balancing sufficiency). *[established in the dive] Under Assumption 8.3, $\hat{\tau}(\pi_B) = \tau(\pi_B)$. In particular, if $\pi_a(X) = g_a(m_A(X), m_B(X))$, the pair of per-creative scores (m_A, m_B) , logged for every user in every arm, is a sufficient logging statistic for $\tau(\pi)$ for every π expressible in the scores. Neither score alone is sufficient.*

A single score fails because within an m_A cell π_B still varies with m_B , and m_B is correlated with δ_A (creative effects share heterogeneity, scores share the baseline), so the cross term does not factor: in the CTR regime, m_A -only cells land at -3.4 against a truth of -1.7 (clean-room -3.44 ± 0.10) and m_B -only cells land on $\tau(\pi_A)$, $+16.4$ —the wrong estimand with the wrong sign. Propensities are not needed because the estimator uses reach *rates* and ITT *rates* by cell, both identified by randomization; π_a ’s individual values never appear. This is the M3 answer: what must be logged is the *sorting variable*, not the propensity, and any coarsening of (m_A, m_B) preserving ratio balancing to the needed tolerance suffices. What must not exist is an unlogged shock that is arm-asymmetric and correlated with δ_A (E14).

8.4.3 Variants, the overlap policy and the logging hierarchy

The *holdout* version above needs a no-ad control cell (Meta’s multi-cell lift design) and only ratio balancing; $Y(0)$ enters only through the randomized arm-versus-holdout contrast, so unlogged confounders of exposure versus $Y(0)$ are harmless. The *no-holdout* version replaces $\tilde{\delta}_a(c)$ by the within-cell exposed-minus-unexposed contrast and needs *full* balancing, $D \perp (Y(0), \delta) \mid c$ in each arm; it fails when unlogged competition selects high-baseline users into exposure. In the CTR regime the two coincide.

Proposition 8.8 (Overlap policy). *[numerically established] For π_{ov} the per-cell weights $\pi_{ov}(c)/r_a(c) \leq 2$ are bounded. In the dive’s DGP its sampling sd equals ITTdiff’s (0.95 versus 0.92×10^{-4} at $N = 3M$) while $\tau(\pi_B)$ costs $+27\%$ (1.17); its coarseness bias is negligible at every K ($0.12 \rightarrow 0.005$ from the coarsest to the finest grid, versus $1.69 \rightarrow 0.07$ for π_B); and it degrades most gracefully under asymmetric shocks.*

The dive recommends $\tau(\pi_{ov})$ as the headline creative effect (“which creative is better as a creative”), with $\tau(\pi_B)$ and $\tau(\pi_A)$ answering the sharper “swap for the same audience” question at a precision cost. Table 8.1 states what each logging level identifies.

Table 8.1: What each logging level identifies (Dive 10 §3.5; design grounded in Theorem 8.7, not a field result). Scores are bucketed quantiles—privacy-compatible, revealing neither the model nor propensities—and the platform already computes both scores in every auction.

Logged	Identifies	Assumption
L0: arm toplines	ITTdiff only	none; $\tau(\pi)$ unidentified, bias sign unknown
L1: advertiser segments + holdout	Λ -bounds on $\tau(\pi_B)$	within-segment ratio divergence $\leq \Lambda$; vacuous in practice (E11)
L2: per-user bucketed (m_A, m_B) at decision time + holdout	$\tau(\pi)$ for any π expressible in scores	ratio balancing (arm-symmetric unlogged shocks allowed)
L2’: L2 without holdout	same	full balancing (no $Y(0)$ confounding of exposure)
L2 + eligibility bit	identified part of $\tau(\pi)$ plus reported unidentified mass	positivity failure made explicit (E7)
L2 + period id	$\tau(\pi)$ under learning or non-stationary delivery	scores comparable within period (E8)

8.5 Numerical evidence

The DGP: $X \in \mathbb{R}^5$ standard normal; baseline $p_0 = \text{base} \cdot \exp(0.5X_1 - 0.125)$; effects $\delta_A = \text{lift} \cdot p_0 \exp(0.8 \text{het} X_2 - 0.32 \text{het}^2)$ and δ_B likewise on $\text{corr} X_2 + \sqrt{1 - \text{corr}^2} X_3$ times 0.9 (B is 10% weaker); defaults $\text{het} = 1$, $\text{corr} = 0.5$. Scores $m_a = \log(p_0 + \delta_a) + 0.3\epsilon_a$; delivery $\pi_a = 0.6 \expit(-0.4 + 1.5z_a - \kappa \text{comp})$ on the standardized score z_a , with unlogged competition $\text{comp} = 0.7X_1 + 0.7X_4$ (reach about 0.26 per arm). Conversion regime: base 2%, lift 15%; CTR regime: base 5×10^{-4} , $\delta \approx \text{CTR}$. All numbers are $\times 10^{-4}$ absolute lift; the seed sweep is 6 seeds at $N = 600\text{k}$, $K = 40$ quantile bins per score, population limits (true conditional means within cells).

Table 8.2: Sign flip, recovery and negative control in the CTR regime (E3, E4, E9; 6 seeds, $N = 600\text{k}$, $K = 40$). Clean-room values are from an independent numpy re-implementation given only the DGP specification.

Quantity	Dive ($\pm$ seed sd)	Clean-room	Note
ITTdiff	$+7.99 \pm 0.06$ –0.08	8.03 ± 0.12	what the platform reports
$\tau(\pi_B)$	–1.69	-1.65 ± 0.09	A in B’s audience
$\tau(\pi_A)$	+16.68	16.74 ± 0.11	A in A’s audience
$\tau(\pi_{\text{ov}})$	+7.10	—	overlap audience
cross-score cells, residual bias	0.010 (E4 table) / 5% (§4)	1.0%	see reviewer’s note
m_A -only cells	–3.4	-3.44 ± 0.10	wrong value
m_B -only cells	+16.4	+16.38	wrong estimand
negative control: ITTdiff	5.48 ± 0.03	5.50 ± 0.01	identical creatives
negative control: cell estimator	0.075 ± 0.002	0.077 ± 0.001	truth 0

The *negative control* (E9) sets $\delta_B \equiv \delta_A$ but scores B with noisier predictions (log-sd 0.8 versus 0.3, a cold-start creative): the A/B test reports a creative effect of 5.48, about 70% of the real effect scale, from delivery alone, while the cell estimator gives 0.075; with equal noise the pure-delivery ITTdiff is 0.01 by symmetry. In the *conversion regime* (E1) ITTdiff overstates $\tau(\pi_B)$ by a factor 1.70 ± 0.03 (1.13 versus 0.67), post-stratifying exposed users on advertiser segments does nothing (1.12), and cross-score cells converge from 0.85 at $K = 3$ to 0.68 at $K = 40$. The finite-sample check (E2; $N = 1\text{M}$, 60 replications) gives an influence-function se of 4.85 against a Monte Carlo sd of 4.53 and 95% coverage 0.93 ± 0.03 . Table 8.3 collects the attack experiments.

Table 8.3: Attack experiments (Dive 10 §4–5). All $\times 10^{-4}$, CTR regime unless noted. Reliability is the squared correlation between logged and delivery score.

Exp.	Attack	Result	Uncertainty
E4	coarseness K	residual bias 36% ($K = 10$), 14% (20), 5% (40); falls $\sim K^{-1.4}$	single seed
E5	unlogged competition (conv.)	holdout error 0.02, no-holdout –0.12 on truth 0.48	single seed
E6	noisy logs	reliability 0.91/0.79/0.48 loses 18%/38%/70% of the correction	single seed
E7	positivity	δ -correlated exclusion: –0.6 vs truth –1.8, identified part +3.1; with eligibility bit 3.18 vs 3.09, missed mass 0.066 exact	single seed
E8	learning delivery	period cells within 0.3 of truth ($K = 15$); pooled cells with shared refit shocks +0.59 vs truth –1.16	3 seeds
E10	bias/sd crossover	CTR 2.4 ($N = 0.3\text{M}$) $\rightarrow$ 8.3 (3M); conversion 0.05 $\rightarrow$ 0.16	20 reps per N
E11	segment Λ -bounds	$\Lambda_{95} = 5.70 \pm 0.01$ (max 219); at $\Lambda = 1.5$: [–18.6, 47.5], width 66.2 ± 0.1	6 seeds
E12	heterogeneity map	flips for $\text{het} \geq 0.5$ and $\text{corr} \leq 0.5$; $ \hat{\tau} - \tau \leq 0.2$ everywhere; $\tau(\pi_{\text{ov}})$ 6.2–8.6	grid
E13	SIMEX repair	quadratic recovers 85/82/62/36% at reliability 0.96/0.91/0.79/0.60	extrapolant ± 1.7
E14	arm-asymmetric shocks	π_B error to –7.9, π_{ov} to –1.5; δ -uncorrelated shocks ≤ 0.25	grid
E15	K selection ($N = 1\text{M}$)	RMSE-optimal $K \approx 30$ for π_B (bias 0.14, sd 1.84); π_{ov} insensitive	12 reps

Two rows need unpacking. The Λ -bound (E11, level L1) uses nine advertiser segments (terciles of X_1 and X_4) and bounds the within-segment ratio by $\pi_B/\pi_A \in [\bar{\rho}_s/\Lambda, \bar{\rho}_s\Lambda]$, $\bar{\rho}_s = r_B(s)/r_A(s)$, with $\delta_A \geq 0$,

giving $\tau(\pi_B) \in [\sum_s p_s \bar{p}_s \Lambda^{-1} \text{ITT}_A(s) - \text{ITT}_B, \sum_s p_s \bar{p}_s \Lambda \text{ITT}_A(s) - \text{ITT}_B]$. The realised Λ (exponentiated 95th percentile of $|\log((\pi_B/\pi_A)/\bar{p}_s)|)$ is 5.7, and even at $\Lambda = 1.5$ the interval is 66 wide on an effect of scale 8; adding the effect driver's tercile leaves $\Lambda_{95} = 5.1$. Score noise alone disperses within-segment delivery ratios. The *crossover* (E10): bias/sd grows as $\sqrt{N}$, so the CTR test is bias-dominated from $N \approx 10^5$ while the conversion test needs $N \approx 10^8$; at typical sizes the conversion creative test is underpowered for ITTdiff and τ alike, and the estimator is a bias fix, not a precision gain. The clean-room agent's finite-sample verdict, adopted verbatim by the dive: at $N = 600\text{k}$ the sd is about 2.4, so the estimator rejects ITTdiff-sized values (about 3 sd) but cannot sign $\tau(\pi_B)$ (10-replicate mean -0.41 ± 0.79).

8.6 Limitations and the devil's advocate

Round 4 attacked the refinements and drew the boundary. *SIMEX is a mitigation, not identification*: its answer depends on the extrapolant by up to 1.7—a reliability-ratio extrapolant from a Gaussian heuristic did *worse* than the quadratic (67/61/53/49% against 85/82/62/36%)—and the real fix is logging the score used at decision time, which makes reliability 1 by construction. *Arm-asymmetric shocks correlated with δ* (E14: competition hits A's auctions only and correlates with the effect driver) break ratio balancing with errors up to -7.9 , larger than the effect; δ -uncorrelated shocks stay ≤ 0.25 . Real auctions have arm-asymmetric unlogged components (creative format changes inventory eligibility, bid landscapes, frequency caps); the theorem says what to log but cannot certify that the unlogged residual is arm-symmetric, and *ratio balancing is untestable from the logs alone*—a specification test (does adding logged features move $\hat{\tau}$?) is the practical guard. Round 5 changed nothing.

The dive's own counterarguments: the *sign flip* is manufactured by $\text{het} = 1$, $\text{corr} = 0.5$ in a lognormal-heterogeneity DGP—at $\text{het} = 0.25$ there is no flip, only a 30% bias—and whether real heterogeneity is that large is an empirical question the dive cannot answer (Pal and Susarla's live campaign found the algorithm moved composition by +2.27 pp female share while the visible creative did not). The *negative control* needs B's scores substantially noisier; for two mature creatives the pure-delivery ITTdiff is about 0. The *Λ -vacuity* is for worst-case bounds ignoring the within-segment distribution of δ_A , under independent score noise across creatives: “segments plus worst-case Λ cannot work”, not “no segment-based bound can”. The *overlap policy's* precision parity was measured with similar reaches (0.26 each); with very different reaches π_{ov} shrinks toward the smaller and answers a narrower question. The 10^8 *crossover* depends on a 2% base rate and 15% lift (about $25\times$ lower at 10% base rates); the portable object is bias/sd, computable ex ante from Equation (8.2). Untested: multiplicative effects, frequency, budget coupling across arms, interference across advertisers (M4).

8.7 Discussion: impacts

For experimentation platforms the message is that divergent delivery is a missing log field, not a nuisance to be harmonized away. The propensities M3 assumed had to be recovered never need to exist in the logs; what identifies the fixed-policy effect is the pair of bucketed per-creative scores the platform computes in every auction anyway, releasable as quantile buckets without exposing its model. A clean room that exports “estimated action rate” deciles for both creatives is already an L2 log; the eligibility bit and the period id are two further one-field additions with precise payoffs (Table 8.1). Against Pal and Susarla's third arm, the dive buys the same estimand for a log field rather than an arm; their natural direct effect is $\tau(\pi_{\text{ctrl-creative}})$ in this notation.

For advertisers the results are cautionary: post-stratifying the exposed audience on visible segments does nothing (1.12 against 1.13), worst-case segment bounds are vacuous, and a cold-start creative can lose an A/B test by 5.5 units on delivery noise alone, so today's honest reading of an A/B winner is “winner under its own optimizer”. For CMO decisions the crossover matters more than the identity: conversion-outcome creative tests at ordinary sizes detect neither the A/B effect nor its correction, so the bias question is moot

there and live for CTR and engagement outcomes.

For theory the chapter reframes M3 as transport across delivery policies (Chapter 2) whose bias has the Jensen/aggregation shape of Chapter 5: a covariance between a weight and a heterogeneous effect, killed by conditioning on the variable that sorts both. Ratio balancing is a weaker relative of the balancing score suited to two-policy problems, and the overlap recommendation imports Li–Morgan–Zaslavsky’s logic that bounded weights buy both precision and a meaningful target population. Chapter 11 picks up the frequency and pacing thread: BL 44 extends the cross-journey table to repeated exposure and pacing-coupled delivery, merging with BL 31.

8.8 Further exploration

BL 29 replaces the worst-case bound with a Dorn–Guo quantile-balancing sharp bound using the within-segment distribution of exposed outcomes, to test whether any advertiser-side bound is informative at $\Lambda_{95} \approx 5$ —the last hope for identification without cross-score logs. BL 30 designs a Hausman-style specification test of ratio balancing across nested cell keys (scores; scores $\times$ placement; scores $\times$ period) with a derived null, the falsifiability hook M3 lacks (linked to BL 4 of Chapter 2). BL 31 drops the independent-arm and single-exposure assumptions: when arms share a budget or pacing controller π_A depends on B’s performance, and with frequency-dependent response the “fixed policy” must fix a distribution over exposure counts; the task is to define $\tau(\pi)$ there and find the extra log key (a pacing multiplier is the candidate). The dive’s other next steps are the Pal–Susarla bridge (the third arm as L2 logging with scores from an arm that never shows A; “extra arm” against “extra log field” at matched budget) and a field pilot with an advertiser whose clean room exports both scores at bucket granularity—the first real $\tau(\pi_{ov})$.

In the compiler’s judgement the most promising extension is BL 30, the only item that can make the theorem’s premise checkable: the estimator is exact under ratio balancing and silently wrong outside it, and a nested-key Hausman test is cheap once L2 logs exist. Its natural companion is the Pal–Susarla bridge, since a platform that has built a third arm holds a validation set for the cheaper route. BL 31 matters more for the book’s other chapters but depends on a frequency-indexed estimand that Chapter 11 is better placed to define.

Reviewer’s note (compiler)

The headline “1% of the divergence bias” is the clean-room’s seed, not the dive’s. Round 2 attack 1 reports “ $K = 40$: 5% (1% in the clean-room’s seed)” and derives $K^{-1.4}$ from 36%/14%/5%; the E4 table row says “ 0.010 ± 0.000 ” and the INDEX line “recover τ to 1%”. A quantity that differs $5\times$ between seeds cannot have seed sd 0.000. E15 (seed 52) gives 0.14 at $K = 30$ and 0.07 at $K = 50$, i.e. 0.7–1.4% of the divergence bias $7.99 + 1.69 = 9.68$; the honest statement is “1–5% at $K = 40$ ”.

Numbers moved between rounds without comment. Round 1 quotes the flip as $+7.8 / -1.8 / +16.5$; E3 gives $+7.99 / -1.69 / +16.68$; E7 uses a truth of -1.8 , E1/E4 “ -1.7 ”. These are different seeds and sizes (drivers use seeds 1, 11, 41, 52 at $N = 600k$ or 1M), consistent with the quoted seed sd of 0.06–0.08 only at $2-3\sigma$ for ITTdiff. The negative control is 0.08 in §4 and the INDEX, 0.075 ± 0.002 in E9.

Single-seed attacks and an unstated regime. E5, E6, E7 and the SIMEX percentages are single-seed population limits with no uncertainty. E2 does not state its regime; its sd (4.85) matches the conversion regime, not the CTR regime whose sd at $N = 1M$ is 1.84 (E15). E1, E2 and E8 have no driver in the shipped code, so the learning-delivery numbers cannot be re-run from it.

Two labels are approximate. E14’s “ δ -uncorrelated” competition variable is $0.7X_1 + 0.7X_4$, and X_1 enters δ_A through the baseline p_0 : the shock is uncorrelated with the effect driver X_2 , not with δ_A . And the overlap cell estimator targets $\sum_c p_c 2r_A r_B / (r_A + r_B)(\dots)$ while the truth uses user-level $2\pi_A \pi_B / (\pi_A + \pi_B)$; they agree only as cells become fine, so “coarseness-immune” means “small at every K ” ($0.12 \rightarrow 0.005$), not exact.

Attribution. The dive credits an “SSRN 2019 w/ Eckles” version of Braun and Schwartz; SSRN 3896024 lists Braun and Schwartz only, and the Eckles critique is Eckles, Gordon and Johnson (2018), cited separately here. Pal and Susarla (2026) was verified by the dive at abstract level only; its $+2.27$ pp figure comes from that abstract.

Consistency checks passed. The INDEX line agrees with the writeup on every other number; $1.17/0.92 = 1.27$ gives

“+27%”; the $\sqrt{N}$ extrapolation of bias/sd from 0.16 at 3M reaches 1 at 1.2×10^8 and from 2.4 at 0.3M at 5×10^4 , matching “10⁸” and “10⁵”; the reliabilities 0.96/0.91/0.79/0.60/0.48 reproduce from log-noise 0.2/0.3/0.5/0.8/1.0 against a true-score variance near 0.98. The file’s last write (23:43 UTC) postdates the log line’s 23:30 by 13 minutes.

Sources

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- [18] Frontier study (internal): the open-questions catalogue `02_open_questions.md`, entry M3; program index `00_INDEX.md`, queue item B4 and backlog entries BL 29–BL 31, BL 44.
- [19] Program (internal): Dive 02, file `02_experiment_transport_map.md` (transport across policies); Dive 06, file `06_aggregation_privacy.md` (Jensen/aggregation algebra); Dive 13, file `13_attribution_impossibility.md` (BL 44).

Chapter 9

The data-archaeology compiler: imputation leakage, certified taxonomies and a provenance-typed likelihood

Provenance

Source dives: Dive 11 *The data-archaeology compiler: imputation leakage, the certify-the-mass audit law, and a provenance-typed likelihood* (B5 / O1 / O3) (run 2026-09-02, file `11_data_archaeology_compiler.md`, written 2026-09-02 03:33 UTC). **Code:** `code/11_data_archaeology_compiler.py` (experiments E1–E20). **Frontier-study problem(s) addressed:** program queue item B5 = adoption barriers O1 (data archaeology is 70–80% of an MMM project) and O3 (the geo tier demands geo-level spend that nationally bought channels never report, so practitioners impute it by population share): a spend-data taxonomy standard, an LLM-assisted extraction pipeline, and geo-imputation whose error enters the MMM likelihood rather than being treated as data. **Backlog items closed:** none; **opened:** BL32–BL36.

Key ideas in this chapter

- Population-share imputation of a nationally bought channel in a geo MMM obeys an exact *leakage law* into the correctly measured channel, $\text{bias}(b_D) / \beta_{TV} = \sum_t s_t d_t \text{Cov}_g(\eta, \zeta) / [\sum_t d_t^2 \text{Var}_g(\zeta) + \sum_t r_t^2 (1 + \tilde{\zeta})^2]$, with the *raw* seasonal cross-product in the numerator: seasonality controls do not protect (digital +17%/TV −7% baseline, +79%/−48% strong; clean-room ± 0.01).
- Imputation manufactures the bias: the national model is unbiased, so a geo model with an imputed channel is worse than no geos; a geo-specific loading on the national path (adstock profiled, 30/30 correct) is unbiased and most precise, but hierarchical shrinkage tighter than the true misallocation re-opens the leak ($+0.05 \rightarrow +0.68$).
- A Hausman-type plug-in-versus-latent difference detects the leak at $|t| = 4\text{--}8$; under Hill the leak flips sign across the knee ($+0.11 \rightarrow -0.05$) and misallocation is observationally equivalent to effectiveness heterogeneity (profile SSR flat to 0.6%), so the imputed channel’s curve can only be bracketed.
- For LLM-classified line items with Pareto spends the certify-top- k MSE law (verified to 1–5%) shows that certifying 2.5% of lines by spend cuts channel-total error from $\sim 5\%$ to 1.5%, while PPI and matrix-calibration rectifiers are MSE-worse than the raw labels at a 20% audit budget.
- Ledger, as allocation regret in percent of optimal revenue: baseline leak 0.28%, strong 8.2%, raw LLM taxonomy 0.13%, certified taxonomy 0.00%, Berkson share noise 0.20%.

9.1 Problem and motivation

Two of the frontier study’s adoption barriers are operational. O1: finding and cleaning spend data is 70–80% of an MMM project—years of weekly history scattered across platforms, agencies and finance systems under chaotic campaign taxonomies. O3: the geo-hierarchical model class, the statistical answer to small T (Chapter 1), needs weekly geo-level spend across 50–100 markets, which national TV buys, sponsorships and some walled gardens never report; practitioners impute it by population share, as Meridian’s documentation

recommends. The queue asked for a taxonomy standard, LLM extraction, and imputation whose error reaches the likelihood.

Dive 11 takes the *compiler* view: every spend series carries a provenance type—*measured*, *classified* (line items assigned to channels by an LLM with a confusion matrix) or *imputed* (a national total split by an assumed share vector)—and the compiler emits, per type, the correct likelihood term and the cheapest acquisition action that makes the residual error decision-negligible. Three questions came first. What does population-share imputation do to a geo MMM—not only to the imputed channel, where Sun et al. (2017) reported positive bias, but to the correctly measured channels beside it? What does LLM misclassification do to channel totals, and how should a finite audit budget be spent, given that prediction-powered inference was built for i.i.d. means rather than Pareto-weighted totals? And which defects are worth fixing, in the decision-loss units of Chapters 3 and 7? Prior art was thin: one Google section on the imputed channels’ own curves, Berkson theory never combined with a geo panel, no audit-size rule for LLM-classified spend, no spend taxonomy at all.

9.2 Background: the ideas this chapter builds on

9.2.1 Geo-hierarchical MMM and population-share imputation (Sun et al., 2017; Meridian)

The geo-level Bayesian hierarchical MMM of Sun, Wang, Jin, Chan and Koehler (2017) models each geo’s per-capita KPI as $y_{gt} = \tau_g + \sum_m \beta_{m,g} H(a_{m,g,t}) + \text{controls} + \varepsilon_{gt}$, adstock and Hill as in Chapter 2, with geo coefficients partially pooled by $\beta_{m,g} \sim \mathcal{N}(\beta_m, \sigma_m^2)$; more observations and more spend variation give tighter credible intervals than a national model. Its §4.4 handles channels known only nationally by allocating national spend to geos in proportion to population—in per-capita units, the same path S_t in every geo—and finds “larger biases and wider credible intervals for the response curves”, positive and worsening as more channels are imputed; the measured channels are not reported. Meridian follows the recommendation, and no package carries a national-only flag with model-side uncertainty. The pooling scale σ_m matters below: fixed small, or estimated from data that cannot push it up, it pulls geo coefficients to their common value.

9.2.2 Classical and Berkson error; weight shifting (Berkson, 1950; Carroll et al., 2006; Abel, 2017)

With *classical* error the observed regressor is $W = X + U$, $U \perp X$: the OLS slope on W is attenuated by the reliability $\lambda = \text{Var } X / (\text{Var } X + \text{Var } U)$, and with a second, correctly measured regressor the lost weight shifts, $\text{plim } b_2 = \beta_2 + \beta_1(1 - \lambda)\gamma_{12}$, γ_{12} the auxiliary-regression coefficient of X_1 on X_2 (Abel, 2017, gives the multi-regressor form). With *Berkson* error the truth scatters around the assigned value, $X = W + U$, $U \perp W$, so $\mathbb{E}[Y | W] = \beta W$ and the linear slope is unbiased; through a nonlinear response $\mathbb{E}[f(W + U) | W] \approx f(W) + \frac{1}{2}f''(W) \text{Var } U$ (Carroll et al., 2006), the Jensen term Chapter 5 works out for Hill. Population-share allocation is Berkson error: S_t is the pivot and the true $S_t(1 + \eta_g + e_{gt})$ scatters around it. Harmlessness requires the deviation to be independent of everything in the design; the persistent part η_g is a geo characteristic (local CPMs, coverage, planners’ weights) correlated with the measured channel’s own loading ζ_g wherever the advertiser’s customers concentrate.

9.2.3 Hausman specification tests (Hausman, 1978)

Given $\hat{b}_0$, efficient under the null but inconsistent under the alternative, and $\hat{b}_1$, consistent under both, $\text{Var}(\hat{b}_1 - \hat{b}_0) = \text{Var}(\hat{b}_1) - \text{Var}(\hat{b}_0)$ under the null, and $(\hat{b}_1 - \hat{b}_0)^T [\text{Var}(\hat{b}_1) - \text{Var}(\hat{b}_0)]^{-1} (\hat{b}_1 - \hat{b}_0)$ is asymptotically χ^2 ; the classic uses are random versus fixed effects and OLS versus IV. The leak test takes the plug-in fit as $\hat{b}_0$ and the latent-loading fit as $\hat{b}_1$.

9.2.4 Prediction-powered inference, matrix calibration, monetary-unit sampling

Prediction-powered inference (Angelopoulos, Bates, Fannjiang, Jordan and Zrnic, 2023) estimates a mean from N predictions f_i and $n \ll N$ gold labels Y_i by correcting the prediction mean with a *rectifier* from the labelled subsample,

$$\hat{\theta}_{PP} = \frac{1}{N} \sum_{i=1}^N f_i - \frac{1}{n} \sum_{i=1}^n (f_i - Y_i), \quad (9.1)$$

unbiased with valid intervals whatever the predictor’s quality, variance $\text{Var}(f)/N + \text{Var}(f - Y)/n$; PPI++ (Angelopoulos, Duchi and Zrnic, 2023) adds a tuning weight so it is never worse than the n -label classical estimator, and DSL (Egami et al., 2023) is the same design-based idea for regressions on LLM-labelled text. *Matrix calibration* (Buonaccorsi, 2010; Gustafson, 2003) estimates the confusion matrix $\hat{M}[c, \hat{c}]$ from the labelled sample and inverts it on predicted-class totals, $T_{\text{true}} = \hat{M}^{-T} T_{\text{pred}}$. All are built for i.i.d. units with bounded contributions. A channel total $\sum_i s_i \mathbf{1}[c_i = k]$ weights units by amounts from a Pareto tail with index $\alpha \approx 1.3$ —infinite variance, a handful of lines carrying most of $\sum s_i^2$ —which auditing practice meets with monetary-unit sampling (PPS, probability proportional to size) and cut-off designs that examine every item above a threshold (PCAOB AS 2315; Amorim et al., 2021). “Certify the mass” is a cut-off design; the dive contributes its MSE law and the head-to-head against the rectifiers.

9.3 Setup and assumptions

Geos $g = 1, \dots, G$ with population shares p_g , weeks $t = 1, \dots, T$, per-capita outcome

$$y_{gt} = \mu_g + \beta_{TV} f_{TV}(a_{gt}^{TV}) + \beta_D f_D(a_{gt}^D) + \text{season}_t + \varepsilon_{gt}, \quad (9.2)$$

geometric adstock $a_t = x_t + \alpha a_{t-1}$, f linear or Hill $H(a; K, s) = a^s / (a^s + K^s)$. National per-capita paths S_t (TV, bought nationally) and D_t (digital, geo-measured). True TV spend $x_{gt}^{TV} = S_t(1 + \eta_g + e_{gt})$, η_g a persistent relative misallocation and e_{gt} time-varying share noise; the imputed regressor is $\hat{x}_{gt}^{TV} = S_t$; digital $x_{gt}^D = D_t(1 + \zeta_g)$. Adstock is linear, so $a(S(1 + \eta_g)) = (1 + \eta_g)a(S)$. Time-demeaned paths $s_t = a(S)_t - \overline{a(S)}$, $d_t = a(D)_t - \overline{a(D)}$; r_t is d residualised on the geo-constant columns of the fitted design (imputed TV path, seasonal basis F_t); Cov_g , Var_g , $\bar{\zeta}$ are unweighted geo moments. Part II: N line items with spends s_i (Pareto, index α), true channel c_i , LLM label $\hat{c}_i$ from a confusion matrix $M[c, \hat{c}]$, audit budget n .

Assumption 9.1 (Persistent, correlated misallocation). η_g, ζ_g are time-invariant with population-weighted mean zero and $\text{Corr}(\eta, \zeta) = \rho_{\eta\zeta} \neq 0$ in the realistic case; e_{gt} is i.i.d. and independent of the design.

Assumption 9.2 (Plug-in design). Geo fixed effects, common β_{TV} on $a(\hat{x}^{TV})$, common β_D on $a(x^D)$, a geo-constant Fourier seasonal basis (three harmonics), equal geo weights in OLS, adstock rates known (the repair profiles α_{TV}).

Assumption 9.3 (Audit model). Off-diagonal confusion rate e ; errors independent across lines or clustered by campaign family; certification exact; spends Pareto with $\alpha \in \{1.3, 2, 3\}$.

Defaults: $G = 50$, $T = 104$, $\alpha_{TV} = 0.5$, $\alpha_D = 0.2$, $\beta_{TV} = \beta_D = 1$, $\sigma = 0.5$; the national paths share a Q4 seasonal ramp and AR(1) log-noise with a common component of weight ρ_{SD} . *Baseline*: $(\rho_{SD}, \rho_{\eta\zeta}, \sigma_\eta = \sigma_\zeta) = (0.6, 0.5, 0.3)$; *strong*: $(0.9, 0.8, 0.5)$. Hill runs: $K = 1.5$, $s = 1$, $\beta_{TV} = 3$, $\beta_D = 1$, $\sigma = 0.3$.

9.4 Main results

9.4.1 The imputation leakage law

The plug-in model of Assumption 9.2, with OLS coefficients b_{TV}, b_D , omits $z_{gt} = \beta_{TV} s_t \eta_g$: a national time shape times a geo loading, which geo-constant controls cannot absorb and geo fixed effects only de-mean.

Theorem 9.4 (Leakage law, linear response). *[established in the dive]* Under ?? 9.1?? 9.2 with linear f ,

$$\frac{\text{bias}(b_D)}{\beta_{TV}} = \frac{\sum_t s_t d_t \text{Cov}_g(\eta, \zeta)}{\sum_t d_t^2 \text{Var}_g(\zeta) + \sum_t r_t^2 (1 + \bar{\zeta})^2}. \quad (9.3)$$

By Frisch–Waugh–Lovell, $\text{bias}(b_D) = \sum_{g,t} \tilde{x}_{gt} z_{gt} / \sum_{g,t} \tilde{x}_{gt}^2$ with $\tilde{x}$ the digital column $d_t(1 + \zeta_g)$ residualised on the geo-constant columns $C_t = (s_t, F_t)$. Because C is identical in every geo and geos are equally weighted, the pooled projection coefficient is $(1 + \bar{\zeta})$ times the time-series projection of d on C , so $\tilde{x}_{gt} = d_t(\zeta_g - \bar{\zeta}) + (1 + \bar{\zeta})r_t$. The cross term in $\sum \tilde{x}^2$ vanishes since $\sum_g (\zeta_g - \bar{\zeta}) = 0$, giving $G[\sum_t d_t^2 \text{Var}_g(\zeta) + (1 + \bar{\zeta})^2 \sum_t r_t^2]$; and $\sum \tilde{x}z = \beta_{TV}G[\sum_t s_t d_t \text{Cov}_g(\eta, \zeta) + (1 + \bar{\zeta})\bar{\eta} \sum_t r_t s_t]$, whose last sum is zero because r is orthogonal to s . The law matches the exact finite-sample projection $(X^T X)^{-1} X^T z$ (the omitted-variable bias, OVB) to three decimals and the Monte Carlo mean to ± 0.01 in every configuration (E1; clean-room ± 0.01 , per-replication correlation 0.86–0.99).

Three consequences. (i) The numerator is the *raw* cross-product of the national paths, not the seasonally residualised one: the leak runs through geo-heterogeneous seasonal *amplitude*, so when the paths share nothing but their Q4 ramp ($\rho_{SD} = 0$) the leak is still $+0.17$. (ii) The leak is largest when the measured channel’s own geo heterogeneity is modest and its time path mostly shared shape (r small)—the ordinary world in which every channel ramps for Q4. (iii) The imputed channel’s own coefficient is pulled the other way, its variance handed to digital: TV -7% , digital $+17\%$ at baseline; TV -48% , digital $+79\%$ strong. The round-1 closed form residualised the numerator on the seasonal basis, under-predicted the leak $4-6\times$ and failed the $\rho_{SD} = 0$ case ($+0.004$ predicted); it is *[refuted within the dive]*.

Corollary 9.5 (Imputation manufactures the bias). *[numerically established]* The population-weighted national model on the same data is unbiased for both channels (E13: $+0.00 / +0.00$), because aggregation kills the geo loading. A geo model with a common coefficient on an imputed channel is therefore strictly worse than no geos at all for the measured channels: strong-scenario bias/sd of b_D is $+0.79 / 0.12$ (geo plug-in), $+0.00 / 0.07$ (national), $+0.00 / 0.04$ (geo-latent, Proposition 9.6).

9.4.2 The latent-loading repair, shrinkage, and the leak test

Proposition 9.6 (Latent-loading repair). *[established in the dive]* Model the imputed channel with a geo-specific loading on the national path, $y_{gt} = \mu_g + \lambda_g a(S)_t + \beta_D a(x^D)_{gt} + F_t^T \gamma + \varepsilon_{gt}$, and report $\beta_{TV} = \sum_g p_g \lambda_g$. It is exactly identified whenever the national path is not collinear with the measured channels’ geo paths, and it is what a hierarchical geo-varying coefficient already does in the linear case—except that the recovered “geo heterogeneity of TV effectiveness” is in truth misallocation. E2: bias 0.00 at every ρ_{SD} from 0 to 0.995, sd rising from 0.024 to 0.058 (oracle with true spend 0.023–0.039).

Three failure modes were measured in the strong scenario. *Shrinkage re-opens the leak* (E3): a hierarchical prior on the λ_g tighter than the true dispersion (0.5) pulls them back toward the plug-in fit—prior sd 0.5, 0.16, 0.05 give bias $+0.05$, $+0.30$, $+0.68$ —so a Meridian-style estimated heterogeneity scale protects only if the data can push it up. *Geo $\times$ season interactions* (E14) remove the purely seasonal leak ($\rho_{SD} = 0$: $0.32 \rightarrow 0.01$) but leave half the correlated-path leak ($\rho_{SD} = 0.9$: $0.77 \rightarrow 0.39$) at a precision cost. *The imputed channel’s adstock must be right* (E11): with α_{TV} fixed at 0.7 or 0.3 instead of 0.5 the latent model leaks $+0.57$ or -0.18 ; profiling α_{TV} on a grid inside the latent model picks the truth 30/30 and restores bias 0.00 (E20), because the loadings absorb any shape and the likelihood then prefers the true adstock.

Construction 9.7 (Leak test). *[established in the dive]* The plug-in and latent fits agree iff $\text{Cov}_g(\eta, \zeta) \sum_t s_t d_t \approx 0$, so the Hausman-type difference $b_D^{\text{plug-in}} - b_D^{\text{latent}}$ is a specification test the compiler runs on every imputed channel. E12: $|t| = 4.0$ at $(G, T) = (20, 52)$, 5.2 at $(50, 104)$, 7.8 at $(100, 156)$ with a leak; $0.0-0.2$ when $\rho_{\eta\zeta} = 0$.

9.4.3 Hill response: sign flip across the knee, and what is not identified

Proposition 9.8 (Sign flip). *[numerically established] Under Hill the omitted term is $\beta_{TV}[H(a_t(1 + \eta_g)) - H(a_t)] \approx \beta_{TV} a_t H'(a_t) \eta_g$, so Equation (9.3) applies with s_t replaced by the time-demeaned elasticity path $a_t H'(a_t)$. For $s = 1$, $a H'(a) = aK / (a + K)^2$ has derivative $K(K - a) / (a + K)^3$: its time variation vanishes at the knee $a = K$ and reverses beyond it. E17: the leak into digital runs from +0.11 at $a/K = 0.2$ through 0.00 at 2.0 to -0.05 at 4.0, and the exact OVB at the true K tracks every point (Table 9.4).*

The imputed channel's own national mROAS under the plug-in runs from -36% ($a/K = 0.2$) to +27% (4.0); the hierarchical- β_g model does much better below the knee but is +25% deep in saturation, and at very large misallocation ($\sigma_\eta = 0.8$) both overstate (plug-in +24%, hierarchical +52%; E5). This reconciles Sun et al.'s positive bias with the negative TV bias here—*demonstrated, not proven*, in the dive's words: persistent misallocation attenuates through the leak below the knee, while *time-varying* share noise is Berkson-through-Hill and inflates (E7: linear b_{TV} unbiased at every σ_e , as Berkson theory requires; Hill plug-in mROAS +1.5% at $\sigma_e = 0.3$, +21% at 0.6). The sign of the bias says which mechanism dominates.

Proposition 9.9 (Non-identification near the knee). *[numerically established] (Demonstrated on one panel, not proven.) With common (β, K) and geo loadings η_g inside the Hill argument, the profiled SSR (sum of squared residuals) is flat in K to 0.6% over $K \in [0.5, 3]$ and $\text{Corr}(\hat{\eta}, \eta)$ ranges from -0.27 to +0.55 (E19): near the knee a geo's response amplitude is stationary in η , so misallocation and effectiveness heterogeneity are observationally equivalent to first order while implying different marginal ROI.*

The linear repair works because amplitude is proportional to $1 + \eta$; the Hill repair fails exactly where the leak is small, so the compiler must record an imputed channel's national curve as *bracketed*, not estimated, unless geo spend is obtained (BL 34).

9.4.4 Taxonomy misclassification: the certify-the-mass law

Law 9.10 (Certify-top- k MSE law). *[established in the dive] Channel k 's total error under LLM labels is $\Delta_k = \sum_i s_i (1[\hat{c}_i = k] - 1[c_i = k])$; certifying a set A zeroes its terms. With off-diagonal rate e ,*

$$\text{Var}(\Delta_k) \approx e(1 - e) \sum_{i \notin A} s_i^2 \text{ (in- and out-flow classes)}, \quad \mathbb{E}[\Delta_k] = e \left(\sum_{i \notin A, \text{inflow}} s_i - \sum_{i \notin A, \text{outflow}} s_i \right), \quad (9.4)$$

verified within 1–5% at every $k \geq 20$ (E16, 400 replications; clean-room 1–3%).

Two facts follow. (a) Under Pareto $\alpha < 2$ the top lines dominate $\sum s_i^2$ —at $\alpha = 1.3$, 90% of it sits in 0.2% of lines and 99% in 12.6% (E8)—so certifying the top 2.5% of lines cuts channel-total RMSE from about 5% to 1.5% and the flow bias from 2.2% to 0.3% (E9: $N = 2000$, $e = 0.10$, asymmetric flow matrix); with 60 lines a week, certifying the top 3 per week takes a national MMM coefficient from -6% bias / 12% RMSE to 0% / 1.3% (E18). At $k = 0$ the RMSE is itself seed-unstable (clean-room 0.165 against the dive's 0.05) because one mislabelled top line moves the total—the strongest argument for certification. (b) *Random-sample rectification is the wrong tool [numerically established]*: the difference estimator Equation (9.1) adds variance $(N/n - 1)V_{\text{tail}}$, so at $n = 400$ of 2000 it is still $2\times$ worse than the raw labels (RMSE 0.10 against 0.05); PPS sampling gives 0.04 and matrix calibration 0.07, since a 10% rate estimated from about 130 labels per class carries 2.6 points of uncertainty times the whole tail spend (the dive's summary calls all three MSE-worse than raw at every budget $n \leq 400$, i.e. $\leq 20\%$ of lines; on PPS see the reviewer's note). PPI's guarantee is validity against the n -label classical estimator, not MSE against raw predictions that are already within a few percent; the round-1 design that used PPI as the taxonomy fix is *[refuted within the dive]*. The ranking survives $\alpha \in \{1.3, 2, 3\}$, size-dependent error and asymmetric flows; with errors clustered by campaign family, certifying by *family mass* beats certifying by line (E15: 0.013 against 0.017 at $n = 50$; 0.000 against 0.009 at $n = 200$). The *audit rule [conjecture / speculative]*: certify down the spend-sorted list of lines or families until $e(1 - e) \sum_{\text{tail}} s_i^2 + [e \cdot \text{net flow}_{\text{tail}}]^2 < (\tau T_k)^2$, T_k the channel total, for the tolerance τ the decision needs (Chapter 7 supplies it; the ledger suggests $\tau \approx 3\%$ is decision-free), using a small random tail sample only to estimate e , never to rectify totals.

9.4.5 The compiler and the ledger

Construction 9.11 (Provenance-typed compiler). [conjecture / speculative] (Design grounded in the theorems, not a field result.) The *taxonomy standard* is minimal and machine-checkable: each line item carries channel, tactic, buying model, geo scope $\in \{\text{geo-measured, national, unknown}\}$, funnel objective, spend basis (gross/net/fee-inclusive), currency, date range, source system, and a *provenance type* $\in \{\text{measured; classified}(e, \text{family id, certified?}); \text{imputed}(\text{share vector, share-uncertainty class})\}$. Extraction is an LLM map from raw campaign strings to this schema with a per-line confidence and a family id. Each provenance type compiles to a likelihood term and an acquisition action (Table 9.1).

Table 9.1: Provenance types and what the compiler emits for each (Dive 11 §3.5).

Provenance type	Likelihood term	Acquisition action
measured	ordinary	none
classified	after certification to τ , ordinary; residual variance inflates the spend measurement covariance (classical, small)	certify by (family) mass down to τ ; small tail sample for e
imputed, persistent misallocation	geo loading λ_g on the national path, un-shrunk or data-scaled prior; report $\sum_g p_g \lambda_g$; run the leak test; curve <i>bracketed</i> under saturation	obtain geo spend or an auxiliary geo share
imputed, time-varying share noise	Berkson term $\mathbb{E}[H(a(1+e))] \approx H + \frac{1}{2} H'' a^2 \sigma_e^2$, $\text{Var} += \beta^2 H'^2 a^2 \sigma_e^2$ (Chapter 5)	only if σ_e is large
imputed with auxiliary geo signal	impressions/GRPs/traffic as share vector; reclassify as measured-with-noise	BL 35 derives the crossover

The *archaeology ledger* (E10, Table 9.2) propagates each defect’s measured biases into a two-channel Hill allocation ($\beta = (3, 1)$, $K = (1.5, 1)$, $s = 1$, budget 2.5, 300 replications) and reports expected regret as a share of optimal revenue. As the dive reads it, for a \$100M budget the baseline leak is worth about \$280k a year and the strong case about \$8M: geo spend for the imputed channel, or the free latent-loading fit, is the highest-value archaeology; certification is cheap and closes its gap; share noise matters only when large.

Table 9.2: Archaeology ledger (E10): expected allocation regret in percent of optimal revenue, $se \leq 0.05$ percentage points; inputs are the biases measured in the experiment named.

Data defect	Input biases (source)	Regret (%)
clean data (sd 3%)	—	0.01
imputation leak, baseline	TV -7% , D $+17\%$ (E1)	0.28
imputation leak, strong	TV -48% , D $+79\%$ (E1)	8.2
taxonomy, raw LLM labels	D $-6\% \pm 12\%$ (E18)	0.13
taxonomy, top 5% certified	sd 1.3% (E18)	0.00
Hill Berkson share noise $\sigma_e = 0.6$	TV $+21\%$ (E7)	0.20

9.5 Numerical evidence

Table 9.3 gives the leakage law’s four E1 configurations with the independent clean-room replication (different seeds, 100 replications) and Table 9.4 the Hill sweep; the strong-scenario estimator ranking (E13, 100 replications) and the repair variants (E3, E11, E14: 60; E20: 30) are quoted in Corollary 9.5 and Proposition 9.6 and the text following them. Negative controls are E1’s third configuration, E12’s $\rho_{\eta\zeta} = 0$ rows ($|t| \leq 0.2$)

and E7’s linear rows; the power control is E12’s (G, T) sweep. Standard errors are reported for E1, E17 ($se \leq 0.01$) and E10 ($se \leq 0.05$ pp); other entries are means and sds.

Table 9.3: Leakage law across the E1 configurations ($G = 50, T = 104, \sigma_\eta = \sigma_\zeta$): Monte Carlo bias of b_D (dive, 60 replications) against the clean-room re-implementation (100); the closed form Equation (9.3) matches the dive column within 0.01 in every row. n/r: not reported.

Configuration	ρ_{SD}	$\rho_{\eta\zeta}$	σ_η	bias b_D (dive)	clean-room	bias b_{TV}
baseline	0.6	0.5	0.3	+0.181	+0.188	−0.07
strong	0.9	0.8	0.5	+0.771	+0.782	−0.48
no correlation (control)	0.6	0.0	0.3	−0.001	+0.001	n/r
seasonal only	0.0	0.8	0.3	+0.168	+0.158	n/r

Table 9.4: Hill operating-point sweep (E17; $\beta_{TV} = 3, K = 1.5$, strong-scenario shares, 15 replications per point, $se \leq 0.01$ on the leaks): leak into b_D under the plug-in, exact OVB at the true K , and relative bias of the imputed channel’s national mROAS.

a/K	0.2	0.8	1.3	2.0	4.0
leak into b_D , plug-in	+0.11	+0.09	+0.03	0.00	−0.05
exact OVB at true K	+0.11	+0.08	+0.04	−0.00	−0.04
mROAS bias, plug-in	−36%	n/r	−4%	n/r	+27%
mROAS bias, hierarchical β_g	−11%	−6%	+1%	n/r	+25%

9.6 Limitations and the devil’s advocate

The dive’s own counterarguments: the DGP puts all of TV’s geo variation into one persistent loading, whereas real misallocation drifts (BL 33) and the outcome may carry geo-varying seasonality correlated with η , adding terms to the law (the leak test remains the honest instrument); the national model is unbiased only because there is no unmodelled national confounder, with one the geo model’s advantage returns for the measured channels; the λ_g are unshrunk, a hierarchical scale estimated from 20 geos $\times$ 52 weeks may not open wide enough, and under Hill nothing repairs; certify-the-mass assumes exact certification and independent errors below the certified set, and a wholesale misreading of a tactic (all “CTV” to “TV”) is a family error E15 handles only if families are identifiable from names; the PPI verdict concerns MSE of spend totals and leaves PPI’s interval validity untouched; the ledger has two channels, one Hill shape and one budget—magnitudes illustrative, ordering the claim.

Stated scope: linear-response law only; two channels (BL 32); Fourier seasonality rather than Meridian’s knot-spline time effects, which E14 suggests leak *more*; measured channel’s adstock known; no multiple-imputation congeniality analysis in Meng’s (1994) sense (the Berkson term is the congenial replacement); spend-basis errors untreated (BL 36). The attack rounds changed three results—the closed form (raw numerator), the taxonomy fix (certification for PPI) and the Hill “repair” (a bracket plus the non-identification statement)—and a fourth round of seed, size, operating-point and shrinkage sweeps changed nothing.

9.7 Discussion: impacts

For practice the message inverts a recommendation. The geo tier is sold as the statistical upgrade and population-share imputation as the harmless workaround that lets national channels ride along; Theorem 9.4 says the workaround converts the *measured* channels’ geo heterogeneity into bias, seasonality controls do not help, and the national model would have been unbiased. The fix is nearly free in a hierarchical package—a

geo-varying coefficient on the national path with a prior the data can widen, adstock profiled—but default shrinkage silently undoes it, so a vendor implementing a `national_media` flag must also change the prior, run the leak test, and under saturation report a bracket rather than a curve. For the pipeline the result is a budget rule: audit the largest lines or families until the tail clears the decision’s tolerance, and do not buy a random-sample rectifier for spend totals. For CMO decisions the ledger orders the archaeology: geo spend for a nationally bought channel (or the loading fit) dominates, and a strong leak costs two orders of magnitude more regret than a raw LLM taxonomy.

For theory the chapter combines Berkson error, weight shifting and a geo panel into one exact projection, separates two imputation mechanisms with opposite signs, and exhibits a non-identification sitting exactly where the leak vanishes. Outward: Chapter 1 inherits the leak as cross-channel shape leakage in Dive 22’s multichannel shape-prior ledger (BL 86); Chapter 7 supplies τ and receives the Hill bracket as a probability-of-regret interval (BL 34); Chapter 6 gains a refresh-flip source, since adding geos to a refresh can create flips through the leak (BL 33); Chapter 5 supplies the Jensen algebra of the Berkson term.

9.8 Further exploration

Five of the dive’s six next steps became backlog items. BL 32 generalises the law to k measured channels (numerator $\sum_t s_t d_t^{(j)} \text{Cov}_g(\eta, \zeta_j)$, denominator the measured channels’ geo-heterogeneity Gram matrix) and tests it against Meridian’s knot-spline time effects. BL 33 asks where between the persistent (attenuating) and i.i.d. (Berkson-upward) limits an AR(1) misallocation η_{gt} changes the leak’s sign and whether a state-space loading λ_{gt} repairs it. BL 34 computes the identified set of an imputed channel’s national mROAS under Hill as the η/β_g exchange curve, in the interval format of Chapter 7. BL 35 derives when a noisy measured share vector (geo impressions, GRPs, traffic; classical error) beats the latent loading (Berkson). BL 36 extends the audit law to gross/net/fee-basis misreads, multiplicative and correlated within source system. The sixth step, a field pilot of the leak test on a real geo MMM with a nationally bought channel, would be the first field measurement of imputation leakage.

In the compiler’s judgement the field pilot is the most valuable next move: everything here rests on a simulated family and the test needs no new data—two fits and a difference—but it first requires a per-dataset variance for the Hausman statistic (see the reviewer’s note). Second is BL 33, since real misallocation is neither fixed nor i.i.d. and the sign-change point decides whether the imputed channel’s ROI should be expected under- or over-stated.

Reviewer’s note (compiler)

Log versus file. The INDEX log row for Dive 11 is timestamped 2026-09-01 ~21:20 and sits after the row for Dive 12 (2026-09-02 ~04:50), while the file was written 2026-09-02 03:33 UTC; the log line otherwise agrees with the writeup on every number.

The PPS row contradicts the headline. The log line and the verification table say PPI/PPS/matrix rectifiers are “MSE-worse than raw labels at any budget $\leq 20\%$ ”, but §3.4 reports PPS at 0.04 against raw 0.05 at $n = 400$ —marginally *better*. As the $k = 0$ RMSE is seed-unstable (0.05 versus the clean-room’s 0.165), the raw baseline is soft in both directions; the safe statement is that top- k certification dominates every rectifier at every budget, which the dive does show.

The leak test’s $|t|$ is not a test statistic. E12 computes $|t|$ as $|\text{mean}|/\text{sd}$ of the plug-in-latent difference across 60 Monte Carlo replications, not a per-dataset Hausman statistic with a variance-difference or bootstrap a practitioner could compute; the power numbers describe detectability in principle.

Numbers that differ across experiments. The headline $+17\%/+79\%$ (§3.1, reused by the ledger) differ from E1’s tabulated $+0.181/+0.771$ for the same configurations; E13 (different seed) gives $+0.79$ for the strong case, and the dive does not say which run the pair comes from. The verification table says E1 has “4 configs” while the code runs five (the fifth, $\rho_{\eta\zeta} = -0.5$, is unreported). The ledger’s dollar reading applies a percentage of optimal *revenue* to a \$100M *budget*.

Thin evidence in places. Proposition 9.9 rests on one simulated panel (E19 has no replication loop); the Hill mROAS biases use 15 (E17) or 40 (E4/E5) replications with standard errors printed by the code but not quoted, and Table 9.4

places the dive's three unlabelled "below the knee" hierarchical- β_g values at $a/K = 0.2, 0.8, 1.3$ by position (the code's grid also has 0.4); the E3 shrinkage entries carry no sds. The closed form uses unweighted geo moments while the DGP renormalises η, ζ to population-weighted zero mean, and the code notes the formula "assumes $\sum_g \eta_g = 0$ "; the derivation above shows the b_D component does not need it (the $\bar{\eta}$ term multiplies $\sum_t r_t s_t = 0$), but the imputed channel's own bias does, so the $-7\% / -48\%$ TV figures are Monte Carlo, not closed-form.

Scope and novelty. Corollary 9.5 holds in a DGP with no national confounder and a linear response; the dive says so, the log line does not. " $\tau \approx 3\%$ is decision-free" is a first-order reading of one ledger. Cut-off sampling of the largest items is standard audit practice (the dive cites PCAOB AS 2315); the new content is the MSE law Equation (9.4) and the Pareto-weighted comparison with PPI. No result depends on a pseudo-inverse or Fisher-information computation of the kind BL 82 flagged.

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- [16] Program (internal): Dive 06 (Jensen/Berkson algebra), Dive 07 (refresh stability), Dive 03/Dive 08 (decision-loss units), Dive 22 (BL 86).

Chapter 10

Horizon bounds for long-run effects under weak surrogacy

Provenance

Source dives: Dive 12 *Long-run effects under weak surrogacy: the price of the long run* (run 2026-09-02, file `12_surrogacy_horizon_bounds.md`, written 2026-09-02 11:34 UTC; spanned two runs—an interrupted run built the constructions, the second re-ran, attacked (E13–E27), verified and wrote up). **Code:** `code/12_surrogacy_horizon_bounds.py`. **Frontier-study problem(s) addressed:** B6/M5 — the surrogate index requires that treatment affect long-run outcomes only through the surrogates, almost surely false for brand advertising whose memory channel bypasses short-run behaviour; develop partial identification of long-run ROAS from a short experiment plus an observational panel, with bounds indexed by a surrogacy-violation parameter in the manner of Cinelli–Hazlett. **Backlog items closed:** none; **opened:** BL 37 (boundary law of the profile-LR), BL 38 (the (G, H) frontier), BL 39 (shape families for de-confounding), BL 40 (onset timing as a second index), BL 41 (interval-valued posteriors in the deadband).

Key ideas in this chapter

- The surrogate-index error obeys the exact identity $SI - \Theta_L = (\gamma - \mathbf{1})^\top \tau_{1:H} - \sum_{t>H} \tau_t$: a negative *bypass tail* and a positive *phantom persistence* term; -27.7% at brand share $\phi = 0.4$, $H = 13$, but $+163\%$ at demand persistence $\rho_d = 0.9$ with *no* long-run effect at all.
- Under complete monotonicity capped at a maximum half-life $hl_{\max}$ the identified set for Θ_L is a convex profile-likelihood programme: noise-free $[13.25, 13.32]$ at $hl_{\max} = 26$, but $1.36\times$ the truth wide with 400 geos and 26 post-weeks.
- The M5 sensitivity curve is $hi \approx 11.5 + 0.235 L_{\text{eff}}(hl_{\max})$ with a flat lower bound (7.7), obeying $hi - \Theta \approx \kappa \sqrt{\text{crit}/a^\top V^{-1}a} c_{\text{slow}}$, $\kappa \in [1.1, 1.6]$; geos buy the upper bound, post-period weeks the lower. Establishing a long-run effect needs about $1600 \text{ geos} \times 52 \text{ weeks}$; ruling one out, $4\times$ more geos.
- One experiment de-confounds a 52-lag observational panel through the bias $s b$ (known shape, unknown scalar): $[10.1, 15.4]$ against $[6.0, 43.3]$ alone, $7\times$ tighter with coverage 1.00—once the non-decaying -0.003 -per-lag bend that geo fixed effects put into the shape is included.
- Misspecified kernels are covered, but a brand component whose onset lies beyond the window is the invisible failure (coverage 0.47, misfit statistic silent); the χ_1^2 profile radius under-covers at the cone boundary and is replaced by an MC-calibrated 6.63.

10.1 Problem and motivation

A geo experiment runs a spend pulse and reads weekly incremental sales for H weeks, typically 8–13. The business needs the *long-run* total $\Theta_L = \sum_{t \leq L} \tau_t$ over $L \approx 104$ weeks, because brand advertising keeps paying through a memory or brand-stock channel long after short-run sales—the natural surrogates—have gone quiet. The standard tool for long-run effects from short experiments, the surrogate index of Athey, Chetty, Imbens and Kang (ACIK), requires that treatment reach the long-run outcome only through the surrogates. Problem M5 observed that this is almost surely false for brand advertising and asked for the missing

instrument: partial identification of Θ_L from a short experiment plus an observational panel, with bounds indexed by an interpretable surrogacy-violation parameter, as Cinelli and Hazlett index omitted-variable bias by a partial R^2 .

Practice fills the gap with heuristics—“cooldown” windows in platform lift tools, Binet and Field’s 60/40 long/short split, Lodish et al.’s long-term multipliers—while the empirical literature reports weekly carryover near 0.9 (Shapiro, Hitsch and Tuchman, 2021; Dubé, Hitsch and Manchanda, 2005, half-life about six weeks), which puts most of a campaign’s effect outside any 13-week window. Dive 12 builds three constructions: an exact identity for the surrogate index’s error under persistent demand, a sensitivity bound indexed by the largest memory half-life the analyst will entertain, and a device by which one experiment de-confounds an entire observational lag structure. The headline is the *price of the long run*: the bound exists, and the data that make it decision-useful cost more than anyone currently pays.

10.2 Background: the ideas this chapter builds on

10.2.1 The surrogate index (Athey, Chetty, Imbens and Kang, 2019)

Given an experimental sample with treatment W and short-run surrogates S but no long-run outcome Y , and an observational sample with both S and Y , the surrogate index is $\mu(s, x) = \mathbb{E}[Y \mid S = s, X = x]$ fitted on the observational sample and applied to the experimental one; the long-run effect estimate is the treatment contrast of $\mu(S_i, X_i)$. Three assumptions deliver unbiasedness: *unconfoundedness* of W in the experiment; *surrogacy* (the Prentice criterion) $W \perp Y \mid S, X$, that treatment moves Y only by moving S ; and *comparability*, that the law of Y given (S, X) is the same in both samples. With a linear index the estimate is $\gamma^\top \hat{\tau}_S$, the surrogate treatment effects weighted by their observational predictive coefficients. When surrogacy fails ACIK write the bias as an average residual direct effect of W given S and discuss bounding it by the explanatory power of W and Y given S ; the dive’s comparator is a partial- R^2 rendering, $|\text{bias}| \leq |\text{SI}| \sqrt{(1 - R_{W|S}^2)(1 - R_{Y|S}^2)}$.

10.2.2 Sensitivity analysis indexed by a partial R^2 (Cinelli and Hazlett, 2020)

For a regression of Y on treatment D and covariates X , omitting a confounder Z biases the coefficient on D by

$$|\widehat{\text{bias}}| = \text{se}(\hat{\tau}) \sqrt{\frac{R_{Y \sim Z|D,X}^2 R_{D \sim Z|X}^2}{1 - R_{D \sim Z|X}^2}} \sqrt{\text{df}}, \quad (10.1)$$

the two partial R^2 values measuring how much of the outcome and of the treatment the unobserved Z would explain. The contribution is a reporting discipline: index the unobservable by parameters on a scale the analyst understands, plot the estimate against them, and read off the *robustness value*—the confounding strength at which the conclusion changes. The dive’s index for surrogacy is $\text{hl}_{\max}$, the largest half-life the memory channel may have: $\text{hl}_{\max} = 0$ is full surrogacy (nothing after the window), $\text{hl}_{\max} = \infty$ imposes nothing beyond monotone decay.

10.2.3 Completely monotone kernels, extrapolation, and shape-restricted bounds

A sequence $g(0), g(1), \dots$ is *completely monotone* if all its finite differences alternate in sign, $(-1)^k \Delta^k g(h) \geq 0$; Hausdorff’s theorem (1921) says this holds if and only if $g(h) = \int_0^1 \rho^h d\mu(\rho)$ for a non-negative measure μ (Bernstein’s theorem is the continuous-time version). Every geometric adstock and every mixture of them belongs to the class; humps, delayed onsets and negative lobes do not. Brown and Grabovsky (2024) show that extrapolating a completely monotone function known on an interval is infeasible to the left but feasible to the right, with error decaying as a power of the distance from the interval—and a response tail

is a right-extrapolation. Mogstad, Santos and Torgovitsky (2018) turn such restrictions into bounds: when target and identified moments are both linear in an unknown function confined to a convex class, the sharp bounds are two linear programmes over a finite basis. Here $\Theta_L = c^\top \mu$ and the observed window $A\mu$ are linear in the mixing weights $\mu \geq 0$ on a grid of decay rates. With Gaussian noise $\hat{t} \sim \mathcal{N}(A\mu, V)$ the natural set is the *profile likelihood-ratio* set $\{\Theta : Q(\Theta) - Q_{\min} \leq \text{crit}\}$, $Q(\Theta)$ the minimum weighted residual sum of squares subject to $c^\top \mu = \Theta$; by Wilks' theorem $\text{crit} = \chi_1^2(0.95) = 3.84$ when the nuisance parameters lie in the interior of their space.

10.2.4 Chi-bar-square laws at a cone boundary (Chernoff, 1954; Self and Liang, 1987)

When the parameter space is a convex cone and the truth lies on its boundary, the likelihood-ratio statistic converges to the squared information-metric distance from a Gaussian draw to the cone, whose law is the mixture $\bar{\chi}^2 = \sum_i w_i \chi_i^2$ with w_i the probability that the projection of the draw lands on a face of dimension i (Chernoff; Self and Liang; Silvapulle and Sen, 2005). The textbook case—one parameter constrained non-negative, truth at zero—gives $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$, so the unadjusted test is conservative; but for a *profile* statistic on a functional whose many nuisance weights sit on the boundary, the weights depend on the covariance and the local cone geometry and the departure from χ_1^2 can go either way. The dive meets the anti-conservative direction.

10.2.5 Known-shape, unknown-scalar de-confounding

A distributed-lag regression of sales on K lags of spend with the demand state d omitted has probability-limit bias $\mathbb{E}[X^\top X]^{-1} \mathbb{E}[X^\top d]$. If spend chases demand, $x_t = \bar{x} + a d_t + u_t$ with d an AR(1) of persistence ρ_d , then $\text{Cov}(x_{t-k}, d_t) = a \text{Var}(d) \rho_d^k$ and the lag covariance is $a^2 \text{Var}(d) R + \sigma_u^2 I$, $R_{kj} = \rho_d^{|k-j|}$: the bias factorises into an unknown scalar $s = a \text{Var}(d)$ times a *shape* b whose every ingredient is estimable from the spend autocovariances. Removing hidden confounding from an observational estimate with an experiment is the programme of Kallus, Puli and Shalit (2018), who learn the bias as a function on the overlap region and extrapolate it, and of Rosenman et al. (2023), who shrink between the two estimates; the dive's variant parametrises the bias *shape* from the spend autocorrelation and lets the experiment fix the one scalar left free.

10.3 Setup and assumptions

Definition 10.1 (Effect dynamics and the long-run target). Per-unit spend produces the impulse response $g(h)$, $h = 0, 1, \dots$; a pulse dx produces the effect curve $\tau_t = \sum_k dx_k g(t-k)$. The benchmark truth is $g(h) = 0.5^h + w_B \rho_B^h$: a fast *activation* component (one-week half-life) plus a slow *brand-stock* component with 26-week half-life, $\rho_B = 0.5^{1/26}$, whose weight w_B makes the brand share of Θ_L equal to ϕ ($\phi = 0.4$ benchmark; $\phi = 0$ is the no-long-run control). The pulse is $dx = (1, 1, 1, 1)$, $L = 104$, and $\Theta_L = 13.32$ at $\phi = 0.4$, 8.00 at $\phi = 0$. Weeks inside the window are $t = 1, \dots, H$, collected in $\tau_{1:H}$.

Definition 10.2 (Experiment and surrogate index). G geos, half treated ($W_g \in \{0, 1\}$), with $y_{gt} = \mu_g + d_{gt} + W_g \tau_t + \varepsilon_{gt}$, d an AR(1) geo demand state with $\rho_d = 0.6$ and unit innovations, $\varepsilon \sim \mathcal{N}(0, 1)$, 52 pre-weeks. Per-week ANCOVA on the pre-period mean gives $\hat{\tau}_{1:H}$ with a Ledoit–Wolf-shrunk $H \times H$ covariance $\hat{V}$; the per-week standard error is about $1.6\sqrt{4/G}$. The surrogate index is fitted on an untreated panel $y_t = d_t + \varepsilon_t$ with $S = (y_1, \dots, y_H)$ and $Y_L = \sum_{t \leq L} y_t$: $\gamma = \text{Var}(S)^{-1} \text{Cov}(S, Y_L)$ and $\text{SI} = \gamma^\top \tau_{1:H}$.

Assumption 10.3 (Capped complete monotonicity). $g(h) = \int \rho^h d\mu(\rho)$ with $\mu \geq 0$ supported on $[0, \rho_{\max}]$, $\rho_{\max} = 0.5^{1/\text{hl}_{\max}}$. In the sharpened form the attack rounds forced: the response has begun by the end of the window and decays no more slowly than half-life $\text{hl}_{\max}$.

Assumption 10.4 (Observational panel for de-confounding). $y_{gt} = \mu_g + \sum_{k \leq K} g_k x_{g,t-k} + d_{gt} + e_{gt}$ with $x_{gt} = \bar{x} + a d_{gt} + u_{gt}$, u iid, a unknown in sign and size, $K = 52$, geo fixed effects, $T_0 = 156$ weeks, $G_0 = 200$

geos. Effects are linear in spend and the panel's confounding runs only through the demand state the spend autocorrelation reveals.

10.4 Main results

10.4.1 The surrogate-index bias identity

Theorem 10.5 (Bias of the linear surrogate index). *[established in the dive] (clean-room exact.) Because the index is linear and treatment enters the surrogates additively,*

$$\text{SI} - \Theta_L = (\gamma - \mathbf{1})^\top \tau_{1:H} - \sum_{t>H} \tau_t. \quad (10.2)$$

The proof is one line ($\Theta_L = \mathbf{1}^\top \tau_{1:H} + \sum_{t>H} \tau_t$); the content is that the two terms have opposite signs and different physics. The *bypass tail* $-\sum_{t>H} \tau_t$ is the effect that never touches the surrogates: negative, growing with ϕ and the brand half-life, shrinking with H . The *phantom persistence* term $(\gamma - \mathbf{1})^\top \tau_{1:H}$ arises because the index learned on the panel that a high y_t predicts high future y —demand persists, so $\gamma_t > 1$ —and reads the treatment-induced shift in S as a demand shift. It is a surrogacy violation *even when the long-run effect is zero*: the surrogate is a noisy proxy for the state driving Y_L , and W shifts the proxy without shifting the state. Conditioning on the pre-period mean does not help (E14: identical bias to three decimals), because the AR(1) state is Markov given y_1 .

At $\phi = 0.4$, $H = 13$: $\Theta_L = 13.32$, $\text{SI} = 9.63$, bias -27.7% ($\sum \gamma = 14.12$; last four γ 's 1.02, 1.06, 1.22, 1.82); a Monte Carlo of the whole pipeline (300-geo panel fit, 200-geo experiment, 40 replications) gives 9.46 ± 0.44 . By horizon at $\phi = 0.4$ the bias is -69% ($H = 2$), -35% (4), -28% (13), -19% (26), -8% (52). The sign flips with demand persistence: at $\phi = 0.2$, $H = 4$ it is -17% for $\rho_d = 0.6$ but $+116\%$ for $\rho_d = 0.9$ and $+295\%$ for 0.95 ; with $\phi = 0$ —no long run at all—and $\rho_d = 0.9$ the index over-states the $H = 4$ effect by $+163\%$. A *crossing horizon* H^* exists at which the terms cancel and the index is accidentally right ($H^* \approx 8$ at $\phi = 0.2$, $\rho_d = 0.9$); it moves with $(\phi, \text{hl}_B, \rho_d)$, none of which the analyst observes. The partial- R^2 comparator (E8, $G = 100$, $H = 13$, from $R_{W|S}^2 = 0.44$, $R_{Y|S}^2 = 0.13$) has width $0.99\times$ the truth but covers only 81%: the R^2 calculus does not know the bias has a term outside the surrogate span.

10.4.2 The completely monotone profile-LR bound indexed by $\text{hl}_{\max}$

Construction 10.6 (Profile-LR horizon bound). *[established in the dive]* Discretise Assumption 10.3 on 200–300 rates $\rho_j \in [0, \rho_{\max}]$; let A ($H \times J$) map weights to the in-window effect curve of the pulse and c to its long-run total, $\Theta_L = c^\top \mu$. Define

$$Q(\Theta) = \min_{\mu \geq 0, c^\top \mu = \Theta} (\hat{\tau} - A\mu)^\top \hat{V}^{-1} (\hat{\tau} - A\mu), \quad \{\Theta : Q(\Theta) - Q_{\min} \leq \text{crit}\}, \quad (10.3)$$

with weeks beyond 13 blocked into 13-week bins to keep $\hat{V}$ well conditioned; the end points are two convex programmes. $\text{crit} = \chi_1^2(0.95) = 3.84$ under-covers on the low side (6–9% one-sided failures at $\phi = 0$); $\text{crit} = 6.63$ is Monte Carlo calibrated ($\leq 3.3\%$ one-sided everywhere tested, E9b).

This is the third generation of the bound: the Round-1 *box* LP ($|\hat{\tau}_t - (A\mu)_t| \leq z \text{se}_t$ simultaneously, Šidák z) covers 1.00 at width $3.8\times$ the truth for $G = 100$, $H = 13$, and the Round-2 *ellipsoid* QCQP ($(\hat{\tau} - A\mu)^\top \hat{V}^{-1} (\hat{\tau} - A\mu) \leq \chi_H^2$) gains little ($3.7\times$) while inheriting Hotelling inflation from a 13-dimensional covariance estimated on 50 geos per arm.

Proposition 10.7 (Noise-free identified set). *[established in the dive] (clean-room exact.) With $\tau_{1:13}$ known exactly at $\phi = 0.4$, Assumption 10.3 leaves $\Theta_L \in [13.25, 13.32]$ for $\text{hl}_{\max} = 26$, $[13.25, 14.77]$ for 52 and $[13.25, 15.53]$ for 104.*

The shape restriction is *very* informative when the curve is known; the whole difficulty is noise.

Law 10.8 (Price of the long run). *[numerically established]* The unidentified direction is the $\rho_{\max}$ component: inside the window it adds an almost constant level per observed week, and its long-run total per unit weight is $c_{\text{slow}} \approx D L_{\text{eff}}$ with D the pulse mass and

$$L_{\text{eff}}(\text{hl}_{\max}) = \frac{1 - \rho_{\max}^L}{1 - \rho_{\max}} \quad (= 56.6 \text{ at } \text{hl}_{\max} = 52). \quad (10.4)$$

With a the in-window signature of that component the profile admits weight $m \leq m_{\max} = \sqrt{\text{crit}/a^T V^{-1} a}$, so

$$\text{hi} - \Theta_{\text{true}} \approx \kappa m_{\max} c_{\text{slow}}, \quad \kappa \in [1.1, 1.6], \quad (10.5)$$

$\kappa > 1$ being the slack from re-fitting the fast components ($\kappa = 1.30, 1.42, 1.57, 1.24, 1.15, 1.09, 1.30$ over seven $(G, H, \text{hl}_{\max})$ configurations, E7b, noise-free $\hat{\tau}$ with the true V).

The earlier run's cruder law $z \cdot \text{se}_{\text{level}} \cdot L_{\text{eff}} \cdot D$ carries a spurious factor D and overshoots about $3\times$ [refuted within the dive]; the naive $z \cdot \text{se}_{\text{GLS}} \cdot L_{\text{eff}}$ is within $\pm 30\%$. The upper bound therefore scales as $1/\sqrt{G}$ and roughly $1/\sqrt{H_{\text{eff}}}$ through the standard error and linearly in $L_{\text{eff}}(\text{hl}_{\max})$ —which is the sensitivity curve.

Proposition 10.9 (The M5 sensitivity curve and the (G, H) asymmetry). *[numerically established]* At $G = 400$, $H = 26$ (E19, 60 replications) the upper bound is $\text{hi} \approx 11.5 + 0.235 L_{\text{eff}}(\text{hl}_{\max})$ while the lower bound is flat at 7.7, independent of $\text{hl}_{\max}$. The lower bound is rate-resolution limited, not long-run limited: at $H = 13$ it sits 5–8 below the truth, improves only $\sqrt{G}$ -slowly ($7.9 \rightarrow 6.1 \rightarrow 5.1$ below truth for $G = 100 \rightarrow 400 \rightarrow 1600$) and barely with H , because a short window cannot tell a 13-week from a 26-week half-life (in-window shapes differ by ρ^{12} : 0.53 against 0.73) and the lower extreme reassigns the slow mass to the fastest rate the noise allows.

So G buys the upper bound, H buys the lower bound's rate resolution, and neither buys the other. The analyst plots the upper bound against the memory half-life they will entertain and reads off the breakdown half-life at which long-run ROAS crosses the decision threshold—the Cinelli–Hazlett plot for surrogacy.

10.4.3 One experiment de-confounds the whole observational lag structure

Construction 10.10 (Joint profile bound with a known-shape bias). *[established in the dive]* Under Assumption 10.4 the distributed-lag OLS $\hat{\beta}$ has probability-limit bias $s b$ with

$$s = a \text{Var}(d), \quad b = (a^2 \text{Var}(d) R + \sigma_u^2 I)^{-1} r, \quad R_{kj} = \rho_d^{|k-j|}, \quad r_k = \rho_d^k, \quad (10.6)$$

where $(\hat{\rho}_d, a^2 \widehat{\text{Var}}(d), \hat{\sigma}_u^2)$ follow from the spend autocovariances $(\hat{\gamma}_0, \hat{\gamma}_1, \hat{\gamma}_2)$. The experiment, which has no s , pins the short lags; the panel's 52 lags, corrected by $\hat{s} b$, deliver the tail. The joint profile is

$$Q = (\hat{\tau} - A_e \mu)^T V_e^{-1} (\hat{\tau} - A_e \mu) + (\hat{\beta} - A_g \mu - s b)^T V_b^{-1} (\hat{\beta} - A_g \mu - s b) \quad (10.7)$$

over $(\mu \geq 0, s \text{ free})$, A_g mapping weights to lag coefficients ρ^k , V_b cluster-robust by geo, and the same $\text{crit} = 6.63$.

The panel alone with s free is unbounded: at $\rho_d = 0.6$ the shape b is collinear with dictionary mass near $\rho = 0.6$, and the experiment breaks the collinearity. The attack rounds forced one correction to the shape itself.

Lemma 10.11 (Fixed effects bend the shape). *[established in the dive]* (clean-room reproduced.) The within transformation over T_o weeks subtracts from every lag's covariance with the omitted state the term $(1/T_o) \sum_j \text{Cov}(x_{t-k}, d_j) \approx a \text{Var}(d)(1 + \rho_d)/((1 - \rho_d)T_o)$, a non-decaying correction that the infinite- T shape Equation (10.6) misses; through $(X^T X)^{-1}$ it becomes a level of about -0.003 per lag (per unit s) at $k = 10, \dots, 52$, cutting $\sum_k b_k$ from 1.18 to 1.02.

Direct simulation (4000 geos, six seeds) gives -0.0029 mean over $k \geq 10$ and $\sum b = 1.016 \pm 0.007$; an independent analytic within-plim gives 1.022 (the verifier's first attempt omitted the cross-lag u covariance and got 0.88; the corrected derivation agrees).

10.5 Numerical evidence

All experiments use Definitions 10.1 and 10.2 with the profile bound at $hl_{\max} = 52$ unless stated. Table 10.1 is the headline sweep (E9, 100 replications); a five-seed sweep at $G = 400$, $H = 26$ (E20) gives width/truth 1.36–1.45 and coverage 0.98–1.00. Table 10.2 is the M5 deliverable.

Table 10.1: The price of the long run: profile-LR bound on Θ_L (truth 13.32, $\phi = 0.4$, $hl_{\max} = 52$, crit = 6.63), E9, 100 replications; bounds are mean $\pm$ sd across replications.

G	H	bound	width/truth	coverage
100	13	$[6.1 \pm 1.5, 42.0 \pm 10.4]$	2.69	1.00
100	52	$[6.7 \pm 1.8, 30.9 \pm 7.9]$	1.82	1.00
400	13	$[7.6 \pm 0.8, 31.1 \pm 5.3]$	1.76	1.00
400	26	$[7.7 \pm 1.2, 25.9 \pm 4.7]$	1.36	1.00
400	52	$[7.7 \pm 1.3, 23.5 \pm 3.8]$	1.19	0.99
1600	13	$[8.5 \pm 0.7, 23.5 \pm 3.4]$	1.12	1.00
1600	52	$[9.2 \pm 0.9, 19.3 \pm 1.8]$	0.76	1.00

Table 10.2: The M5 sensitivity curve (E19, $G = 400$, $H = 26$, 60 replications, truth 13.32): mean upper bound against the index $hl_{\max}$; the lower bound is 7.7 throughout. L_{eff} from Equation (10.4); the fit is $hi \approx 11.5 + 0.235 L_{\text{eff}}$.

$hl_{\max}$ (weeks)	4	8	13	26	52	104	208
L_{eff}	6.3	12.0	19.2	35.6	56.6	75.3	88.0
upper bound	13.0	15.1	17.2	21.2	25.9	29.7	32.2
coverage	0.35	0.87	≥ 0.98	≥ 0.98	≥ 0.98	≥ 0.98	≥ 0.98

Power control (E13, 100 replications). The probability that the lower bound excludes the no-long-run total 8.0 when $\phi = 0.4$ is 0.12 ($G = 100$, $H = 13$), 0.39 (400, 26), 0.93 (1600, 52), 1.00 (6400, 52); the probability that the upper bound excludes 13.3 when $\phi = 0$ is 0.00, 0.03, 0.47, 1.00 on the same grid. The instrument discriminates, but establishing a long-run effect needs about 1600 geo-equivalents times 52 weeks and ruling one out four times as many geos. Demand persistence $\rho_d = 0.9$ or 0.95 doubles to triples every width because the per-week standard error doubles; t_3 shocks and national common shocks change nothing (E11).

Misspecified kernels (E10, E16). An Erlang-2 hump, an 8-week delayed onset and a pull-forward negative lobe are all covered by the geometric-only dictionary at $G = 400$, $H = 26$ (coverage 0.993–1.00) and at $G = 1600$, $H = 52$ (0.99–1.00), with widths 0.7–1.0 $\times$; enlarging the dictionary with Erlang and lagged kernels costs 1.5–2.5 $\times$ width for no coverage gain, and $Q_{\min}$ barely moves (13.5–15.9 against $k = 16$ blocks): the width is dominated by the tail extrapolation, which does not care about in-window shape. The counterexample (E16b) is a brand component with onset at lag 30 and $H = 26$: coverage 0.47, and $Q_{\min} = 11.2$ on $k = 14$ does not flag it; with $H = 52$ the onset is inside the window, $Q_{\min}$ rises to 22.9 and coverage returns to 0.95.

De-confounding (Table 10.3). With a $G = 100$, $H = 13$ experiment and a 200×156 panel (E12, 40 replications) the joint bound is 7 $\times$ tighter than the experiment alone, with $\hat{s} = 0.69 \pm 0.26$ against the true 0.78. The value persists with better experiments (E23: joint width 0.25 $\times$ at (400, 26) and 0.18 $\times$ at (1600, 52) against 1.40 $\times$ and 0.75 $\times$ alone) and under strong confounding $a = 1.5$ (E24: [12.6, 15.5], coverage 1.00). The audit at $G = 400$, $H = 26$ (E26, 60 replications) found 10% high-side failures with the infinite- T shape—15% even with the true ($\rho_d, a^2 \text{Var}(d), \sigma_u^2$)—and 3% (coverage 0.95, width 0.23 $\times$) with the within- T_o shape of Lemma 10.11.

Table 10.3: Experiment-grounded de-confounding of a 52-lag panel (E12, $G = 100$, $H = 13$, panel 200×156 , 40 replications, truth 13.32) and the two repairs (E21 with 25, E22 with 20 replications).

Estimator	bound	width/truth	coverage
experiment alone	[6.0, 43.3]	2.80	—
naive panel (no correction)	[15.0, 17.1]	—	0.00
panel alone, s free	unbounded	—	—
joint, single shape b	[10.1, 15.4]	0.39	1.00
joint, lead family $\ell \in \{0, 1, 2, 3, 4, 6\}$ (true lead 0/2/4)	—	0.44–0.46	0.88–0.92
joint, Hill truth, fixed scale	—	—	0.85
joint, Hill truth, free scale κ	—	1.49	1.00

10.6 Limitations and the devil’s advocate

Three attack rounds changed results. *The under-coverage of the χ_1^2 radius is not a $\hat{V}$ artifact*: with the true covariance the one-sided low failure is still 6–7% (E15), and it grows with dictionary richness—2.7% at $hl_{\max} = 1$, 4.7% (3), 5.0% (8), 6.7% (52) at $G = 400$, $H = 26$, against 1.3% for the unconstrained one-sided sum test (E15b). The diagnosis is a chi-bar-square in which the nuisance cone over-fits noise, so $Q_{\min}$ is too low; the calibrated 6.63 stands and the exact law is BL 37. *Anticipatory spend* ($x_t = \bar{x} + a d_{t+\ell} + u_t$) makes the single-shape joint bound confidently wrong—[26, 30] against 13.3, coverage 0—but loudly, $Q_{\min} = 1600$ –2600 against about 63 degrees of freedom (E17); a family of shapes b_ℓ with $r_{\ell,k} = \rho_d^{|k-\ell|}$, each with its own free scalar, restores coverage 0.88–0.92 at width 0.44–0.46 $\times$ and $Q_{\min} = 68$ –75 (E21). *Saturation*: when both sources obey $\sum_k g_k \text{Hill}(x_{t-k})$ and the experiment measures the finite difference at its operating point, the panel’s linear lag coefficients are Hill’s average derivative times g , a common scale across lags: coverage 0.85 at fixed scale, and 1.00 at width 1.49 $\times$ with a free panel/experiment scale κ profiled on a grid—still 4 $\times$ tighter than the experiment alone (5.67 $\times$)—with $\hat{\kappa} = 1.50 \pm 0.29$ against about 1.2 expected (E22). *Two-component demand* (E18) degrades gracefully: [11.8, 17.4], coverage 0.90, $Q_{\min} = 74$. A residual $Q_{\min} \approx 80$ against 63 degrees of freedom persists after the shape fix, falling to 71 with $G_o = 800$ but not with $T_o = 400$ (E27)—small-sample bias of CR0 cluster-robust covariance at 53 parameters and 200 clusters plus roughly +8 unexplained; coverage is unaffected.

The dive’s own devil’s advocate fixes the scope. The -28% holds for a linear index without covariates; a nonlinear index, or one trained on a panel containing past campaigns, changes γ (the identity still holds), and the phantom term assumes the panel’s demand persistence equals the experiment’s—comparability, which ACIK assume anyway. The 1.36 $\times$ width is one noise calibration (per-week se about 0.16 on a peak effect near 2); every width shrinks with the standard error, but the *ratio* to what a naive analyst reports is set by L_{eff}/H . The linearity of hi in L_{eff} is measured on seven points and its slope mixes $z \cdot \text{se}$, κ and the truth’s own brand level: empirical, not derived. The 7 $\times$ needs spend exogenous except through the AR(1) state, a common effect scale between panel and experiment (κ absorbs saturation, not heterogeneity) and the fixed-effects shape at the right T_o ; and crit = 6.63 was calibrated on construction 2’s boundary geometry, not the joint problem’s. Structural limits: linear-in-spend effects, a single pulse, additive near-Gaussian geo noise, no competitive response or price co-movement.

10.7 Discussion: impacts

For practice the first message is a warning about the surrogate index in dynamic panels: its error is the sum of two structural terms of opposite sign, one present with no long-run effect whatsoever. A platform that trains an index on persistent-demand panels and applies it to a 4-week test over-states by +116% to +295% at $\rho_d = 0.9$ –0.95, and the horizon at which it is accidentally right depends on quantities it does not observe; the partial- R^2 interval inherits the problem (81% coverage). The second is the sensitivity curve as a reporting

standard: report $[lo, hi](hl_{\max})$, read off the breakdown half-life, and state *both* assumptions—decay no slower than $hl_{\max}$ and onset within the window (BL 40)—since the second is the one $Q_{\min}$ cannot police. The third is the cost ledger: a 13-week, 100-geo test bounds Θ_L within $2.7\times$ its value, and establishing a long-run effect at all needs 1600×52 . For MMM vendors the de-confounding construction is the more immediately usable result: any advertiser with one geo test and a three-year weekly spend panel can compute $\hat{s}b$ and $Q_{\min}$ today, tighten the long-run bound $4\text{--}7\times$, and treat an exploding $Q_{\min}$ as the first field detection of anticipatory spend. It is a concrete form of the experiment-as-likelihood-row idea of Chapter 2: the experiment constrains the short lags, the panel the tail, and the shared object is the kernel μ .

For theory the chapter joins three literatures the dive found unjoined—surrogate-index sensitivity (ACIK and the recent identified-set, persistent-confounding and efficiency work the dive surveys, none specialised to a dynamic panel with a brand-stock channel), completely monotone extrapolation (Brown and Grabovsky) and LP shape-restriction bounds (Mogstad, Santos and Torgovitsky)—and adds the observation that the confounding bias of a distributed lag has a shape identified by the spend autocorrelation. Outward links: Chapter 7 receives $[lo, hi](hl_{\max})$ as an interval-valued posterior (BL 41); Chapter 3 supplies the multi-pulse and switch designs that would resolve the lower bound’s rate limit (BL 38); Chapter 14 measures the memory kernel directly and could calibrate this chapter’s bound.

10.8 Further exploration

The dive’s next steps became five backlog items. BL 37 asks for the exact chi-bar-square weights of the profile statistic on the completely monotone cone as a function of dictionary richness, so that 6.63 becomes a computed radius. BL 38 is the (G, H) frontier: since the lower bound is rate-resolution limited, two-pulse and switch designs and longer post-periods with *fewer* geos at matched cost may buy it more cheaply than geos do. BL 39 generalises the shape idea to AR(2) and two-state demand and lead-plus-lag families, with a $Q_{\min}$ -driven selection rule and a derived null (linking BL 4 and BL 30). BL 40 adds a maximum-onset-lag index beside $hl_{\max}$. BL 41 feeds the interval into the act/hold protocol of Chapter 7. The dive also proposes a field pilot with any advertiser holding one geo test and a three-year spend panel.

In the compiler’s judgement the most promising extension is BL 38 joined to BL 39: the asymmetry result says the two ends of the interval are bought with different currencies, and the de-confounding construction says an observational panel is by far the cheapest currency for the tail. A design study treating post-period weeks, geos and panel length as one budget, with the shape family chosen by $Q_{\min}$, would turn the price of the long run from a lament into a purchase order. Second is BL 37, because every coverage number here rests on a constant whose theoretical status is unknown.

Reviewer’s note (compiler)

Provenance. The INDEX log line is stamped 2026-09-02 ~04:50 but the file was written at 11:34 UTC, seven hours later, without explanation; Round 1 (box LP, ellipsoid, the refuted width law) was “reconstructed from the code” rather than from a writeup. Log line and writeup agree on every headline number I checked (-28% , $+163\%$, $11.5 + 0.235 L_{\text{eff}}$, κ 1.1–1.6, 1600×52 , $4\times$, 0.47 , $7\times$, 1.00 , -0.003).

Replication counts. Coverage “1.00” in Table 10.3 rests on 40 replications (E12); E22 (free-scale coverage 1.00), E23 and E24 use 20; E17, E18 and E21 use 25–30; the E13 power control uses one seed. Coverage 1.00 on 20 replications has a lower 95% binomial limit near 0.83, indistinguishable from the 0.95 target. The headline width $1.36\times$ at (400, 26) is the low end of the seed sweep’s 1.36–1.45.

The calibrated radius. $\text{crit} = 6.63$ equals $\chi_1^2(0.99)$; it was calibrated on four (ϕ, G, H) configurations of construction 2 and applied unchanged to the joint problem Equation (10.7), whose boundary geometry differs (the dive concedes this); the E26 high-side failures were cured by the shape fix, not the radius, and 6.63 was never tested separately for the joint problem. “ $4\times$ more” for ruling out a long-run effect is the grid step from (1600, 52) at power 0.47 to (6400, 52) at 1.00, not a matched-power computation.

Width law and sensitivity curve. Law 10.8 was verified with noise-free $\hat{\tau}$ and the true V from 1500 Monte Carlo draws—a setting no practitioner has—and each of the seven κ values is a single deterministic evaluation (one V per configuration), not a distribution. Recomputing L_{eff} from Equation (10.4), the fit $11.5 + 0.235 L_{\text{eff}}$ leaves

residuals of +1.1 to +1.3 at $hl_{\max} = 13\text{--}52$ (largest 1.3 at 26) and the seven points are visibly concave; the slope is an approximation.

The ACIK comparator. The code implements the partial- R^2 bound as $|SI|\sqrt{(1 - R_{W|S}^2)(1 - R_{Y|S}^2)}$, attributed to “ACIK eq. 4.3”, with $R_{Y|S}^2$ at its population value. I could not confirm this form in the NBER version I checked, whose Theorem 4 states the bias as an average residual direct effect; the comparator’s 81% coverage is a property of the dive’s rendering.

Small discrepancies. $\hat{\kappa} = 1.50 \pm 0.29$ against “about 1.2” expected (E22) is a one-sigma miss the dive does not discuss; $Q_{\min} \approx 80$ against 63 degrees of freedom contains “roughly +8 unexplained”. The surrogate-index algebra indexes weeks $1, \dots, H$ while the code applies γ to the first H entries of the curve from week 0; the identity is unaffected. No FIM or pseudo-inverse enters the chapter, so BL 82 does not apply.

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- [15] Frontier study (internal): 02_open_questions.md, entry M5; 00_INDEX.md, queue item B6 and

backlog entries BL 37–BL 41.

- [16] Program (internal): Dive 02, file `02_experiment_transport_map.md` (operator likelihood); Dive 04, file `04_decision_voi_ext.md` (switch designs); Dive 08/Dive 09 (deadband layer).

Chapter 11

An impossibility theorem for causal attribution, and the designs that escape it

Provenance

Source dives: Dive 13 *An impossibility theorem for causal attribution: the cross-journey table* (run 2026-09-02, file 13_attribution_impossibility.md, written 15:40 UTC); Dive 14 *EXTEND of 13: the experimental route to causal Shapley — a proof of the design theorem, the holdout-only design, and the precision-lift-aliasing frontier* (run 2026-09-02, file 14_attribution_impossibility_ext.md, written 19:44 UTC). **Code:** code/13_attribution_impossibility.py, code/14_attribution_impossibility_ext.py. **Frontier-study problem(s) addressed:** M8 — characterise the class of journey-value functions for which Shapley attribution equals an average of per-touch causal effects, and prove that no observational attribution rule can jointly satisfy efficiency, symmetry and a counterfactual-consistency axiom under confounded exposure. **Backlog items closed:** BL 42 (opened by Dive 13, resolved by Dive 14); **opened:** BL 43–BL 45 (from Dive 13), BL 46–BL 50 (from Dive 14).

Key ideas in this chapter

- Journey-level Shapley on observational conversion rates (Google-style DDA) is the efficient, causal-null allocation of realised lift *iff* exposure is unconfounded; its bias is the Shapley value of a bias-game family on the off-diagonal of the cross-journey table $c(S, A) = \mathbb{E}[Y(\mathbf{1}_A) \mid Z = \mathbf{1}_S]$, the only unidentified object. Shao–Li population Shapley fails even unconfounded, by ignoring reach.
- Impossibility: no rule computed from the observational law satisfies causal efficiency and causal null (symmetry not needed). Two smooth monotone models at $n = 2$ agree on every observable cell to 10^{-17} and on the total lift 0.0502 yet assign all credit to different channels; at $n \geq 3$ parametric families are identified by functional form alone.
- In a three-channel benchmark a causally null retargeting channel earns 30% of DDA credit; a cross-journey selection odds ratio of $\Gamma \approx 1.10$ – 1.26 overturns DDA’s ranking while the true channel odds ratios run 1.28–2.54.
- The escape: under a k -way interaction cap, expose- $\leq m_b$ / hold-out- $\leq m_t$ cells identify the causal Shapley allocation *iff* $m_b + m_t \geq \min(k, n - 1)$ (proved; 1,070/1,070 clean-room): $\text{Sh}_j = \frac{1}{2}(\text{solo} + \text{removal})$ from $2n + 2$ cells under pairwise interactions, exact aliasing $|\text{leak}| = 1/((k + 1)\binom{k}{m_b})$; global plus single holdouts never suffice for $n \geq 3$.
- Every exactly-capped design carries one over-identifying restriction, “credits add up to the global-holdout lift”: a df-1 cap test (power 82–100%) and a 2.3 – $7\times$ variance gain for the BLUE, paid in aliasing onto innocent channels. The precision-lift frontier obeys $N\ell \text{ tr } V \rightarrow \kappa$; the $2n + 2$ design is within 5–16% of optimal, holdout-only designs 2.8 – $5.6\times$ off and collapsing like n^4 .

11.1 Problem and motivation

Multi-touch attribution allocates each conversion among the channels that touched the converting user, from journey logs alone. Last touch, equal split, Markov removal effects and the “data-driven” Shapley allocations of Google Analytics all take the observed conversion rate of each journey as input and return

credits summing to observed conversions. Practitioners suspect the credits are not causal—a retargeting channel that fires at users about to buy collects credit for conversions it did not cause—and observational lift is known to fail against randomised ground truth (Gordon et al., 2019, 2023). Attribution has nevertheless escaped a formal verdict because its object is an *allocation*, whose axioms live on a cooperative game rather than on potential outcomes. Problem M8 asked for the verdict: characterise the journey-value functions for which Shapley attribution is an average of per-touch causal effects, and prove that no observational rule satisfies efficiency, symmetry and counterfactual consistency under confounded exposure. Berman (2018) had shown the divergence as an equilibrium phenomenon; the study wanted an identification theorem.

Dive 13 answers with one object, the cross-journey table $c(S, A)$: what users who received journey S would have converted at had only the sub-journey A reached them. Its diagonal is what attribution tools see; every causal axiom pulls on its off-diagonal, which nothing observational identifies. This yields the characterisation, the impossibility theorem (symmetry dropped; an exact swap inside a smooth family), sharp sensitivity bounds with attribution robustness values, and a first-order law for how targeting manufactures credit. It also opens the escape: randomised multi-cell holdouts identify the causal Shapley allocation if the menu is rich enough, and the ghost-ads menu is not. Dive 14 turns that into a design theory: a proved identification theorem, a closed-form estimator with an exact aliasing coefficient, a df-1 test of the interaction cap, and a convex precision–lift frontier pricing every design in one currency.

11.2 Background: the ideas this chapter builds on

11.2.1 The Shapley value and Harsanyi dividends (Shapley, 1953; Harsanyi, 1963; Grabisch, 1997)

A cooperative game on $N = \{1, \dots, n\}$ is a set function $v : 2^N \rightarrow \mathbb{R}$. The Shapley value averages marginal contributions over all arrival orders,

$$\text{Sh}_j(v) = \sum_{B \subseteq N \setminus \{j\}} \frac{|B|! (n - |B| - 1)!}{n!} [v(B \cup \{j\}) - v(B)], \quad (11.1)$$

and is the unique allocation satisfying *efficiency* ($\sum_j \text{Sh}_j = v(N) - v(\emptyset)$), *null player* ($v(B \cup \{j\}) = v(B)$ for all B implies $\text{Sh}_j = 0$), *symmetry* (interchangeable players get equal credit) and *additivity* ($\text{Sh}(v + w) = \text{Sh}(v) + \text{Sh}(w)$): Shapley is linear on the 2^n -dimensional game space). The Möbius inverse on the subset lattice gives the *Harsanyi dividends*

$$D_v(T) = \sum_{B \subseteq T} (-1)^{|T \setminus B|} v(B), \quad v(A) = \sum_{T \subseteq A} D_v(T), \quad \text{Sh}_j(v) = \sum_{T \ni j} \frac{D_v(T)}{|T|}, \quad (11.2)$$

the part of $v(T)$ not explained by proper subsets (main effects, pairwise interactions, and so on), each shared equally among its members. Grabisch (1997) calls v *k-additive* if $D_v(T) = 0$ for $|T| > k$; the chapter’s “interaction cap” is *k-additivity*.

11.2.2 Data-driven attribution: Shapley on observational rates (Shao and Li, 2011, and successors)

Shao and Li (2011) estimate from logs the conversion rates $P(y \mid x_i)$ of users who saw channel i and $P(y \mid x_i, x_j)$ of users who saw both, crediting i with its own rate plus half of each pairwise excess: a truncated Shapley value of the game whose value at a channel set is the observed conversion rate of users exposed to it. Dalessandro et al. (2012) argued Shapley is the “causally motivated” allocation and computed it on the same rates; Zhao et al. (2018) extended it to ordered journeys; Anderl et al. (2016) fitted a Markov chain to journey states and defined a channel’s *removal effect* as the drop in the chain’s conversion probability when its state is deleted. Google’s DDA works per realised journey: it compares the conversion probability of each observed

path with that of the path minus each touchpoint, distributes the path's conversions by the Shapley formula, and sums over paths. None states an identification condition; the value function is observational and the causal reading is asserted.

11.2.3 The causal critiques: Berman (2018) and Singal et al. (2022)

Berman (2018) treats the attribution rule as an incentive contract between an advertiser paying per attributed conversion and publishers choosing whom to show ads to. Under last touch, publishers “exploit the baseline” by advertising to users who would convert anyway, so attributed and causal contribution diverge in equilibrium; Shapley on the *interventional* revenue function—each publisher's marginal contribution to conversions that would not otherwise occur—is the benchmark restoring efficient incentives. Singal, Besbes, Desir, Goyal and Iyengar (2022) give the only axiomatic treatment with a counterfactual flavour: in a Markov journey model, *counterfactual efficiency* (credits sum to the model's incremental conversions) and a *counterfactual null player* axiom characterise $CASV = SV(M) - SV(M_{cf})$, the Shapley value of the fitted model minus that of the model with advertising effects removed. The counterfactual is *model-internal*, so the axioms are satisfiable exactly because exposure is unconfounded within the model. The explanation literature has analogues for a fixed function—conditional Shapley violates the dummy axiom (Sundararajan and Najmi, 2020); complete linear attributions cannot test counterfactual behaviour (Bilodeau et al., 2024)—not for a confounded treatment.

11.2.4 Multi-channel potential outcomes and the evidence against observational lift

A unit's potential outcomes are indexed by the full exposure vector, $Y(z)$ for $z \in \{0, 1\}^n$, so “the effect of channel j ” is a contrast $Y(z + e_j) - Y(z)$ depending on the other channels present: a per-touch causal effect is a marginal contribution in the game $A \mapsto Y(\mathbf{1}_A)$, which is why Shapley is the natural bridge. Exposure is *unconfounded* if Z is independent of the potential outcomes; targeting on purchase intent violates this by construction. Gordon et al. (2019) compared 15 large Facebook experiments with observational methods on the same data: observational lift overstated RCT lift by about $3\times$ in half the studies, and one study with a 6.6% exposure rate reported 1,306% against 2.4%. Gordon, Moakler and Zettelmeyer (2023), on 663 experiments, found double machine learning and stratified propensity matching giving median lifts of 83% and 173% against 29%.

11.2.5 Sensitivity analysis: Rosenbaum's Γ , Tan's marginal model, robustness values

Rosenbaum (2002) parametrises hidden bias by one odds ratio $\Gamma \geq 1$: two units alike on observables may differ in their odds of treatment by at most Γ ; one computes the inferences consistent with each Γ and reports where the conclusion first becomes ambiguous. Tan (2006) states the *marginal sensitivity model*, bounding the odds ratio between the propensity given the potential outcome and the propensity given observables, which yields sharp bounds for effects linear in the unobserved cells. Cinelli and Hazlett (2020) package the output as a *robustness value*: the minimal confounding strength that brings an estimate to zero or overturns a conclusion.

11.2.6 Ghost ads, multi-cell holdouts and optimal design (Johnson, Lewis and Nubbemeyer, 2017; Zhou et al., 2025)

A lift experiment randomises users into an exposed group and a holdout never shown the ad; the classic holdout wastes public-service impressions and cannot tell which holdout users *would* have been exposed. *Ghost ads* (Johnson, Lewis and Nubbemeyer, 2017) fix both: the platform logs when a holdout user would have won the auction and serves the next-best ad, identifying the counterfactually exposed subgroup at no media cost. Multi-cell versions (Gordon et al., 2019; Waisman and Gordon, 2025) vary intensity or creative of one campaign, never the *subset* of channels. Here a “cell” is a random slice in which a specified channel

set is suppressed; the ghost-ads menu is the global holdout plus one single-channel holdout per channel. Allocating users to cells is *L-optimal* design—minimise $\text{tr}(L \text{ Var } \hat{\theta})$ for a linear criterion, here the summed variance of the Shapley vector—and is convex because the information matrix is linear in cell shares. Zhou, Mee, Hamers and Zheng (2025) recover Shapley values from regular fractional factorials under a low-order truncation with classical aliasing tables; the designs below are Hamming-weight shells, not regular fractions, so their aliasing must be derived afresh. Feit and Berman (2019) price a single holdout’s opportunity cost in a Bayesian “test and roll”.

11.3 Setup and assumptions

Definition 11.1 (Exposure, the cross-journey table, three games). Unit i has exposure set $Z_i \in \{0, 1\}^n$ and potential outcomes $\{Y_i(z)\}$; $\mathbf{1}_S$ is the indicator of $S \subseteq N$ and $Z \wedge \mathbf{1}_A$ the coordinatewise minimum. The observational law P is summarised by $P(S) := \mathbb{P}(Z = \mathbf{1}_S)$ and $v_{\text{obs}}(S) := \mathbb{E}[Y \mid Z = \mathbf{1}_S]$. The *cross-journey table* is

$$c(S, A) := \mathbb{E}[Y(\mathbf{1}_A) \mid Z = \mathbf{1}_S], \quad A \subseteq S, \quad (11.3)$$

with observed diagonal $c(S, S) = v_{\text{obs}}(S)$ and unobserved off-diagonal. Three games: $v_*(A) := \mathbb{E}[Y(\mathbf{1}_A)]$ (everyone exposed to exactly A); $v_{\text{pop}}(A) := \mathbb{E}[Y(Z \wedge \mathbf{1}_A)] = \sum_S P(S) c(S, S \cap A)$ (channels in A keep their realised delivery, the rest are switched off); the bias game $b := v_{\text{obs}} - v_*$. The *realised lift* is $v_{\text{pop}}(N) - v_{\text{pop}}(\emptyset) = \mathbb{E}[Y(Z) - Y(0)]$.

Definition 11.2 (Rules and axioms). An *observational rule* is any map $\varphi(P) \in \mathbb{R}^n$. Rules studied: population-game Shapley $\text{Sh}(v_{\text{obs}})$ (Shao–Li style); journey-level DDA, $\text{DDA}_j := \sum_S P(S) \text{Sh}_j(v_{\text{obs}}|_S)$; the removal effect $\tau_j := \mathbb{E}[Y(Z) - Y(Z - e_j)]$, what a single-channel ghost-ad holdout measures; equal split. Axioms: (E) *causal efficiency*, $\sum_j \varphi_j = \mathbb{E}[Y(Z) - Y(0)]$; (N) *causal null*, $\varphi_j = 0$ whenever $Y_i(z)$ does not depend on z_j for every unit; (S) symmetry; (L) linearity.

Assumption 11.3 (Scope). Journeys are sets, not ordered or repeated paths; outcomes are binary; in experimental cells suppression is perfect and cells do not interfere. The theorems are nonparametric; the numerical DGPs use one latent confounder with logistic links.

The *benchmark DGP* has $n = 3$, $U \sim \mathcal{N}(0, 1)$, $Z_j \mid U \sim \text{Bern}(\sigma(a_j + g_j U))$ with σ the logistic function, $a = (-0.5, -1.0, -1.5)$ and targeting slopes $g = (0.4, 1.0, 1.6)$, and $Y(z) \mid U \sim \text{Bern}(\sigma(\beta_0 + \beta U + \theta^\top z))$ with $\beta_0 = -2.5$, $\beta = 1$, $\theta = (0.5, 0.5, 0)$: channel 3 is the “retargeting” channel, most targeted, lowest reach, causally null. Population quantities are exact by Gauss–Hermite quadrature (error $< 10^{-12}$).

Definition 11.4 (Cells, caps and two-layer designs (Dive 14)). A cell A exposes a random slice to the channels in A and suppresses the rest; its mean is $v(A) := v_{\text{pop}}(A)$, with dividends $D(T)$ and causal Shapley $\text{Sh}_j = \sum_{T \ni j} D(T) / |T|$. A *k-way cap* imposes $D(T) = 0$ for $|T| > k$. The two-layer design $B(m_b, m_t) := \{A : |A| \leq m_b \text{ or } |N \setminus A| \leq m_t\}$ holds the expose-at-most- m_b (bottom) and hold-out-at-most- m_t (top) cells; Dive 13’s layered design is $L_m = B(m, m)$, the ghost-ads menu is $B(0, 1)$, the *holdout-only* design $B(0, k)$ never deprives a user of more than k channels. A linear functional of D is identified iff its coefficient vector lies in the row space of the incidence matrix $X[A, T] = \mathbf{1}[T \subseteq A]$ restricted to $|T| \leq k$. Cell A gets share f_A of N users, has variance $s_A^2 = v(A)(1 - v(A))$ and forgoes lift $\ell_A = [v(N) - v(A)] / [v(N) - v(\emptyset)]$ per user; $\ell = \sum_A f_A \ell_A$. The *leak* of a dividend T , $|T| > k$, into an estimator with cell weights c is $\sum_A c_{jA} \mathbf{1}[T \subseteq A] - \mathbf{1}[j \in T] / |T|$.

11.4 Main results

11.4.1 Decomposition identities and the journey-level lemma

By linearity, $\text{Sh}(v_{\text{obs}}) = \text{Sh}(v_*) + \text{Sh}(b)$, and $\text{Sh}(v_*)$ is by construction a Shapley-weighted average of per-touch causal effects.

Proposition 11.5 (The class asked for in M8(i)). *[established in the dive] Shapley on the observational game equals a Shapley average of per-touch causal effects exactly on $v_* + \ker \text{Sh}$, an affine subspace of codimension n ; in dividend coordinates $\ker \text{Sh} = \{b : \sum_{T \ni j} D_b(T)/|T| = 0 \ \forall j\}$, containing every symmetric bias $b(S) = h(|S|)$ with $h(n) = h(0)$ and every constant (uniform-selection) bias. Targeting-generated confounding lands in it only by measure-zero cancellation.*

For the journey-level rule the causal target is v_{pop} and the bias is a *family* indexed by the realised journey. Since $B \mapsto c(S, S \cap B)$ is the subgame on S with outside players null, $\text{Sh}_j(v_{\text{pop}}) = \sum_S P(S) \text{Sh}_j(A \mapsto c(S, A))$, whereas DDA replaces $c(S, A)$ by $c(A, A)$:

$$\text{DDA}_j - \text{Sh}_j(v_{\text{pop}}) = \sum_S P(S) \text{Sh}_j(b_S), \quad b_S(A) := c(A, A) - c(S, A), \quad A \subseteq S. \quad (11.4)$$

Because $b_S(S) = 0$, efficiency gives $\sum_j \text{Sh}_j(b_S) = c(S, \emptyset) - c(\emptyset, \emptyset)$: DDA's total credit exceeds the realised lift by exactly $\mathbb{E}[Y(0)] - \mathbb{E}[Y(0) \mid Z = \emptyset]$, the classic baseline selection bias.

Lemma 11.6 (Journey-level lemma). *[established in the dive] If exposure is unconfounded ($c(S, A) = c(A, A)$) for all $A \subseteq S$) then $\text{DDA} = \text{Sh}(v_{\text{pop}})$ exactly (E1: $\max |\text{DDA} - \text{Sh}(v_{\text{pop}})| = 2.8 \times 10^{-17}$ at $g = 0$): journey-level Shapley on observational rates is the efficient, symmetric, causal-null allocation of realised lift. The population-game rule is not: $\text{Sh}(v_*)$ allocates the lift of a world in which everyone sees every channel. With equal per-exposure effects and reach (0.38, 0.18, 0.08) its shares are $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ against causal shares (0.586, 0.291, 0.123) (E22).*

11.4.2 The impossibility theorem

Theorem 11.7 (Impossibility of causal attribution from observational data). *[established in the dive] Let $n \geq 2$ and $\mathcal{D}$ be any class of DGPs closed under the swap construction below. No observational rule $\varphi(P)$ satisfies (E) and (N) on $\mathcal{D}$. Symmetry and linearity are not needed, and the conclusion survives (i) restriction to monotone targeting ($g > 0$), monotone confounding ($\beta > 0$) and non-negative effects, (ii) a global holdout (realised total lift known), and (iii) restriction to a smooth single-index parametric family.*

The nonparametric proof is in principal-strata form: let the latent type be the exposure set itself, so that any table $\{c(S, A)\}$ with the observed diagonal is realised by some DGP with law P ; for each j choose the off-diagonal so that j is inert ($c(S, A) = c(S, A \setminus j)$) and all n tables share the same total lift $\Delta > 0$. Then $\varphi(P)$ is one vector, (N) forces every $\varphi_j(P) = 0$ and (E) forces $\sum_j \varphi_j(P) = \Delta$. The identified part of the table is its diagonal and every causal axiom pulls on the off-diagonal; the Shapley operator is not at fault—Singal et al.'s axioms are all satisfiable given the off-diagonal. The failure is identification, not axiomatics.

Proposition 11.8 (Exact swap inside a smooth family, $n = 2$). *[established in the dive] With Gaussian U and logistic links, DGP A ($a = (-0.5, -1.5)$, $g = (0.6, 1.8)$, $\beta_0 = -2$, $\beta = 1.2$, $\theta = (0.8, 0)$: channel 2 null) and DGP B ($a = (-0.947, -1.036)$, $g = (2.779, 0.517)$, $\beta_0 = -1.724$, $\beta = 1.063$, quadratic confounder term $\beta_2 = -0.193$, $\theta = (0, 1.001)$: channel 1 null) have identical $\mathbb{P}(Z = \mathbf{1}_S, Y = y)$ on all eight cells (maximum difference 5.6×10^{-17} ; clean-room 1.4×10^{-17}) and identical realised lift 0.0502. Under A , $\text{Sh}(v_{\text{pop}}) = (0.0502, 0)$; under B , $(0, 0.0502)$. Both satisfy $g > 0$, $\beta > 0$, $\theta \geq 0$; every observational rule—DDA (0.0664, 0.0523), population Shapley (0.200, 0.213), Markov, last touch, equal split—returns the same answer for both (E15).*

Remark 11.9 (The parametric boundary at $n \geq 3$). *[numerically established] At $n = 3$ the rigid logit/logit/-Gaussian family (11 parameters against 15 observable degrees of freedom) admits no exact swap: the best channel-1-null fit to the channel-3-null truth leaves a maximum cell mismatch of 7.0×10^{-3} (about 2% of the cell), with or without a two-component mixture for U (E19). Functional form identifies the allocation at $n \geq 3$ the way Heckman selection is identified without an exclusion restriction—detectably only with of order 10^5 users per journey cell, and only if the family is right; with nonparametric U the swap always exists.*

11.4.3 Sensitivity bounds and attribution robustness values

$\text{Sh}_j(v_{\text{pop}}) = \sum_{S, A \subseteq S} \text{coef}_j(S, A) c(S, A)$ is linear in the table with signed coefficients (Shapley weights times $P(S)$), so any cell-wise sensitivity model on the unidentified cells gives *sharp* bounds by plugging the upper

cell bound where the coefficient is positive and the lower where it is negative.

Construction 11.10 (Cross-journey odds-ratio models and robustness values). [\[transported from the literature\]](#) Three models on the off-diagonal: the *global box*, odds $c(S, A) \in \text{odds } c(A, A) \cdot [1/\Gamma, \Gamma]$ for every $A \subsetneq S$; the *channelwise* model, odds $c(S, A) \in \text{odds } c(A, A) \cdot \prod_{k \in S \setminus A} [1/\Gamma_k, \Gamma_k]$, in which the log-odds selection shift is additive in the *missing* channels (what the first-order law below implies) and the analyst sets one number per channel; and *monotone* (positive) selection, $c(S, A) \geq c(A, A)$. The smallest channelwise parameters consistent with a DGP solve an LP (minimise $\sum_k \log \Gamma_k$ subject to every cell's log-odds ratio being at most $\sum_{k \in S \setminus A} \log \Gamma_k$). The *attribution robustness values* are the Γ at which channel j 's interval first contains zero and the Γ at which the top-two DDA ordering becomes ambiguous, with DDA as the $\Gamma = 1$ point.

For the benchmark $\Gamma_k^* = (1.28, 1.82, 2.54)$ while the global $\Gamma^* = 5.31$ (one extreme cell dominates); Table 11.2 gives widths and robustness values. The ranking is overturned at $\Gamma = 1.10$ (channelwise), 1.12 (box) or 1.20–1.26 (monotone): a 10–25% cross-journey selection odds ratio makes DDA's ranking uninterpretable while the true Γ_k^* run 1.3–2.5. The intervals are conservative—the truth sits 0.22–0.44 of the way up the channelwise intervals across four targeting regimes (0.37–0.60 for the box), a further factor 2–3 against the single-index truth (BL 43).

11.4.4 The targeting-intensity bias law

With Gaussian U and a log-linear tilt, $\mathbb{P}(Z = \mathbf{1}_S \mid U) / P(S) \approx 1 + U L(S)$, $L(S) = \sum_k g_k (s_k - \bar{e}_k)$, $\bar{e}_k = \sigma(a_k)$, so to first order in the targeting slopes

$$b(S) \approx \beta_S L(S), \quad \beta_S := \mathbb{E}[U p(\mathbf{1}_S, U)]; \quad b_S(A) \approx \beta_A [L(A) - L(S)] = -\beta_A \sum_{k \in S \setminus A} g_k, \quad (11.5)$$

with $p(z, U) := \mathbb{P}(Y(z) = 1 \mid U)$ and β_S the outcome's U -slope under exposure S .

Law 11.11 (Targeting-intensity bias: operator form holds, closed form refuted). [\[numerically established\]](#) $\text{Sh}(b) \approx \text{Sh}(\beta \cdot L)$ and $\text{DDA} - \text{Sh}(v_{\text{pop}}) \approx \sum_S P(S) \text{Sh}(b_S)$ with b_S from Equation (11.5). The operator predictions are within 1–4% of the exact bias for targeting scale ≤ 0.2 ($g \leq 0.32$), within 10% up to $g \approx 0.6$, then over-predict by up to $2\times$ at the benchmark scale as the logistic tilt saturates (E3); across probit links, bimodal and skewed U and U -dependent persuasion they hold to $\pm 5\%$ at scale 0.1 and $\pm 10\%$ at 0.25 (E11). [\[refuted within the dive\]](#) Were the outcome slope exposure-invariant ($\beta_S \equiv \bar{\beta}$) the law would collapse to $\text{Sh}_j(b) \approx \bar{\beta} g_j$ and $\text{DDA}_j - \text{Sh}_j(v_{\text{pop}}) \approx \bar{\beta} g_j \mathbb{P}(j \in Z)$ —credit inflation equal to a channel's own targeting slope times the confounder's outcome slope, independent of its true effect and of the other channels. Round 2 refuted this even as $g \rightarrow 0$: the ratios exact/closed-form converge to 1.37, 1.08, 1.01, not 1, because in a logistic model β_S varies $1.7\times$ across exposure sets, so channel 1's bias carries the others' targeting through β_S . The repaired form uses the Shapley-weighted slope $\bar{\beta}_j = \sum_B w_j(B) \beta_{B \cup j}$, exact for the strongly targeted channel (ratio 1.01–1.05 at scale ≤ 0.2).

Sign and ordering survive: the null channel's DDA credit rises monotonically from 0 at $g_3 = 0$ through 0.0271 at the benchmark $g_3 = 1.6$ to 0.0394 at $g_3 = 2.5$ (41% of total credit), and its population-Shapley rank goes from third to first at $g_3 \approx 2$ (E2). A 50% gap in targeting slope ($g_2 = 1.5$ against $g_1 = 1.0$) overturns a 50% gap in true effect (0.6 against 0.4) (E13). The exchange rate $\partial \text{DDA}_j / \partial g_j \div \partial \text{DDA}_j / \partial \theta_j$ is 0.45–0.59: a unit of targeting slope buys about half the credit of a unit of log-odds causal effect (E14; BL 45).

11.4.5 The design theorem

A cell suppressing H identifies $v_{\text{pop}}(N \setminus H) = \sum_{T \subseteq N \setminus H} D(T)$. Dive 13 found by rank computation that the ghost-ads menu identifies only τ_j and the total, never Sh for $n \geq 3$ under any cap (rank 4 of 7 at $n = 3$), and that L_m identifies Sh under a k -way cap iff $m \geq \min(\lceil k/2 \rceil, \lfloor n/2 \rfloor)$, exhaustively for $n \leq 8$, proved only at $k = 2$ (BL 42). Dive 14 proved the general statement.

Theorem 11.12 (Identification of causal Shapley from two-layer designs). *[established in the dive]* Under a k -way cap the causal Shapley allocation is identified from $B(m_b, m_t)$ iff $m_b + m_t \geq \min(k, n - 1)$. (Exhaustive rank check $n = 3, \dots, 8$: 1,042 cases, 0 mismatches; clean-room 1,070 cases, $n \leq 7$, 0 mismatches.)

The proof fixes j and works in the $G = \text{Stab}(j) \subset S_n$ symmetric sector: design, cap and Sh_j are G -invariant, so Sh_j is identified iff it lies in the span of the G -averaged cell functionals. With $\sigma_t := \sum_{T \ni j, |T|=t} D(T)$ and $\omega_t := \sum_{T \not\ni j, |T|=t} D(T)$, the averaged bottom cells span exactly $\{D_\emptyset, \sigma_t, \omega_t : t \leq m_b\}$, and the top layer, after a Möbius transform on the holdout lattice ($Q(H) := \sum_{T \supseteq H} D(T)$, an alternating sum of the cells $v((N \setminus H) \cup B)$, $B \subseteq H$), gives the averaged co-dividend sums

$$R_h := \sum_{H \ni j, |H|=h} Q(H) = \sum_t \binom{t-1}{h-1} \sigma_t, \quad h = 1, \dots, m_t. \quad (11.6)$$

Modulo the bottom layer the unknowns are σ_t, ω_t for $t = m_b + 1, \dots, k$ and the target is $1/t$ on the σ 's, 0 on the ω 's. *Sufficiency*: if $m_b + m_t \geq n - 1$ the design is the full factorial; otherwise $k - m_b \leq m_t$ and the rows R_h are polynomials $\binom{t-1}{h-1}$ of degrees $0, \dots, m_t - 1$ at $k - m_b \leq m_t$ points, which interpolate $1/t$. *Necessity*: a degree- $\leq (m_t - 1)$ polynomial cannot equal $1/t$ at $k - m_b \geq m_t + 1$ points, since the m_t -th divided difference of $1/t$ over positive points is $(-1)^{m_t} / \prod t_i \neq 0$. Explicit witness at $n = 4, k = 3, L_1$: a 3-additive game with every L_1 cell zero and $\text{Sh} = (\frac{1}{6}, -\frac{1}{6}, 0, 0)$.

Corollary 11.13 (Pairwise interactions: $2n + 2$ cells). *[established in the dive]* Under $k = 2$, $B(1, 1)$ —global holdout, no holdout, n single holdouts, n single-channel-only exposures—identifies Sh for every n through

$$\text{Sh}_j = \frac{1}{2} [v(\{j\}) - v(\emptyset)] + \frac{1}{2} [v(N) - v(N \setminus j)] = \frac{1}{2} (\text{solo} + \text{removal}), \quad (11.7)$$

since for 2-additive games $\text{Sh}_j = D_j + \frac{1}{2} \sum_i D_{ij}$ and both brackets are dividend sums (Dive 13; clean-room proved, checked to 4×10^{-16}). Its error under a true 3-way dividend is $D_{123}/6$ per channel, uniform.

Corollary 11.14 (Holdout-only designs). *[established in the dive]* $B(0, k)$ identifies Sh under a k -way cap; for $k = 2$,

$$\text{Sh}_j = [v(N) - v(N \setminus j)] - \frac{1}{2} \sum_{i \neq j} [v(N) - v(N \setminus i) - v(N \setminus j) + v(N \setminus \{i, j\})], \quad (11.8)$$

the removal effect minus half the pairwise removal interactions (exact to 10^{-15}). The “expose only j ” cells are not necessary for identification; what they buy is precision.

11.4.6 The closed-form estimator and the aliasing law

Construction 11.15 (Symmetrised estimator). *[established in the dive]* With $r = k - m_b$ and λ solving the $r \times r$ binomial-Vandermonde system $\sum_h \lambda_h \binom{t-1}{h-1} = 1/t$, $t = m_b + 1, \dots, k$,

$$\hat{\text{Sh}}_j = \sum_{t \leq m_b} \frac{\hat{\sigma}_t}{t} + \sum_{h=1}^r \lambda_h \left[\hat{R}_h - \sum_{t \leq m_b} \binom{t-1}{h-1} \hat{\sigma}_t \right], \quad (11.9)$$

with $\hat{\sigma}_t$ from Möbius inversion of bottom cells and $\hat{R}_h$ from alternating sums of top cells. For $(m_b, m_t, k) = (1, 1, 2)$, $\lambda = \frac{1}{2}$ gives Equation (11.7); for $(0, 2, 2)$, $\lambda = (1, -\frac{1}{2})$ gives Equation (11.8). Exact on random k -additive games in nine configurations up to $n = 8$ (error $\leq 4 \times 10^{-15}$).

Law 11.16 (Aliasing of uncapped dividends). *[established in the dive]* Applied to a game with a dividend of order $t' > k$ on T , the cap- k estimator's coefficient on $\sigma_{t'}$ is $p(t')$, the degree- $(r - 1)$ polynomial interpolating $1/t$ at $t = m_b + 1, \dots, k$, so for $j \in T$

$$\text{leak} = p(t') - \frac{1}{t'} = (-1)^{r+1} \frac{\prod_{t=m_b+1}^k (t' - t)}{t' \prod_{t=m_b+1}^k t}, \quad |\text{leak}|_{t'=k+1} = \frac{m_b! (k - m_b)!}{(k + 1)!} = \frac{1}{(k + 1) \binom{k}{m_b}}, \quad (11.10)$$

with sign $(-1)^{k-m_b+1}$; for $j \notin T$ the leak is exactly zero.

The leak is minimised by the *balanced* design $m_b = \lfloor k/2 \rfloor$ and worst, $1/(k+1)$, for the one-sided $B(0, k)$ and $B(k, 0)$: $1/6$ against $1/3$ at $k = 2$, $1/12$ against $1/4$ at $k = 3$ (verified to machine precision, $k \leq 5$, $n \leq 7$; E3). Higher orders extrapolate: for $B(0, 3)$ the order-5 and order-6 leaks are 0.8 and 1.67, for $B(1, 2)$ and $B(2, 1)$ at most $1/3$. Summed over $j \in T$, one order- $(k+1)$ dividend moves the *total* credit by $(-1)^{k-m_b+1} D(T) / \binom{k}{m_b}$, a sign-known diagnostic of cap violation.

11.4.7 The one over-identifying restriction: add-up test and BLUE

Proposition 11.17 (Credits add up). *[numerically established] Every exactly-capped design $B(m_b, m_t)$ with $m_b + m_t = k \leq n - 2$ has rank $X = \#cells - 1$ (E14, $n = 4, \dots, 8$, seven families; clean-room $n \leq 7$), and the one restriction is $\sum_j \hat{Sh}_j = v(N) - v(\emptyset)$: the closed-form credits sum to the global-holdout lift (the discrepancy's weight vector spans $\text{null}(X^T)$, cosine 1.000000 in every case). For $2n + 2$ cells it reads $\sum_i [v(N \setminus i) - v(\{i\})] = (n - 2)[v(N) - v(\emptyset)]$.*

In the S_n -symmetric sector the top layer supplies $m_t + 1$ equations for m_t unknowns, one more than needed; the general df-1 theorem is BL 47. Two consequences. The *add-up test*: the df-1 lack-of-fit χ^2 of the cap is the standardised add-up gap, whose mean under one order- $(k+1)$ dividend is $D(T) / \binom{k}{m_b}$ (E15: gap $+0.0037$ against $D_{123}/2 = +0.0045$ at $n = 5$, the shortfall being other 3- and 4-way dividends). And the cap-unbiased estimators form a *line* $c(\alpha) = c_{\text{closed}} + \alpha v$, v the add-up discrepancy's weight vector, on which variance is quadratic and aliasing affine. The BLUE $c = \Sigma^{-1} X (X^T \Sigma^{-1} X)^+ w$ (Σ the diagonal cell covariance, w the Shapley coefficient vector on capped dividends) cuts the closed form's $\sum_A c_A^2$ from 1.000 to 0.437, 0.429, 0.443 for $B(1, 1)$ at $n = 4, 5, 8$, from 4.0 to 0.86 for $B(0, 2)$ at $n = 5$, from 15.9 to 2.27 for $B(0, 3)$ at $n = 6$ (E4, E14): a $2.3\text{--}7\times$ gain, of which the hand rule—subtract the add-up gap equally from every channel—captures 85–99% (0.500, 0.480, 0.469). The price is unstructured aliasing: under the BLUE a 3-way dividend biases channels outside it, including a null one (Section 11.5).

11.4.8 The precision–lift frontier and the κ law

For a fixed design $\text{tr Var}_{\text{BLUE}}(f)$ is matrix-convex in the allocation f , so $\min_f \text{tr } V + \mu \ell(f)$ is a convex programme: L-optimal design with a lift penalty.

Law 11.18 (The lift price of a Shapley allocation). *[established in the dive] As $\ell \rightarrow 0$ (mass concentrating on the no-holdout cell),*

$$N \ell \text{ tr } V \rightarrow \kappa := \min_{C \text{ cap-unbiased}} \left(\sum_{A \neq N} \|C_A\|_2 s_A \sqrt{\ell_A} \right)^2, \quad (11.11)$$

with optimal shares $g_A \propto \|C_A\| s_A / \sqrt{\ell_A}$ (Cauchy–Schwarz on $\sum_A \|C_A\|^2 s_A^2 / f_A$ against $\sum_A f_A \ell_A$; the no-holdout cell is free). Verified to three decimals in 17 (n, k, design) cases (E13; clean-room 0.71041 against 0.71044, under 0.01%).

κ prices an allocation in users times lift forgone per unit of Shapley variance: $\sum_j \text{Var}(\hat{Sh}_j) = \delta^2$ needs $N \ell \geq \kappa / \delta^2$. With five channels, $\hat{Sh}_j \approx 0.01\text{--}0.035$ and target sd 0.002 per channel, $\delta^2 = 2 \times 10^{-5}$ and $N \ell \approx 55,000$: with 5M users the layered design costs 1.1% of the slice's lift, the holdout-only design 2.9%. The frontier product is within 3% of κ already at $\ell = 5\text{--}10\%$.

11.5 Numerical evidence

Table 11.1 gives the exact benchmark allocations (identities to $\leq 3 \times 10^{-17}$, reproduced clean-room to all digits): the null channel takes 30% of DDA credit and ranks second on population Shapley, both observational rules invert the two real channels, and removal effects over-allocate by double-counting the small positive interaction. Finite-panel DDA is unbiased for its population value (E9: $N = 10^4, 10^5, 10^6$, 30 replications), so

this is bias, not noise—it exceeds the sampling sd by $1.6\text{--}7.9\times$ at $N = 10^4$ and $16\text{--}103\times$ at 10^6 ; at $n = 6$ with 44% of 64 journey cells under 30 users ($N = 3 \times 10^4$) unconfounded DDA bias is at most 11% of the credit and vanishes by $N = 10^6$ (E21). The power control is a global-holdout Hausman test of “DDA total equals incremental total” (statistic $\tilde{Y}_0 - \hat{v}(\emptyset)$; E10, 200 replications, $N = 2 \times 10^5$): rejection 0.04 ± 0.014 under no confounding, $t = 29.5 \pm 1.1$ under benchmark and 16.9 ± 1.1 under half confounding.

Table 11.1: Benchmark DGP ($n = 3$, channel 3 causally null): exact allocations by rule (Dive 13 E1, quadrature error $< 10^{-12}$).

Rule	Ch. 1	Ch. 2	Ch. 3 (null)	Sum
$\text{Sh}(v_{\text{pop}})$, causal target	0.0230	0.0220	0.0000	0.04497 (realised lift)
Removal effects τ_j	0.0241	0.0230	0	0.0471
Journey-level DDA	0.0274	0.0365	0.0271	0.0911 ($= \mathbb{E}Y - v_{\text{obs}}(\emptyset)$)
Population Shapley $\text{Sh}(v_{\text{obs}})$	0.104	0.154	0.133	—

Table 11.2: Bounds on $\text{Sh}_j(v_{\text{pop}})$ at the true sensitivity parameters ($\Gamma^* = 5.31$, $\Gamma_k^* = (1.28, 1.82, 2.54)$) and robustness values from the DDA point (Dive 13 E5, E16; grid resolution 0.01). RV_0 : Γ at which each credit’s interval touches 0; RV_{rank} : Γ at which the top-two ordering is ambiguous.

Model	Widths (ch. 1, 2, 3)	RV_0	RV_{rank}
Global box	(0.16, 0.15, 0.13)	(1.8, 2.3, 2.0)	1.12
Channelwise	(0.054, 0.063, 0.066)	(1.6, 1.85, 1.65)	1.10
Channelwise + monotone	(0.035, 0.042, 0.045)	—	1.20–1.26

The experimental route recovers the truth: at $n = 3$, $N = 2\text{M}$ users over 8 cells, the pairwise-cap design gives mean (0.0230, 0.0220, 0.0001) with sd 0.0005, equal to a two-arm lift test with the same N (0.90–0.98 $\times$; E8); under strong synergy ($\theta_{12} = \theta_{23} = 2$, 3-way dividend -0.019) its bias is 0.0016 on every channel, uniform as predicted (E18). At $n = 4$ the layered design (10 cells) beats the size- ≤ 2 -holdout design (12 cells) on precision, sd 0.00051 against 0.00056, but forgoes more lift, 51.5% against 43.6% (E20).

Dive 14’s Monte Carlo uses binomial cells, $N = 2\text{--}4\text{M}$, 40–60 replications (sd of a reported sd about 10%; biases carry sd ≈ 0.00006); empirical sds match predictions (0.00034–0.00043 against 0.00037–0.00038) and identified designs are unbiased to 1.7×10^{-4} when the cap holds. Table 11.3 collects the comparison. The negative control is the ghost-ads menu $B(0, 1)$ with an additive cap, which under pairwise synergy at $n = 5$ is biased $+0.0087$ on $\text{Sh}_1 = 0.036$, -0.0084 on $\text{Sh}_3 = 0.009$ and -0.0047 on the null channel, 12–29 sd (E6). The misspecification control adds a 3-way term ($\theta_3 = 0.8$, $D_{123} = +0.009$): the closed form on $B(1, 1)$ biases channels 1–3 by $+0.0013$ each and channel 5 by exactly 0, while the BLUE biases channels 1–3 by $+0.0002\text{--}0.0003$ but channels 4 and 5 by -0.0013 and -0.0011 , the latter causally null (3–4 sd at sd 0.0003–0.0004 in E6; the dive’s headline “5.7 sd on the null channel at $N = 4\text{M}$ ” is E15’s exact computation, quoted for $\theta_3 = 1.5$); $B(1, 2)$ under a 3-way cap is unbiased to 1.7×10^{-4} . The add-up test rejects 2–3/60 under the null and 49/60 ($B(1, 1)$), 57/60 ($B(1, 2)$), 60/60 (full) against $D_{123} = 0.009$ at $N = 2\text{M}$, 18–29/60 against 0.0042 (E7). Non-identified designs have row-space residuals 0.12–0.35 and pseudo-estimates erring by 0.07–0.38 on unit-scale games; identified ones err by 0 (E10). A KernelSHAP-style size-uniform allocation over all 32 cells is 18% less precise than the 12-cell $B(1, 1)$ at equal lift (E12); Neyman allocation gains only 3–10% over equal (E8).

On the frontier at $n = 5$, $k = 2$ (E5) the variance-optimal allocation forgoes 54% of the slice’s lift; moving to 10% forgone costs $4.3\times$ in variance for the full factorial and $B(1, 1)$ alike; $B(0, 2)$ sits 2.8–3.7 $\times$ above the full-factorial frontier at every lift level, $B(2, 0)$ 3–5 $\times$; $B(1, 1)$ is within 11% of the 32-cell optimum; and at $k = 3$ the 22-cell $B(1, 2)$ is the optimum, the programme putting zero mass on the ten pairs-exposed cells. Cap-raising is nearly free at $n = 5$ ($B(1, 2)$ under a 3-way cap: $\text{tr } V \cdot N = 2.29$ against 2.20, +4%) and $B(2, 2)$ under a 4-way cap beats $B(1, 1)$ at $n = 7$ (3.68 against 4.40), at 2–4 $\times$ the cells (E11). Scaling in n is decisive for holdout-only designs: $\text{tr } V \cdot N$ grows $0.81 \rightarrow 6.1$ for $B(1, 1)$ from $n = 3$ to 8 (about n^2) but $0.81 \rightarrow 156$ for

Table 11.3: Design comparison under a pairwise cap unless stated (Dive 14 E3, E4, E9, E11, E13, E14). $\sum c^2$: squared weight sum, closed form $\rightarrow$ BLUE (equal cell variances); κ : lift price of Equation (11.11); last column: Neyman-allocated BLUE $\text{tr } V \cdot N$ on a monotone pairwise game ($n = 3 \rightarrow 8$, E9), and for $B(1, 2)$ the $n = 5$ logistic DGP against $B(1, 1)$ (E11). Ratios are exact given the game and vary $\pm 15\%$ across the six games tried.

Design	Cells, $n=5$	Leak, order $k+1$	$\sum c^2$	κ , $n=5$	κ , $n=6$	$\text{tr } VN$, $n=3 \rightarrow 8$
Full factorial	32	—	—	1.05	1.29	—
$B(1, 1)$, $2n + 2$ cells	12	1/6	1.00 $\rightarrow$ 0.43	1.11	1.50	0.81 $\rightarrow$ 6.1
$B(0, 2)$, holdout-only	17	1/3	4.0 $\rightarrow$ 0.86	2.91	7.26	0.81 $\rightarrow$ 156
$B(2, 0)$, bottom-only	17	1/3	—	5.01	16.3	—
$B(1, 2)$, 3-way cap	22	1/12	—	optimum	—	2.29 vs 2.20 ($n=5$)
$B(0, 3)$, 3-way cap, $n=6$	—	1/4	15.9 $\rightarrow$ 2.27	—	—	—

$B(0, 2)$ (about n^4 : $\binom{n}{2}$ cells each with weight $\frac{1}{2}$), $25\times$ worse at $n = 8$, $3.5\times$ at $n = 5$ (E9).

11.6 Limitations and the devil’s advocate

The dives’ strongest counterarguments: “*Confounding breaks identification—trivial.*” As a slogan; the content is survival under monotone restrictions, a known total and a smooth family at $n = 2$, the sharpness that follows from the obstruction being one table’s off-diagonal, and the honest boundary that at $n \geq 3$ single-index families are identified by the Gaussian-logistic shape, which the same data cannot test. “*Tools use paths and regularised models.*” Ordering and repetition add unobserved cells but no off-diagonal source; regularisation changes $\hat{v}_{\text{obs}}$, not its estimand. “*The bias law is first order.*” Agreed; it over-predicts by up to $2\times$ at the benchmark and serves to motivate the channelwise model and fix sign and ordering. “*Bounds at Γ_k^* are tautologically covering.*” The deliverable is the robustness value; the intervals are conservative by $2\text{--}3\times$. “*The design is commercially unrealistic.*” “Expose only j ” cells cost about 50% of a slice’s lift at equal allocation; Dive 14’s answer is that the lift budget, not the cell count, binds, and it can be 1–3% by concentrating mass on the no-holdout cell—though cross-platform holdout coordination exists today only inside walled gardens. “*Advertisers want removal effects.*” But $\sum_j \tau_j \neq$ total lift: -20% under overlapping reach ($\sum(|T| - 1)D(T) = -0.079$ on 0.39) and $+47\%$ under synergy (E6). “*Caps are arbitrary.*” Testable at 82–100% power, but with df 1: a 3-way dividend cancelling in the symmetric sector is invisible. “*The κ law is an asymptotic.*” Within 3% at $\ell = 5\text{--}10\%$.

What the attack rounds changed: the closed-form bias law was refuted and replaced by the operator law; box- Γ bounds gave way to the channelwise model ($3\times$ tighter); a discrete- U swap that would not converge under the total-lift constraint was replaced by the continuous- U construction with a quadratic confounder term; Dive 13’s “costs the precision of one lift test” was found to *overstate* the cost, the closed form paying $2.3\times$ in variance against the BLUE; and the aliasing law was found to describe the closed form only, the BLUE’s aliasing being design-dependent and only computed. Remaining limits: binary outcomes (revenue changes s_A only; the Γ models need Tan-style density-ratio bounds); perfect suppression (BL 48); pilot-rate plug-ins; the trace criterion; the df-1 result proved only for the symmetric sector; no dynamic or adaptive delivery, whose divergence bias (Chapter 8) compounds this one (BL 44); and a cost model counting lift forgone but not coordination cost or option value.

11.7 Discussion: impacts

For practice the first message is a verdict: DDA credits cannot be causal, not because Shapley is the wrong operator but because the data lack the off-diagonal cells every causal axiom needs. The benchmark prices this at realistic scale—30% of credit to a null channel, a ranking that flips under a 10% selection odds ratio—and Construction 11.10 gives any attribution vendor a sensitivity column to add today: one Γ_k per channel and the Γ at which the ranking dissolves. The global-holdout Hausman test is a cheap companion, since DDA’s

total minus a global holdout's lift is exactly $\mathbb{E}[Y(0)] - \mathbb{E}[Y(0) \mid Z = \emptyset]$. For platforms whose delivery creates the confounding, Law 11.11 says inflation scales with targeting slope at an exchange rate of about one half against causal effect (BL 45), the identification-side counterpart of Berman's incentive result.

The second message is constructive. An advertiser already running ghost-ad holdouts can add “this channel only” cells and obtain from $2n + 2$ cells a Shapley allocation that adds up to the global-holdout lift, with a free specification test whose sign names the missing interaction, priced in users times lift forgone. Dive 14's protocol: choose k from a pilot's add-up test (or start at 2) and run $B(\lfloor k/2 \rfloor, \lceil k/2 \rceil)$, never holdout-only for $n \geq 5$; put nearly all users in the no-holdout cell and spread the lift budget with $g_A \propto \|C_A\| s_A / \sqrt{\ell_A}$, sizing by $N\ell \geq \kappa/\delta^2$; estimate three ways (closed form, null channels protected; add-up-projected; BLUE) and report the add-up χ_1^2 ; feed $\hat{Sh}_j \pm \text{sd}$ into the act/hold layer of Chapter 7. Inside one walled garden, with placements as channels, it is implementable now.

For theory the chapter settles M8 nonparametrically and sharpens it parametrically: the class in M8(i) is $v_* + \ker Sh$, the impossibility in M8(ii) needs neither symmetry nor linearity, and the one $\text{Stab}(j)$ -sector lemma that proves the design theorem also yields the estimator, the exact aliasing coefficients and the single over-identifying restriction. It touches the k -additive approximation literature (Grabisch; Pelegrina et al., 2026, who state that no guarantee exists when only a subset of coalitions is observed) and the odd-part representation of Fumagalli et al. (2026), whose $f(S) - f(S^c)$ is the closest existing statement to “bottom and top layers enter additively” (BL 50). Outward: Chapter 8 supplies the fixed-policy estimand with which the table must merge for paths and pacing (BL 31, BL 44); Chapter 3 owns the EVSI machinery that closes the cost loop (BL 46 merges with BL 7); Chapter 4's scheduler hosts adaptive cap-raising (BL 49); Chapter 7 consumes the intervals.

11.8 Further exploration

Dive 13's next steps were the general design proof (now Theorem 11.12); single-index LP bounds replacing the cell-wise model by the identified set of (β, g, a, θ) under a smooth family with free U -distribution, to see how much of the 2–3 $\times$ conservatism disappears (BL 43); the table for ordered and repeated exposure, with the bias family indexed by policy as well as journey (BL 44); a field run of the $\frac{1}{2}$ (solo + removal) rule reporting $D_{123}/6$ as the residual; the Berman-style equilibrium with the exchange rate (BL 45); and the decision layer (BL 41). Dive 14 adds the multi-channel Test & Roll maximising EVSI minus κ -priced lift (BL 46); the df-1 theorem via the S_n -isotypic decomposition of Hamming-weight shells (BL 47); compliance-aware cells with an exposure-weighted incidence matrix (BL 48); an always-on $B(1, 1)$ whose add-up statistic triggers the $B(1, 2)$ cells, with Chapter 4's scheduler and Chapter 7's threshold machinery (BL 49); and the paired-allocation conjecture $f_A = f_{N \setminus A}$, which might give κ a closed form (BL 50).

In the compiler's judgement the most promising extension is BL 48 together with the field protocol: ghost-ad suppression is never perfect, and if logged compliance can enter the incidence matrix without losing the add-up restriction, one walled-garden campaign could deliver the first causal Shapley allocation with a specification test attached. Second is BL 43, because the 2–3 $\times$ gap between cell-wise bounds and the single-index truth is exactly the price of the parametric assumption that Remark 11.9 says does all the identifying work in any “causal attribution” model fitted to logs alone.

Reviewer's note (compiler)

The exact swap needs a term the family description omits. DGP B of Proposition 11.8 carries a quadratic confounder coefficient $\beta_2 = -0.193$ while A has none; “smooth single-index family” means the family with U^2 in the outcome index, introduced in Round 3 after the discrete- U swap failed (max table difference 6×10^{-3}). Eight free parameters against seven cell probabilities plus the total-lift constraint make the swap exactly determined by counting, so 10^{-17} exactness is a solved square system; the content is existence under the sign restrictions.

Parameter count at $n = 3$. Remark 11.9 quotes “11 parameters”; the E19 code optimises 10 (three a , three g , β_0 , β , two free θ). The conclusion is unaffected.

Two θ_3 values for one dividend. Dive 14 §1 attributes $|D_3| \leq 0.009$ to $\theta_3 = 0.8$ and E6 runs at 0.8, but the Round-2

attack and the INDEX log line cite “ $D_{123} = +0.009 \dots (5.7 \text{ sd at } N = 4M \text{ for } \theta_3 = 1.5)$ ”. The E15 code computes exact biases and sds at $N = 4M$ for $\theta_3 \in \{0, 0.8, 1.5\}$, so the 5.7 sd almost certainly belongs to the larger $\theta_3 = 1.5$ dividend, not to $D_{123} = 0.009$; the E6 bias of $-0.0013 / -0.0011$ at sd 0.0003–0.0004 (3–4 sd) is the number to trust for the 0.009 case, and Section 11.5 reports both.

Clean-room case counts are different enumerations. “1,042 cases ($n \leq 8$)” against “1,070 cases ($n \leq 7$)” is not a contradiction: the code enumerates $(m_b, m_t) \in \{0, \dots, n-1\}^2$ with $m_b + m_t \leq n+1$, the clean room evidently $\{0, \dots, n\}^2$ ($48 + 100 + 180 + 294 + 448 = 1,070$). Neither set contains the other; both report zero mismatches.

Ranges from different experiments. “ $B(1, 1)$ within 5–16%” is the κ ratio at $n = 5, 6$ ($1.11 / 1.05, 1.50 / 1.29$) while “within 11%” is the E5 frontier at $n = 5$; “2.8–5.6 $\times$ ” for $B(0, k)$ is κ -based and “2.8–3.7 $\times$ ” frontier-based. “85–99%” of the BLUE gain captured by the add-up projection is not reproducible from the three quoted triples (89–95%). “ $\approx n^4$ ” for holdout-only variance is a heuristic: $0.81 \rightarrow 156$ is a factor 193 against $(8/3)^4 = 51$, so the exponent is if anything understated.

Corrections across dives and log lines. Dive 13’s log line says the $2n + 2$ design identifies Shapley “at the precision of one lift test”; Dive 14 E4 shows that closed form pays $2.3\times$ in variance against the BLUE on the same cells, so the true cost is lower. The same log line states the layered rule as “ $m = \lceil k/2 \rceil$ ” without the $\lfloor n/2 \rfloor$ boundary the verifier added in Round 4.

Monte Carlo widths and pinv. The rejection rate 0.04 ± 0.014 (200 replications) and the power counts 49/60, 57/60 carry binomial uncertainty of about ± 0.05 . Identification checks are exact ranks of 0/1 incidence matrices, so no BL 82-type FIM dependence; but the BLUE uses a pseudo-inverse of $X^T \Sigma^{-1} X$ (rcond 10^{-12}), and the verifier found the frontier optimiser collapsing onto $f_N = 1$, where that matrix is singular and the pseudo-inverse silently returns $\text{tr } V = 0$ —the dive therefore trusts only the fixed- ℓ and group- L_1 forms. Log timestamps ($\sim 15:40, \sim 19:45$) match the file writes (15:40, 19:44 UTC).

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- [25] Program (internal): Dive 10, file 10_algorithmic_delivery_estimands.md (BL 31); Dive 03/Dive 04 (EVSI, BL 7); Dive 05 (scheduler); Dive 08/Dive 09 (act/hold protocol, BL 41).

Chapter 12

The calibrated simulator: certifying an agent-based market model against held-out experiments

Provenance

Source dives: Dive 15 *The calibrated simulator: a certification harness, the certification law, and the quantitative counterfactual bar* (run 2026-09-02, file `15_calibrated_simulator.md`, written 2026-09-02 23:32 UTC). **Code:** `code/15_calibrated_simulator.py`. **Frontier-study problem(s) addressed:** E9 — a calibrated simulator whose counterfactual predictions match held-out geo-experiment results at scale, “the necessary bar before simulation can substitute for any experiment”; H7 — a theory of *when* synthetic consumers are valid (contamination, distribution shift, homogeneity). Grand challenge 3. **Backlog items closed:** none; **opened:** BL 51–BL 54.

Key ideas in this chapter

- The *counterfactual bar* is a residual-variance threshold $\omega_\star^2 = \gamma c + \sqrt{\gamma^2 c^2 + 2\gamma c v_x}$, the point at which an experiment’s value of information stops covering its cost c given the simulator as prior; for go/no-go decisions it is per decision, $\sigma_{pp} \psi(|m - \tau_0|/\sigma_{pp}) \leq c$ (clean-room exact to 0.0004).
- The certificate is the exactly unbiased excess variance $\hat{\omega}^2 = [r^\top r - \sum_e (1 - h_{ee}) v_e] / (E - 2)$ (chi-square UCB coverage 0.949–0.955 for $E = 8$ –120); certification is a variance-ratio test in one gap ratio $\kappa = (\omega_\star^2 + \bar{v}) / (\omega^2 + \bar{v})$, $E_{\text{exact}} = \min\{E : \chi_{0.8, E-2}^2 / \chi_{0.05, E-2}^2 \leq \kappa\}$: 157 experiments certify a simulator half as bad as the bar with tests as noisy as the bar (MC 157 ± 2). Library *precision* should match the bar ($\nu_\star \approx 0.85$, cost flat within 1%, 2.6 – $2.8\times$ dearer at $\nu = 0.1$), refuting the dive’s Round-1 rule “match the simulator”.
- The noise-floor contamination test has blind spot $\lambda < \omega^2 / (\omega^2 + v)$ and is evaded by a noise-adding memoriser, which a raw-data re-estimation test catches (power 0.71); outcome-selected libraries make a simulator look better than perfect ($\hat{\omega}^2 = -0.08$).
- Variance collapse in *levels* is repaired exactly by one latent-scale slope (additive Platt recalibration errs -15.7% / $+14.5\%$ out of hull); heterogeneity in *sensitivity* needs a mixture parameter (latent family -12.7% at rare baselines). Certifying at every new experiment falsely certifies 28% of simulators at the bar; a prequential e-process holds 3.3% at power 0.86.

12.1 Problem and motivation

The frontier study’s third grand challenge is a market simulator—LLM agents or hybrid structural—held to a *counterfactual bar*: predictions validated against a held-out library of real experiments before any measurement claim rests on it. E9 names the bar without defining it; H7 asks when synthetic consumers are valid and lists the failure modes: training contamination, distribution shift for genuinely new products, and homogeneity that understates the human variance which is itself the central measurement obstacle. The literature reports correlations, sign-agreement rates and “corrected” mean-squared errors, none of which says whether a simulator is good *enough*, enough for what, or how many experiments it takes to find out.

Dive 15 defines the bar, derives what a library of E experiments can certify and at what cost, and specifies the harness. The simulator is a black box emitting a counterfactual lift prediction; nothing depends on its being an LLM, and that is the point: the certificate is earned against experiments, not argued from behavioural realism. Five claims were reproduced clean-room to Monte Carlo precision by an independent subagent given only their statements. The harness is a derived specification, not a fielded system, and the mapping from an agent population to a single residual-variance parameter is the speculative step; the dive labels it so.

12.2 Background: the ideas this chapter builds on

12.2.1 Generative agents and synthetic consumers (Park et al., 2023, 2024; Horton, 2023; Brand et al., 2023)

Park et al. (2023) introduced *generative agents*: LLM-backed characters with a memory stream of natural-language observations, retrieval scored by recency, importance and relevance, periodic *reflection* that abstracts memories into higher-level inferences, and planning that turns them into behaviour. Park et al. (2024) grounded the architecture in people: two-hour interviews with 1,052 U.S. respondents became the agents' memories, and the agents reproduced each respondent's General Social Survey answers at 85% of the respondent's own two-week test-retest accuracy (*genagents* ships the agent bank). Horton (2023) framed LLMs as *homo silicus*, models of humans endowed with preferences by prompt and run through classic behavioural-economics experiments, recovering the qualitative human patterns; Brand, Israeli and Ngwe (2023) showed GPT answers to conjoint-style choices yield downward-sloping demand curves and willingness-to-pay that track human values. This line supplies the object certified here: a synthetic population that, given a pre-period description and an intervention, emits a predicted lift.

12.2.2 Validating simulators against experiments (Hut and Masoero, 2026; Hewitt et al., 2026; Bisbee et al., 2024)

Hewitt, Ashokkumar, Ghezze and Willer (2026) had GPT-4 predict 476 survey-experiment effects: $r = 0.85$ (0.90 on unpublished studies), with systematic over-estimation of magnitudes and $r = 0.27$ on field experiments. Hut and Masoero (2026, Amazon) is the closest object to this chapter: 67 historical marketing A/B tests each simulated by 1,000 customer-mapped agents, sign agreement 0.70, launch-decision alignment 0.41, and a *corrected MSE* that subtracts each test's sampling variance from its squared residual (the moment idea below), which a Platt-scaling recalibration on 16 tests reduces about $77\times$ "to near the noise floor". Against this, Bisbee et al. (2024) measure *variance collapse* in silicon samples (dispersion 16.1 against the human 31.4) and instability across prompts and time, and two 2026 preprints find descriptive fit does not predict causal-effect error. Missing from all of it: a bar on the residual tied to cost and decision value, a sample-size law for the library, a contamination test formalised against *noisy* labels, and a structural recalibration validated for transport—the dive's four constructions.

12.2.3 Excess variance and prediction intervals (DerSimonian and Laird, 1986; Riley, Higgins and Deeks, 2011)

Random-effects meta-analysis sees k estimates $y_i = \theta_i + \varepsilon_i$ with known sampling variances v_i and $\theta_i \sim (\mu, \tau^2)$. DerSimonian and Laird estimate the between-study variance by moments: with $Q = \sum_i w_i(y_i - \bar{y}_w)^2$, $w_i = 1/v_i$, $\mathbb{E}[Q] = (k-1) + \tau^2 C$ for a known C , so $\hat{\tau}^2 = \max\{0, (Q - (k-1))/C\}$ —variance *in excess of* sampling noise (Veroniki et al., 2016, review sixteen variants). Riley, Higgins and Deeks (2011) insist the user's object is the *prediction interval* for a new study, $\hat{\mu} \pm t_{k-2} \sqrt{\hat{\tau}^2 + \text{se}(\hat{\mu})^2}$. The chapter's certificate is the excess variance of a *regression* of experimental estimates on simulator predictions, with one refinement: residual r_e has variance $(1 - h_{ee})(\omega^2 + v_e)$, h_{ee} the leverage (diagonal of $H = X(X^T X)^{-1} X^T$), so the noise to subtract is $\sum_e (1 - h_{ee})v_e$, not $\sum_e v_e$.

12.2.4 Variance-ratio tests and their power

If n residual degrees of freedom of a Gaussian regression have variance σ^2 , then $ns^2/\sigma^2 \sim \chi_n^2$. Testing $H_0 : \sigma^2 \geq \sigma_0^2$ rejects when $ns^2/\sigma_0^2 \leq \chi_{a,n}^2$ (lower a -quantile); against $\sigma_1^2 < \sigma_0^2$ the power is $\mathbb{P}(\chi_n^2 \leq (\sigma_0^2/\sigma_1^2)\chi_{a,n}^2)$, a function of the ratio and n only, and the sample size for power $1 - b$ is the smallest n with $\chi_{1-b,n}^2/\chi_{a,n}^2 \leq \sigma_0^2/\sigma_1^2$. With $\chi_{p,n}^2 \approx n + z_p\sqrt{2n}$ this is $n \approx 2(z_b + \kappa z_a)^2/(\kappa - 1)^2$ for ratio κ ; the asymmetric κz_a is what the dive's Round-1 law omitted.

12.2.5 Platt scaling versus latent-scale recalibration (Platt, 1999)

Platt scaling fits a monotone map from a raw score f to a calibrated probability, $\mathbb{P}(y = 1 \mid f) = 1/(1 + \exp(Af + B))$, on held-out data: two parameters applied *after* the model; Hut and Masoero use the additive lift-scale analogue $\tau \approx \alpha + \beta s$. The dive contrasts this with recalibration *inside the link*: if purchase is a probit event, one slope on the latent index reproduces lifts at every baseline and dose, which no affine map on the lift scale can.

12.2.6 Recursive residuals and e-processes (Brown, Durbin and Evans, 1975; Ramdas et al., 2023)

In a Gaussian linear model fitted sequentially, the standardised one-step-ahead *recursive residuals* $u_t = (y_t - x_t^\top \hat{\beta}_{t-1}) / \sqrt{1 + x_t^\top (X_{t-1}^\top X_{t-1})^{-1} x_t}$ are i.i.d. $\mathcal{N}(0, \sigma^2)$ under the model (the *prequential* property: forecast, then observe). An *e-process* for H_0 is a nonnegative process with $\mathbb{E}_{H_0}[M_t] \leq 1$ at every stopping time; Ville's inequality gives $\mathbb{P}_{H_0}(\sup_t M_t \geq 1/a) \leq a$, so “reject when $M_t \geq 1/a$ ” is valid however often one looks; the running likelihood ratio of i.i.d. observations is the canonical case.

12.2.7 Expected value of sample information as the bar (Chapter 3)

An experiment's value to a decision is its expected value of sample information: expected Bayes loss under the current posterior minus expected Bayes loss after the experiment, averaged over outcomes not yet seen (Raiffa and Schlaifer, 1961; Chapter 3). The dive treats the certified simulator as the *prior*: it substitutes for an experiment when the experiment's EVSI against the simulator's recalibrated predictive distribution falls below c . The bar is therefore a variance, and it depends on the decision.

12.3 Setup and assumptions

The library is E past randomized experiments (geo lift tests, ghost-ad holdouts) with true lift τ_e and estimate $\hat{\tau}_e = \tau_e + \varepsilon_e$; before each reads out the simulator emits a prediction s_e . The recalibration regression is $\hat{\tau}$ on $(1, s)$ with design matrix X , hat matrix H , leverages h_{ee} , residuals r , and $\bar{v} = E^{-1} \sum_e v_e$; $\rho = \omega^2/\omega_\star^2$ and $\nu = \bar{v}/\omega_\star^2$ are simulator quality and library noise relative to the bar $\omega_\star^2$ (Theorem 12.6).

Assumption 12.1 (A1–A3). (A1) $\varepsilon_e \sim \mathcal{N}(0, v_e)$, v_e known from the experiment's own analysis, independent across experiments. (A2) s_e is fixed before $\hat{\tau}_e$ is observed; its violation is *contamination*, a fraction λ of items on which the simulator returns the published $\hat{\tau}_e$. (A3) $\tau_e = \alpha + \beta s_e + u_e$, $u_e \sim \mathcal{N}(0, \omega^2)$ independent of s_e ; ω^2 is the simulator's *counterfactual residual variance* after the best affine recalibration.

Definition 12.2 (Target decision). A new decision has prediction $m = \hat{\alpha} + \hat{\beta} s_{\text{new}}$ and predictive variance $V = \hat{\omega}^2 + (\hat{\omega}^2 + \bar{v})[1/E + (s_{\text{new}} - \bar{s})^2/S_{ss}]$, $S_{ss} = \sum_e (s_e - \bar{s})^2$. Under *quadratic* loss (profit $\tau x - \gamma x^2/2$; acting on $\hat{\tau}$ loses $(\tau - \hat{\tau})^2/2\gamma$) or *threshold* loss (go/no-go at τ_0 , losing $|\tau - \tau_0|$ on the wrong side), an experiment costs c and returns $x = \tau + \mathcal{N}(0, v_x)$.

The structural sub-model for homogeneity: consumer i buys iff $U_i + a_i d + \eta_i > 0$, $U_i \sim \mathcal{N}(\mu, \sigma_H^2)$, $\eta_i \sim \mathcal{N}(0, 1)$, dose d , sensitivity $a_i = a$ for a responsive fraction q and 0 otherwise; with $\varsigma = \sqrt{1 + \sigma_H^2}$ (the dive's s) the true lift is $L(\mu, d) = q[\Phi((\mu + ad)/\varsigma) - \Phi(\mu/\varsigma)]$. A *homogeneous* simulator has no idiosyncratic dispersion: it matches the baseline through $\mu' = \Phi^{-1}(p_0)$ and predicts $\Phi(\mu' + a'd) - \Phi(\mu')$.

12.4 Main results

12.4.1 The excess-variance certificate and the certification law

Proposition 12.3 (Exactly unbiased excess variance). *[established in the dive]* Under A1–A3,

$$\hat{\omega}^2 = \frac{r^\top r - \sum_e (1 - h_{ee}) v_e}{E - 2} \quad (12.1)$$

has $\mathbb{E}[\hat{\omega}^2] = \omega^2$ exactly. For constant v , $r^\top r = (\omega^2 + v)\chi_{E-2}^2$, so $\text{sd}(\hat{\omega}^2) = (\omega^2 + v)\sqrt{2/(E-2)}$ and

$$\text{UCB}_{1-a} = \frac{(E-2)(\hat{\omega}^2 + \bar{v})}{\chi_{a,E-2}^2} - \bar{v} \quad (12.2)$$

is an exact upper confidence bound. The certificate “ $\omega^2 \leq \omega_\star^2$ ” is issued iff $\text{UCB} \leq \omega_\star^2$.

Since $r = (I - H)(u + \varepsilon)$, $\mathbb{E}[r^\top r] = \omega^2(E-2) + \sum_e (1 - h_{ee})v_e$. Hut and Masoero's corrected MSE is the same object without the leverage term and the two recalibration degrees of freedom. E1 (Table 12.1) confirms mean, sd and 0.95 coverage from $E = 8$ to 120 and shows the practical fact: the estimator is honest but *noisy*—at $E = 30$ its sd (0.275) exceeds the estimand (0.25), and it is negative 41% of the time at $E = 8$, 18% at $E = 30$.

Law 12.4 (Certification law). *[established in the dive]* Let $\kappa = (\omega_\star^2 + \bar{v})/(\omega^2 + \bar{v}) = (1 + \nu)/(\rho + \nu)$. With constant v the event $\text{UCB}_{1-a} \leq \omega_\star^2$ is $\chi_{E-2}^2 \leq \kappa \chi_{a,E-2}^2$, so

$$\text{power} = \mathbb{P}(\chi_{E-2}^2 \leq \kappa \chi_{a,E-2}^2), \quad E_{\text{exact}} = \min \left\{ E : \frac{\chi_{1-b,E-2}^2}{\chi_{a,E-2}^2} \leq \kappa \right\}, \quad (12.3)$$

with $E - 2 \approx 2(z_b + \kappa z_a)^2 / (\kappa - 1)^2 = 2(z_b + \kappa z_a)^2 (\rho + \nu)^2 / (1 - \rho)^2$.

Rearranging Equation (12.2), certification is $(\hat{\omega}^2 + \bar{v})/(\omega^2 + \bar{v}) \leq \kappa \chi_{a,E-2}^2 / (E - 2)$, and the left side is $\chi_{E-2}^2 / (E - 2)$: the variance-ratio test with ratio κ . *Everything depends on κ alone.* A simulator half as bad as the bar ($\rho = 0.5$) with experiments as noisy as the bar ($\nu = 1$) needs 157 experiments; at $\nu = 0.5$, 82 (and, by the law alone, ≈ 55 with experiments four times more precise, $\nu = 0.25$); a simulator at $\rho = 0.7$ with $\nu = 1$ needs 480 (E2, Table 12.2; twelve cells within Monte Carlo resolution, power at E_{exact} 0.79–0.83). Hut and Masoero's 67 tests can certify only a bar the simulator beats by a wide margin.

Corollary 12.5 (Match library precision to the bar). *[numerically established]* If experiment cost is $\propto 1/v$ (geo-weeks), total certification cost is $\propto E(\rho, \nu)/v$; its minimiser is $\nu_\star \approx 0.81$ – 0.91 for $\rho = 0.25$ – 0.7 , the surface is flat there ($\text{cost}(\nu_\star)/\text{cost}(1) = 0.99$ – 1.00), and over-precise libraries waste money ($\nu = 0.1$ costs 2.6 – $2.8\times$ more). Run library experiments at the precision of the bar, $\bar{v} \approx \omega_\star^2$.

12.4.2 The counterfactual bar

Theorem 12.6 (Quadratic-loss bar). *[established in the dive]* With prior $\mathcal{N}(m, V)$ from the recalibrated simulator and an experiment of variance v_x , the posterior variance is $(1/V + 1/v_x)^{-1}$ and $\text{EVSI}_{\text{quad}}(V) = V^2/[2\gamma(V + v_x)]$, so the simulator substitutes ($\text{EVSI} \leq c$) iff

$$V \leq \omega_\star^2 := \gamma c + \sqrt{\gamma^2 c^2 + 2\gamma c v_x}. \quad (12.4)$$

The variance reduction is $V - Vv_x/(V + v_x) = V^2/(V + v_x)$; equating EVSI to c gives the positive root of $V^2 - 2\gamma cV - 2\gamma cv_x = 0$ (Monte Carlo agrees within 0.8%; E3a). In units of v_x with $\tilde{c} = c/(v_x/2\gamma)$ (cost relative to the loss a one-experiment-wide prior carries), $\omega_*^2/v_x = 0.25/0.37/0.64/1.00/1.62$ at $\tilde{c} = 0.05/0.1/0.25/0.5/1$; the simulator must be worth 4.0/2.7/1.6/1.0/0.6 *experiment-equivalents* $EE := v_x/\omega^2$. A simulator exactly as informative as one experiment clears the bar only when the experiment costs half the loss its own width carries.

Proposition 12.7 (Per-decision bar for threshold decisions). *[established in the dive] For go/no-go at τ_0 with loss $|\tau - \tau_0|$ on the wrong side, the exact preposterior value of the experiment is*

$$\text{EVSI}_{\text{thresh}}(m, V) = \sigma_{pp} \psi\left(\frac{|m - \tau_0|}{\sigma_{pp}}\right), \quad \sigma_{pp} = \frac{V}{\sqrt{V + v_x}}, \quad \psi(z) = \phi(z) - z(1 - \Phi(z)), \quad (12.5)$$

so the harness should emit the per-decision verdict “experiment iff $\sigma_{pp} \psi(|m - \tau_0|/\sigma_{pp}) > c$ ”.

The updated posterior mean $m' = m + [V/(V + v_x)](x - m)$ is preposteriorly $\mathcal{N}(m, \sigma_{pp}^2)$, and the value is $\mathbb{E}[(m' - \tau_0)^+] - (m - \tau_0)^+$, the normal loss integral—Chapter 7’s probability-of-regret logic with the simulator as prior. The quadratic bar is wrong for thresholds by a factor depending on where the prediction sits ($\text{EVSI}_{\text{thresh}}/\text{EVSI}_{\text{quad}}$ from 3.5 at $m = 0$ to 0.00 at $m = 1$, $V = 0.25$; E3c). The closed form checks at $(m, V) = (0, 1), (0.5, 1), (1, 2)$: 0.2821/0.0998/0.1234 against Monte Carlo 0.2800/0.0989/0.1237 and clean-room 0.2822/0.1000/0.1230 ± 0.0003 (E12a).

12.4.3 Contamination: the noise-floor test and its blind spot

Proposition 12.8 (Noise-floor test and blind spot). *[established in the dive] An honest simulator cannot predict ε_e , so $\mathbb{E}[r^T r] \geq \sum_e (1 - h_{ee})v_e$; under $H_0 : \omega = 0$ with homoscedastic v , $r^T r / \sum_e (1 - h_{ee})v \sim \chi_{E-2}^2/(E - 2)$. Reject “honest” when $r^T r < [\chi_{a,E-2}^2/(E - 2)] \sum_e (1 - h_{ee})v$: residual variance below the label noise proves memorisation. A memoriser of fraction λ has expected residual variance $(1 - \lambda)(\omega^2 + v)$, so the test has power only when $(1 - \lambda)(1 + \omega^2/v) < 1 - z_a \sqrt{2/(E - 2)}$; the blind spot is $\lambda < \omega^2/(\omega^2 + v)$.*

A simulator with real error ω^2 can hide a memorised fraction up to $\omega^2/(\omega^2 + v)$: at $E = 30$, $v = 1$, $\omega = 0.5$ (boundary 0.2), rejection at $\lambda = 0/0.2/0.4/0.6/0.8/1$ is 0.01/0.09/0.32/0.70/0.95/1.00 (clean-room 0.011/0.073/0.303/0.714/0.976/1.000), and at $\omega = 1$ (boundary 0.5) 0.00/0.01/0.06/0.32/0.79/1.00 (E6a). The boundary is a soft transition with power about equal to size (the clean-room subagent’s caveat). Worse, the test is *strategically evadable*: a memoriser adding $\mathcal{N}(0, v + \omega^2)$ noise to the published estimate passes it (rejection 0.18) and presents as an honest $\omega = 0.5$ simulator (E6b). The floor test is necessary, never sufficient.

Construction 12.9 (Re-estimation test). *[established in the dive] With raw data, re-estimate every library lift by a second valid estimator $\hat{\tau}'_e = \tau_e + \varepsilon'_e$ (different control geos, different covariate adjustment, CUPED against raw), $\text{Corr}(\varepsilon, \varepsilon') = r < 1$, and form $D_e = (s_e - \hat{\tau}'_e)^2 - (s_e - \hat{\tau}_e)^2 - (v'_e - v_e)$. An honest simulator has $\mathbb{E}[D_e] = 0$ whatever its ω ; a memoriser has $\mathbb{E}[D_e] = 2\lambda v(1 - r) > 0$, and adding noise does not help because its output still tracks ε , not ε' . Test by a one-sided t -test on D_e .*

At $E = 30$ the noise-adding memoriser is caught with power 0.71 at $r = 0.5$ (0.40 at $r = 0.8$; E6d); against partial contamination power is weak (0.35 at $\lambda = 0.4$, $r = 0.5$; E6c) and scales with $E(1 - r)^2$, so the re-estimator should minimise r (disjoint control geos).

12.4.4 Which recalibration family transports

Proposition 12.10 (Latent-scale exactness under level collapse). *[established in the dive] With $q = 1$, $\Phi(\mu/\zeta + (a/\zeta)d) - \Phi(\mu/\zeta) \equiv \Phi((\mu + ad)/\zeta) - \Phi(\mu/\zeta)$: a homogeneous simulator recalibrated on the latent scale with one slope $a'' = a/\zeta$ is exact at every baseline and dose (clean-room maximum error 1.1×10^{-16}). With $q < 1$ (heterogeneity in sensitivity) the one-slope family fails and only the mixture family $q'[\Phi(\mu' + a''d) - \Phi(\mu')]$ transports.*

Level collapse is harmless once recalibration happens inside the link, but the additive family $\alpha + \beta s$ is not: on a 24-experiment library (doses 0.3–1.0, baselines $\mu \in [-1.5, -0.5]$, 200k users per arm, 300 replications; E5) its in-range relative rmse is 0.022 against 0.002 for the latent family, but at dose 2.5 it errs -15.7% and at the rare baseline $\mu = -3$ by $+14.5\%$, while the latent family stays within 0.3%. With $q = 0.4$ the latent family errs $+3.8\%$ at dose 2.5 and -12.7% at $\mu = -3$ (clean-room $-13.0\% / +3.5\%$); the mixture transports (rmse 0.033/0.043; clean-room recovers $q' = 0.400$, $a'' = 0.500$ exactly), and the additive family with the same two parameters fails at $-15.9\% / +15.2\%$. For H7: *what* the simulator gets wrong determines the family, and a certificate is valid only inside the library’s (dose, baseline) hull unless the family is structural. E4 makes the factual/counterfactual point directly: $\text{Corr}(\text{level error}, \text{lift error}) = +0.02$, and $6\times$ worse level fit with the right sensitivity gives $4.6\times$ smaller lift error (0.019 against 0.087).

12.4.5 Shift, selection and sequential certification

Shift (E7, $E = 60$). With $\omega^2(x) = 0.4^2 + 0.6x^2$ in a novelty covariate x and a library concentrated near $x = 0$, the global prediction interval covers 0.999/0.94/0.71/0.54 at $x = 0/1/2/3$; a kernel-local excess variance restores 0.93/0.92/0.86/0.80 with effective sample sizes 36/47/18/4. Certification is local; H7’s “genuinely new products” is $x = 3$, where the harness has no certificate to give.

Selection (E8). Selection on the simulator’s prediction leaves $(\alpha, \beta, \omega^2)$ unbiased (0.002/1.001/0.249 against 0/1/0.25) because OLS conditions on the regressor. Selection on the *outcome* is fatal: keeping only positive results gives $\alpha = 0.71$, $\beta = 0.56$, $\hat{\omega}^2 = -0.08$ (better than perfect); keeping only $|t| > 1.64$ gives $\beta = 1.33$, $\hat{\omega}^2 = 0.50$, twice too pessimistic. The library must ingest every pre-registered experiment, nulls included.

Construction 12.11 (Prequential e-process certificate). [\[established in the dive\]](#) Form recursive residuals $u_E = (\hat{\tau}_E - x_E^\top \hat{\beta}_{E-1}) / \sqrt{1 + h_E}$, i.i.d. $\mathcal{N}(0, \omega^2 + \bar{v})$ under the Gaussian model, and the running likelihood ratio of the design alternative $\rho = 0.5$ against the bar $\sigma_0^2 = \omega_\star^2 + \bar{v}$; certify when it reaches 20.

Certifying at every new experiment with the fixed- E bound falsely certifies a simulator sitting exactly at the bar with probability 0.284 ± 0.010 over looks at every E from 8 to 150 (0.16 when ω^2 is 20% above the bar; E10, $\bar{v} = 0.5$). The e-process holds the false rate at 0.033 ± 0.004 and 0.009 with power 0.86 (median E 67) at $\rho = 0.5$; five geometric looks at $a = 0.01$ each do about as well (0.037/0.013; 0.86 at median E 80). This is the simplest instance of Chapter 13’s confidence-sequence problem.

12.5 Numerical evidence

All libraries are synthetic draws from A1–A3 with $s_e \sim \mathcal{N}(0, 1)$, $\alpha = 0$, $\beta = 1$; E1 uses $\omega = 0.5$, $v = 0.8$, 4,000 replications; E2 sets $\omega_\star^2 = 1$ and finds the Monte Carlo E for power 0.8 by bisection at 3,000 replications. Clean-room figures come from an independent re-implementation given only the claims.

Table 12.1: E1: the excess-variance estimator Equation (12.1) at $\omega^2 = 0.25$, $v = 0.8$ (4,000 replications; MC se of the mean ≤ 0.009). Clean-room values in parentheses.

E	8	15	30	60	120
mean $\hat{\omega}^2$	0.2395 (0.2558)	0.2530	0.2496 (0.2543)	0.2478	0.2494 (0.2483)
sd, Monte Carlo	0.594 (0.603)	0.418	0.275 (0.281)	0.196	0.137 (0.136)
sd, formula	0.606	0.412	0.281	0.195	0.137
$\mathbb{P}(\hat{\omega}^2 < 0)$	0.41	–	0.18	–	–
UCB _{0.95} coverage	0.950	0.949	0.955	0.949	0.950

The decision duel (E12b) closes the loop on 20,000 go/no-go decisions with $c = 0.05$, $v_x = 1$, $E = 60$, global bar 0.37. At $\omega^2 = 0.5$ (above the bar, so the global rule runs every experiment at loss 0.1122) the

Table 12.2: E2: library size for power 0.8 at level 0.05, exact law Equation (12.3) / Monte Carlo bisection, by $\rho = \omega^2/\omega_\star^2$ and $\nu = \bar{v}/\omega_\star^2$. Clean-room: $(0.5, 0.5) \rightarrow 82$, MC power 0.798; $(0.25, 1) \rightarrow 62$, MC power 0.807.

$\rho \backslash \nu$	0.1	0.5	1	2
0.25	14 / 13	31 / 31	62 / 62	157 / 155
0.5	39 / 40	82 / 81	157 / 157	383 / 397
0.7	129 / 131	258 / 261	480 / 465	1130 / 1109

per-decision rule runs 14% of experiments at loss 0.0831 ± 0.001 , against an oracle knowing ω^2 at 0.0805 and simulator-only at 0.0964; at $\omega^2 = 0.2$ both rules certify and match the oracle (0.0393 against 0.0390); at $\omega^2 = 1$ –2 both fall back to always-experiment (0.139/0.140), 12–15% above an oracle that still skips 17% of experiments. The *certification drag* (E13b) prices not yet knowing: at $\omega^2 = 0.2$ against a bar of 0.558 ($\rho = 0.36$, $\nu = 1.79$, $E_{\text{exact}} = 190$) the certification probability rises 0.10/0.17/0.27/0.46/0.74 at $E = 10/20/40/80/160$, and the bar rule loses 0.020–0.060 per decision more than the oracle.

12.6 Limitations and the devil’s advocate

Three headline results changed under attack. *Round 1* derived the law $E \approx 2 + 2(z_a + z_b)^2(\rho + \nu)^2/(1 - \rho)^2$ and from it the rule “match the library precision to the simulator”, $\nu_\star = \rho$. [refuted within the dive] *Round 2* found it under-predicted E by 1.5–3 $\times$ (power 0.23–0.76 at the predicted E): approximating the chi-square quantile in the denominator of Equation (12.2) by mean-minus-z-sd is wrong; the exact event is the variance-ratio test and the correct approximation carries κz_a . The rule moved to $\nu_\star \approx 0.85$ and flat—match the *bar*. *Second*, *Round 1* took the quadratic bar as *the bar*; for go/no-go its error runs from +250% to –100% (E3c), and *Round 3* replaced it with Proposition 12.7. *Third*, heavy tails and heteroscedastic v break the chi-square bound: t_3 recalibration errors inflate $\text{sd}(\hat{\omega}^2)$ 2.7 $\times$ (0.77 against 0.28 at $E = 30$) and cut coverage to 0.92; log-sd 0.8 dispersion in v with v estimated on 20 df cuts it to 0.89 (E9). A symmetric t -interval under-covered even under normality (0.875; the statistic is right-skewed); a Satterthwaite-matched chi-square with effective df from the empirical fourth moment, capped at $E - 2$, recovers about half the shortfall (0.93/0.93 at $E = 30$, 0.92/0.91 at $E = 60$) at a 15–50% wider bound (E13a). Attacks that did not land: selection on the prediction, v_e estimated on 20 df (coverage 0.948), seed sensitivity (± 0.006).

The dive’s own counterarguments: under-reported v_e makes $\hat{\omega}^2$ conservative, but *over*-reported v_e —a real risk with optimistic geo-test analyses—biases it by exactly $-(v_{\text{reported}} - v_{\text{true}})$, so a 20% over-statement of the se at $\nu = 1$ shifts $\hat{\omega}^2$ by $-0.44 \omega_\star^2$ and can certify a bad simulator. Shared shocks across experiments are absorbed into $\hat{\omega}^2$ (conservative) unless the simulator was conditioned on the same period. The bar is scalar: multichannel decisions need Chapter 7’s ellipsoid, and EVSI omits an experiment’s value as trap insurance (Chapter 4), so the bar is *lenient* toward the simulator. A memoriser trained on the full library including the internal re-estimates is invisible to both tests; only post-cutoff experiments catch it. Latent-scale exactness is specific to the probit-normal structure, and the mixture family assumes the persona set spans the responsive segment at all. *Round 4* flagged two untested optimisms: e-process residuals are independent only under Gaussianity and homoscedasticity, and the per-decision bar plugs in $\hat{\omega}^2$ clipped at zero, optimistic at small E .

12.7 Discussion: impacts

For practice the chapter turns E9’s slogan into a specification. The harness (Dive 15 §6) ingests every pre-registered experiment, nulls included, at precision $v_e \approx \omega_\star^2$, keeping raw data so a low- r second estimator exists; hashes and time-stamps predictions before readout and logs the training-cutoff split; chooses the recalibration family by the failure mode the library reveals (additive in-hull only, latent-scale for level collapse, mixture for sensitivity heterogeneity) and stamps the certificate with its hull; reports $\hat{\omega}^2$, its chi-

square or Satterthwaite bound and $EE = v_x / \text{UCB}$, certifying against Equation (12.4) for continuous decisions and emitting Equation (12.5) per threshold decision; gates on the floor, re-estimation and temporal-split tests; and updates by the e-process, re-certifying locally as new categories or doses enter. For vendors of synthetic-consumer products, a sign-agreement rate is not a credential, and the number of validation experiments a claim needs is set by Law 12.4—in the hundreds for a simulator anywhere near the bar. For a CMO the per-decision rule is the actionable object: it recovers most of the oracle’s savings (0.0831 against 0.0805) where the global bar recovers none (0.1122).

For MMM vendors the link runs through Chapter 3 and Chapter 2: a certified simulator is a prior of known variance on the lift a design would measure and can enter a Meridian, Robyn or PyMC-Marketing calibration through the operator likelihood exactly as an experiment would, with ω^2 in place of the experiment’s variance—inside the certificate’s hull, and only after the library is checked for outcome selection. For theory, the bar, the certificate and the sample-size law are one object seen three ways: EVSI sets the variance threshold, the leverage-corrected excess variance measures the distance from it, and the variance-ratio test in κ prices the library. The honest reading of the field numbers (sign agreement 0.70; $r = 0.27$ on field experiments) is $\rho \gg 1$ for in-market lift today, in which case the law says a library will correctly *refuse* to certify—a useful output, but not the one E9 hopes for.

12.8 Further exploration

Four backlog items opened. BL 51: a trimmed or sub-Gaussian excess-variance certificate with a valid bound under heavy tails and the E penalty of a t -family law; the same tails degrade the experiments themselves in Chapter 15. BL 52: the temporal-split contamination test—the pre/post-cutoff difference in $\hat{\omega}^2$ as an F -type test with a power law in $(E_{\text{pre}}, E_{\text{post}}, \nu)$ —the only test that catches a whole-library memoriser. BL 53: the multichannel bar, Chapter 7’s ellipsoid loss plus Chapter 4’s trap-insurance value, making the certificate a matrix inequality on the residual covariance. BL 54: apply the law to Hut and Masoero’s 67 tests—the largest ω_x^2 certifiable at power 0.8 and the marginal value of the next 20 experiments. The dive also proposes structural family discovery (link-scale families chosen by a $Q_{\min}$ -style rule from Chapter 10) and the join to Chapter 13, where the adaptive-budget e-process is a confidence sequence for a scale parameter. In the compiler’s judgement BL 54 should run first because it is cheap and decisive: with ν computed from a real library’s reported standard errors, Law 12.4 says in one line whether 67 experiments can certify any bar a business would accept, and the answer calibrates how much of the harness is worth building now. Second is BL 52, since the whole-library memoriser is the one adversary the harness cannot currently see.

Reviewer’s note (compiler)

Checked. Every number was traced to the writeup (§3–§8) and, where ambiguous, to `15_calibrated_simulator.py`; the closed forms Equations (12.1) to (12.5) match the code, and the bars 0.37 (E12, $c = 0.05$) and 0.558 (E13b, $c = 0.1$) reproduce from Equation (12.4). The three E12a check-points were identified as $(m, V) = (0, 1), (0.5, 1), (1, 2)$ by recomputation; the writeup does not say which cells they are.

INDEX versus writeup. The log line’s “2.7 $\times$ ” for over-precise libraries is the writeup’s range 2.6–2.8 $\times$. BL 51 in the INDEX says heavy tails cut coverage to “0.87–0.92”; the writeup reports 0.92 (t_3) and 0.89 (heteroscedastic, estimated ν), and 0.87 appears nowhere in it (the code has a combined t_3 + dispersion + 20-df cell the writeup omits). The claim that $\nu = 0.25$ takes “ ≈ 55 ” experiments is computed from the law, not a Monte Carlo cell (evaluating Equation (12.3) at $\rho = 0.5, \nu = 0.25$ gives 53).

Under-specified or under-supported. The mixture family’s “rmse 0.033/0.043” does not say which two of the three targets it refers to. The E1 mean at $E = 8$ (0.2395) is 1.1 MC standard errors below 0.25—unremarkable. The claim that Hut and Masoero’s corrected MSE “is the same object without the leverage term” is the dive’s reading and was not verified here. The e-process is validated only at $\omega^2 = 1.0$ and 1.2 under exact Gaussianity; its per-step factor is decreasing in the true variance, so it is a supermartingale over the whole composite null $\omega^2 \geq \omega_x^2$, but the heavy-tailed case is untested. **Scope.** No LLM appears in the dive: every “simulator” is a Gaussian or probit generator with a chosen ω^2 , and the harness has not been run against real experiments. The log timestamp ($\sim 23:30$) and file time (23:32 UTC) agree.

Sources

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Chapter 13

Inference inside the measurement–allocation feedback loop

Provenance

Source dives: Dive 16 *Inference in the measurement–allocation feedback loop: the loop’s pathology is frozen learning, not false confidence* (run 2026-09-03, file `16_feedback_loop_inference.md`, written 03:44 UTC); Dive 19 *EXTEND of dive 16: detecting the stall, the multichannel stall geometry, and what survives adstock and confounding* (run 2026-09-03, file `19_feedback_loop_inference_ext.md`, written 21:41 UTC; results `code/19_results_all.json`). **Code:** `code/16_feedback_loop_inference.py`, `code/19_feedback_loop_inference_ext.py`. **Frontier-study problem(s) addressed:** M7 — Hadad et al. solve inference under a *known* adaptive assignment rule; in practice budgets adapt to interim estimates, so assignment depends on past estimators; wanted: valid confidence sequences for ROAS under endogenous, estimator-dependent adaptive designs (the statistical twin of E2). **Backlog items closed:** BL 56 (partly), BL 57, BL 58 (by Dive 19); **opened:** BL 55–BL 58 (from Dive 16), BL 68–BL 72 (from Dive 19; BL 70 later priced in Chapter 1).

Key ideas in this chapter

- A policy-agnostic anytime-valid confidence sequence is the RSS contour $\{\theta : \text{RSS}_t(\theta) \leq R_t^*\}$ with a mixture radius: valid under estimator-dependent spend and unknown σ (coverage 0.998–1.000 in every policy cell; clean-room 0.9999 under a CI-triggered policy) at $2.3\text{--}3\times$ the naive width.
- Smooth feedback is second-order for the naive interval: $O(g)$ bias exactly zero, $O(g)$ variance $\propto \kappa_3 g$ with coefficient -0.00156 matched to three digits, coverage 0.95 at any gain. The only first-order failure is drift $\times$ trigger (0.90/0.88/0.86 at $T = 104/208/416$; fixed hold 0.96); peeking dominates everything ($0.66 \rightarrow 0.40$).
- A certainty-equivalent loop without exogenous exploration stalls (Keskin–Zeevi transported): D frozen at 23.9 from $T = 104$ to 832, manifold identity 0.028 vs 0.375; in the Hill MMM loop the mROAS half-width stays $\approx 40\%$ forever and the model’s own mROAS at current spend echoes the breakeven target (1.000 ± 0.005 vs true 0.81–1.32). $t^{-1/4}$ exploration beats the stall by 27% in profit; constant exploration loses.
- The echo at the set-point is an identity (sd 10^{-15}); at realised spend its false-alarm law is a scale mixture (0.22/0.043/0.012 at $k = 1/3/6$) the Gaussian theory misses by $5\text{--}150\times$. The exogenous clock $D_t^\perp = vt$ alarms sharply but needs logged set-points; no log-only statistic separates exploration from demand chasing, which is the real pathology: spend +158%, true mROAS 0.27 reported as 1.01, naive coverage 0.000 and the anytime-valid CS 0.007–0.017.
- Multichannel: the budget direction is never identified under budget-neutral variation (rmse 5.6 vs 0.13); optimal passive exploration is $V \propto H_f^+$ with curvature-free cost (1.77 vs 1.75, 5.2 vs 6.25), exactly a mixture of pairwise switches with $w_{ij} = 1/(c_i c_j S)$, but bent inside the loop. An AR(1)-mixture CS restores validity at AR = 0.8 ($0.927 \rightarrow 1.00$) and is 42% narrower.

13.1 Problem and motivation

Every operating measurement system is a closed loop: a model reports a marginal return, the advertiser moves budget, and next week’s data arrive from the moved budget. M7 frames this as a *validity* problem—the model’s intervals were calibrated for a design that does not depend on the estimator, yet the design does, through an allocation rule that is usually unlogged and often a threshold (“scale when the lift is significant”). The adaptive-inference literature (Hadad et al., 2021; Zhang, Janson and Murphy, 2020) handles designs that depend on past *outcomes* through a known, logged rule; the marketing loop’s rule depends on past *estimators* and is neither.

Dive 16 argues the framing is mostly wrong. Validity under arbitrary predictable designs is not open: a mixture martingale over the regression likelihood gives an anytime-valid set that needs no knowledge of the policy. The naive interval is damaged only at second order by smooth feedback, at first order by one combination (a drifting hold level plus estimate-triggered jumps), and far more by weekly peeking than by adaptivity. The real pathology is that a certainty-equivalent allocator *stops learning*: spend converges to where the fitted and true curves cross, the design information freezes, and the reported marginal ROAS at current spend becomes an exact echo of the target that set the spend—Keskin and Zeevi’s (2014) incomplete learning transported from price to spend, with a corollary pricing does not need: “mROAS ≈ 1.0 at current spend, stable for two years” can be the loop closing on itself. Dive 19 then asks whether the stall can be detected from logs, what the echo’s false-alarm rate is against a channel genuinely at breakeven, what the stall looks like with several channels under a budget, and what survives adstock, a demand confounder that also drives spend, and autocorrelated residuals. Its central finding: the dangerous regime is the loop reacting to a demand state the model does not see, and there the anytime-valid interval fails as completely as the naive one.

13.2 Background: the ideas this chapter builds on

13.2.1 Confidence sequences and test martingales; why fixed- n intervals fail

A *confidence sequence* at level $1 - \delta$ satisfies $\mathbb{P}(\theta_0 \in \text{CS}_t \forall t) \geq 1 - \delta$: one guarantee for every look, hence for any data-dependent stopping time. The construction (Ville, 1939; Robbins, 1970; Lai, 1976) starts from a *test martingale*, a nonnegative process with $M_0 = 1$ and $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \leq M_{t-1}$ under the null; *Ville’s inequality* $\mathbb{P}(\exists t : M_t \geq 1/\delta) \leq \delta$ is the time-uniform Markov inequality. *Robbins’ method of mixtures* builds $M_t(\theta_0) = \int \prod_{s \leq t} p_{\theta}(y_s \mid \mathcal{F}_{s-1}) d\Pi(\theta) / \prod_{s \leq t} p_{\theta_0}(y_s \mid \mathcal{F}_{s-1})$ from a proper prior Π on the alternative, and $\text{CS}_t = \{\theta_0 : M_t(\theta_0) < 1/\delta\}$. Howard, Ramdas, McAuliffe and Sekhon (2021) generalise to *sub- ψ processes* S_t with variance process V_t for which $\exp(\lambda S_t - \psi(\lambda) V_t)$ is a supermartingale; mixing over λ with a $\mathcal{N}(0, \rho^2)$ density gives the normal-mixture boundary

$$|S_t| \leq \sqrt{(V_t + \rho^2) \log((V_t + \rho^2)/(\rho^2 \delta^2))} \quad \text{for all } t, \quad (13.1)$$

whose width grows like $\sqrt{V_t \log V_t}$ against the fixed- n $\sqrt{V_t}$: the $\sqrt{\log}$ is the price of time-uniformity. A non-negative process dominated by a test martingale under every null is an *e-process* (Ramdas, Grünwald, Vovk and Shafer, 2023); Lai (1976) handles unknown variance by mixing over both parameters, and Wasserman, Ramdas and Balakrishnan (2020) show that profiling a nuisance parameter out of a likelihood-ratio process preserves validity.

Fixed- n intervals fail in two distinct ways. Under monitoring, the law of the iterated logarithm guarantees the z-statistic eventually crosses 1.96, so a 5% test looked at repeatedly rejects a true null with probability one; Armitage, McPherson and Rowe (1969) computed type-I error about 0.19 at 10 looks and 0.37 at 100. Under *adaptive* collection the failure exists at fixed T : if x_t depends on past estimates it depends on past errors, $\text{Cov}(x_t, \varepsilon_u) \neq 0$ for $u < t$, and the exogeneity least squares needs is lost (Nie, Tian, Taylor and Zou, 2018: arm means from a bandit that samples the apparently better arm are negatively biased).

13.2.2 Least squares under adaptive designs (Lai–Wei, 1982; Abbasi-Yadkori et al., 2011; Hadad et al., 2021; Zhang–Janson–Murphy, 2020)

Lai and Wei (1982) prove that least squares with martingale-difference errors and $\mathcal{F}_{t-1}$ -measurable regressors is strongly consistent whenever $\lambda_{\min}(X_t^\top X_t) \rightarrow \infty$ and $\log \lambda_{\max} = o(\lambda_{\min})$, but asymptotic normality needs the much stronger *stability* condition that $X_t^\top X_t/t$ converge to a deterministic limit—which a spend-follows-estimate loop can violate. For a *known* noise scale R , Abbasi-Yadkori, Pál and Szepesvári (2011) give the self-normalised ellipsoid $\|\hat{\theta}_t - \theta\|_{\tilde{V}_t} \leq R\sqrt{2\log(\det(\tilde{V}_t)^{1/2}\det(\lambda I)^{-1/2}/\delta)} + \sqrt{\lambda}S$ uniformly in t for every predictable design, $\tilde{V}_t = \lambda I + X_t^\top X_t$: the same $\sqrt{d \log t}$ price as Equation (13.1).

Hadad, Hirshberg, Zhan, Wager and Athey (2021) evaluate a policy after a bandit with *known* assignment probabilities $e_t(w)$: they average the doubly robust score $\hat{\Gamma}_t(w) = \hat{\mu}_t(w) + \mathbf{1}\{W_t = w\}(Y_t - \hat{\mu}_t(w))/e_t(w)$ with *adaptive weights* h_t that are $\mathcal{F}_{t-1}$ -measurable, $\hat{Q}^h(w) = \sum_t h_t \hat{\Gamma}_t(w) / \sum_t h_t$, chosen (for instance $h_t \propto \sqrt{e_t(w)}$) so that the conditional variances are stabilised and a martingale central limit theorem applies even when $e_t(w) \rightarrow 0$. Zhang, Janson and Murphy (2020) show that least squares on batched-bandit data is asymptotically *non-normal* when arms are indistinguishable, because allocation probabilities fail to concentrate, and standardise within each batch before combining as a martingale sum. Both need a logged assignment rule with known probabilities—what the marketing loop lacks.

13.2.3 Incomplete learning under certainty equivalence (Keskin and Zeevi, 2014, 2018)

A seller faces demand $D_t = a + bp_t + \varepsilon_t$ with unknown (a, b) and each period sets the revenue-maximising price under the current least-squares fit: greedy iterated least squares, the certainty-equivalent policy. Keskin and Zeevi (2014) prove that with positive probability the prices converge to an *uninformative price*, where the fitted and true demand lines intersect, so that further observations confirm the fit and never move it; regret is then linear. Their fix is a *dispersion condition*, $\sum_{s \leq t} (p_s - \bar{p}_t)^2 \geq \kappa\sqrt{t}$, forcing deviations of order $t^{-1/4}$ per period and recovering regret $O(\sqrt{T} \log T)$ against the $\Omega(\sqrt{T})$ lower bound; Lai and Wei (1987) add probing noise of decaying variance to a self-tuning regulator for the same reason, and Keskin and Zeevi (2018) treat general certainty-equivalence control. Ziemann and Sandberg (2021, 2024) give the multidimensional version: an optimal policy can be *uninformative*—Fisher information singular—and since $\text{Fisher}(\pi) = \text{Fisher}(\pi^*) + O(\text{regret})$, low regret forces near-singular information along a subspace whose dimension sets a regret lower bound.

13.2.4 Feedback detection and curvature-shaped exploration

y *Granger-causes* x (Granger, 1969) if lagged y improves the prediction of x_t given lagged x ; the test is the F -statistic for $y_{t-1}, \dots, y_{t-p}$ in the regression of x_t on its own lags and theirs. Caines and Chan (1975) prove that two stationary processes are feedback-free iff neither Granger-causes the other, so “is spend policy-driven?” is a Granger test of outcome $\rightarrow$ spend. Gustavsson, Ljung and Söderström (1977) show that a system under a fixed low-order feedback law is not identifiable from closed-loop data—the regressors are collinear—unless the regulator is complex or time-varying or an external excitation is present; the certainty-equivalent allocator is such a regulator. On the exploration side, Levine and Koltun’s (2013) maximum-entropy controller has action covariance Q_{uu}^{-1} , the inverse curvature of the value function, and Aoyama et al. (2026) note that inverse-Hessian exploration costs $\tau n/2$ per step independently of the Hessian’s eigenvalues. Dive 19 derives the *learning-side* corollary: Hessian-whitened exploration makes estimation precision isotropic in the profit metric, so the cost of a target decision precision is curvature-free.

13.3 Setup and assumptions

Definition 13.1 (The single-channel loops). Weekly data follow the *linear* model $y_t = \alpha + \beta x_t + \varepsilon_t$ (marginal ROAS β) or the *Hill* model $y_t = \alpha + \beta H(x_t; K, s) + \varepsilon_t$, $H(x) = x^s / (x^s + K^s)$, $\text{mROAS}(x) = \beta H'(x)$; ε_t independent, mean zero, variance σ^2 , third moment κ_3 . The estimator is OLS with intercept on all data through t (linear) or profile-OLS over a K grid (Hill), refreshed every R weeks. Spend follows a *predictable policy* $x_t = \pi(\hat{\theta}_{t-1}, x_{t-1}) + \xi_t$, $\xi_t \sim \mathcal{N}(0, v)$ exogenous *exploration*. Families: *smooth*, $x_t = x_0 + g(\hat{\beta}_{t-1} - \beta_{\text{ref}}) + \xi_t$, the optimum of a linear-quadratic advertiser with profit $\beta x - x^2 / (2g) - mx$, $x^* = g(\beta - m)$, *loop gain* $g = 1/\text{curvature}$; *threshold*, jump to x_{hi} or x_{lo} when $\hat{\beta}$ (deadband) or the naive CI (CI-triggered) crosses β_{ref} , else hold at the last *realised* spend (which inherits ξ : a random-walk hold) or the last *base* level; *self-driven*, smooth with $v = 0$. The Hill loop sets $x_t = x^*(\hat{\beta}, \hat{K}) = \arg \max_x \hat{\beta} \hat{H}(x) - x$ (target mROAS 1) every R weeks with partial adjustment ρ . Every run opens with $T_0 = 8$ burn-in weeks at a fixed $\pm 20\text{--}30\%$ pulse. Working points: linear $\alpha = 10$, $\beta = 2$, $x_0 = 5$, $\sigma = 1$, $v = 1$, $x_{\text{hi}}/x_{\text{lo}} = 7.5/2.5$; Hill $\alpha = 20$, $\beta = 40$, $K = 5$, $s = 1$, $\sigma = 2$, $R = 4$, $x^* = \sqrt{\beta K} - K = 9.14$.

Definition 13.2 (Information clocks, targets, and the multichannel loop). The *design information* is $D_t = \sum_{u \leq t} (x_u - \bar{x}_t)^2$ (linear; $\text{Var}(\hat{\beta}) = \sigma^2 / D_t$ for a fixed design) or $D_H = \sum_u (H_u - \bar{H})^2$ at the profiled K (Hill); Dive 19's *exogenous clock* is $D_t^\perp = \sum_{u \leq t} (x_u - x^*(\hat{\theta}_{u-1}))^2$, the dispersion of realised spend around the logged set-point. Targets are the fixed-time coverage of the naive 95% t (linear) or profile-likelihood-ratio (Hill) interval at T , its *time-uniform* coverage over weekly looks (what a “scale when significant” protocol consumes), the growth of D_t , and the profit cost of exploration. The multichannel loop is $y = \alpha + \sum_j (b_j x_j - c_j x_j^2 / 2) + \varepsilon$ under $\mathbf{1}^\top x = B$ —the second-order expansion of any separable Hill loop at its optimum, c_j the profit curvature—with the allocator equalising fitted marginal returns $\hat{b}_j - \hat{c}_j x_j$. With $P = I - \mathbf{1}\mathbf{1}^\top / n$, $H = \text{diag}(c)$, $H_f = PHP$ and pseudo-inverse H_f^+ , the decision loss from a feasible-gradient error e is $\frac{1}{2} e^\top H_f^+ e$ and budget-neutral exploration $\xi \sim \mathcal{N}(0, s^2 V)$, $V\mathbf{1} = 0$, costs $\tau = \frac{1}{2} s^2 \text{tr}(HV)$ per week. Dive 19 adds adstock $a_t = x_t + \lambda a_{t-1}$, an unobserved AR(1) demand state d_t ($\rho_d = 0.8$) entering as γd_t , and *demand chasing* $x_t = x^*(\hat{\theta}) + \varphi d_t + \xi_t$.

Assumption 13.3 (Independent errors; no omitted regressor). ε_t is independent across weeks (the one assumption Construction 13.4 needs; Construction 13.5 relaxes it), and the likelihood the estimator mixes over is correctly specified: nothing drives both spend and outcome outside the model (Proposition 13.14 is what happens otherwise).

13.4 Main results

13.4.1 A policy-agnostic anytime-valid confidence sequence

Construction 13.4 (RSS-contour confidence sequence). [established in the dive] (Assembly of known parts; Dive 16.) For any proper mixing prior Π on (θ, σ^2) and true (θ_0, σ_0^2) , $M_t = \int p(y_{1:t} | x_{1:t}, \theta, \sigma^2) d\Pi / p(y_{1:t} | x_{1:t}, \theta_0, \sigma_0^2)$ is a nonnegative martingale with $M_0 = 1$ for *every* predictable design, because the likelihood factorises as $\prod_s p(y_s | x_s, \theta, \sigma^2)$ with x_s any function of the past—including past estimators and past intervals—so Ville gives $\mathbb{P}(\exists t : M_t \geq 1/\delta) \leq \delta$. Projecting $\{M_t < 1/\delta\}$ onto θ by the sup over σ^2 , $\sup_\sigma p(y | \theta, \sigma) = (2\pi \text{RSS}_t(\theta)/t)^{-t/2} e^{-t/2}$, gives

$$\text{CS}_t = \{\theta : \text{RSS}_t(\theta) \leq R_t^*\}, \quad R_t^* = \frac{t}{2\pi} \exp\left(-\frac{2(\log m_t + \log \delta)}{t} - 1\right), \quad (13.2)$$

with m_t the normal-inverse-gamma marginal likelihood in closed form ($\theta | \sigma^2 \sim \mathcal{N}(\mu_0, \sigma^2 \Lambda_0^{-1})$, $\sigma^2 \sim \text{IG}(a_0, b_0)$; in the Hill case mixed with a uniform prior on the K grid).

The set has exactly the shape of the classical likelihood-ratio set—the ellipsoid $(\theta - \hat{\theta})^\top X^\top X (\theta - \hat{\theta}) \leq R_t^* - \text{RSS}_{\min}$ around OLS—with a data-dependent radius. Its projected half-width on β is $\text{se}_\beta \cdot \rho_t$, $\rho_t^2 =$

$(R_t^* - \text{RSS}_{\min})/s^2$, and expanding the exponential gives the *radius law*

$$\rho_t^2 \approx (t-2)(e^{2(\Delta_t + \log(1/\delta))} - 1) \approx 2\log(1/\delta) + 2\Delta_t, \quad \Delta_t = \log p(y \mid \hat{\theta}, \hat{\sigma}) - \log m_t, \quad (13.3)$$

against $1.96^2 = 3.84$ for the naive interval, $\Delta_t \approx \frac{1}{2} \log \det(I + \Lambda_0^{-1} X^\top X)$ plus prior terms being the Occam penalty. Under a vague prior the measured ρ^2 is 38.1/33.5/32.4/32.8/34.0/35.5 at $T = 26\text{--}832$ ($g = 3$, $v = 1$): a half-width $2.9\text{--}3.0\times$ naive with time-uniform coverage 0.9993–1.000 (clean-room: 0.9999 under the CI-triggered policy at width ratio 2.98). The refinement CS-B takes the burn-in posterior as the mixing prior for the adaptive phase—legitimate because it is fixed before any adaptive week—and drops the ratio to $2.34\text{--}2.48\times$ at $T = 104\text{--}832$, against $1.78\text{--}2.05\times$ for Bonferroni over weekly looks, which needs a fixed horizon and a non-adaptive design. The martingale is Robbins’ and Lai’s; the assembly—the RSS-contour form, the fact that it needs no knowledge of the policy, and the price—is the contribution.

Construction 13.5 (AR(1)-mixture confidence sequence). [\[numerically established\]](#) (Dive 19.) The conditional density $p(y_t \mid y_{t-1}, x_{1:t}, \theta, \sigma^2, \phi) = \mathcal{N}(\alpha(1-\phi) + \beta(x_t - \phi x_{t-1}) + \phi y_{t-1}, \sigma^2)$ is a valid one-step likelihood for every predictable x , so a mixture over (θ, σ^2, ϕ) with ϕ on a grid is again a martingale, and the sup over (σ^2, ϕ) gives $\text{CS}_t = \{\theta : \min_\phi \text{RSS}_{\phi,t}(\theta) \leq R_t^*\}$. In the smooth loop ($g = 10$, $v = 1$) it has time-uniform coverage 1.00 at AR = 0/0.6/0.8 against 1.00/0.99/0.927 for the iid mixture (naive fixed-time 0.947/0.80/0.66) and is narrower where it matters—median half-width 0.355 vs 0.445 at AR = 0.6, 0.204 vs 0.352 (42% narrower) at AR = 0.8 (the GLS efficiency of prewhitening)—at a 6% price at AR = 0.

13.4.2 Smooth feedback is second-order

Write $x_t = \xi_t + g d_t$ with ξ the exogenous design and $d_t = \hat{\beta}_{t-1} - \beta_{\text{ref}}$ the estimator the policy reacts to. The OLS error is N/D , $N = \sum (x_t - \bar{x})\varepsilon_t$, $D = \sum (x_t - \bar{x})^2$; to first order $N = N_0 + g N_1$, $D = D_0 + g D_1$ with $N_1 = \sum (d_t - \bar{d})\varepsilon_t$, $D_1 = 2 \sum (\xi_t - \bar{\xi})(d_t - \bar{d})$, and $d_t = \sum_{u < t} w_{u,t} \varepsilon_u$ with centred weights $w_{u,t} = (\xi_u - \bar{\xi}_{t-1})/S_{t-1}$.

Theorem 13.6 (First-order benignity). [\[established in the dive\]](#) Conditional on ξ , for independent mean-zero errors: (i) the $O(g)$ bias of $\hat{\beta}$ vanishes for any noise distribution, so bias is $O(g^2)$; (ii) the $O(g)$ variance term is

$$\begin{aligned} \frac{d \text{Var}}{dg} \Big|_0 &= 2 \left[\frac{\mathbb{E}(N_0 N_1)}{D_0^2} - \frac{\mathbb{E}(N_0^2 D_1)}{D_0^3} \right], \\ \mathbb{E}(N_0 N_1) &= -\frac{\kappa_3}{T} \sum_s (\xi_s - \bar{\xi}) \sum_{u > s} w_{s,u}, \quad \mathbb{E}(N_0^2 D_1) = 2\kappa_3 \sum_t (\xi_t - \bar{\xi})^2 \sum_{s > t} (\xi_s - \bar{\xi}) w_{t,s}, \end{aligned} \quad (13.4)$$

proportional to κ_3 , zero for symmetric noise, with sign $-\text{sign}(\kappa_3 g)$.

The bias proof is two lines: $\mathbb{E}[N_1] = 0 - (\sigma^2/T) \sum_t \sum_{u < t} w_{u,t} = 0$ by centring, and $\mathbb{E}[N_0 D_1] = 2\sigma^2 \sum_t (\xi_t - \bar{\xi}) \cdot 1 = 0$. Positive skew with positive gain therefore *shrinks* the true variance below the naive se^2 (conservative); negative skew, or a policy that cuts spend when the estimate rises, inflates it. With common random numbers ($T = 40$, $v = 1$, $R = 40,000$) the bias shift is $+0.00001$ to $+0.00007$ at $g = 0.5\text{--}4$ ($\text{sd}(\hat{\beta}) = 0.15$); Gaussian $d \text{Var} / g^2$ is $-0.00060 / -0.00055 / -0.00046 / -0.00030$ (quadratic); at $\kappa_3 = 2$, $d \text{Var} / g$ is $-0.00186 / -0.00198 / -0.00204 / -0.00189$ (linear) against Equation (13.4) on the realised designs, -0.00156 , which matches $-0.00186 + 0.00030$ (the g^2 term at $g = 0.5$) exactly; clean-room $-0.00181 / -0.00195 / -0.00203$, flipping to $+0.00118 / +0.00079 / +0.00013$ under negative skew. At $T = 104$ the naive coverage is 0.950–0.957 for $g = 0\text{--}100$; skew $+2$ pushes it to 0.968/0.978 at $g = 10/30$, skew -2 to 0.931/0.920.

Proposition 13.7 (Failure taxonomy). [\[numerically established\]](#) At $T = 104$ with weekly looks the naive interval’s failures rank peeking (uniform coverage 0.66 fixed design, 0.55 CI-triggered; 0.40–0.56 by $T = 416$) $\gg$ drift $\times$ trigger (fixed-time 0.902/0.881/0.855 at $T = 104/208/416$ for the CI-triggered policy with random-walk hold, the se under-estimating the true sd by 35–96%) $\gg$ smooth adaptivity (0.950–0.957). A pure random walk with no adaptivity covers at 0.943–0.946 and the same triggers with a fixed hold at 0.946–0.967 (Table 13.1).

The anti-conservatism needs both ingredients—the jump’s lever arm is selected on the estimate’s error, and the drift lets that selection dominate the design as T grows—a distinction the clean-room agent flagged independently, turning Round 1’s “threshold policies fail” into “drift $\times$ trigger fails”. CS-B removes all three sources at once (0.998–1.000 in every cell).

13.4.3 The learning stall and the policy echo

Proposition 13.8 (Stall of the certainty-equivalent loop). *[transported from the literature] (Keskin–Zeevi transported to spend; by simulation.) In the self-driven linear loop ($v = 0, g = 10$) the design information freezes—median $D_T = 22.9/23.8/23.9/23.9$ at $T = 30/104/416/832$ —and with it everything: $\text{sd}(\hat{\beta}_T) = 0.106/0.099/0.098/0.098$, naive se 0.217 and half-width 0.425 at $T = 30$ and at $T = 832$. Spend converges (mean $|x_T - x_{T-1}| = 3 \times 10^{-5}$) to a level x_∞ at which the fitted line crosses the true line, the stall manifold*

$$(\hat{\alpha} - \alpha) = -(\hat{\beta} - \beta) x_\infty, \quad (13.5)$$

verified as mean $|(\hat{\alpha} - \alpha) + (\hat{\beta} - \beta)x_\infty| = 0.028$ against the unconstrained scale $|\hat{\beta} - \beta| x_0 = 0.375$ (clean-room 0.027 vs 0.371), with $\text{sd}(x_\infty) = g \cdot \text{sd}(\hat{\beta}_\infty)$ exactly (0.9799 = 0.9799).

Two things pricing does not emphasise follow. The naive interval is *conservative and frozen* (fixed-time coverage 0.96–0.98 across $g = 1$ –100), because the loop’s negative feedback corrects the estimate without adding dispersion: the stall is invisible to a coverage audit and visible only as an interval that stops shrinking. And gain trades slope error for spend error one-for-one ($\text{sd}(\hat{\beta}_\infty) = 0.290/0.186/0.097/0.058/0.041$ against $\text{sd}(x_\infty) = 0.29/0.56/0.97/1.74/2.94$ at $g = 1/3/10/30/100$) with the per-period loss $g \text{Var}(\hat{\beta}_\infty)/2$ nearly flat at 0.042–0.083, about 1% of profit. *[refuted within the dive]* Round 1 had predicted from the recursion $D' = g^2 \sigma^2 / D$ that information grows like $g \sigma \sqrt{2t}$; the attack refuted it, because a frozen slope error contributes an *offset* the sample mean absorbs, not dispersion; a one-shot cubic stall law reproduces the scaling of $\text{sd}(\hat{\beta}_\infty)$ but misses the constants by up to 40%.

Proposition 13.9 (The policy echo). *[established in the dive] In the Hill loop the model’s own mROAS at the set-point, $\hat{\beta} \hat{K} / (x^* + \hat{K})^2$, equals the breakeven target by construction at every refit, in every regime: 1.0000 with $\text{sd} \leq 10^{-15}$ across all of Dive 19’s configurations (0.003 with a grid optimiser; Dive 16’s 1.0000 ± 0.005 reflects the refit lag). The true mROAS at that spend has sd 0.16 and 5–95% range $[0.81, 1.32]$, and the parameters wander along the manifold ($\hat{\beta}$ 5–95% $[35, 220]$ against 40; $\hat{K}$ $[0.5, 47]$ against 5; 37% of runs pin $\hat{K}$ at a grid edge).*

The estimand the loop reports converges to the policy’s set-point. The Hill stall survives every realism knob Dive 16 tried (refresh 1–13 weeks, partial adjustment, $\sigma, s = 2$, gain): mROAS-error sd 0.179/0.164/0.158/0.155 at $T = 52/104/416/832$, naive half-width $0.44 \rightarrow 0.38$ ($\approx 40\%$ of mROAS) forever, at a loss of only 0.08–0.10 per week ($\approx 0.5\%$ of profit) because the profit curve is flat at its optimum. In three channels the fitted marginal returns are equal across channels by construction (fitted spread 0, sd 6×10^{-16} , against true spread 0.46 quadratic and $0.127 \rightarrow 0.120$ Hill).

Proposition 13.10 (Restoring learning). *[transported from the literature] Keskin–Zeevi’s dispersion condition transports unchanged: exploration $\xi_t \sim \mathcal{N}(0, t^{-2d})$ with $d = 0.25$ ($\sum \xi^2 \propto \sqrt{t}$) keeps D growing ($57.6 \rightarrow 163$ from $T = 104$ to 832 ; $\text{sd}(\hat{\beta})$ $0.105 \rightarrow 0.071$), $d = 0.5$ barely ($36.9 \rightarrow 66$), $d = 1$ not at all ($27 \rightarrow 32$). In the Hill loop decaying exploration $2t^{-1/4}$ (20% of spend in week 8, 6% by week 832) is profitable—cumulative loss 60.4 ± 1.4 vs 82.6 ± 3.1 for the stall at $T = 832$ (27% lower), mROAS sd $0.157 \rightarrow 0.092$ —while constant exploration at 11–16% of spend loses more than the stall (95.8, 162.7; Table 13.2).*

Law 13.11 (Profit price of a valid interval). *[established in the dive] (Weak feedback; qualified at strong feedback.) A half-width w needs $D_T \geq \rho^2 \sigma^2 / w^2$ ($\rho = 1.96$ naive, ρ_t for the CS); exogenous exploration of variance v costs $v / (2g)$ per week and buys $D \approx Tv$, so*

$$C(w) = \frac{\rho^2 \sigma^2}{2gw^2}, \quad (13.6)$$

independent of the horizon: the price of inference is paid once, in profit, and only its timing depends on T .

At $g = 1$, $w = 0.15$ (theory 85.3) the measured exogenous cost is 73.2/76.3/91.4/89.2 at $T = 52/104/208/416$ on a $\pm 14\%$ grid. At $g = 10$ (theory 8.5) the exogenous cost *falls* with T (7.3/6.3/5.3/4.3) because the loop's own wobble supplies dispersion, but the total deviation cost *rises* (9.0/10.1/12.3/17.2): the loop's persistent offset from x^* costs profit without adding dispersion. Offset costs, dispersion informs—and Remark 13.18 shows that even the dispersion the loop generates is worth $4\text{--}20\times$ less than the same dispersion supplied exogenously.

13.4.4 Detecting the stall

Construction 13.12 (The exogenous clock). [\[established in the dive\]](#) (Dive 19.) Under a pure certainty-equivalent loop $D_t^\perp \equiv 0$; under exploration of variance v it grows at v per week (0.0096/0.038/0.086/0.239/0.956 at $v = 0.01/0.04/0.09/0.25/1$; raw spend variance overstates it by 3–30% because the set-point wanders). A book-keeping identity, not a test, but the only clock guaranteed *unconfounded*: it counts dispersion that no belief and no demand state chose. Its threshold is Law 13.11 transported locally, $v_{\min} = (1.96\sigma/w)^2/T_h$ for half-width w by horizon T_h , and the alarm “ $D^\perp/\text{week} < v_{\min}$ ” over W weeks has false-alarm rate $\mathbb{P}(\chi_W^2/W < v_{\min}/v)$ and power one against the exact stall: with $W = 26$ the alarm rate is 1.00/1.00/0.83/0.02/0.00/0.00 at $v = 0/0.01/0.04/0.09/0.25/1$, firing exactly where exploration has stopped buying anything (true-mROAS sd at the set-point $0.169 \rightarrow 0.101$, loss $0.093 \rightarrow 0.036$ across the sweep).

Without set-points an auditor has the *Lai–Wei clock* ($\lambda_{\min}$ of the design at T vs $T/2$: 1.01 stall, 1.47 exploration, 1.40 demand chasing, 1.72 fixed spend plus exploration) and the *Caines–Chan test* (Granger 5% rejection 0.19 stall, 0.54 at weekly refits; 0.10 exploration; 0.30 chasing; 0.04 fixed). [\[refuted within the dive\]](#) as a hope: the eigenvalue clock separates the stall from everything else but not learning from pseudo-learning, and the Granger test has power 0.2–0.5 and is structurally silent on confounding—once lagged spend, which carries the state, is conditioned on, lagged outcomes add little. No log-only statistic in the dive separates exogenous exploration from demand-driven spend (BL 68).

Law 13.13 (Echo false-alarm law). [\[numerically established\]](#) Let $\hat{m}_j$ be the reported mROAS at realised spend $x^* + \xi_j$, $\xi_j \sim \mathcal{N}(0, v)$, at refit j . Then $\hat{m}_j - 1 = e_j + \hat{m}'_j \xi_j$, with e_j the slow model error at x^* (adjacent-refit correlation 0.99) and $\hat{m}'_j \xi_j$ the fitted slope times the fresh draw, and the false-alarm rate of “ $|\hat{m}_j - 1| \leq \epsilon$ for k consecutive refits” against a breakeven, non-stalled channel is the scale mixture

$$\text{FA}_k(\epsilon) = \mathbb{E}_{e, \hat{m}'} \left[\left(\Phi\left(\frac{\epsilon - e}{|\hat{m}'| \sqrt{v}}\right) - \Phi\left(\frac{-\epsilon - e}{|\hat{m}'| \sqrt{v}}\right) \right)^k \right], \quad e \sim \mathcal{N}(0, \text{se}^2 - \mathbb{E}[\hat{m}'^2]v), \quad (13.7)$$

$\hat{m}'$ drawn from the fitted slopes. The stall hits every cell with probability one.

At $T = 208$, $\text{sd}(\hat{m} - 1) = 0.21$ with components 0.13 and 0.16 (true $m'(x^*) = -0.141$) and adjacent-refit correlation only 0.36. The obvious Gaussian theory— $\mathcal{N}(0, \Sigma)$ with the empirical sd and correlation—is wrong by $5\text{--}150\times$ at $k \geq 3$ (Table 13.3), because $\hat{\beta}$ and $\hat{K}$ wander along the ridge so $|\hat{m}'|$ varies $3\times$ across runs, and scale mixtures put far more mass near zero than a normal of the same variance; Equation (13.7) is within $2\times$. Operationally, “mROAS = 1.00 ± 0.02 for three consecutive quarterly refreshes” has a false-alarm rate below 1% against a truly breakeven channel with 10% spend dispersion and a hit rate of one against the stall; with no dispersion the test is vacuous.

13.4.5 Confounded pseudo-learning: anytime validity is not identification

Proposition 13.14 (Demand chasing). [\[established in the dive\]](#) (Negative result; clean-room exact.) With an unobserved demand state and chasing $\varphi = 1.5$, $\gamma = 6$ in the Hill loop, every clock reads the opposite of the stall— D_x grows $55\times$ faster (7,650 vs 138 at $T = 208$), the eigenvalue clock reads 1.40, $\hat{\beta}$ concentrates (5–95% [83, 134] vs [35, 238])—on the wrong value: spend is 158% above the true optimum, the true mROAS there is 0.27, and the model reports 1.01 ± 0.13 . In the linear loop ($\varphi = 1$, $\gamma = 1$, $v = 0.3$) the naive interval's coverage is 0.000 (bias +0.66 on $\beta = 2$) and the confidence sequence of Construction 13.4 has time-uniform coverage 0.007–0.017 (clean-room 0.016). Observing the demand state removes all of it (spend bias +2%, mROAS bias 0.004).

The CS is valid under *any* predictable spend policy and useless under a spend that carries an omitted regressor: its guarantee is exactly as strong as the likelihood it mixes over. This is Heusch’s (2026) endogeneity bias with a sharper diagnosis—“performance chasing” is harmless if it chases the model’s own optimum and catastrophic if it chases realised outcomes that contain demand. At $\varphi = 0.5$, $\gamma = 2$ the bias is still +0.46.

13.4.6 Multichannel geometry, the whitened exploration law and its switch representation

Proposition 13.15 (The budget direction). *[numerically established] In the quadratic loop ($n = 3$, $B = 10$) the feasible-direction gradient error is frozen (rmse 1.469 at $T/2$, 1.466 at T ; loss 1.77 per week) and budget-neutral isotropic exploration ($s = 0.5$) unfreezes it ($0.311 \rightarrow 0.225$; loss 0.174). But with spend always on $\mathbf{1}^\top x = B$ the regressors $(1, x_1, \dots, x_n)$ are collinear and the total marginal value $\mathbf{1}^\top g/n$ is identified by the quadratic form only: rmse 5.6 (median 3.4, true level 2.3) at the stall, 1.06 under budget-neutral exploration, 0.13 only when exploration is not budget-neutral.*

A loop that explores within a fixed budget can learn how to split it and never whether the budget is right; the “total mROAS” such a model reports is a functional-form artefact. This is Chapter 7’s budget coupling seen from the identification side and Ziemann–Sandberg’s singular subspace made concrete: under budget coupling the singular direction is **1**. Chapter 1 later proves the exact rank theorem and prices the escape (BL 70).

Law 13.16 (Passive curvature-whitened exploration). *[established in the dive] (Derived; verified to its constant.) With spend $x^* + \xi_t$, $\xi \sim \mathcal{N}(0, s^2 V)$ on the feasible subspace, $\text{Cov}(\hat{e}) = \sigma^2 (Ts^2 V)^+$ and $\mathbb{E}[L] = \sigma^2 \text{tr}(HV) \text{tr}(H_f^+ V^+) / (4T\tau)$ at fixed cost τ , minimised by Cauchy–Schwarz at*

$$V \propto H_f^+, \quad \mathbb{E}[L]^* = \frac{\sigma^2(n-1)^2}{4T\tau}, \quad (13.8)$$

independent of the curvature spectrum. Isotropic $V = P$ costs $\text{tr}(H_f) \text{tr}(H_f^+) / (n-1)^2$ times more, $V \propto H_f$ (stiff directions) $\sum \lambda^2 \sum \lambda^{-2} / (n-1)^2$. At $c = (0.1, 0.4, 1.6)$, $b = (3, 3, 3.5)$, $\tau = 0.2$, $T = 208$: mean loss 0.046/0.026/0.134 for isotropic/ H^+ / H , ratios 1.77 vs theory 1.75 and 5.2 vs 6.25 (the shortfall is the burn-in’s information; clean-room 1.76 and 5.39, derivation confirmed).

Theorem 13.17 (Switch representation). *[established in the dive] For a separable Hessian $H = \text{diag}(c)$, with $S = \sum_k 1/c_k$ and $d_{ij} = e_i - e_j$,*

$$H_f^+ = \sum_{i < j} w_{ij} d_{ij} d_{ij}^\top, \quad w_{ij} = \frac{1}{c_i c_j S} > 0, \quad (13.9)$$

exactly: the optimal exploration is a rotating schedule of Chapter 3’s budget-neutral pairwise switches with squared amplitudes $\propto 1/(c_i c_j)$ —switch most between the two flattest channels.

The proof is the constrained inverse $M = C^{-1} - C^{-1} \mathbf{1} \mathbf{1}^\top C^{-1} / S$, whose off-diagonals $-1/(c_i c_j S)$ are the negatives of the weights (residual 10^{-15} over 200 random curvature vectors at each $n = 3, \dots, 6$; $c = (0.1, 0.4, 1.6)$ gives $w = 1.905, 0.476, 0.119$). It is not generic: with cross-channel interactions the nonnegative-least-squares residual is nonzero in 4.5%/28%/65% of random Hessians at $n = 3/4/5$ —the pairwise dictionary has exactly $n(n-1)/2$ elements, the dimension of symmetric forms on the feasible subspace, so representability is a sign condition, not a rank condition (BL 71).

Remark 13.18 (The law bends inside the loop). *[refuted within the dive] in part. With the same three designs at the same τ inside the certainty-equivalent loop, the ranking survives by *median* with attenuated ratios (isotropic/ H^+ 1.07, H/H^+ 1.75 vs 1.63/3.78 passive) and *reverses* by mean (0.76, 0.91), because the loss is heavy-tailed (mean/median 4–6; q_{99} 2.1–2.6 against a median 0.05). The natural refinement—the loop’s own wobble is already H^{-1} -shaped (whitened eigenvalues 0.38, 1.28), so water-fill the complement—was built and refuted (median 0.068 vs 0.045). A passive design given an *exogenous* wobble of the loop’s measured*

covariance reaches loss 0.012 against the loop’s 0.19–0.25 mean / 0.045 median: the same dispersion is worth 4–20 $\times$ less when the loop generates it, because the loop samples where its own error is ($x_{\text{dev}} = H^+e$) and at certainty-equivalent gain the $O(g^2)$ terms are not small ($g \cdot \text{sd} / \sqrt{v} \approx 7$). Dive 16’s “dispersion informs” is a weak-feedback statement (BL 69).

Proposition 13.19 (Adstock leaves the stall intact). *[numerically established] With true $\lambda = 0.5$ and a 9-point λ grid profiled at each refit ($T = 312$): own long-run mROAS 1.000 (sd 10^{-16}), true-mROAS sd at the set-point 0.248 $\rightarrow$ 0.240, $\hat{\lambda}$ frozen at burn-in precision (sd 0.092 at both times, mean 0.478). Decaying exploration $1.5t^{-1/4}$ cuts the mROAS error (0.240 $\rightarrow$ 0.184) and the loss (0.124 $\rightarrow$ 0.076) but leaves $\text{sd}(\hat{\lambda})$ at 0.092: iid weekly dispersion does not identify carryover, as Chapter 1 found for the passive design (the transient needs block pulses of length $\approx 3/(1 - \lambda)$).*

13.5 Numerical evidence

Linear-loop cells use $R = 2,000$ – $8,000$ replications (± 0.005 on a coverage), re-implemented from the claim text by a clean-room agent at $R = 8,000$ – $60,000$; Table 13.1 collects the taxonomy with its controls. The CS-B power control ($g = 10$, $v = 1$, $T = 104$) excludes $\beta + 0.4$ (a 20% error) in 68% of runs at median week 79 and $\beta + 0.8$ always at median week 24, while the naive interval excludes the *true* value in 55% of runs by week 33. The Hill loop (Table 13.2) uses $R = 1,000$; Dive 19’s diagnostics $R = 300$ – 400 (± 0.02 – 0.03 on a rate) with a clean-room at $R = 1,600$ for the echo table, whose values Table 13.3 reports because two of the dive’s own $R = 400$ cells (0.015 and 0.003) were 2–5 $\times$ high on 6 and 1 hits. The multichannel evidence is $R = 300$ passive and $R = 600$ in-loop at $T = 208$: passive mean loss 0.046 ± 0.003 / 0.026 ± 0.001 / 0.134 ± 0.011 for isotropic/ H^+ / H ; in-loop means 0.190/0.250/0.227 (se 0.016–0.021), medians 0.048/0.045/0.078 (se $\approx 10\%$). Five seeds give true-mROAS sd 0.155–0.170 and D_x 129–152 for the Hill stall.

Table 13.1: Coverage of the naive 95% interval for β in the linear loop (Dive 16 E1, E4, E7, E9). “Uniform” is coverage at every weekly look through T ; CS-B is the burn-in-prior sequence of Construction 13.4. Monte Carlo ± 0.005 ; clean-room values in parentheses.

Policy	T	fixed-time	uniform, naive	uniform, CS-B
fixed design	104	0.950	0.66 (0.658)	0.998–1.000
smooth $g = 10$, $v = 1$	104	0.950–0.957 (0.953–0.955)	0.67–0.69	0.9992
smooth $g = 100$, $v = 0.09$	104	0.94–0.97 (0.950–0.953)	—	1.000
smooth, skew -2 , $g = 30$ – 100	104	0.920–0.924	—	—
self-driven $g = 10$, $v = 0$	104	0.96–0.98 (0.981)	0.77	1.000
pure random walk	104–416	0.943–0.946	—	—
CI-triggered, random-walk hold	104/208/416	0.902/0.881/0.855 (0.907)	0.55/—/0.40	0.9980
CI-triggered, fixed hold	104–416	0.963–0.967	—	—
deadband 0.1, random-walk hold	104/208/416	0.915/0.909/0.929	—	0.9988
smooth $g = 10$, AR(1) 0.3 / 0.6	104	— / 0.826	—	0.9982 / 0.9778
smooth $g = 10$, heteroskedastic	104	—	—	0.9985

Table 13.2: Hill MMM loop ($\beta = 40$, $K = 5$, $s = 1$, $\sigma = 2$, refit every 4 weeks, spend at the fitted optimum; Dive 16 E12, $R = 1,000$). sd_m is the sd of the mROAS error at current spend; the half-width is the naive one for mROAS (the CS is 2.3–2.6 $\times$ wider at coverage 1.000); loss is cumulative profit loss with its standard error.

Exploration	T	sd_m	naive half-width	fixed-time cov.	cum. loss
none	52	0.179	0.44	0.975–0.979	—
none	832	0.155	0.38	0.975–0.979	82.6 ± 3.1
constant 1.0 (11% of spend)	832	—	—	—	95.8
constant 1.5 (16%)	832	—	—	—	162.7
decaying $2t^{-1/4}$	832	0.092	—	—	60.4 ± 1.4

Table 13.3: False-alarm rate of the echo test at realised spend against a genuinely breakeven, non-stalled channel with 10% spend dispersion, $T = 208$, 4-weekly refits (Dive 19 E3; clean-room $R = 1,600$ with binomial se). Hit rate against the stall is 1 in every cell.

ϵ	$k = 1$	$k = 3$	$k = 6$	Gaussian theory, $k = 3 / 6$
0.05	0.223 ± 0.010	0.043 ± 0.005	0.012 ± 0.003	$0.0085 / 8 \times 10^{-5}$
0.02	0.094	0.0063 ± 0.002	< 0.003	—
0.01	0.048	0.0006 ± 0.0006	≈ 0	—
scale-mixture law Equation (13.7), $\epsilon = 0.05$	—	0.056	0.026	—

13.6 Limitations and the devil’s advocate

Against “second-order”: Theorem 13.6 is conditional on a persistent exogenous design; as $v \rightarrow 0$ the expansion parameter $g \cdot \text{sd}(\hat{\beta}) / \sqrt{v}$ is not small—at $g = 100$, $v = 0.09$ the bias reaches 0.24 sd and coverage 0.92 under negative skew, and Dive 19 measures the parameter at ≈ 7 for a certainty-equivalent loop. It also assumes independent errors; at $\text{AR}(1) = 0.6$ the fixed-design naive interval already fails (0.83). *Against the stall:* real spend is not the fitted optimum—finance cuts, seasonal budgets and channel caps are exploration in disguise; the stall is the limit of a *disciplined* certainty-equivalent allocator, which is what automated MMM-to-budget pipelines (Pathak et al., 2026; Chapter 4) aim to build, and drift in the true β does not rescue it. *Against the echo:* exact only at the fitted optimum (with a target $\neq 1$ or a budget the echo is of that target); at realised spend the test needs dispersion, and its false-alarm law is an approximation within $2\times$. *Against the diagnostics:* the exogenous clock is only as good as the set-point log, which pipelines commonly overwrite with realised spend, and implementation noise is informative only if independent of demand, which finance-driven cuts are not. *Against the CS:* 2.3–3 $\times$ naive is a heavy price at $T \approx 100$; Ville’s slack (0.998 at $\delta = 0.05$) means a calibrated δ could be larger (BL 55); a practitioner who never peeks and uses a smooth policy is entitled to the naive interval. *Against the pricing law:* LQ only; its strong-feedback failure (17 vs 8.5 at $T = 416$) shows the loop’s regret cannot be booked as inference spend. *Against the confounding result:* $\varphi = 1.5$, $\gamma = 6$ is a strong chaser and the numbers are illustrative. *Against the exploration law:* local quadratic, scalar noise, one gain, a heavy tail unexplained, a switch theorem needing a separable Hessian. *Against the budget-direction claim:* curvature does identify $\mathbf{1}^\top g$, just $8\times$ worse than a total pulse, and a fixed-form model will report a confident number. *Against novelty:* every ingredient is known; the contributions are the transport, the taxonomy with sizes, the first-order coefficient, the echo and its false-alarm law, the learning-side whitening corollary with its switch representation and in-loop breakdown, the budget-direction non-identification, the validity-is-not-identification demonstration, and the $\text{AR}(1)$ projection.

What the attack rounds changed: the $\sqrt{t}$ growth law was refuted and replaced by the manifold; “threshold policies fail” became “drift $\times$ trigger”; a pulseless burn-in that collapsed spend in week one was replaced; the Gaussian echo theory was refuted by its own Monte Carlo; the in-loop exploration law came out *reversed* in Round 2, was partly rescued by fixing a clipped channel and a tiny exploration budget, then held by median only; water-filling was refuted; “frozen” three-channel Hill spread became “slowly shrinking” (0.127 $\rightarrow$ 0.120).

13.7 Discussion: impacts

For practice the first message is a reassurance with a boundary. A vendor whose users move budget smoothly in response to model output—a Meridian or Robyn optimiser applied with partial adjustment—owes them no new interval: Theorem 13.6 says the naive interval keeps its level at any gain and the sign of its one first-order term is known. What the vendor owes them is a warning about the two things that do break it, looking every week (0.66) and scaling on a trigger from a drifting hold level (0.86). Both are removed by Construction 13.4, which any Bayesian MMM can emit almost for free—the marginal likelihood m_i is what the sampler already computes and the RSS contour is the model’s own likelihood-ratio set with a different

radius—and which, unlike the sequential tests in production at experimentation platforms, needs no logged assignment rule.

The second message is for the CMO. A stable, precise marginal ROAS of 1.0 at current spend, produced by a pipeline that sets spend where the model says the marginal ROAS is 1.0, is not a measurement of the channel; it is the set-point returned as a measurement (Proposition 13.9). The remedy is not more model but a small amount of deliberate, *logged*, decaying dispersion—20% of spend early, 6% after a decade in the dive’s units—which pays for itself in profit (27% lower cumulative loss) and turns on the only unconfounded clock. The information architecture this needs is a set-point log, which MMM-to-budget pipelines discard; the exogenous clock, the echo test at realised spend and the switch schedule of Theorem 13.17 all consume it. Chapter 7’s deadband protocol gains an “information clock stopped” trigger with a computable false-alarm rate (BL 72); Chapter 3’s switch experiments acquire an optimal schedule.

The third message is a warning about the limits of the fix. Proposition 13.14 shows that the case practitioners actually fear—spend chasing demand the model does not observe—defeats naive and anytime-valid intervals alike (0.000 and 0.007–0.017) while every clock reads “learning”. This is Chapter 9’s leakage in a closed loop and Chapter 4’s traps with a statistic attached; no log-only statistic in the two dives separates it from healthy exploration.

For theory the chapter moves M7 from “find a valid interval” to “the valid interval exists and is cheap; the problem is identification”. It transports Keskin–Zeevi’s incomplete learning to saturating response with a corollary pricing does not need, because pricing reports prices, not measurements; gives the loop-gain question a coefficient (Equation (13.4)); and shows that under budget coupling the uninformative subspace is exactly the budget direction—the seed of Chapter 1’s rank theorem, which proves the sum direction lies in the null space of any number of budget-neutral tones and prices the escape at $4\text{--}615\times$.

13.8 Further exploration

Of Dive 16’s next steps four became backlog items: a calibrated- δ or stitched boundary tuned to the burn-in posterior, to take the CS from $2.3\times$ toward the $1.8\times$ Bonferroni figure without a fixed horizon (BL 55); the stall statistic and echo test (BL 56, largely done by Dive 19 but without a log-only confounding detector); the multichannel geometry (BL 57, done); and adstock, confounding and autocorrelation (BL 58, done). Dive 19 opened five more: a log-only confounding detector, with a Hausman contrast between pulse-window and loop-window slopes or the sign of Equation (13.4)’s skew coefficient as a feedback fingerprint (BL 68); the $O(g^2)$ covariance correction and the optimal design given the loop’s error-correlated wobble, where the answer may be to explore where the wobble is most error-correlated rather than least dispersed (BL 69); the budget-direction escape, since priced in Chapter 1 (BL 70); switch cones under synergy, with triple switches $d_{ij} + d_{jk}$ (BL 71); and an exact echo law from the ridge distribution of $(\hat{\beta}, \hat{K})$, wired into Chapter 7’s protocol with a derived α (BL 72). Both dives propose an empirical run of the clocks on Heusch’s endogenous-spend benchmark, where set-points are known by construction, and Dive 16 notes that Chapter 12’s prequential e-process is the scale-parameter special case of Construction 13.4.

In the compiler’s judgement the most valuable extension is BL 68, because it alone addresses the regime the dives found actually dangerous, and its failure would itself be a theorem worth having: if no function of spend and outcome logs separates exploration from demand chasing, the set-point log is not a convenience but a necessary condition for auditing any closed-loop measurement system. Second is BL 69: the $4\text{--}20\times$ gap between exogenous and loop-generated dispersion is the largest unexplained number in the chapter and decides how much of the “exploration in disguise” in real budgets is worth anything.

Reviewer’s note (compiler)

Timestamps. The INDEX log line for Dive 19 is stamped “2026-09-03 ~14:45” while the file was last written at 21:41 UTC, a seven-hour gap with no stated reason; the Dive 16 pair (03:30 log, 03:44 file) is consistent.

“Nine realism configurations”. Dive 16 §3.3 lists “refresh 1 or 13 weeks, $\rho = 0.5/0.25$, $\sigma = 1/4$, $s = 2$, half or double

gain” and says the stall holds “in all nine configurations”. The shipped E13 () runs seven (base, refresh 1, refresh 13, $\rho = 0.5$, $\sigma = 1$, $s = 2$, gain $\times 2$); $\rho = 0.25$, $\sigma = 4$ and half gain are not in the code. Dive 19’s E10 independently covers $\sigma = 4$ and $s = 2$, so the conclusion is safe, but the count is not reproducible from the file.

“**Matched to three digits**”. The -0.00156 match subtracts a g^2 term ($+0.00030$) estimated from the *Gaussian* run at $g = 0.5$ from the *skewed* run’s -0.00186 ; that the $O(g^2)$ coefficient is the same under skew is assumed, not shown. The raw skewed measurements (-0.00186 to -0.00204 ; clean-room -0.00181 to -0.00203) sit 16–31% above the theory. “Three digits” describes the corrected number only.

Gaussian false-alarm constants differ between text and JSON. Dive 19 §3.1 quotes the Gaussian theory as 0.0085 ($k = 3$) and 8×10^{-5} ($k = 6$), whence “5–150 $\times$ ”; `19_results_all.json` records 0.0069 and 5.5×10^{-5} for the same cells, which would give 6 $\times$ and 220 $\times$. The conclusion is unaffected; the quoted constants do not reproduce from the shipped results.

The exogenous-clock “identity”. $D_t^\perp = vt$ is true by construction—the code sums the injected ζ^2 —so “holds to Monte Carlo precision” checks bookkeeping, not a result; and all five E11 cells share one seed, so $D^\perp / (vW) = 0.956$ in every cell: one realisation rescaled, not five confirmations, and the ledger’s “ $\pm 3\%$ ” should be read accordingly. The threshold and the alarm rates are the genuine content.

Pseudo-inverses (BL 82). The INDEX names dives 16 and 19 among those reporting `pinv`-based numbers without the $s_{\min}/s_{\max}$ audit Dive 21 made mandatory. Dive 19’s pseudo-inverses $(H_f^+, (Ts^2V)^+)$ act on a feasible subspace of *known* rank $n - 1$, so the risk is far lower than for a singular FIM and the switch weights are verified in closed form; the passive-law ratios (1.77, 5.2) and the loss $\frac{1}{2}e^T H_f^+ e$ have nonetheless not had the audit. Dive 16 reports no FIM-based standard errors.

Grids, labels, rounding. Dive 16’s code uses an 80-point K grid on $[0.5, 60]$, Dive 19 a 40-point grid; Dive 16’s burn-in pulse is 0.2 (linear) and 0.3 (Hill), Dive 19’s 0.25. The INDEX rounds the pure random walk to “0.95” (writeup 0.943–0.946), and “coverage 0.95 at any gain” covers a band 0.943–0.957 that includes one seed at 0.943. Dive 16’s claim text mis-stated $\text{sd}(\hat{\beta}_T)$ at $T = 40$ as 0.24 (it is 0.15) until the clean-room caught it. The pricing law is called “verified” on a $\pm 14\%$ grid where the measured values (73–91) straddle theory (85.3) by up to 14%.

Consistency checks passed. $27\% = (82.6 - 60.4)/82.6$; $42\% = 1 - 0.204/0.352$; $55\times = 7,650/138.5$; 158%, 0.27, [83, 134], [35, 238], the switch weights ($S = 13.125$), the passive ratios 1.77/5.18, the in-loop medians and means, the water-filling cells, the exogenous-wobble 0.012, the AR-CS widths and coverages, the E5 rmse values and the adstock cells all reproduce from the JSON to the stated digits.

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Chapter 14

The human-mechanism panel: attention, memory, purchase

Provenance

Source dives: Dive 17 *The human-mechanism panel: what a single-source panel can identify, the two-instrument design, and the SNR ladder* (run 2026-09-03, file `17_human_mechanism_panel.md`, written 2026-09-03 14:24 UTC; results `code/17_results_quick.json`). **Code:** `code/17_human_mechanism_panel.py`. **Frontier-study problem(s) addressed:** H1 — the chain attention → memory (category-entry-point links) → later purchase “has never been measured end to end”; design the minimal longitudinal single-source panel (instruments, power mathematics, cost model). Touches H3 and H5. **Backlog items closed:** none; **opened:** BL 59–BL 62.

Key ideas in this chapter

- With person-randomised ad suppression and repeated memory surveys, the exposure-to-memory kernel (κ, λ) is identified from wave contrasts at $20\text{--}70\sigma$ (half-life 6.6 ± 0.75 weeks from $N = 3,000$; clean-room $\hat{\lambda} = 0.900 \pm 0.007$) while the same experiment measures the purchase link at $t = 1.4$ over two years: the purchase half of the chain sits in Lewis–Rao’s regime.
- Tracking-only decay is biased upward (0.93 versus 0.90) and $3\text{--}7\times$ noisier: persistent heavy attenders masquerade as slow memory.
- Sales-fitted geometric adstock is not the memory half-life: a 20% direct share collapses the pseudo-true α from 0.90 to 0.55 (half-life $6.6 \rightarrow 1.2$ weeks); agreement is uninformative and the gap measures the direct share.
- A rotating half-sample survey turns panel conditioning into a second randomised memory instrument that point-identifies the mediated share (otherwise only upper-bounded); the ads = survey equality test is powerless at $N \leq 10k$, so direct paths are detected by timing (35% share at 65/95/100% power for $N = 3k/10k/30k$).
- Within-person OLS attenuates β_Y $8\text{--}10\times$ (within-person reliability 0.19) and sign-flips under feedback while both instruments hold; design laws (spacing $\approx$ one time constant, span $\approx$ two, a 26-week pre-period worth $2.3 \times N$) give a memory-only panel of $N \approx 500$ (\$0.3M) and the H1 panel of $N \approx 3,500$ (\$2.2M).

14.1 Problem and motivation

Every long-term effect in marketing measurement is inferred through a convention—an adstock decay fitted to aggregate sales, an extrapolation, a 60:40 rule—rather than observed. Problem H1 names the mechanism these conventions proxy for (attention at exposure, encoding into memory linked to category entry points, retrieval at a purchase occasion months later) and the asset that would observe it, a longitudinal single-source panel with attention, memory and purchase on the same people, “arguably the field’s deepest empirical gap”. It does not say what such a panel would *identify* that geo experiments and MMMs cannot, how large it must be, or what it costs.

Dive 17 treats the panel as a measurement instrument: which objects in the chain are identified, under which randomisation, by which contrast; the precision of each as a function of panel size N , waves W , spacing

Δ , survey reliability and purchase frequency f ; and a priced minimal design. The answer is asymmetric. The memory half of the chain is identified an order of magnitude more precisely than sales data allow, because a survey contrast between randomised arms sees a memory difference directly; the purchase half sits exactly where Lewis and Rao (2015) put it and is rescued only by design. Along the way the dive poses a test nobody had run—whether an MMM’s sales-fitted adstock and an independently measured memory decay should agree—and shows that in general they should not.

14.2 Background: the ideas this chapter builds on

14.2.1 Mental availability and category entry points (Sharp, 2010; Romaniuk and Sharp, 2016; Romaniuk, 2023)

The Ehrenberg-Bass account of advertising is a memory account. When a category need arises buyers retrieve a few brands from memory; a brand’s *mental availability* is its propensity to be thought of in such situations, and the retrieval cues are *category entry points* (CEPs), the needs, occasions and companions that define a purchase situation. Mental availability is the breadth and freshness of the brand’s brand-CEP links; Romaniuk’s trackers ask, per CEP, which brands the respondent associates with it, and report the brand’s share of all links (mental market share). Advertising builds and refreshes links (Ehrenberg’s “reinforcement, not persuasion”); because links decay, continuous reach matters. The dive’s latent memory state M_{it} is this construct on one dimension: link strength, refreshed by attention, decaying between exposures.

14.2.2 Attention as a dose (Nelson-Field, 2020; ARF, 2024)

Viewability is not attention. Nelson-Field’s panels record gaze with the device camera during natural media use and classify each second of an exposure as active (eyes on the ad), passive or none; attentive seconds per exposure vary several-fold across platforms at equal viewability, and short-term advertising strength and its decay track attentive seconds, not impressions. Hence the dive’s dose is attentive seconds A_{it} . The construct is not standardised (the ARF’s 2024 validation across 12 providers found convergent validity inconsistent across method classes), so the dive checks that only the instrument’s relative timing, not its level, matters.

14.2.3 Single-source panels (Lodish et al., 1995; Jones, 1995)

IRI’s BehaviorScan split-cable panels (about 3,000 households per market, exposure randomised by cable address) yielded Lodish et al.’s meta-analysis of 389 TV tests, roughly half with a significant sales effect, 55 estimates of persistence, and more than 150 cessation tests of which just over half showed no significant loss. Jones’s STAS panel (2,000 households, meters plus scanners) compared brand share among households exposed in the seven days before purchase with the unexposed, a design Ehrenberg, Broadbent and McDonald contested because heavy buyers are heavy viewers. Modern assets (NCS, Kantar Worldpanel, Numerator, Amplified Intelligence) each cover at most two of the three nodes. Randomisation made BehaviorScan credible; co-observation without it does not separate effect from selection.

14.2.4 The unfavourable economics of ad measurement (Lewis and Rao, 2015)

From 25 large randomised display experiments Lewis and Rao showed why returns to advertising are nearly unmeasurable: individual purchases have a coefficient of variation $CV = \sigma/\mu$ near 10 for a typical retailer while a profitable campaign moves them by a few percent. For a two-arm test of relative lift ℓ ,

$$n_{\text{per arm}} \approx 2 (z_{1-\alpha/2} + z_{1-\beta})^2 (CV/\ell)^2, \quad (14.1)$$

so $CV = 10$ and $\ell = 0.05$ need about 6×10^5 people per arm at 80% power; their median ROI confidence interval exceeded 100 percentage points. The dive’s simulated brand is kinder ($CV \approx 0.93$ over two years)

but Equation (14.1) still governs it: the purchase rung scales as $\sqrt{Nf}$, the memory rung is independent of f .

14.2.5 Heterogeneity versus state dependence (Heckman, 1981; Kenny and Zautra, 1995)

In panel data past outcomes predict current ones both under true persistence (state dependence) and under a persistent individual effect (heterogeneity); a covariance estimator reads both as persistence, and only structure or exogenous variation separates them. Kenny and Zautra's STARTS model (stable trait plus autoregressive trait plus state) gives the covariance between waves k apart as

$$c(k) = V_m + V_x \phi^k, \quad k \geq 1, \quad (14.2)$$

identifiable from four or more waves; a tracker estimating memory decay from its own autocorrelations fits it.

14.2.6 Causal mediation and encouragement designs (Imai, Keele and Yamamoto, 2010; Imai, Tingley and Yamamoto, 2013)

With treatment Z , mediator M and outcome Y , the average causal mediation effect $\mathbb{E}[Y(z, M(1)) - Y(z, M(0))]$ is identified under *sequential ignorability*: treatment ignorable given covariates, and the mediator ignorable given treatment and covariates. Randomising Z buys the first, never the second; the sensitivity analysis parameterises the violation by $\rho = \text{Corr}(\varepsilon_M, \varepsilon_Y)$ and reports the ρ at which the effect vanishes. Imai, Tingley and Yamamoto showed that a randomised *encouragement* of the mediator with an exclusion restriction (it affects Y only through M) identifies the effect without the second assumption, by a Wald ratio of outcome contrast to mediator contrast. The dive's rotating survey is such an encouragement design; the exclusion restriction is its price.

14.2.7 The mere-measurement effect (Morwitz, Johnson and Schmittlein, 1993; Chandon, Morwitz and Reinartz, 2005)

Households asked about their intention to buy a car or a computer subsequently bought at higher rates than unasked households. Chandon, Morwitz and Reinartz attribute this to *self-generated validity* (measurement makes the attitude more accessible, so it drives behaviour more) and find repeated measurement polarises; survey methodologists call it panel conditioning (Warren and Halpern-Manners, 2012). For the dive this is prior art that a survey is a memory intervention: a CEP battery refreshes links by some δ , a silent bias in a fixed panel and, once randomised, an instrument.

14.2.8 Adstock, half-life, pseudo-true values, exponential design (Broadbent, 1979; Jin et al., 2017; Box and Lucas, 1959)

Broadbent's adstock $a_t = x_t + \alpha a_{t-1}$ was introduced as an advertising *stock* in memory decaying geometrically and is still reported as a half-life $\ln \frac{1}{2} / \ln \alpha$ (6.6 weeks at $\alpha = 0.90$). The Bayesian MMM of Jin et al. fits α to weekly aggregate sales, where Chapter 1 shows a nearly flat likelihood ridge. When the true impulse response is not geometric, least squares returns a *pseudo-true* α , the closest member of the geometric family. Box and Lucas showed that the D-optimal observation time for the rate θ of $D_0 e^{-\theta t}$ is the time constant $1/\theta$; and Chapter 10 (Dive 12) established the known-shape/unknown-scalar trick, regressing a noisy response on a temporal shape known from elsewhere so only a scalar is estimated.

14.3 Setup and assumptions

Persons $i = 1, \dots, N$, weeks $t = 1, \dots, T$ ($T = 104$); half randomised to a holdout $Z_i = 0$, half exposed. Impressions per exposed person-week are $\text{Poisson}(\mu_t)$ on a flighted schedule (6 weeks on, 7 off, $\mu = 3$); attentive seconds A_{it} sum Gamma-distributed dwell over impressions with a persistent propensity $a_i \sim \text{Gamma}(2, 1)$ (mean 2 s per impression). Memory is $M_{it} = m_i + x_{it}$, a trait $m_i \sim \mathcal{N}(0, 1)$ plus a transient

$$x_{it} = \lambda x_{i,t-1} + \kappa A_{it} + \eta_{it}, \quad \eta_{it} \sim \mathcal{N}(0, 0.15^2), \quad \lambda = 0.90, \kappa = 0.012. \quad (14.3)$$

The survey $S_{it} = M_{it} + e_{it}$, $e_{it} \sim \mathcal{N}(0, 0.7^2)$, is observed at wave weeks ($\Delta = 8, W = 12$), cross-sectional reliability ≈ 0.72 , sd $\sigma_S \approx 1.3$; optionally the survey adds δ to x_{it} ($\delta = 0.3$ in attacked runs) and a *rotating* design surveys a random half at each wave, $R_{iw} \in \{0, 1\}$. Category occasions are $\text{Poisson}(f)$ per week ($f = 0.5$; swept 0.05–1.5), the brand chosen with probability $\text{expit}(\alpha_i + bM_{it} + dA_{it})$, $b = 0.25, d = 0$ (a *direct* path $d > 0$ is the alternative), $\alpha_i = -2 + 0.5z_1 + \sqrt{0.75}z_2$ so that purchase propensity correlates 0.5 with the memory trait; feedback ψ optionally adds ψ to memory per purchase. Base numbers: 4.4 brand purchases per person-year, exposed-arm lift $\approx 5.4\%$, two-year purchase CV ≈ 0.93 . Targets: (T1) (κ, λ) ; (T2) $\beta_Y := \partial \mathbb{E}[Y_{it}] / \partial M_{it} = fb p(1-p)$ averaged over persons (truth 0.0152 per week per unit); (T3) the mediated share s ; (T4) the sales-fitted α against λ .

Assumption 14.1 (Person-level randomised exposure). Exposure can be suppressed for randomly chosen panelists on their own devices; without it nothing below the memory contrast is identified.

Assumption 14.2 (Known attention path, linear memory). The exposed arm's expected attention path $\mathbb{E}[A_t]$ is known up to scale; memory is the linear AR(1) state Equation (14.3) with one decay.

Assumption 14.3 (Survey exclusion, Level C only). Taking the survey affects purchase only through memory.

14.4 Main results

14.4.1 The identification ladder

Construction 14.4 (Four contrasts, four objects). [established in the dive] *Level A* (memory kernel): the between-arm wave contrast is a convolution of the known attention path with the kernel,

$$\mathbb{E}[S_{iw} \mid Z = 1] - \mathbb{E}[S_{iw} \mid Z = 0] = \kappa \sum_{j \geq 0} \lambda^j \mathbb{E}[A_{w-j}], \quad (14.4)$$

and weighted nonlinear least squares on the W contrasts recovers (κ, λ) with no stationarity assumption, no natural variation and no purchase data. *Level B* (purchase scale): under pure memory mediation the weekly purchase contrast $\text{diff}_t = \mathbb{E}[Y_t \mid Z = 1] - \mathbb{E}[Y_t \mid Z = 0]$ equals $\beta_Y \cdot \text{shape}_t$ with shape_t the Level-A memory path, so only a scalar is estimated from purchases, by a matched filter $\hat{c} = \sum_t w_t \text{diff}_t$ —a difference of arm means of $q_i = \sum_t w_t Y_{it}$, so its standard error is exact. *Level C*: Construction 14.5. *Level D*: Proposition 14.6.

The Level-A numbers are the headline (Table 14.1): $\hat{\lambda} = 0.896$ with Monte Carlo sd 0.012 from 3,000 people, a half-life of 6.6 ± 0.75 weeks, a single wave's contrast at $|t| = 6.7$ on average. Level B from the same experiment gives $t = 1.58$ for the matched filter and 1.39 for the plain two-year total: memory at 7σ per wave, purchases at 1.4σ over two years.

Construction 14.5 (The rotating survey as an ad-free memory instrument). [conjecture / speculative] (The dive's speculative step.) If the survey refreshes memory by δ , randomising which half is surveyed at each wave creates $NW/2$ randomised memory bumps carrying no ad content. The bump is identified from the next wave, $\delta\lambda^\Delta = \mathbb{E}[S_{i,w+1} \mid R_{iw} = 1] - \mathbb{E}[S_{i,w+1} \mid R_{iw} = 0]$, and the purchase response by the Wald ratio

$$\hat{\beta}_Y^{\text{survey}} = \frac{\hat{g}}{\hat{\delta} \sum_{k=1}^{\Delta-1} \hat{\lambda}^k}, \quad (14.5)$$

with $\hat{g}$ the person $\times$ wave two-way demeaned contrast of purchases in the $\Delta - 1$ weeks after a wave between surveyed and unsurveyed waves.

At $\delta = 0.3$, $N = 3,000$ (E4) the instrument returns $\hat{\delta} = 0.307$ and $\hat{\beta}_Y^{\text{survey}} = 0.0147$ ($t = 2.2$) against the ad instrument's 0.0162 ($t = 2.0$) and truth 0.0152; at $\delta = 0$ it returns $\hat{\delta} = 0.003$ (sd 0.059) and is void. The half-sample halves survey cost, and the conditioning bias a fixed panel suffers silently becomes an estimated quantity.

Proposition 14.6 (Partial identification of the mediated share). *[established in the dive]* Write the purchase impulse response as $h_Y = \beta_Y h_M + h_\perp$, with $h_\perp$ unmeasured parallel paths. Level A gives h_M , Level B gives h_Y ; proportionality $h_Y \propto h_M$ is necessary for pure mediation but not sufficient, since a parallel path with the same decay is invisible. From Levels A–B the share $s = \beta_Y \sum h_M / \sum h_Y$ is bounded below by 0 and above by the proportional part of h_Y . Under Assumption 14.3, Level C point-identifies β_Y and s , and $\beta_Y^{\text{survey}} = \beta_Y^{\text{ads}}$ becomes a testable implication of pure mediation.

14.4.2 Tracking-only decay: heterogeneity masquerading as persistence

Proposition 14.7 (Tracking-only decay is biased and noisy). *[numerically established]* Estimating λ from the tracker's between-wave covariances by Equation (14.2) without an experiment is biased upward and 3–7 $\times$ noisier than Level A (E2, 20 replications): pooled $\hat{\phi}^{1/8} = 0.929$ (sd 0.035), arm-demeaned 0.915 (0.037), exposed-only 0.926 (0.056), holdout-only 0.889 (0.078), experimental 0.892 (0.010); clean-room 0.938 (0.033) pooled, 0.902 (0.064) holdout.

Heavy attenders accumulate persistently higher memory across flights and the covariance decomposition reads that as slower decay; the holdout arm is unbiased but has so little transient variance that its sd balloons. A tracker without randomised exposure cannot separate memory persistence from audience persistence at any N (the H5 debate). The sign depends on the heterogeneity being in exposure response; what survives is that the bias is unsigned and unbounded without an experiment.

14.4.3 Sales-fitted adstock is not the memory half-life

Proposition 14.8 (Pseudo-true adstock under a direct path). *[established in the dive]* (Clean-room exact.) If the true sales impulse response is $h_k = \omega \mathbf{1}[k = 0] + (1 - \omega)(1 - \lambda)\lambda^k$, a share ω instantaneous and the rest memory, the least-squares geometric adstock $\alpha^* = \arg \min_a \min_c \sum_k (h_k - c a^k)^2$ takes the values of Table 14.2: at $\lambda = 0.90$, $\omega = 0.2$ gives $\alpha^* = 0.548$, a fitted half-life of 1.2 weeks against a true 6.6.

The geometric family cannot represent a spike plus a tail, and least squares, dominated by lag 0, sacrifices the tail. In the finite-sample MMM version (E7b: $N = 20,000$ aggregate weekly sales, flighted spend) the pure-memory world returns $\alpha = 0.904 \pm 0.018$ but the 20%-direct world (a setting E15 measures at a 19.6% share of the lift) returns 0.80 ± 0.10 , between the two pseudo-true values on the ridge of Chapter 1. The sales-fitted α is thus neither the memory half-life nor a stable pseudo-true value; only the panel's λ is a property of humans rather than of an estimator. The H1 test of the adstock convention is $\hat{\alpha}_{\text{MMM}}$ versus $\hat{\lambda}_{\text{panel}}$: a gap is expected whenever a direct path exists, and its size measures ω .

14.4.4 Design laws

Law 14.9 (Wave spacing and span). *[established in the dive]* (Fisher law clean-room exact.) After a burst the memory contrast decays as $D_0 \lambda^T$; with W waves at lags $0, \Delta, \dots, (W - 1)\Delta$ and noise $2\sigma_S^2 / N$ per wave, the linearised se of $\hat{\lambda}$ is minimised at $\Delta = 11, 8, 6, 4$ weeks for $W = 2, 3, 4, 6$ (se 0.0092, 0.0073, 0.0064, 0.0055 at $D_0 = 1$): the span converges to about two time constants ($2 / (-\ln \lambda) = 19$ weeks), the Box–Lucas point 9.5 weeks is the $W = 2$ optimum, and extra waves inside the span buy little ($W: 2 \rightarrow 6$ cuts the se 1.7 $\times$ for 3 $\times$ the surveys).

The attack round found the single-burst law optimistic (E3, three waves at the realistic $D_0 = 0.34$: Monte Carlo sd 0.034/0.041/0.050/0.168 at $\Delta = 3/6/10/16$ against theory 0.030/0.022/0.022/0.029, i.e. $1.5\text{--}2\times$ the Fisher value for $\Delta \geq 6$, with heavy tails at $\Delta = 16$); the refinement (E3b) compares repeated flights with regular waves at fixed budget $N(300 + 4W)$ dollars: $W = 4/6/12/24$ ($N = 3,303/3,222/3,000/2,636$) gives $\text{sd}(\hat{\lambda}) = 0.054/0.033/0.019/0.016$ and survey-instrument $t = 0.5/1.0/2.2/2.2$. The knee is $\Delta \approx$ one time constant; halving it again is neutral.

Proposition 14.10 (A 26-week pre-period is worth $2.3 \times N$). *[numerically established] Because purchase heterogeneity (α_i) is persistent, each panelist's pre-campaign purchase rate is the strongest covariate available: with flights from week 30 (E14) the total-effect t rises $1.47 \rightarrow 1.67 \rightarrow 1.86 \rightarrow 2.22$ with 0/4/13/26 pre-weeks, $1.51\times$ in t and about $2.3\times$ in N -equivalent (clean-room $1.96 \rightarrow 2.70, 1.38\times$).*

The matched filter, by contrast, gains only $1.58/1.39 = 1.14\times$ under the base flighting; its gain, $\sqrt{T \sum_t s_t^2 / \sum_t s_t}$, grows with sparser flights.

14.4.5 Detecting a direct path by timing, not scale

The equality test of Proposition 14.6 has no power at feasible N (E6): with a 20% direct share the ad instrument reads 0.0197 against the survey's 0.0147 at $N = 3,000$, a 25–34% gap, but the mean z is 0.49 and the rejection rate 0–2.5% at $N \leq 10,000$.

Proposition 14.11 (Shape-transfer test). *[numerically established] Regress the weekly purchase contrast on the Level-A memory shape and the contemporaneous attention path $\mathbb{E}[A_t]$ (a lag-0 term) with the exact person-level standard errors of Construction 14.4; a direct path loads on the lag-0 term. Against a 17% direct share the test rejects at 15/40/88% for $N = 3k/10k/30k$, against 35% at 65/95/100%; the null rate is 0.05–0.10 (E5b, 60 replications: 0.067 at 3k, 0.10 at 30k), mildly anti-conservative because the shape is estimated.*

Shape misspecification manufactures a direct path: with heterogeneous decay ($\lambda \in \{0.7, 0.97\}$, half each) the one-exponential fit follows the slow component ($\hat{\lambda} = 0.962$) and the fast component's residual loads on lag 0, a false rejection rate of 0.183 at $N = 30,000$. Fitting a two-exponential kernel to the wave contrasts (identifiable at 30,000, components 0.69–0.79 and 0.97 recovered; weakly at 3,000) brings false rejection to 0.05, with the base-model null rate 0.117 versus 0.10. The mediated-share estimator (E13) is point-identified but expensive: pure mediation gives median 1.01 (IQR 0.81–1.29) at $N = 10,000$ and 0.97 (0.88–1.09) at 30,000; a 20% direct share (truth 0.80) gives 0.84 (0.65–0.94) and 0.78 (0.70–0.87).

14.4.6 The within-person regression is worthless

Proposition 14.12 (Attenuation and sign reversal of within-person OLS). *[numerically established] OLS of purchases on the measured memory score with person and wave fixed effects gives 0.0016 against 0.0152, a $9.5\times$ attenuation (clean-room $7.75\times$), because the within-person reliability of the score is 0.19 although its cross-sectional reliability is 0.72. With purchase-to-memory feedback $\psi = 0.3/0.6$ the OLS slope turns negative ($-0.0018/ -0.0037$; on the true M , $-0.007/ -0.0095$) while both instruments hold (survey 0.0140/0.0153, ads 0.0156/0.0153).*

Within a person the transient memory variance (sd 0.34 plus the exposure lift) is small against survey error (0.7). The panel's value is in its randomisations, not in co-observation—the individual-level form of Gordon et al.'s (2019) finding that observational methods fail against RCT ground truth.

14.5 Numerical evidence

All experiments run the simulator of the Setup at base parameters, with 40 replications (E1, E14), 20 (E2, E8–E10), 30 (E3b, E4), 60 (E5b), 16 (E13), 12 per cell (E11) and 10 (E7b).

Table 14.1: The identification ladder at $N = 3,000$, two years, $\Delta = 8$, $W = 12$ (E1, E4, E8). Monte Carlo means (MC sd); truth $\lambda = 0.90$, $\kappa = 0.012$, $\beta_Y = 0.0152$; “lin. se” is the dive’s conservative linearised standard error.

Level	Estimand	Estimate (MC sd)	t	Clean-room
A	λ , wave contrasts	0.896 (0.012); lin. se 0.017	54	0.8998 (0.0072)
A	κ , wave contrasts	0.0124 (0.0015)	7.0	0.01226 (0.00136)
B	β_Y^{ads} , matched filter	0.0133 (0.0074)	1.58	—
B	total effect, purchases per 2 yr	0.42 (se 0.30)	1.39	se 0.29–0.30, t 1.74
C	δ , rotating survey ($\delta = 0.3$)	0.307 (0.080)	—	—
C	β_Y^{survey}	0.0147 (0.0057)	2.2	—
—	within-person OLS on S	0.0016	—	0.0019

Table 14.2: Pseudo-true geometric adstock α^* fitted by least squares to a spike-plus-memory impulse response with direct share ω and memory decay λ (E7; clean-room identical to 3–4 digits in all 15 cells).

$\lambda \backslash \omega$	0	0.1	0.2	0.3	0.5
0.80	0.80	0.73	0.60	0.40	0.17
0.90	0.90	0.847	0.548	0.253	0.096
0.95	0.95	0.915	0.238	0.124	0.050

Table 14.3: SNR ladder (E11): t -statistics after two years with 12 rotating waves at $\Delta = 8$, by category frequency f (occasions per week; brand purchases per year in parentheses). Cells are $N = 1,000 / 3,000 / 10,000$; 12 replications per cell, t uncertain by $\pm 15\%$.

f (purch./yr)	$t(\lambda)$	$t(\beta_Y^{\text{ads}})$	$t(\beta_Y^{\text{survey}})$	$t(\text{total})$
0.05 (0.46)	23 / 38 / 72	0.5 / 1.0 / 2.1	0.7 / 0.8 / 1.0	0.4 / 0.9 / 2.0
0.2 (1.8)	24 / 39 / 73	0.8 / 1.6 / 2.7	0.9 / 1.4 / 2.1	0.8 / 1.6 / 2.6
0.5 (4.6)	23 / 38 / 72	1.1 / 1.9 / 3.3	1.4 / 2.1 / 4.1	1.0 / 1.9 / 3.0
1.5 (13.7)	22 / 40 / 71	1.4 / 2.5 / 4.1	1.8 / 3.5 / 6.3	1.3 / 2.4 / 3.9

In Table 14.3 the memory kernel is category-independent at $20\text{--}70\sigma$; the purchase link scales as $\sqrt{Nf}$, and the survey instrument overtakes the ad instrument as f rises because it repeats $NW/2$ times within person. For 80% power ($t \approx 2.8$) at a 5% lift, $\sqrt{N}$ scaling gives frequent CPG ($f = 0.5$) $N \approx 7,400$ (ads) or 4,800 (survey), about 3,300 with a 26-week pre-period; weekly categories 2,000–4,700; infrequent categories ($f = 0.05$) 18,000 (8,000 with pre-period), where the survey instrument is useless (72,000).

Controls: attrition (E9) with per-wave hazards of about 10% depending on the memory trait or purchase propensity (retention 28–34%) leaves (κ, λ) unbiased ($\hat{\lambda} = 0.891\text{--}0.898$) because dropout is independent of Z . Misspecification (E10): saturating encoding gives $\hat{\lambda} = 0.883$ (sd 0.035) and $\hat{\kappa} = 0.0086$ (–30%, the small-dose tangent) with both β 's intact; an attention level mis-estimated $\times 1.3$ rescales $\hat{\kappa}$ exactly ($0.0089 = 0.0116/1.3$) and nothing else; survey noise $\times 2$ moves the λ sd from 0.022 to 0.035, $\div 2$ to 0.018. Cost (E12; assumed prices: recruit \$40, attention device \$120/yr, purchase capture \$60/yr, incentive \$60/yr, survey \$8): \$568 per person for two years, so $N = 1,000/3,000/10,000/30,000$ costs \$0.57M/\$1.7M/\$5.7M/\$17M. Two designs follow. *Design M* (memory only): $N \approx 500$, one year, 6–8 waves at $\Delta = 8$ after two flights, randomised holdout; $\text{se}(\hat{\lambda}) \approx 0.03$ (half-life ± 2 weeks), κ at 3σ , the $\hat{\alpha}_{\text{MMM}}$ -versus- $\hat{\lambda}$ test for one brand, \$0.2–0.3M, nothing on purchase. *Design P* (the H1 panel): $N \approx 3,500$ in a frequent CPG category (8–10k in an infrequent one), a 26-week purchase-only pre-period, two years of flighted exposure with holdout, 12 rotating half-sample waves, device attention on the exposed arm, about \$2.2M; delivers (κ, λ) at 40σ , β_Y at $\approx 2.8\sigma$ per instrument (4σ pooled), 65–95% power against a 35% direct path and a mediated share with IQR ± 0.15 . A ± 0.1 share or 88% power against a 17% path needs $N \approx 30,000$ (\$17M).

14.6 Limitations and the devil's advocate

The dive attacks its own headlines. The 40σ kernel needs person-randomisable exposure and an attention path known up to scale; a panel that measures attention but cannot suppress ads is back to Proposition 14.7. Memory is one-dimensional and linear; if CEP links are many and saturate, κ is a small-dose tangent and λ a dominant-mode decay (0.962 for a mixture's slow component), a property of humans but not what a single- λ adstock means. The adstock table assumes white-noise spend weighting; under flighted spend the MMM sits on a ridge and might land near λ by luck, so only "agreement is uninformative, disagreement expected under a direct path" survives. The survey instrument needs a measurable δ (the mere-measurement evidence concerns intentions questions and durables; at $\delta = 0.1$ the instrument's t falls $3\times$) and an exclusion restriction that self-generated validity arguably violates in the *same* direction as memory. "3,500 people and \$2.2M" holds for a 5% lift in a frequent category with a pre-period: halve the lift and N quadruples, drop the pre-period and it doubles, move to a quarterly category and it triples; the device price may be optimistic by $2\times$.

The attack rounds changed four things: the single-burst optimum gave way to the fixed-budget flighted comparison; the equality test was demoted to a consistency check and the shape-transfer test built; the two-exponential shape was added after heterogeneous decay produced false direct paths; and in Round 4 the pre-period gain fell from $2.5\times$ to $1.51\times$ after the clean-room disagreed (the unadjusted arm had summed pre-period weeks into the outcome). Unmodelled: inter-purchase timing and stockpiling, interference, ghost-ad non-compliance (BL 48), seasonality, attention measurement error, and creative (one brand, so κ is confounded with H4).

14.7 Discussion: impacts

For practice the chapter reorders what a single-source panel is for. The industry's reason to build one, linking exposure to purchase on the same people, is what it does worst: at any affordable N the purchase link is a $1\text{--}3\sigma$ object rescued by a pre-period and category choice, as Equation (14.1) predicts. What it does uniquely well is measure memory dynamics, and Design M already tests a convention every MMM vendor ships. Proposition 14.8 tells Meridian, Robyn and PyMC-Marketing users what the test means: a fitted α far below

a measured λ is an estimate of the direct share, not a contradiction, and a fitted α near λ is not confirmation. For brand-tracking vendors Proposition 14.7 is the uncomfortable one: decay from tracker autocorrelations cannot be trusted without exposure randomisation, whatever the sample. For experimentation teams the design laws are directly usable: a memory wave every time constant, a span of two, a rotating half-sample that halves cost and turns the conditioning artefact into an instrument, and a purchase-only pre-period.

For theory, the ladder makes precise what “measuring the mechanism” buys: an upper bound on the mediated share from kernel proportionality for free, a point at the price of an exclusion restriction, and the lesson that co-observation of mediator and outcome is worth nothing without randomisation. Outward links: the Level-A kernel is the known shape Chapter 10 needs to de-confound long-run effects from observational sales; BL 61 transports (κ, λ) into an adstock prior through the operator likelihood of Chapter 2 and prices it in the decision loss of Chapter 3; the unstable sales-fitted α lives on the ridge of Chapter 1; and Lodish’s cessation tests, the only existing “ λ from purchases”, are under Proposition 14.8 consistent with either fast memory decay or a large direct share—Design M discriminates.

14.8 Further exploration

Four next steps became backlog items. BL 59 replaces the AR(1) with a bounded count of decaying CEP links and asks what the Level-A contrast then estimates and whether the α -versus- λ test survives a non-exponential kernel. BL 60 is the δ pilot: a rotating, memory-only design at $N \approx 500$ to measure the CEP-survey refresh before Design P is funded, with an Imai–Keele–Yamamoto ρ -sensitivity for the exclusion. BL 61 is the value-of-information case for Design M; BL 62 randomises format (skippable versus not) within the exposed arm for a second dose level, testing whether encoding is linear in attentive seconds (H3). Unlisted: a heavy-tail ladder at Lewis–Rao’s CV ≈ 10 , and the join to Chapter 10’s frontier with a measured shape.

In the compiler’s judgement the most valuable extension is BL 60: everything distinguishing this design from BehaviorScan-with-a-survey rests on δ being usable and excludable, a \$0.3M pilot settles the first and bounds the second, and if δ is small the panel reduces to a memory-kernel instrument plus a Lewis–Rao experiment. Second is BL 61, since $\text{se}(\hat{\lambda}) \approx 0.03$ from 500 people is worth exactly what it moves in an MMM’s decisions.

Reviewer’s note (compiler)

What “20–70 σ ” means. The ladder’s $t(\lambda)$ is $\hat{\lambda}/\text{se}(\hat{\lambda})$, a test against $\lambda = 0$; against the alternatives that matter ($\lambda = 0.8$, say) the contrast is $(0.90 - 0.80)/0.017 \approx 6\sigma$ at $N = 3,000$. The honest headline is $\text{se}(\hat{\lambda}) \approx 0.01\text{--}0.02$, and Design P’s “40 σ ” is not a 40 σ discrimination between half-lives.

Numbers that differ between writeup and results file. E3’s Monte Carlo row is sd 0.034/0.041/0.050/0.168 with $\Delta = 16$ mean 0.845 in the writeup but 0.037/0.045/0.060/0.188 and 0.821 in `17_results_quick.json`; no re-run is recorded. The clean-room’s λ sd (0.0072) is 30–40% tighter than the dive’s (0.012), attributed to weighting; by the delta method the half-life uncertainty is ± 0.8 weeks from 0.012 and ± 0.5 from 0.0072. The INDEX log stamps the dive “~10:30” while the file was written 14:24 UTC.

The 9.5 $\times$ attenuation is two factors. In E8 the within-person OLS on the *true* memory gives 0.0087, already 1.7 $\times$ below 0.0152, because purchases in the seven weeks after a wave are regressed on memory at the wave; times the within-person reliability 0.193 this is 0.0017, the reported 0.0016. Reliability explains a factor of about 5; the writeup attributes the whole to it.

Labels and single-seed numbers. The code prints “50% direct share” for the setting $d = 0.5b \cdot 0.30/6$ (JSON key `direct50`); E15 measures its realised share at 19.6% and the writeup correctly says 20%; E5’s 17% and 35% are likewise realised shares of nominal 0.3 and 0.5. The “ $\approx 10\%$ ” lift in the claim sheet was a single-seed number corrected to 5.4% (E15) at verification (§5 of the writeup); the 80%-power sizes rest on $\sqrt{N}$ scaling from 12-replication t -values ($\pm 15\%$), so $N \approx 7,400$ is roughly $\pm 30\%$. The mere-measurement link in the dive’s sources carries the title of Morwitz and Fitzsimons (2004), not Morwitz, Johnson and Schmittlein (1993).

Softened or under-supported claims. Level C is labelled speculative in §1 and §7 of the writeup, but the INDEX line says “point-identifies the mediated share” without the caveat. The two-exponential fix’s base-model null rate (0.117) is slightly worse than the one-exponential’s (0.10); “no loss” is generous. Design M costs “\$0.2–0.3M” in the

writup, “\$0.3M” in the INDEX. On BL 82: E1’s linearised se comes from `pinv` of a 2×2 Gauss–Newton matrix; the problem is well conditioned and the Monte Carlo sd is reported alongside, so the audit does not bite, but the ladder’s $t(\lambda)$ column uses the linearised se. Checks passed: $\ln \frac{1}{2} / \ln 0.9 = 6.58$, $\ln \frac{1}{2} / \ln 0.548 = 1.15$, $1.51^2 = 2.28$, $0.0777 / 0.0104 = 7.5$; Table 14.2 and E7b, E8, E9, E10, E12, E14, E15 match the JSON.

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Program (internal): Dive 01 (adstock ridge), Dive 02 (operator likelihood), Dive 12 (known-shape trick), files 01_mmm_identified_set.md, 02_experiment_transport_map.md, 12_surrogacy_horizon_bounds.md.

Chapter 15

Lift under heavy tails: the price of a margin restriction

Provenance

Source dives: Dive 18 *Semiparametric efficiency for lift under zero-inflated heavy tails: the margin decomposition, the tail-free bound, and the winsorization–regret calculus* (run 2026-09-03, file `18_heavy_tail_efficiency.md`, written 2026-09-03 16:01 UTC). **Code:** `code/18_heavy_tail_efficiency.py`; **results** `code/18_results_quick.json` (the short `quick` run; the writeup’s tables come from the longer `all` run, not on disk). **Frontier-study problem(s) addressed:** M6 — the semiparametric efficiency bound for the ATE of ad exposure with zero-inflated, heavy-tailed outcomes; how close CUPED, latent stratification and trimmed estimators come to it; a principled trade of identified trimming bias against variance minimising *decision regret* rather than MSE. **Backlog items closed:** none; **opened:** BL 63–BL 67.

Key ideas in this chapter

- The unrestricted bound for heavy-tailed lift is Hahn’s, the difference in means (DIM) attains it, and it is Lewis–Rao’s floor: when the tail is idiosyncratic basket size, covariates buy at most 4% in standard error even at CUPED $R^2 = 0.12$.
- Every gain beyond it is a *margin restriction*, priced by the variance-ratio law $V_E/V_U \rightarrow e^{-s^2}$, $V_S/V_U \rightarrow (1 + s^2)e^{-s^2}$ (Hodges–Lehmann: $(1 + \pi s^2/3)e^{-s^2}$) for lognormal baskets of log-sd s ; clean-room 0.187/0.531 against 0.189/0.536.
- Under a restriction the *relative* lift is tail-free ($\sqrt{n}$ at Pareto index 1.5 where DIM decays as $n^{-1/3}$; se ratio $12.6\times$ at 800k) but paid for by the exact composition-bias law $(1 - \kappa)/\kappa$ (+100% at $\kappa = 0.5$), undetectable by the rank test at 200k (power 0.40).
- Flat-prior regret is exactly $\text{MSE}/2$ (additive bias) or $\sigma^2/(2R^2)$ (attenuation), so the winsorisation cap is chosen by the retention index $R(c) = e(c)\sqrt{v(\infty)/v(c)}$ under an effect-location profile: incidence effects want $c \rightarrow 0$ ($2.6\times$), scale effects the 95th percentile (+21%), tail-located effects no cap; per-arm caps inflate the false-positive rate to 0.10–0.12.
- The rank-identified basket ratio $\log \hat{\kappa} = (\text{mean-log shift})/\hat{w}_1$ is composition-robust ($1.6\text{--}3.6\times$ se gains) and cuts go/no-go regret $3\times$ against DIM at 1M (0.75 versus 2.25×10^{-3} ; at 200k DIM is worse than not experimenting, 3.83 versus 2.62), but breaks under a truncated-selection stratum (+122%, coverage 0.64) and is blind, like every restricted estimator, to tail-located effects.

15.1 Problem and motivation

Lewis and Rao (2015) showed that per-user retail sales are so variable relative to any plausible advertising effect that experiments with millions of users cannot separate profitable campaigns from ruinous ones: the standard deviation of sales is about ten times its mean, so a 5% lift is 0.005 of a standard deviation. Problem M6 asked whether this is a *floor*: what is the semiparametric efficiency bound for the average treatment effect (ATE) of exposure when per-user revenue is zero-inflated and heavy-tailed, how close do CUPED, latent stratification and trimmed estimators come to it, and can an *identified* trimming bias be traded against variance so as to minimise the loss of the go/no-go decision rather than the mean squared error?

Dive 18’s first finding is that M6 is posed as if one bound existed. The unrestricted bound is Hahn’s, DIM attains it, and if the outcome variance is infinite so is the bound. Everything the toolkit does better than DIM—dropping the basket from the incidence effect, fitting lognormal strata, capping the top percentile—works by restricting how treatment acts on the two *margins* of the outcome, incidence and basket, whether or not the restriction is stated. The dive makes it explicit, prices it in closed form, prices what it costs when false, shows the false case is undetectable at Lewis–Rao’s sample sizes, and builds an estimator that keeps most of the gain while dropping the worst of the assumption. The regret question has a short exact answer.

15.2 Background: the ideas this chapter builds on

15.2.1 The efficiency bound for the ATE (Hahn, 1998; Hirano, Imbens and Ridder, 2003)

For (Y, D, X) with binary treatment D , potential outcomes $Y(0), Y(1)$ and unconfoundedness given X (automatic in an experiment), let $e(X) = \mathbb{P}(D = 1 \mid X)$, $\sigma_d^2(X) = \text{Var}(Y(d) \mid X)$ and $\tau(X) = \mathbb{E}[Y(1) - Y(0) \mid X]$. Hahn showed that no regular estimator of $\tau = \mathbb{E}[\tau(X)]$ from N observations has asymptotic variance below

$$V_{\text{eff}} = \mathbb{E} \left[\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1 - e(X)} + (\tau(X) - \tau)^2 \right], \quad (15.1)$$

that knowing the propensity score does not lower it, and Hirano, Imbens and Ridder gave an estimator attaining it. “Unrestricted” means the model constrains the law of $(Y(0), Y(1), X)$ only by unconfoundedness and finite second moments: the bound is over every outcome shape. Without covariates and with n users per arm at $e = \frac{1}{2}$, Equation (15.1) is $[\text{Var } Y(1) + \text{Var } Y(0)]/n$, exactly DIM’s variance; when $\text{Var } Y = \infty$ no regular $\sqrt{N}$ -consistent estimator exists.

15.2.2 The unfavourable economics of lift (Lewis and Rao, 2015)

Across 25 large display-advertising experiments for retailers the median ratio of the standard deviation of per-user sales to its mean was 9.9; a campaign lifting sales by a few percent explains $R^2 \approx 5.4 \times 10^{-6}$ of outcome variance, and the median confidence interval on return on investment exceeded 100 percentage points. Lagged sales as a covariate cut residual variance by at most 40% (standard error -23%). A one-sentence remark that decomposing sales into incidence, frequency and basket “could reduce variance” was not developed.

15.2.3 Regression adjustment and its heavy-tailed descendants (Deng et al., 2013; Wang and Burke, 2019; Wicker, 2026)

CUPED replaces Y by $\tilde{Y} = Y - \theta(P - \mathbb{E}P)$ with P a pre-experiment metric and $\theta = \text{Cov}(Y, P) / \text{Var}(P)$ fitted on the pooled sample; a common θ keeps DIM on $\tilde{Y}$ unbiased and cuts its variance by $1 - \text{Corr}(Y, P)^2$. Its heavy-tailed descendants document a pathology reproduced below: winsorising each arm at its own quantile makes the caps arm-dependent random variables and inflates the false-positive rate (Wang and Burke); estimates move 53% between the pooled and arm-stratified conventions (Wicker).

15.2.4 Trimming, two-part models and robust means

Winsorising at a cap c estimates $\tau_c = \mathbb{E}[\min(Y(1), c)] - \mathbb{E}[\min(Y(0), c)]$, identified but not τ ; the classical calculus trades this bias against the variance the tail no longer contributes. Lee (2009) bounds an intensive-margin effect under monotone selection by trimming the arm with surplus positives; Chen and Roth (2024) argue that $\log(1 + y)$ outcomes should give way to an explicit split into an extensive margin (did the user buy) and an intensive one (how much, given buying). Devroye, Lerasle, Lugosi and Oliveira (2016) prove

that with only a $(1 + \alpha)$ -th moment, $\alpha \leq 1$, the minimax deviation rate of any mean estimator is $n^{-\alpha/(1+\alpha)}$, the sample mean’s rate: median-of-means and Catoni estimators buy sub-Gaussian concentration, never a rate.

15.2.5 Trimmed Match (Chen and Au, 2022)

In a randomised paired geo experiment with per-pair response and spend differences $(\Delta Y_i, \Delta X_i)$ the iROAS θ solves $\sum_i (\Delta Y_i - \theta \Delta X_i) = 0$. Under a homogeneous iROAS the residuals $\varepsilon_i(\theta) = \Delta Y_i - \theta \Delta X_i$ are *randomisation-symmetric*—swapping which geo of a pair was treated flips the sign of ε_i —so trimming the largest and smallest residuals, not the outcomes, leaves an exactly unbiased estimating equation; the trim rate minimises the confidence-interval width. This is what M6 meant by “identified trimming bias”; its transport to unpaired user-level tests was not attempted (BL 65).

15.2.6 Latent stratification (Berman and Feit, 2024; Johnson, Lewis and Reiley, 2017)

Write $Y = B V$, a purchase indicator times a basket. Berman and Feit assume monotonicity—the ad never stops a purchase—so users fall into three principal strata: always-buyers, influenced buyers who purchase only if treated, and never-buyers. With a lognormal basket per stratum the two arms form a finite mixture fitted by maximum likelihood; on five catalog experiments the stratified ATE had 30–60% lower variance than DIM. The model is parametric and states no bound. Johnson, Lewis and Reiley gained 31% precision by discarding sales demonstrably unaffected by the ad.

15.2.7 Adaptive estimation of a shift (Stein, 1956; Bickel, 1982; Hodges and Lehmann, 1963)

If two samples share an unknown density f up to a location shift Δ , Bickel proved Stein’s conjecture that Δ can be estimated as well as if f were known: the efficient variance is $(1/n_0 + 1/n_1)/I_f$, $I_f = \int (f'/f)^2 f$ the Fisher location information. The Hodges–Lehmann (HL) estimator, the shift at which the Mann–Whitney statistic takes its null value, has variance $(1/n_0 + 1/n_1)/(12(\int f^2)^2)$, asymptotic relative efficiency (ARE) at least 0.864 against the t -test for any f and $3/\pi \approx 0.955$ at the normal, and needs no moments. A *scale* effect on a positive basket is a location shift of $\log V$; for lognormal baskets $I_f = 1/s^2$.

15.2.8 MSE against rollout regret (Parikh et al., 2026)

Across families of Amazon experiments, Parikh et al. report that the MSE of the effect estimate is convex in the winsorisation level and minimised by covariate adjustment with an outcome-specific cap, yet plain DIM minimises *rollout regret*—“a dramatic divergence between MSE and regret rankings”—with no theory. The dive derives a candidate explanation, a prediction about their data rather than a test of it.

15.3 Setup and assumptions

Two arms of n users each ($\pi = \frac{1}{2}$), user-level randomisation, no interference; each user’s revenue is $Y = B V$ with $B \in \{0, 1\}$ and $V > 0$, observed once.

Assumption 15.1 (Monotone strata). Users are always-buyers (mass p_0 ; $B = 1$ in both arms), influenced buyers (mass $p_1 - p_0$; $B = 1$ only if treated) or never-buyers; no defiers. Write $\ell = p_1/p_0 - 1$ for the relative incidence lift (base 0.05), $w_0 = p_0/p_1$ and $w_1 = 1 - w_0$ for the influenced share of treated positives ($w_1 = 0.048$ at $\ell = 0.05$).

Assumption 15.2 (Basket laws). Always-buyers’ control basket has law F_A with mean m_A and log-sd s ; under treatment it is scaled by $(1 + \gamma)$ (base $\gamma = 0$). Influenced buyers’ basket has law F_I with mean κm_A

(base $\kappa = 1$, $F_I = F_A$). Optionally only the top decile of always-buyers is scaled by $(1 + g_{\text{tail}})$. Base F_A is lognormal with $s \approx 1.4$ after user heterogeneity, $m_A \approx \$389$, $\text{CV}_V \approx 2.7$, hence $\text{CV}_Y \approx 12.7$; the alternative is a Pareto tail of index $\alpha \in (1, 2)$, so $\text{Var } V = \infty$.

The targets are the ATE $\tau = m_A[(p_1 - p_0)\kappa + p_0\gamma]$ (plus a tail term), the relative lift $L = \tau/(p_0 m_A) = \ell\kappa + \gamma$, and the decision “scale iff $L > L_0$ ” with loss $|L - L_0|$ when wrong. A 26-period pre-period revenue P sharing the user’s latent propensity is the covariate ($R^2 \approx 0.12$). Three nested models: $\mathcal{M}_U$ (unrestricted), $\mathcal{M}_S$ (scale: $\kappa = 1$, baskets scaled by $1 + \gamma$) and $\mathcal{M}_E$ (extensive-only: $\kappa = 1$, $\gamma = 0$).

Definition 15.3 (Estimators). DIM; CUPED on P ; winsorised DIM at a pooled cap c fixed by a quantile of the pooled positives; EXT = $(\hat{p}_1 - \hat{p}_0) \hat{m}$, $\hat{m}$ the pooled positive mean (assumes $\mathcal{M}_E$); SCALE = $\hat{m}_0(\hat{p}_1(1 + \hat{\gamma}) - \hat{p}_0)$, $\hat{\gamma}$ from a median or HL shift of $\log V$ between the arms’ positives (assumes $\mathcal{M}_S$); κ -ISO, EXT with a prior on κ ; RANK- κ (Construction 15.11) and its grid-posterior version.

15.4 Main results

15.4.1 The margin decomposition of the bound

Theorem 15.4 (The unrestricted bound is Lewis–Rao’s floor). *[transported from the literature]* Under $\mathcal{M}_U$ without covariates the bound is $V_U = [\text{Var } Y(1) + \text{Var } Y(0)]/n \approx 2p_0 m_A^2 (1 + \text{CV}_V^2)/n$, attained by DIM; for lognormal baskets $1 + \text{CV}_V^2 = e^{s^2}$. Covariates lower it by the share of σ^2 they explain, but the heavy part of σ^2 is the idiosyncratic basket $\text{Var}(V \mid \text{user})$, which no pre-period covariate predicts.

In the base design the covariate bound is 4% below the no-covariate bound in standard error (0.664 against 0.693 on $\tau = 0.973$) although $R^2 = 0.12$. The general claim is structural: the bound retains within-user basket variance, and if $\text{Var } V = \infty$ it is infinite.

Under $\mathcal{M}_S$, $\log V$ is a two-sample shift problem, so by Bickel’s theorem γ is adaptively estimable with information I_f through a moment-free score, and to first order in ℓ

$$V_S \approx \frac{2p_0 m_A^2}{n} \left[(1 - p_0) + \frac{1}{I_f} \right], \quad V_E \approx \frac{2p_0(1 - p_0) m_A^2}{n}, \quad (15.2)$$

the second because under $\mathcal{M}_E$ only two binomials carry sampling error.

Law 15.5 (Variance-ratio law). *[established in the dive]* For lognormal baskets with log-sd s , to first order in ℓ and p_0 ,

$$\frac{V_E}{V_U} \rightarrow (1 - p_0) e^{-s^2}, \quad \frac{V_S^{\text{HL}}}{V_U} \rightarrow \left(1 + \frac{\pi}{3} s^2\right) e^{-s^2}, \quad \frac{V_S^{\text{eff}}}{V_U} \rightarrow (1 + s^2) e^{-s^2}. \quad (15.3)$$

The law is p_0 -independent to first order and sees the tail only through s : at $s = 0.5$ there is nothing to gain (0.78, 0.98); at $s = 1.27$ ($\text{CV}_V = 2$) EXT has one fifth of DIM’s variance and SCALE one half; at $s = 2$ ($\text{CV}_V \approx 7$) 1.8% and 9%. The clean-room re-derivation gave 0.187 ± 0.008 (predicted 0.189) and 0.531 ± 0.018 (0.536) at $s = 1.27$ with 2000 replications; “efficient” rests on Bickel, the HL and efficient constants being 3% apart there.

Proposition 15.6 (Tail-free relative lift). *[established in the dive]* Under $\mathcal{M}_S$ or $\mathcal{M}_E$ the relative lift $L = \ell\kappa + \gamma$ involves no moment of F_A : ℓ comes from binomials, γ from ranks; the absolute ATE keeps m_A only as a common scale of weight τ/m_A . Under a Pareto tail of index $\alpha \in (1, 2)$ DIM’s error decays as $n^{-(1-1/\alpha)}$ while EXT and SCALE decay as $n^{-1/2}$.

At $\alpha = 1.5$ (E3; 50k–800k per arm, 100 replications) DIM’s error scale ($1.4826 \times \text{MAD}$) shrinks with exponent -0.328 (theory $-1/3$), its ordinary sd is erratic (0.66 at 200k, 0.67 at 800k; kurtosis 5–28) and its normal intervals over-cover (0.98–0.99) because the plug-in se is inflated by whichever outlier is in sample; EXT and SCALE shrink at -0.46 and -0.50 with coverage 0.93–0.98, and at 800k the se ratio is $12.6 \times$ (0.67 against 0.053). Clean-room: -0.311 and -0.465 . By Devroye et al. the *unrestricted* rate is $n^{-(1-1/\alpha)}$ for every estimator: the $\sqrt{n}$ rate is bought by the restriction, not by robustness.

15.4.2 What the restriction costs when it is false

Proposition 15.7 (Composition-bias law). *[established in the dive]* If influenced buyers spend κ times what always-buyers spend, EXT's relative bias is $(1 - f_I(1 - \kappa)) / \kappa - 1$, f_I the influenced share of pooled positives; in the small- ℓ limit, $(1 - \kappa) / \kappa$.

At $\kappa = 0.5$ this is +100% (+103% observed; clean-room $+99.6 \pm 1.7\%$); κ -ISO with prior 1 ± 0.25 is +95%; SCALE-HL is -32%, its rank shift half-absorbing the composition, and -133% (a sign flip) at $\kappa = 0.3$; DIM is unbiased. The tail-free gain is an *assumption purchase*, and the assumption is about people, not tails. It is untestable where it matters: the Mann-Whitney test of $F_1^+ = F_0^+$ (treated against control positives) is tail-free but has power 0.40 at $\kappa = 0.5$ and 0.77 at $\kappa = 0.3$ with 200k per arm (null 0.02–0.05), reaching 0.925 at $\kappa = 0.5$ only with 1M.

15.4.3 The winsorisation calculus

Definition 15.8 (Effect-location profile and retention index). For a pooled, pre-fixed cap c let $v(c)$ be the per-user variance of $\min(Y, c)$ under control, $e(c) = \tau_c / \tau$ the *effect-location profile*, and $R(c) = e(c) \sqrt{v(\infty) / v(c)}$ the *retention index*. The path $c \downarrow 0$ is EXT ($\min(Y, c) \rightarrow c \mathbf{1}\{Y > 0\}$); $c = \infty$ is DIM.

Lemma 15.9 (Additive bias, flat prior). *[established in the dive]* For “go iff $\hat{\tau} > 0$ ” with loss $|\tau|$ when wrong, $\hat{\tau} \sim \mathcal{N}(\tau + b, \sigma^2)$ and a flat prior on τ , the Bayes regret is $(b^2 + \sigma^2) / 2$: exactly half the MSE.

Lemma 15.10 (Proportional bias, flat prior). *[established in the dive]* If $\hat{\tau} \sim \mathcal{N}(e\tau, \sigma^2)$ the regret is $\sigma^2 / (2e^2) \propto v(\infty) / R^2$; the MSE is improper ($b^2 \propto \tau^2$ does not integrate) and attenuation is free whenever it buys more than proportional variance.

Both were verified by quadrature to six decimals. For a fixed additive bias “minimise regret rather than MSE” is therefore empty; the substantive question is which prior and which *kind* of bias. For a threshold decision $\hat{L} > L_0 > 0$ the attenuated estimator must first be de-biased by $e(c)$, under $\mathcal{M}_S$ a tail-free functional of $\hat{F}_A$; un-debiased attenuation biases toward “no-go”, helping when the prior sits below the threshold (regret 0.058 against 0.103) and hurting above it (0.301 against 0.103).

The profile decides the cap (Table 15.3): for an incidence-driven effect R rises as c falls and the best cap is $c \rightarrow 0$, a gain of $e^{s^2/2} \approx 2.6\times$ in se; for a 5% scale effect it is the 95th percentile, worth +21%; for an effect in the top decile every cap below the 99.9th percentile *loses*. The industry’s “99th–99.9th percentile” is right for scale-type effects only. Hence a derived reading of Parikh et al.: MSE-optimal winsorisation is aggressive when effects are small relative to noise ($\text{MSE} \approx v(c)$) while regret follows $R(c)$, and the rankings diverge when $e(c)$ falls faster than $\sqrt{v(c)}$ —a tail-located profile. The cap must also be a fixed functional of the pooled data: under the null with a 99th-percentile cap (E11, 200 replications) a pooled cap keeps the two-sided false-positive rate at 0.055 for lognormal and Pareto baskets, per-arm caps give 0.095 and 0.12.

15.4.4 The rank-identified basket ratio (Round 3)

Construction 15.11 (RANK- κ). *[conjecture / speculative]* Under Assumption 15.1 with $\gamma = 0$ the treated positives are the mixture $F_1^+ = w_0 F_A + w_1 F_I$, so $F_I = (F_1^+ - w_0 F_0^+) / w_1$ is nonparametrically identified, but its mean is a heavy-tailed functional (DIM in disguise). Impose the *scale-mixture shape* $F_I(y) = F_A(y / \kappa)$. Then, exactly,

$$\mathbb{E}[\log V \mid \text{treated positive}] - \mathbb{E}[\log V \mid \text{control positive}] = w_1 \log \kappa, \quad (15.4)$$

so $\log \hat{\kappa} = (\text{mean-log difference}) / \hat{w}_1$ needs only $\text{Var}(\log V) < \infty$, true for lognormal and Pareto alike, and $\hat{L} = \hat{\ell} \hat{\kappa}$ is a tail-free relative lift built from two binomials and two log-means.

With an intensive effect the shift is $w_0 \log(1 + \gamma) + w_1 \log \kappa$; computing $\hat{\kappa}$ as if $\gamma = 0$ gives $\hat{\kappa} \rightarrow \kappa(1 + \gamma)^{w_0/w_1}$, exponent 20 at $\ell = 0.05$, but $\hat{L} = \ell \hat{\kappa} \rightarrow \ell \kappa + \kappa \gamma + O((\gamma/\ell)^2)$, the true $\ell \kappa + \gamma$ to first order when

$\kappa \approx 1$: the rank shift measures the *total* intensive-margin lift and the κ/γ split is irrelevant at first order (clean-room: $\hat{\kappa} \rightarrow 1.2202$ and $\hat{L} = 0.0610$ against 0.0600 at $\gamma = 0.01$; the expansion breaks at $\gamma = \ell$, where $\hat{L} = 0.133$ against 0.100). Precision: $\text{se}(\log \hat{\kappa}) \approx s\sqrt{1/n_0^+ + 1/n_1^+}/w_1 \approx 0.46$ at 200k per arm, 0.19 at 1M; the estimator is median-unbiased but mean-biased by about $\exp(\text{se}^2/2) - 1$ (+11% at 200k), and a grid posterior over $\log \kappa$ with prior $\mathcal{N}(0, 0.5^2)$ stabilises it when $\Delta \hat{p} = \hat{p}_1 - \hat{p}_0$ is noisy. Dividing by $\hat{w}_1 \propto \Delta \hat{p}$, it is undefined without an incidence effect—exactly where the intensive margin carries everything—so the scale and rank constructions are complementary, not nested.

15.4.5 The blind spot and its guard

Proposition 15.12 (Tail-located effects). *[numerically established] When only the top decile of always-buyers spends +50% ($\tau = 5.04$, no incidence effect; E9) DIM recovers 105% of it at $t = 6.3$, winsor-99 72%, EXT 0.8%, SCALE 5%, and RANK- κ is undefined: every restricted estimator is blind to a tail-located effect by construction. The tail-free detector is the exceedance count $\mathbb{P}(Y > c)$ by arm on a grid of c : $z = 5.5$ (power 1.0) at the 99th percentile, $z = 0.4$ at the 90th, since a scale effect on the top decile does not move $\mathbb{P}(Y > c_{90})$.*

A grid of exceedance tests is thus the pre-registrable guard that switches from restricted to unrestricted analysis (BL 66).

15.5 Numerical evidence

All experiments simulate ?? 15.1?? 15.2 with $p_0 = 0.05$ and a 5% incidence lift unless stated. Table 15.1 is the Round-1 benchmark (E1: 200k per arm, 200 replications, $\tau = 0.973$, Monte Carlo error on each sd about 5%): DIM sits on Hahn’s bound with $t = 1.36$ (Lewis–Rao’s $t \approx 1.0$ –1.3), CUPED buys the 4% available, the 99th-percentile cap buys 27% in se at -9% attenuation, EXT has $t = 3.63$.

Table 15.1: E1 benchmark, 200k per arm, 200 replications, $\tau = 0.973$, base DGP. Hahn’s no-covariate bound is se 0.693, the covariate bound 0.664; the law predicts se ratios 0.27 (EXT) and 0.73 (SCALE-HL).

Estimator	Bias	sd	$t = \tau/\text{sd}$
DIM	−0.01	0.716	1.36
CUPED ($R^2 = 0.12$)	−0.01	0.679	1.43
Winsor 99th / 99.9th / 99.99th	−0.089 / −0.032 / −0.019	0.526 / 0.643 / 0.695	1.85 / 1.51 / 1.40
SCALE (median / HL)	−0.02 / −0.02	0.594 / 0.517	1.64 / 1.88
EXT	+0.01	0.268	3.63
κ -ISO (prior 1 ± 0.25 ; posterior sd 0.357)	+0.01	0.278	3.50
RANK- κ / posterior	+0.08 / +0.01	0.541 / 0.386	1.80 / 2.52

Table 15.2 verifies Law 15.5 (E2: 200k per arm, 60 replications, $\pm 18\%$ Monte Carlo error per ratio). At $s = 2$ the clean-room’s numerators were within 5% of theory but $\text{Var}(\text{DIM})$ was under-measured by 13% even with 2000 replications (its fourth moment is e^{6s^2}): the law is asymptotic and at $s = 2$ hard to *see*. E14 ($p_0 \in \{0.01, 0.05, 0.2\}$) gave EXT ratios 0.21/0.24/0.15 against 0.20/0.19/0.16 predicted; E12 found HL attaining the bound on a log-logistic basket (ARE 1.15 ± 0.14) and 0.83 ± 0.07 on the lognormal (theory 0.955).

Table 15.4 is the composition attack and its repair (E5/E6; 200k per arm, 100 replications). RANK- κ ’s bias is the Jensen term at every κ and its se gain shrinks as κ grows because $\text{se}(\hat{\tau}) \propto \kappa \text{se}(\log \hat{\kappa})$; the clean-room reproduced its median 0.498 ± 0.008 at $\kappa = 0.5$. At 1M, $\kappa = 0.5$ gives bias -0.012 ± 0.019 and sd 0.121 against DIM 0.354 ($2.9\times$), $\kappa = 1$ gives 0.206 against 0.355 ($1.7\times$). At 200k the Wald interval on L covers 0.87–0.93, a log-scale interval 0.94 (E7).

Table 15.2: E2: observed / predicted variance ratios against DIM over the basket log-sd s (200k per arm, 60 replications, $\pm 18\%$ per ratio). The code’s EXT prediction omits the $(1 - p_0)$ factor of Equation (15.3).

s	0.5	0.9	1.27	1.6	2.0
V_E/V_U	0.777 / 0.779	0.485 / 0.445	0.221 / 0.199	0.082 / 0.077	0.018 / 0.018
V_S^{HL}/V_U	1.04 / 0.98	0.96 / 0.82	0.63 / 0.54	0.31 / 0.28	0.093 / 0.095

Table 15.3: E4: effect-location profile $e(c)$ and retention index $R(c)$ by cap (pooled positive quantile; $R(\infty) = 1$), population values from a 3M-user always-buyer sample. The best cap maximises R .

Effect model		0.5	0.9	0.95	0.99	0.999	0.9999
extensive, $\kappa = 1$	$e(c)$	0.28	0.69	0.80	0.93	0.99	1.00
	$R(c)$	2.31	1.77	1.60	1.32	1.13	1.06
extensive, $\kappa = 0.5$	$e(c)$	0.44	0.83	0.90	0.97	1.00	1.00
	$R(c)$	3.56	2.12	1.81	1.39	1.14	1.06
scale, $\gamma = 5\%$	$e(c)$	0.08	0.46	0.60	0.83	0.96	0.99
	$R(c)$	0.68	1.17	1.21	1.18	1.10	1.05
tail, top decile +50%	$e(c)$	0	0	0.15	0.61	0.89	0.97
	$R(c)$	0	0	0.30	0.87	1.02	1.03

Table 15.4: E5/E6 composition attack, 200k per arm, 100 replications: relative bias of the restricted estimators; RANK- κ ’s bias ($\pm$ Monte Carlo se), sd (DIM’s is 0.76 throughout) and median $\hat{\kappa}$; power of the Mann–Whitney test of $F_1^+ = F_0^+$ at 5%. The law $(1 - \kappa)/\kappa$ gives +233%, +100%, 0, -33% .

κ	EXT	SCALE-HL	RANK- κ bias	RANK- κ sd (gain)	median $\hat{\kappa}$	MW power
0.3	+237%	−133%	+0.05 $\pm$ 0.02 (+17%)	0.21 (3.6 $\times$)	0.317	0.77
0.5	+103%	−32%	+0.06 $\pm$ 0.03 (+12%)	0.29 (2.6 $\times$)	0.514	0.40
1.0	≈ 0	≈ 0	+0.10 $\pm$ 0.05 (+10%)	0.47 (1.6 $\times$)	1.01	0.02–0.05 (null)
1.5	–	–	+0.16 $\pm$ 0.07 (+11%)	0.69 (1.1 $\times$)	1.46	–

The *power control* (E8; 200k per arm, $\kappa = 0.7$, one-sided 5%, 200 replications, Monte Carlo se 0.035) gives null rejection rates 0.05 (DIM), 0.025 (EXT), 0.025 (incidence z), 0.05 (winsor-99), and power against a 5% incidence lift (a 3.5% revenue lift) of 0.255, 0.355 and 0.955 (EXT and incidence z). Seed sensitivity (E13; three 60-replication seeds) spans DIM 0.56–0.69, EXT 0.22–0.26, RANK 0.39–0.55.

Table 15.5 is the decision duel M6 asked for (E10): breakeven $L_0 = 0.03$, $\kappa \sim U[0.3, 1.2]$ per replication so that $L = 0.05\kappa$ with 67% of prior mass above L_0 , loss $|L - L_0|$ when wrong, in units of baseline revenue per user. At Lewis–Rao’s 200k a DIM-based decision has *higher* regret than not experimenting (3.83 against always-go’s 2.62): its noise switches off more good campaigns than bad ones. MSE and regret rank differently exactly as Lemmas 15.9 and 15.10 predict when bias is neither additive nor flat-prior: EXT’s MSE on L is $2.5\times$ better than DIM’s (5.2 against 12.9×10^{-4}) yet its regret is no better, since assuming $\kappa = 1$ it says “go” almost always; the posterior has the best MSE at 200k (2.7) but RANK- κ the best regret. At 1M RANK- κ cuts regret $3\times$ relative to DIM (0.75 against 2.25, five Monte Carlo se apart) while DIM is only 14% below always-go: the decision value comes almost entirely from the margin decomposition, not from sample size.

Table 15.5: E10 go/no-go regret ($\times 10^3$, baseline revenue per user) at $L_0 = 0.03$, $\kappa \sim U[0.3, 1.2]$; $\pm$ Monte Carlo se. Always-go and never-go need no experiment.

n per arm	DIM	EXT	RANK- κ	posterior	κ -ISO	always-go	never-go
200k (600 reps)	3.83 ± 0.28	3.24 ± 0.22	2.21 ± 0.22	2.54 ± 0.22	3.26 ± 0.23	2.62	9.81
1M (300 reps)	2.25 ± 0.29	2.60 ± 0.25	0.75 ± 0.15	0.79 ± 0.15	2.46 ± 0.25	2.61	9.98

15.6 Limitations and the devil’s advocate

Round 1 built the decomposition, the calculus and EXT/SCALE. Of Round 2’s four attacks, composition and tail-located effects landed (Propositions 15.7 and 15.12), infinite variance was survived (Proposition 15.6), and testability failed where it matters (power 0.40). Round 3 built RANK- κ and attacked it. (i) *Shape*. If marginal buyers are the *small* buyers by selection— F_I is F_A truncated below its median, effective $\kappa = 0.17$ —the mean-log shift identifies the geometric-mean ratio, not the arithmetic one: median $\hat{\kappa} = 0.345$ (twice too high), τ bias +122% at bias/sd 0.9, log-interval coverage 0.64 (truncation at the 80th percentile: $\hat{\kappa}$ 0.64 against 0.38, coverage 0.72). A second-log-moment correction assuming F_I lognormal, $\kappa_{\text{arith}} = \kappa_{\text{geo}} \exp((\sigma_I^2 - \sigma_A^2)/2)$ with both variances identified from the mixture, moves the median to 0.22 (truth 0.169) with interquartile range 0.16–0.37: a partial, noisy repair. (ii) *Unmodelled* γ . A 1% basket effect on always-buyers moves $\hat{\kappa}$ to 1.22 (predicted exactly), $\hat{L}$ to 0.069 against 0.060 and τ by +14% at bias/sd 0.3; at $\gamma = 3\%$, $\hat{\kappa} \rightarrow 1.74$ and the sd explodes to 5.9, so the posterior version is required. (iii) *Heavier tail* (Pareto $\alpha = 1.5$, $\kappa = 0.7$): RANK- κ sd 0.32 against DIM 1.57 (4.8 $\times$), bias -0.03 ± 0.03 , coverage 0.94; EXT +29% against the law’s +42%, reconciled through the finite- ℓ form and the downward finite-sample bias of Pareto sample means. Round 4 changed no construction; Round 5 (clean-room) confirmed the five headline claims and supplied the finite- ℓ bias form, the γ/ℓ expansion and the exponential-of-log mean bias.

The dive’s own devil’s advocate: the 4% is a property of a DGP in which the heavy tail is idiosyncratic basket size; with persistent baskets $E[\sigma^2(X)]$ shrinks more and nonparametric adjustment could beat CUPED’s linear projection. Law 15.5 is first order in ℓ and p_0 at $\pi = \frac{1}{2}$ for lognormal baskets. “Tail-free” does not mean the tail has gone: it has been moved into κ . Both lemmas assume a flat prior and normal errors, and under infinite variance DIM’s errors are not normal at any feasible n . The $3\times$ regret gain is at 1M under a prior uniform on $[0.3, 1.2]$, where EXT’s $\kappa = 1$ is wrong most of the time; under a prior concentrated at $\kappa \approx 1$, EXT or κ -ISO would win. At 200k RANK- κ ’s gain over always-go is 16%: it makes a 200k-user experiment *barely* worth running, where DIM makes it worth *not* running. The exceedance guard had power 1.0 because the effect was +50% on a decile; +5% on the top percentile would be invisible to it and to DIM alike. Structural limits: single-period outcomes (no frequency margin), no interference (Chapter 8), monotone strata (defiers break the mixture algebra), a stylised covariate, no real experiment library.

15.7 Discussion: impacts

For experimentation platforms the choice of *estimator family* is a choice of which margin to restrict and should be reported as such: a lift report showing the incidence lift, the rank-based basket shift and DIM side by side exposes the restriction the headline rests on, with the Mann–Whitney statistic on positives and the exceedance grid as its tail-free diagnostics (the first weak at Lewis–Rao’s sample sizes). The winsorisation level should be a pre-registered function of an effect-location hypothesis through $R(c)$, and the cap must be pooled: per-arm caps double the false-positive rate. For MMM calibration (Chapter 2) the lift test’s standard error is what enters the likelihood; under an incidence-driven effect the tail-free estimator shrinks it by $e^{s^2/2}$, the difference between a prior that moves the model and one that does not, but the calibration then inherits an assumption about marginal buyers the MMM cannot check.

For CMO decision-making the sharpest result is Table 15.5: at 200k users per arm, deciding on DIM is worse than not measuring, and the experiment becomes worth running only through a restriction whose failure the experiment cannot detect. Lemma 15.10 with an informative prior belongs in the decision loss of Chapter 7 and the value-of-information machinery of Chapter 3 (BL 67); the exceedance grid is the trigger Chapter 7’s protocol lacks for switching from restricted to unrestricted analysis (BL 66). Chapter 12’s certification law loses coverage under t_3 recalibration noise (BL 51); this chapter says where the heaviness comes from and which functional of it is Gaussian.

For theory the chapter settles M6’s first part negatively (the unrestricted bound is Hahn’s and can be infinite) and reframes the rest: the interesting bounds are the restricted, tail-free ones, and the regret question is not “regret versus MSE” but “which prior and which kind of bias”. The composition law and the untestability result are why latent stratification’s 30–60% gains should carry a κ sensitivity band.

15.8 Further exploration

The dive’s next steps became five backlog items. BL 63 fits the scale mixture to Berman–Feit-style catalog data for the cross-category distribution of κ —the prior Table 15.5 needs and the empirical answer to “do marginal buyers spend less”—and tests truncation against scale with the second-log-moment statistic. BL 64 extends the decomposition to Lewis–Rao’s three margins, incidence $\times$ frequency $\times$ basket: counts are tail-free, only the basket is heavy. BL 65 transports Trimmed Match’s symmetric residual to user-level tests by pairing within pre-period strata. BL 66 derives the joint null of the exceedance z ’s over a quantile grid (a Brownian-bridge functional) and its power. BL 67 replaces Lemma 15.10’s flat prior by the decision loss of Chapter 3 and Chapter 7, giving $c^*(\text{prior}, e(\cdot), v(\cdot))$ with a composition band, and tests the effect-location reading of Parikh et al. on their data if obtainable. Not adopted: recomputing Chapter 11’s precision–lift frontier and Chapter 12’s certification law under $\mathcal{M}_E$ (the user count changes by e^{s^2} for incidence-driven effects; the κ of Chapter 11 is a different quantity), and a semiparametric strata model replacing Berman–Feit’s lognormal components by the mixture-subtraction identification of F_I .

In the compiler’s judgement BL 63 dominates: every headline here is conditional on κ and on the shape of F_I , both of which five catalog experiments would measure directly, and a measured κ prior would turn Table 15.5 from a stylised duel into a sizing rule. Second is BL 64, because the frequency margin is where Lewis–Rao placed the recoverable variance and it is tail-free by the same argument as incidence.

Reviewer’s note (compiler)

The results file on disk is the quick run. `18_results_quick.json` has 60 replications for E1 and fewer for E5/E8/E9/E11/E14; its numbers differ from the writeup’s 200-replication run, which is not on disk: E1 DIM sd 0.676 against 0.716, RANK- κ 0.485 against 0.541, posterior 0.326 against 0.386; E4 retention values differ by up to 6% (2.18 against 2.31 at the median cap, 3.37 against 3.56, scale-model best 1.15 against 1.21). The differences sit within E13’s seed spread and change no ordering, but Tables 15.1, 15.3 and 15.4 cannot be reproduced from the file shipped. **Numbers the writeup does not report.** The quick-run E8 gives the posterior RANK- κ a null rejection rate of 0.48

under the criterion “posterior lower bound on L above zero” (100 replications); the writeup’s E8 lists the other four estimators only. Since the posterior version is “required” at $\gamma = 3\%$ and has the best MSE at 200k, a reader should check whether this is the criterion (the posterior is formed only when $\hat{p}_1 > \hat{p}_0$) or a calibration failure. The quick-run E14’s SCALE-HL ratios (0.47/0.95/0.58 against 0.536 predicted, $\pm 23\%$) are unreported in the writeup; the $p_0 = 0.05$ value is 78% above prediction. E11’s quick run gives pooled-cap false-positive rates of 0.08 for both laws against the writeup’s 0.055 (per-arm 0.12/0.15 against 0.095/0.12); at 100 replications the Monte Carlo se is 0.027, so “0.055” is not a demonstrated nominal rate.

Small inconsistencies. E2’s coded EXT prediction is e^{-s^2} without the $(1 - p_0)$ factor that Equation (15.3), E14 and the clean-room (0.189) include; 5% at $p_0 = 0.05$, inside the $\pm 18\%$ error; the INDEX log line also drops it. “Covariates buy $\leq 4\%$ ” is in standard error (0.664/0.693), 8% in variance, and CUPED’s observed gain (0.716 to 0.679) is 5% in se, so “the 3% seen here” in the writeup’s devil’s advocate is not reconstructible from its table. The writeup calls RANK- κ ’s 16% gain over always-go at 200k “1.4 MC se”; from Table 15.5 it is $0.41/0.22 = 1.9$ se. E9’s t is 6.3 in the writeup, 7.7 in the 50-replication quick run.

Novelty and support. Lemma 15.9 is elementary (at $b = 0$ it is $2\sigma^2 \int_0^\infty u\Phi(-u) du = \sigma^2/2$); its value is as a corrective to M6’s phrasing. The “efficient” constant of Equation (15.3) is not separated from the HL constant by any simulation, and E12’s lognormal ARE of 0.83 ± 0.07 is 1.8 se below theory. The “12.6 $\times$ ” divides EXT’s se by DIM’s *ordinary* sd, which the dive itself calls erratic. The 2026 preprints (Parikh et al., Wicker) are absent from the program’s bibliographies and could not be checked.

Consistency checks passed. The INDEX log line agrees with the writeup on every number checked (0.187/0.531, 12.6 $\times$, 0.40, 2.6 $\times$, +21%, 0.10–0.12, 0.75/2.25, 3.83/2.62, +122%, 0.64, $z = 5.5$); the log timestamp ($\sim 16:10$) follows the file write (16:01). Table 15.2 matches the JSON to three decimals; $1.01^{20} = 1.2202$, $0.05 \times 1.2202 = 0.0610$ and $0.05 \times 1.05^{20} = 0.133$ reproduce; BL 63–BL 67 match the INDEX.

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- [17] Frontier study (internal): 02_open_questions.md, entry M6; 01_state_of_the_field.md on Lewis–Rao and Trimmed Match.
- [18] Program (internal): Dive 18, file 18_heavy_tail_efficiency.md; 00_INDEX.md log row 18 and backlog BL 63–BL 67; Dive 15 for BL 51.

Appendix A

The open-problem ledger

The program’s rule 5 requires every dive that discovers a new open problem to append it to the backlog with a one-line rationale. Eighty-eight such items existed at compilation. Table A.1 lists them in numerical order with the dive that opened each, its status on 2026-09-07 as recorded in the index (“resolved”, “priced” and “discharged” are the index’s own words; “partly” marks items the index says were advanced but not closed), and the chapter of this compendium in which the item is discussed. The one-line rationales are in the chapters’ “Further exploration” sections; the index file carries the full text.

Fourteen items were closed by later dives: BL 1, BL 2, BL 73 and BL 75 by the Phase D identification chain; BL 5 and BL 6 by Dive 04; BL 20 by Dive 08; BL 21 by Dive 09; BL 42 by Dive 14; BL 57 and BL 58 by Dive 19 (BL 56 partly); BL 70 was priced rather than solved by Dive 21; and BL 82 was discharged for dives 20–22 only. The index names three candidates for run 23: computing the shape-prior ledger on a real fitted MMM’s Hessian together with the flighting-calendar rank audit (BL 81 + BL 74 + BL 86), the response-curve estimand for prior pricing (BL 85), and intermodulation under synergy (BL 79).

Table A.1: The open-problem ledger: every backlog item (BL) spawned by the program, in numerical order, with the dive that opened it, its status as of 2026-09-07, and the chapter of this compendium in which it is discussed.

BL	Problem	Opened by	Status	Chapter
1	Seasonality-orthogonal perturbation design	01	resolved by Dive 20	1
2	The price of identification	01	resolved by Dive 20	1
3	Calibration under model misspecification	02	open	2
4	A specification test for transport	02	open	2
5	The EVSI power plateau — boundaries	03	resolved by Dive 04	3
6	Second-order EVSI scoring	03	resolved by Dive 04	3
7	Power–cost EVSI frontier	03	partly advanced by Dive 04	3
8	Optimal switch-experiment sizing	04	open	3
9	The k-experiment portfolio law	04	open	3
10	LQC robustness under misspecification	04	open	3
11	Belief-independent risk ledgers	05	open	4
12	The low-drift trap paradox	05	open	4
13	Filter-class dependence of trap economics	05	open	4
14	DP-noised covariates through the Hill link	06	open	5
15	Endogenous share drift	06	open	5
16	Optimal release granularity G	06	open	5
17	Analytic transport de-biasing	06	open	5
18	Rotation theory of refresh decorrelation	07	open	6
19	Posterior mean vs MAP flip rates	07	open	6
20	Optimal hysteresis deadband h^*	07	resolved by Dive 08	6
21	Multichannel radial deadband	08	resolved by Dive 09	7
22	Estimating (q, K) in the field	08	partly advanced by Dive 09	7
23	Robust filtering under heavy-tailed refresh noise	08	open	7
24	Trust as a stock	08	open	7
25	General-n tiered reallocation	09	open	7
26	Adaptive-M stability	09	open	7
27	The discrete-monitoring cost constant	09	open	7

BL	Problem	Opened by	Status	Chapter
28	Cadence under measured refresh-error correlation	09	open	7
29	Sharp segment-level Λ -bounds	10	open	8
30	Ratio-balancing specification test	10	open	8
31	Fixed-policy estimands under budget-coupled arms and frequency	10	open	8
32	n-channel leakage matrix under spline time effects	11	open	9
33	Drifting misallocation η_{gt}	11	open	9
34	The Hill bracket for imputed channels	11	open	9
35	Auxiliary-share instruments	11	open	9
36	Audit law under scale (basis) errors	11	open	9
37	Boundary distribution of the profile-LR on the completely-monotone cone	12	open	10
38	The (G, H) frontier of the long-run identified set	12	open	10
39	Shape families for experiment-grounded de-confounding	12	open	10
40	Onset-timing as a second surrogacy index	12	open	10
41	Interval-valued posteriors in the deadband	12	open	10
42	Proof of the layered-design theorem	13	resolved by Dive 14	11
43	Single-index identified set for attribution	13	open	11
44	Cross-journey table for paths and frequency	13	open	11
45	Attribution-induced targeting equilibrium	13	open	11
46	Test & Roll for attribution	14	open	11
47	The $df = 1$ theorem	14	open	11
48	Compliance-aware cell algebra	14	open	11
49	Adaptive cap-raising	14	open	11
50	Paired (odd-part) allocation	14	open	11
51	Heavy-tailed certification	15	open	12
52	Temporal-split contamination power law	15	open	12
53	Multichannel counterfactual bar	15	open	12
54	Certifiable bar for an existing library	15	open	12
55	Calibrated- δ policy-agnostic CS	16	open	13
56	Stall detection statistic	16	partly resolved by Dive 19	13
57	Multichannel stall geometry	16	resolved by Dive 19	13
58	Stall under adstock and spend-demand confounding	16	resolved by Dive 19	13
59	Bounded CEP-link memory state	17	open	14
60	The survey-refresh δ pilot and its exclusion	17	open	14
61	Panel-measured kernel as MMM prior	17	open	14
62	Attention as an instrumented dose	17	open	14
63	The κ library	18	open	15
64	Three-margin bound	18	open	15
65	User-level Trimmed Match	18	open	15
66	Exceedance-grid guard	18	open	15
67	Regret-optimal cap under an informative prior	18	open	15
68	Log-only confounding detector	19	open	13
69	In-loop exploration under error-correlated wobble	19	open	13
70	The budget-direction escape	19	priced by Dive 21	13
71	Switch cones under synergy	19	open	13
72	Exact echo false-alarm law	19	open	13
73	The n-channel spectral comb	20	resolved by Dive 21	1
74	Estimating Ψ and $S_{\text{eps}}(\omega)$ in the field	20	open	1
75	Price the prior	20	resolved by Dive 22	1
76	Drift-capped identification budget	20	open	1
77	Planner-feasible probes	20	open	1
78	The window-contrast term in curvature information	20	open	1
79	Intermodulation under synergy	21	open	1
80	Bin assignment as an optimisation	21	open	1

BL	Problem	Opened by	Status	Chapter
81	Audit a real flighting calendar	21	open	1
82	The estimability audit	21	discharged for dives 20–22 by Dive 22	1
83	Closed form for the full-model optimal phase	21	open	1
84	Delayed-peak adstock and the DC null	21	open	1
85	The response-curve estimand for prior pricing	22	open	1
86	The multichannel shape-prior ledger	22	open	1
87	The EB widening rule as a shipped diagnostic	22	open	1
88	The nonlinear ladder gap	22	open	1

Appendix B

The program's protocol

The rules below are those recorded in the program's index file; they are reproduced so that the epistemic labels used throughout this compendium can be read against the standard each dive was held to.

1. Work the queue in order, one item per run. Sequence numbers are permanent and zero-padded.
2. Every fourth run is an EXTEND run: instead of a new item, revisit the prior dive with the most promising "next steps" section and push it further (deeper derivation, better simulation, stronger counterexample, more complete design). Extensions get their own sequence number with the same slug and the suffix `_ext`.
3. Depth standard (full protocol, as of run 03): formalise mathematically with explicit assumptions; run a parallel literature subagent at dive start; a minimum of three construct-attack-refine rounds, iterating until two consecutive rounds produce no material revision; at least eight numerical experiments including at least two robustness sweeps and at least one negative or power control; uncertainty estimates on every headline number and a devil's-advocate paragraph against every headline claim; a clean-room verification subagent re-derives at least one core result before write-up; established results are separated from speculation with clear labels. Write-ups include the attack-and-refine history.
4. Each dive ends with "Next steps for a future session"—the hook extensions build on.
5. If a dive discovers a new open problem, append it to the queue backlog with a one-line rationale.

Two consequences for the reader. Dives 01 and 02 predate the full protocol (rule 3 was adopted "as of run 03"): they have no attack-and-refine history and no clean-room verification, which the reviewer notes of Chapters 1 and 2 take into account. And the "clean-room" verifier is a second automated agent given only the claim and the model, not an independent human; its reproductions to machine precision establish that a derivation is internally consistent under the stated model, not that the model describes any market.

Appendix C

File inventory

Everything in this compendium derives from the following files, as they stood on the user's disk at 2026-09-07 (timestamps in Table 1). Paths are relative to `marketing_measurement/frontier_study/`.

- **Framing documents:** `README.md`, `01_state_of_the_field.md`, `02_open_questions.md`, `03_mmm_adoption_barriers.md`.
- **Bibliographies:** `resources/math_stats.md`, `resources/engineering_scale.md`, `resources/human_wisdom.md`, `resources/mmm_adoption_barriers.md`, `resources/repos_and_tools.md`, `resources/books_and_institutes.md`.
- **Program index:** `deep_dives/00_INDEX.md` (queue, backlog, log).
- **Dive reports:** `deep_dives/01_mmm_identified_set.md` through `deep_dives/22_price_of_the_prior.md` (twenty-two files; see Table 1).
- **Code and results:** `deep_dives/code/NN_<slug>.py` for each dive, with result files `17_results_quick.json`, `18_results_quick.json`, `19_results_all.json`, `22_results.json`, the `20_*.json` experiment outputs, and the scratch directories `_scratch19/` and `_scratch22/`. The code was consulted to disambiguate formulas and numbers; nothing was re-run for this compendium.